One-parameter Semigroups of Positive Operators, 2. Aufl.

Formelkorrekturen nach Freigabe durch Günther Greiner

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Inhalt: Teil I, Kap. 3; Teil II, Kap. 3; Teil II, Abschnitt 4.2 und die Notes zu Kap. 4 (ein geänderter Verweis); Teil III, Kap. 3.

Darstellung: Neuer Text erscheint blau, gestrichener Text rot und klein. Bei geänderten Formeln steht die alte Fassung rot, die neue blau. Reine Quelltextänderungen ohne Wirkung im Druck (z. B. entfernte `\label`) sind nicht markiert. Die Seitenzahlen weichen durch die Markierungen etwas von der Springer-Fahne ab.

Ausnahmen nach Greiners Entscheidung (Teil II):

- Kap. 3, S. 196, Z. 567: nicht geändert.
- Kap. 3, Notes (Z. 2547): Verweis auf Example 3.3.10 (ii).
- Abschnitt 4.2, Z. 1039: Alternative gewählt; der Verweis in den Notes zu Kap. 4 zeigt jetzt auf Proposition 4.2.13.

Abweichungen vom Wortlaut der Listen:

- Teil III, Kap. 3, Z. 115 und 684: Komma statt Punkt, weil der Satz mit "which shows" bzw. "which implies" weitergeht.
- Teil II, Kap. 3, Z. 1611: im Fließtext zwei Formeln $C(G)$, $G \coloneqq \dots$ statt `\quad`.
- Teil II, Kap. 3, Z. 2063–2071: eine `multline*`-Umgebung statt vier einzelner abgesetzter Formeln.
- Teil I, Kap. 3, Z. 1134, und Teil III, Kap. 3, Z. 1649: zweizeilig umbrochen, weil die Formeln in normaler Größe zu breit sind.

Abschnitt 4.2 (Teil II): Bei Listenzeilen aus Abschnitt 4.1 mit weiteren Fundstellen ("ebenso Z. . . . ") sind nur die Stellen in Abschnitt 4.2 geändert.

Chapter 3

Spectral Theory

by
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3.1 Introduction

In this chapter, we begin a systematic analysis of the spectrum of a strongly continuous semigroup $\mathcal{T} = (T(t))_{t\geqslant 0}$ on a complex Banach space E. By the spectrum of the semigroup we mean the spectrum $\sigma(A)$ of its generator A. In particular, we are interested in the precise relations between $\sigma(A)$ and $\sigma(T(t))$. The heuristic formula

$$T(t) = \mathrm{e}^{tA}$$

serves as a leitmotif and suggests relations of the form

$$\sigma(T(t)) = \mathrm{e}^{t\sigma(A)} = \{\mathrm{e}^{t\lambda} \colon \lambda \in \sigma(A)\},$$

called *spectral mapping theorem*. These — or similar — relations will be of great use in Chapter 4 enabling us to determine the asymptotic behavior of the semigroup $\mathcal{T}$ by examining the spectrum of its generator.

As motivation and also as a preliminary step, we focus here on the *spectral radius*

$$r(T(t)) := \sup\{|\lambda| \colon \lambda \in \sigma(T(t))\}, \quad t \geqslant 0, \tag{3.1.1}$$

and demonstrate its relationship with the *spectral bound*

$$s(A) := \sup\{\mathrm{Re}(\lambda) \colon \lambda \in \sigma(A)\} \tag{3.1.2}$$

of the generator A and to the *growth bound*

$$\omega_0 := \inf\{\omega \in \mathbb{R}\colon \|T(t)\| \leqslant M_\omega \cdot \mathrm{e}^{\omega t} \text{ for all } t \geqslant 0 \text{ and suitable } M_\omega\} \tag{3.1.3}$$

of the semigroup $\mathcal{T} = (T(t))_{t\geqslant 0}$. (Note that we sometimes write $\omega_0(\mathcal{T})$ or $\omega_0(A)$ instead of ω_0). The Examples 3.1.3 and 3.1.4 below illustrate the main difficulties to be encountered.

Proposition 3.1.1 *Let ω_0 be the growth bound of the strongly continuous semigroup $\mathcal{T} = (T(t))_{t\geqslant 0}$. Then*

$$r(T(t)) = \mathrm{e}^{\omega_0 t} \tag{3.1.4}$$

for every $t \geqslant 0$.

Proof. From Eq. (1.1.1) on page 4 we know that

$$\omega_0(\mathcal{T}) = \lim_{t\to\infty} \frac{1}{t} \log \|T(t)\|.$$

Since the spectral radius of $T(t)$ is given as

$$r(T(t)) = \lim_{n\to\infty} \|T(nt)\|^{1/n},$$

we obtain for $t > 0$

$$r(T(t)) = \lim_{n\to\infty} \exp\Big(\frac{t}{nt} \log \|T(nt)\|\Big) = \mathrm{e}^{\omega_0 t}. \qquad \square$$

It was shown in Proposition 1.1.11 that the spectral bound $s(A)$ is always dominated by the growth bound ω_0 and therefore $\mathrm{e}^{s(A)t} \leqslant r(T(t))$. If the above mentioned spectral mapping theorem holds—as is the case for bounded generators (e.g., see Theorem VII.3.11 of Dunford & Schwartz [5])—we obtain the equality

$$\mathrm{e}^{s(A)t} = r(T(t)) = \mathrm{e}^{\omega_0(\mathcal{T})t},$$

hence $s(A) = \omega_0(\mathcal{T})$. Therefore, the following corollary is a consequence of the definitions of $s(A)$ and $\omega_0(\mathcal{T})$.

Corollary 3.1.2 *Consider the semigroup $\mathcal{T} = (T(t))_{t\geqslant 0}$ generated by some bounded linear operator $A \in \mathcal{L}(E)$. If $\mathrm{Re}(\lambda) < 0$ for each $\lambda \in \sigma(A)$, then $\lim_{t\to\infty} \|T(t)\| = 0$.*

Through this corollary we have re-established Lyapunov's famous result, which assures that the solutions of the linear Cauchy problem

$$\dot{x}(t) = Ax(t), \quad x(0) = x_0 \in \mathbb{C}^n \quad \text{and} \quad A = (a_{ij})_{n\times n}$$

are *stable*, i.e., they converge to zero as $t \to \infty$ if (and only if) the real parts of all eigenvalues of the matrix A are less than zero.

The situation is much more difficult for unbounded generators, and $s(A)$ may differ drastically from $\omega_0(\mathcal{T})$ as can be seen in the following example (see Greiner et al. [11]).

Example 3.1.3 (Banach function space) Consider the Banach space E of all complex valued continuous functions on $\mathbb{R}_+$ which vanish at infinity and are integrable for $e^x\,dx$, i.e.,

$$E := C_0(\mathbb{R}_+) \cap L^1(\mathbb{R}_+, e^x\,dx)$$

endowed with the norm

$$\|f\| := \|f\|_\infty + \|f\|_1 = \sup\{|f(x)| : x \in \mathbb{R}_+\} + \int_0^\infty |f(x)|e^x\,dx.$$

The translation semigroup

$$T(t)f(x) := f(x+t)$$

is strongly continuous on E and one shows as in Section 1.2.4 that its generator is given by

$$Af = f', \quad D(A) = \{f \in E : f \in C^1(\mathbb{R}_+), f' \in E\}.$$

First we observe that $\|T(t)\| = 1$ for every $t \geqslant 0$, hence $\omega_0(\mathcal{T}) = 0$. Moreover it is clear that λ is an eigenvalue of A as soon as $\mathrm{Re}(\lambda) < -1$. In fact, the function

$$x \mapsto e_\lambda(x) := e^{\lambda x}$$

$$\mathsf{x} \mapsto \varepsilon_\lambda(\mathsf{x}) := e^{\lambda \mathsf{x}}$$

belongs to $D(A)$ and is an eigenvector of A, hence $s(A) \geqslant -1$. For $f \in E$ and $\mathrm{Re}(\lambda) > -1$

$$\|\cdot\|_1\text{-}\lim_{t\to\infty} \int_0^t e^{-\lambda s}T(s)f\,ds$$

exists since $\|T(s)f\|_1 \leqslant e^{-s}\|f\|_1$ for $s \geqslant 0$, and

$$\|\cdot\|_\infty\text{-}\lim_{t\to\infty} \int_0^t e^{-\lambda s}T(s)f\,ds$$

exists since $\int_0^\infty e^x|f(x)|\,dx < \infty$. Therefore $\int_0^\infty e^{-\lambda s}T(s)f\,ds$ exists in E for every $f \in E$ such that $\mathrm{Re}(\lambda) > -1$.

As we observed in Proposition 1.1.11, this implies $\lambda \in \varrho(A)$. Therefore $\mathcal{T} = (T(t))_{t\geqslant 0}$ is a semigroup having $s(A) = -1$, but $\omega_0(\mathcal{T}) = 0$.

Example 3.1.4 (Hilbert space, Zabczyk [27]) For every $n \in \mathbb{N}$ consider the n-dimensional Hilbert space $H_n := \mathbb{C}^n$ and operators $A_n \in \mathcal{L}(H_n)$ defined by the matrices

$$A_n = \begin{pmatrix} 0 & 1 & \cdots & 0 \\ \cdot & 0 & 1 & \cdot \\ \cdot & \cdot & \cdot & 1 \\ 0 & \cdot & \cdot & 0 \end{pmatrix}_{n\times n}.$$

These matrices are nilpotent and therefore $\sigma(A_n) = \{0\}$. The elements

$$x_n := n^{-1/2}(1, \dots, 1) \in H_n$$

satisfy the following properties.

(i) $\|x_n\| = 1$ for every $n \in \mathbb{N}$,

(ii) $\lim_{n\to\infty} \|A_n x_n - x_n\| = 0$,

(iii) $\lim_{n\to\infty} \|\exp(tA_n)x_n - \mathrm{e}^t x_n\| = 0$.

Consider now the Hilbert space

$$H := \bigoplus_{n\in\mathbb{N}} H_n \text{ and the operator } A := (A_n + 2\pi \mathrm{i} n)_{n\in\mathbb{N}}$$

with maximal domain in H.

Analogously we define a semigroup $\mathcal{T} = (T(t))_{t\geqslant 0}$ by

$$T(t) := (\mathrm{e}^{2\pi \mathrm{i} n t} \exp(tA_n))_{n\in\mathbb{N}}.$$

Since $\|\exp(tA_n)\| \leqslant \mathrm{e}^t$ for every $n \in \mathbb{N}$, $t \geqslant 0$, and since $t \mapsto T(t)x$ is continuous on each component $E_n H_n$, it follows that $\mathcal{T}$ is strongly continuous. Its generator is the operator A as defined above.

For $\lambda \in \mathbb{C}$, $\mathrm{Re}(\lambda) > 0$, we have $\lim_{n\to\infty} \|R(\lambda - 2\pi \mathrm{i} n, A_n)\| = 0$, hence

$$(R(\lambda, A_n + 2\pi \mathrm{i} n))_{n\in\mathbb{N}} = (R(\lambda - 2\pi \mathrm{i} n, A_n))_{n\in\mathbb{N}}$$

is a bounded operator on H representing the resolvent $R(\lambda, A)$. Therefore we obtain $s(A) \leqslant 0$. On the other hand, each $2\pi \mathrm{i} n$ is an eigenvalue of A, hence $s(A) = 0$.

Take now $x_n \in H_n$ as above and consider the sequence $(x_n)_{n\in\mathbb{N}}$. From (iii) it follows that for $t > 0$ the number e^t is an approximate eigenvalue of $T(t)$ with approximate eigenvector $(x_n)_{n\in\mathbb{N}}$ (see Definition 3.2.1 below). Therefore

$$\mathrm{e}^t \leqslant r(T(t)) \leqslant \|T(t)\|$$

and hence $\omega_0(\mathcal{T}) \geqslant 1$. On the other hand, it is easy to see that $\|T(t)\| = \mathrm{e}^t$, hence $\omega_0(\mathcal{T}) = 1$.

Finally, if we take $S(t) := \mathrm{e}^{-t/2} T(t)$, we obtain a semigroup $\mathcal{S}$ having spectral bound $-\frac{1}{2}$ but satisfying $\lim_{t\to\infty} \|S(t)\| = \infty$ in contrast with Corollary 3.1.2.

These examples show that neither the conclusion of Corollary 3.1.2, i.e., "$s(A) < 0$ implies stability", nor the "spectral mapping theorem"

$$\sigma(T(t)) = \exp(t \cdot \sigma(A))$$

is valid for arbitrary strongly continuous semigroups. A careful analysis of the general situation will be given in Section 3.6 below, but we will first systematically develop the necessary spectral theoretic tools for unbounded operators.

3.2 The Fine Structure of the Spectrum

As usual, to a closed linear operator A with dense domain $D(A)$ in a Banach space E, we associate its spectrum $\sigma(A)$, its resolvent set $\varrho(A)$ and its resolvent

$$\lambda \mapsto R(\lambda, A) := (\lambda - A)^{-1}$$

which is a holomorphic map from $\varrho(A)$ into $\mathcal{L}(E)$. In contrast to the finite dimensional situation, where a linear operator fails to be surjective if and only if it fails to be injective, we now have to distinguish different cases of *non-invertibility* of $\lambda - A$. This distinction gives rise to a subdivision of $\sigma(A)$ into different subsets. We point out that these subsets need not be disjoint, but our definitions are justified by the fact that for each of the following subsets of $\sigma(A)$ there exist canonical constructions converting the corresponding spectral values into eigenvalues (see Proposition 3.2.2 (ii) and Proposition 3.4.4 below).

Definition 3.2.1 For a closed, densely defined, linear operator A with domain $D(A)$ in the Banach space E we define the

(i) *point spectrum* $P\sigma(A)$ the set of all $\lambda \in \mathbb{C}$ such that $A - \lambda$ is not injective.

(ii) *approximate point spectrum* $A\sigma(A)$ the set of all $\lambda \in \mathbb{C}$ such that $A - \lambda$ is not injective or $(A - \lambda)D(A)$ is not closed in E.

(iii) *residual spectrum* $R\sigma(A)$ the set of all $\lambda \in \mathbb{C}$ such that $(A - \lambda)D(A)$ is not dense in E.

From these definitions it follows that $\lambda \in P\sigma(A)$ if and only if there exists a non-zero *eigenvector* $f \in D(A)$ such that $Af = \lambda f$, i.e., λ is an *eigenvalue*. It follows from the *Open Mapping Theorem* that $\lambda \in A\sigma(A)$ if and only if λ is an *approximate eigenvalue*, i.e., there exists a sequence $(f_n)_{n\in\mathbb{N}}$ in $D(A)$, called an *approximate eigenvector*, such that $\|f_n\| = 1$ and $\lim_{n\to\infty} \|Af_n - \lambda f_n\| = 0$.

Clearly we have $P\sigma(A) \subseteq A\sigma(A)$ and $\sigma(A) = A\sigma(A) \cup R\sigma(A)$ where the union need not be disjoint.

The following proposition is a first indication that the subdivision we made yields nice properties.

Proposition 3.2.2 *For a closed, densely defined, linear operator $(A, D(A))$ in a Banach space E the following holds.*

(i) *The topological boundary $\partial\sigma(A)$ of $\sigma(A)$ is contained in $A\sigma(A)$.*

(ii) *$R\sigma(A) = P\sigma(A')$ for the adjoint operator A' on E'.*

Proof. (i) Take $\lambda_0 \in \partial\sigma(A)$ and $\lambda_n \in \varrho(A)$ such that $\lambda_n \to \lambda_0$. Since

$$\|R(\lambda_n, A)\| \geqslant r(R(\lambda_n, A)) = (\mathrm{dist}(x, \sigma(A)))^{-1}$$

$$\|\mathsf{R}(\lambda_\mathsf{n}, \mathsf{A})\| \geqslant \mathsf{r}(\mathsf{R}(\lambda_\mathsf{n}, \mathsf{A})) = (\mathrm{dist}(\lambda_\mathsf{n}, \sigma(\mathsf{A})))^{-1}$$

(see Proposition 3.2.5 (ii)), by the uniform boundedness principle we find $f \in E$ such that

$$\lim_{n\to\infty} \|R(\lambda_n, A)f\| = \infty.$$

Define $g_n \in D(A)$ by

$$g_n := \|R(\lambda_n, A)f\|^{-1} R(\lambda_n, A)f$$

and use the identity

$$(\lambda_0 - A)g_n = (\lambda_0 - \lambda_n)g_n + (\lambda_n - A)g_n$$

to show that $(g_n)_{n\in\mathbb{N}}$ is an approximate eigenvector corresponding to λ_0.

(ii) This is a simple consequence of the Hahn-Banach theorem. □

In order to illuminate the above definitions we now return to the Standard Examples introduced in Section 1.2 and discuss the fine structure of the spectrum of these strongly continuous semigroups, i.e., of their generators and their semigroup operators.

Example 3.2.3 (The Spectrum of Multiplication Semigroups) Take $E = C_0(X)$ for some locally compact space X and take a continuous function $q : X \mapsto \mathbb{C}$ $q : X \to \mathbb{C}$ whose real part is bounded above. As observed in Section 1.2.3 the multiplication operator

$$M_q : f \mapsto q \cdot f$$

with maximal domain $D(M_q)$ generates the multiplication semigroup

$$T(t)f := \mathrm{e}^{tq} \cdot f\,,\ f \in E.$$

$$\mathsf{T(t)f} := \mathrm{e}^{\mathsf{tq}} \cdot \mathsf{f},\quad \mathsf{f} \in \mathsf{E}.$$

Since M_q is bounded if and only if q is bounded, we conclude that M_q is invertible (with bounded inverse $M_{1/q}$) if and only if

$$0 \notin \overline{\{q(x) : x \in X\}}.$$

Therefore we obtain

$$\sigma(M_q) = \overline{q(X)} = \overline{\{q(x) : x \in X\}},$$

and

$$\sigma(T(t)) = \overline{\{\exp(tq(x)) : x \in X\}}.$$

In particular the following *weak spectral mapping theorem* is valid

$$\sigma(T(t)) = \overline{\exp(t\sigma(M_q))}.$$

In addition, we observe that to each spectral value of A M_q (resp. of $T(t)$) there exists an approximate eigenvector and hence

$$\sigma(A) = A\sigma(A) \text{ and } \sigma(T(t)) = A\sigma(T(t)).$$

$$\sigma(\mathsf{M_q}) = \mathsf{A}\sigma(\mathsf{M_q}) \text{ and } \sigma(\mathsf{T(t)}) = \mathsf{A}\sigma(\mathsf{T(t)}).$$

Since each Dirac functional is an eigenvector for the adjoint multiplication operator, we obtain

$$q(X) \subseteq R\sigma(M_q) \text{ and } \mathrm{e}^{tq(X)} \subseteq R\sigma(T(t)).$$

The eigenvalues of M_q can be characterized as follows.

> $\lambda \in P\sigma(M_q)$ if and only if the set $\{x \in X : q(x) = \lambda\}$ has non-empty interior (analogously for $P\sigma(T(t))$).

For example, it follows that $P\sigma(M_q) = \emptyset$ for $E = C_0(\mathbb{R}_+)$ and $q(x) = -x$, $x \in \mathbb{R}_+$.

On $E = L^p(X, \Sigma, \mu)$ analogous results are valid, but their exact formulation—using the notion *essential range*, see Goldstein [8]—is left to the reader.

Example 3.2.4 (The Spectrum of Translation Semigroups)

We consider the translation semigroup

$$T(t)f(x) := f(x + t)$$

on $E = C_0(\mathbb{R}_+)$ (or $L^p(\mathbb{R}_+)$, see Section 1.2.4). Its generator A is the first derivative and for every $\lambda \in \mathbb{C}$, $\mathrm{Re}(\lambda) < 0$, the function $\varepsilon_\lambda : x \mapsto \mathrm{e}^{\lambda x}$ belongs to $D(A)$ and satisfies

$$A\varepsilon_\lambda = \lambda\varepsilon_\lambda,$$

hence $\lambda \in P\sigma(A)$.

Since $\mathcal{T} = (T(t))_{t \geqslant 0}$ is a contraction semigroup, it follows that

$$\sigma(A) = \{\lambda \in \mathbb{C} : \mathrm{Re}(\lambda) \leqslant 0\} \text{ and } \mathrm{i}\mathbb{R} \subseteq A\sigma(A)$$

(use Proposition 3.2.2 (i) or show directly that $f_n(x) = \mathrm{e}^{\mathrm{i}\alpha x}\mathrm{e}^{-x/n}$ defines an approximate eigenvector for $\mathrm{i}\alpha$, $\alpha \in \mathbb{R}$). Using the same functions one obtains

$$P\sigma(T(t)) = \{e^{\lambda t} : \operatorname{Re}(\lambda) < 0\} = \{z \in \mathbb{C} : |z| < 1\},$$
$$\sigma(T(t)) = \{z \in \mathbb{C} : |z| \leqslant 1\} \text{ for every } t > 0.$$

In the case of the translation group on $E = C_0(\mathbb{R})$ one has $\sigma(A) \subseteq i\mathbb{R}$. As above one obtains approximate eigenvectors for every $\alpha \in \mathbb{R}$ from $f_n(x) = e^{i\alpha x}e^{-|x|/n}$, hence

$$\sigma(A) = A\sigma(A) = i\mathbb{R}.$$

The generator A of the nilpotent translation semigroup from Section 1.2.6 has empty spectrum by Proposition 1.1.11. The resolvent is given by

$$R(\lambda, A)f(x) = e^{\lambda x}\int_x^\infty e^{-\lambda s}f(s)\,ds \quad (f \in L^p([0,\tau]), \lambda \in \mathbb{C}).$$

Finally, the generator of the periodic translation group from Section 1.2.5 on

$$E = \{f \in C([0,1]) : f(0) = f(1)\}$$

has point spectrum

$$P\sigma(A) = 2\pi i\mathbb{Z}$$

with eigenfunctions $\varepsilon_n(x) := \exp(2\pi i n x)$. In Section 3.5 we show that $\sigma(A) = 2\pi i\mathbb{Z}$.

We now return to the general theory and recall from Corollary 3.1.2 that it is very useful (e.g., for stability theory) to be able to convert spectral values of the generator A into spectral values of the semigroup operator $T(t)$ and vice versa. As shown in Examples 3.1.3 and 3.1.4 this is not possible in general. Therefore we tackle first a much easier *spectral mapping theorem*: the relation between $\sigma(A)$ and $\sigma(R(\lambda_0))$, where $R(\lambda_0) := R(\lambda_0, A)$ for some $\lambda_0 \in \varrho(A)$.

Proposition 3.2.5 *Let $(A, D(A))$ be a densely defined closed linear operator with non-empty resolvent set $\varrho(A)$. For each $\lambda_0 \in \varrho(A)$ the following assertions hold.*

(i) $\sigma(R(\lambda_0))\setminus\{0\} = (\lambda_0 - \sigma(A))^{-1}$, *in particular,* $r(R(\lambda_0)) = (\operatorname{dist}(\lambda_0, \sigma(A)))^{-1}$.

(ii) *Analogous statements hold for the point-, approximate point-, residual spectra of A and $R(\lambda_0, A)$.*

(iii) *The point α is isolated in $\sigma(A)$ if and only if $(\lambda_0 - \alpha)^{-1}$ is isolated in $\sigma(R(\lambda_0))$. In that case the residues (resp. the pole orders) in α and in $(\lambda_0 - \alpha)^{-1}$ coincide.*

Proof. (i) This is well known and can be found, for example, in Dunford & Schwartz [5, VII.9.2].

(ii) We show that $\alpha \in A\sigma(A)$ if $(\lambda_0 - \alpha)^{-1} \in A\sigma(R(\lambda_0))$ and leave the proof of the remaining statements to the reader.

Take $(f_n)_{n\in\mathbb{N}}$ a sequence in E such that $\|f_n\| = 1$,

$$\|(\lambda_0 - \alpha)^{-1}f_n - R(\lambda_0, A)f_n\| \to 0$$

and

$$\|R(\lambda_0, A)f_n\| \geqslant \frac{1}{2}|\lambda_0 - \alpha|^{-1}.$$

Define

$$g_n := \|R(\lambda_0, A)f_n\|^{-1} R(\lambda_0, A)f_n \in D(A)$$

and deduce from

$$\begin{aligned}(\alpha - A)g_n &= \|R(\lambda_0, A)f_n\|^{-1} \cdot [(\lambda_0 - A) - (\lambda_0 - \alpha)]R(\lambda_0, A)f_n \\ &= \|R(\lambda_0, A)f_n\|^{-1} \cdot (\lambda_0 - \alpha)[(\lambda_0 - \alpha)^{-1} - R(\lambda_0, A)]f_n\end{aligned}$$

that (g_n) is an approximate eigenvector of A to the eigenvalue α.

(iii) First we recall the well known *resolvent equation*.

For any $z, \lambda_0 \in \varrho(A)$ we have

$$R(\lambda_0, A) - R(z, A) = -(\lambda_0 - z)R(\lambda_0, A)R(z, A).$$

From this it follows that

$$(\lambda_0 - z)^2 \cdot R(z, A) = \Big((\lambda_0 - z)^{-1} - R(\lambda_0, A)\Big)^{-1} - (\lambda_0 - z).$$

If we now take a circle Γ with center α and sufficiently small radius, then the residue P of $R(\cdot, A)$ at α is

$$\begin{aligned}P &= \frac{1}{2\pi \mathrm{i}} \int_\Gamma R(z, A)\,\mathrm{d}z = \\ &= \frac{1}{2\pi \mathrm{i}} \left[\int_\Gamma (\lambda_0 - z)^{-2} R\Big((\lambda_0 - z)^{-1}, R(\lambda_0, A)\Big)\,\mathrm{d}z - \int_\Gamma (\lambda_0 - z)^{-1}\,\mathrm{d}z \right].\end{aligned}$$

$$\begin{aligned}\mathsf{P} &= \frac{1}{2\pi \mathrm{i}} \int_{\Gamma} \mathsf{R}(\mathsf{z}, \mathsf{A})\,\mathrm{d}\mathsf{z} \\ &= \frac{1}{2\pi \mathrm{i}} \left[\int_{\Gamma} (\lambda_0 - \mathsf{z})^{-2} \mathsf{R}\Big((\lambda_0 - \mathsf{z})^{-1}, \mathsf{R}(\lambda_0, \mathsf{A})\Big)\,\mathrm{d}\mathsf{z} - \int_{\Gamma} (\lambda_0 - \mathsf{z})^{-1}\,\mathrm{d}\mathsf{z} \right].\end{aligned}$$

If λ_0 lies in the exterior of Γ, the second integral is zero. The substitution

$$\tilde{z} := (\lambda_0 - z)^{-1}$$

yields a path $\tilde{\Gamma}$ around $(\lambda_0 - \alpha)^{-1}$ and we obtain

$$P = \frac{1}{2\pi \mathrm{i}} \int_{\tilde{\Gamma}} R(\tilde{z}, R(\lambda_0, A))\,\mathrm{d}\tilde{z}$$

which is the residue of $R(\cdot, R(\lambda_0, A))$ at $(\lambda_0 - \alpha)^{-1}$. The final assertion on the pole order follows from the identities

$$V_{-n} = ((\lambda_0 - \alpha)^{-1} R(\lambda_0, A))^{n-1} U_{-n} \quad (n \in \mathbb{N}),$$

where U_{-n}, resp. V_{-n} stand for the n-th coefficient in the Laurent series of $R(\cdot, A)$, resp. $R(\cdot, R(\lambda_0, A))$ at α, resp. $(\lambda_0 - \alpha)^{-1}$. This has already been proved for $n = 1$ and follows for $n > 1$ by induction using the relations

$$U_{-n-1} = (A - \alpha) U_{-n} \quad \text{and} \quad V_{-n-1} = \Big(R(\lambda_0, A) - (\lambda_0 - \alpha)^{-1} \Big) V_{-n}. \qquad \square$$

3.3 Spectral Decomposition

In the next two sections we develop some important techniques for our further investigation of semigroups and their generators. Even though these methods are well known (compare, e.g., Section VII.3 of Dunford & Schwartz [5]) and rather technical, it is useful to present them in a coherent way.

Our interest in this section is the following: Let E be a Banach space and $\mathcal{T} = (T(t))_{t \geqslant 0}$ a strongly continuous semigroup with generator A. Suppose that the spectrum $\sigma(A)$ splits into the disjoint union of two closed subsets σ_1 and σ_2. Does there exist a corresponding decomposition of the space E and the semigroup $\mathcal{T}$?

In the following definition, we explain what we understand by "corresponding decomposition".

Definition 3.3.1 Assume that $\sigma(A)$ is the disjoint union

$$\sigma(A) = \sigma_1 \cup \sigma_2$$

of two non-empty closed subsets σ_1, σ_2. A decomposition

$$E = E_1 \oplus E_2$$

of E into the direct sum of two non-trivial closed $\mathcal{T}$-invariant subspaces is called a *spectral decomposition* corresponding to $\sigma_1 \cup \sigma_2$ if the spectrum $\sigma(A_i)$ of the generator A_i of $\mathcal{T}_i := (T(t)_{|E_i})_{t \geqslant 0}$ coincides with σ_i for $i = 1, 2$.

For a better understanding of the above definition we recall that to every direct sum decomposition $E = E_1 \oplus E_2$ there corresponds a continuous projection $P \in \mathcal{L}(E)$ such that $PE = E_1$ and $P^{-1}(0) = E_2$. Moreover, the subspaces E_1, E_2 are $\mathcal{T}$-invariant if and only if P commutes with the semigroup $\mathcal{T}$, i.e., $T(t)P = PT(t)$ for every $t \geqslant 0$. In this case it follows that the domain $D(A)$ of the generator A splits analogously and $D(A) \cap E_i$ is the domain $D(A_i)$ of the generator A_i of the restricted semigroup $\mathcal{T}_i$, $i = 1, 2$. We write

$$A = A_1 \oplus A_2$$

and say that A *commutes with* P and call P a *spectral projection*. In terms of the generator A this means that for $f \in D(A)$ we have $Pf \in D(A)$ and $APf = PAf$.

The existence of such projections reduces the semigroup $\mathcal{T}$ into two (possibly simpler) semigroups $\mathcal{T}_1, \mathcal{T}_2$ such that

$$\sigma(A) = \sigma(A_1) \cup \sigma(A_2) \quad \text{and} \quad \sigma(T(t)) = \sigma(T_1(t)) \cup \sigma(T_2(t)).$$

For example, in some cases (see Theorem 3.3.3 below) it can be shown that one of the reduced semigroups has additional properties.

In order to achieve such decompositions we will assume that $\sigma(A)$ decomposes into sets σ_1 and σ_2 and will then try to find a corresponding spectral projection. Unfortunately such spectral decompositions do not exist in general.

Example 3.3.2 Take the rotation semigroup from Section 1.2.5 on the Banach space $L^p(\Gamma)$, $1 \leqslant p < \infty$, $\tau = 2\pi$. It was stated in Example 3.2.4 and will be proved in Section 3.5 that its generator A has spectrum

$$\sigma(A) = P\sigma(A) = \mathrm{i}\mathbb{Z}$$

where $\varepsilon_k(z) := z^k$ spans the eigenspace corresponding to $\mathrm{i}k$, $k \in \mathbb{Z}$.

Now, $\sigma(A)$ is the disjoint union of $\sigma_1 := \{0, \mathrm{i}, 2\mathrm{i}, \ldots\}$ and $\sigma_2 := \{-\mathrm{i}, -2\mathrm{i}, \ldots\}$. By a result of M. Riesz there is no projection $P \in \mathcal{L}(L^1(\Gamma))$ satisfying $P\varepsilon_k = \varepsilon_k$ for $k \geqslant 0$, $P\varepsilon_k = 0$ for $k < 0$, hence there is no spectral decomposition of $L^1(\Gamma)$ corresponding to σ_1, σ_2 (Lindenstrauss & Tzafriri [19, p.165]).

On the other hand, for $L^p(\Gamma)$, $1 < p < \infty$, such a spectral projection exists (l.c., 2.c.15). As long as $p \neq 2$ we can always decompose $\sigma(A)$ into suitable subsets admitting no spectral decomposition (l.c., remark before 2.c.15). Clearly, for $p = 2$ such spectral decompositions always exist.

In the above example both subsets σ_1, σ_2 of $\sigma(A)$ are unbounded. But as soon as one of these sets is bounded a corresponding spectral decomposition can always be found.

Theorem 3.3.3 *Let $\mathcal{T}$ be a strongly continuous semigroup on a Banach space E and assume that the spectrum $\sigma(A)$ of the generator A can be decomposed into the disjoint union of two non-empty closed subsets σ_1 and σ_2.*

If σ_1 is compact, then there exists a unique corresponding spectral decomposition $E = E_1 \oplus E_2$ such that the restricted semigroup $\mathcal{T}_1$ has a bounded generator.

Proof. We assume the reader to be familiar with the spectral decomposition theorem for bounded operators (see, e.g., Dunford & Schwartz [5, p.572]) and apply

the spectral mapping theorem for the resolvent (Proposition 3.2.5 (i)) in order to decompose $R(\lambda, A)$ instead of A.

For $\lambda_0 > \omega_0(\mathcal{T})$ it follows from Proposition 3.2.5 that $\sigma(R(\lambda_0, A)) \setminus \{0\} = (\lambda_0 - \sigma(A))^{-1}$. From $\sigma(A) = \sigma_1 \cup \sigma_2$ we obtain a decomposition of $\sigma(R(\lambda_0, A)) \setminus \{0\}$ into

$$\tau_1 := (\lambda_0 - \sigma_1)^{-1}, \quad \tau_2 := (\lambda_0 - \sigma_2)^{-1}.$$

Since σ_1 is compact, the set τ_1 is compact and does not contain 0. Only in the case that σ_2 is unbounded, the point 0 will be an accumulation point of τ_2. Therefore $\sigma(R(\lambda_0, A)) \cup \{0\}$ is the disjoint union of the closed sets τ_1 and $\tau_2 \cup \{0\}$.

Take now P to be the spectral projection of $R(\lambda_0, A)$ corresponding to this decomposition. Then P commutes with $R(\lambda_0, A)$ (by definition), with $R(\lambda, A)$ for every $\lambda > \omega_0(\mathcal{T})$ (use the series representation of the resolvent), with $T(t)$ for each $t \geqslant 0$ (use Proposition 2.1.10) and therefore with the generator A (in the sense explained above). In particular, we obtain

$$R(\lambda_0, A)P = R(\lambda_0, A_1), \quad R(\lambda_0, A)(\mathrm{Id} - P) = R(\lambda_0, A_2)$$

for the generator A_1 of $\mathcal{T}_1 = (T(t)P)_{t\geqslant 0}$ $\mathcal{T}_1 = (T(t)P)_{t\geqslant 0}$ and A_2 of $\mathcal{T}_2 = (T(t)(\mathrm{Id} - P))_{t\geqslant 0}$ $\mathcal{T}_2 = (T(t)(\mathrm{Id} - P))_{t\geqslant 0}$. Applying the spectral mapping theorem for the resolvent (Proposition 3.2.5) we conclude

$$\sigma(A_1) = \sigma_1 \text{ and } \sigma(A_2) = \sigma_2,$$

i.e., P is a spectral projection corresponding to σ_1, σ_2. Finally, the above spectral decomposition of $R(\lambda_0, A)$ is unique and satisfies $0 \notin \sigma(R(\lambda_0, A_1))$. Therefore $R(\lambda_0, A_1)^{-1} = (\lambda_0 - A_1)$ is bounded. □

Example If we do not require $\mathcal{T}_1$ to be uniformly continuous, the above spectral decomposition need not be unique, as can be seen from the following example.

Consider a decomposition $E = E_1 \oplus E_2$ and add a direct summand E_3 with a strongly continuous semigroup $\mathcal{T}_3$ $\mathcal{T}_3$ whose generator A_3 has empty spectrum (e.g., Section 1.2.6). Then still $\sigma(A) = \sigma_1 \cup \sigma_2$, but $E_1 \oplus (E_2 \oplus E_3)$ and $(E_1 \oplus E_3) \oplus E_2$ are two different spectral decompositions corresponding to σ_1, σ_2.

The importance of the above theorem stems from the fact that $\mathcal{T}_1$ has a bounded generator and therefore is easy to deal with. In particular, the asymptotic behavior of $\mathcal{T}_1$ can be deduced from the location of σ_1.

Corollary 3.3.4 *Assume that $\sigma(A)$ splits into non-empty closed sets σ_1, σ_2 where σ_1 is compact and consider the corresponding spectral decomposition $E = E_1 \oplus E_2$ for which $\mathcal{T}_1$ is uniformly continuous.*

For all constants ν, $\omega \in \mathbb{R}$ satisfying

$$\nu < \inf\{\mathrm{Re}(\lambda) \colon \lambda \in \sigma_1\} \leqslant \sup\{\mathrm{Re}(\lambda) \colon \lambda \in \sigma_1\} < \omega$$

there exist $m \leqslant 1$, $M \geqslant 1$ *such that*

$$m \cdot \mathrm{e}^{\nu t}\|f\| \leqslant \|T_1(t)f\| \leqslant M \cdot \mathrm{e}^{\omega t}\|f\|$$

for every $f \in E_1$, $t \geqslant 0$.

Proof. Since the generator A_1 of $\mathcal{T}_1$ is bounded, we have $T_1(t) = \exp(tA_1)$ and $\sigma(T_1(t)) = \exp(t\sigma(A_1))$. Therefore by the remark following Proposition 3.1.1, the spectral bound $s(A_1)$ coincides with the growth bound $\omega_0(T_1)$ $\omega_0(\mathcal{T}_1)$ and we have the upper estimate. The lower estimate is obtained by applying the same reasoning to $-A_1$ which generates the semigroup $(T_1(t)^{-1})_{t\geqslant 0}$ on E_1. □

It is clear from Examples 3.1.3 and 3.1.4 that no norm estimates for $(T_2(t))_{t\geqslant 0}$ can be obtained from the location of σ_2. Only by adding appropriate hypotheses we will achieve spectral decompositions admitting norm estimates on both components (see Theorem 3.6.6 below).

Another way of obtaining such norm estimates is by constructing spectral decompositions starting from a semigroup operator $T(t_0)$ (instead of A, and $R(\lambda, A)$, resp., as in Theorem 3.3.3).

Corollary 3.3.5 *If* $\sigma(T(t_0)) = \tau_1 \cup \tau_2$ *for two non-empty, closed, disjoint sets* τ_1, τ_2 *and if* P *is the spectral projection corresponding to* $T(t_0)$ *and* τ_1, τ_2, *then* $\sigma(A)$ *splits into closed subsets* σ_1, σ_2 *and* P *is the corresponding spectral projection for* $\mathcal{T}$ *and* σ_1, σ_2.

Proof. The spectral projection P of $T(t_0)$ is obtained by integrating $R(\lambda, T(t_0))$ (see, e.g., Dunford & Schwartz [5, Section VII.3]). Since every $T(t)$, $t \geqslant 0$, commutes with $T(t_0)$, it must commute with $R(\lambda, T(t_0))$, hence with P. The statement on the decomposition $\sigma(A) = \sigma_1 \cup \sigma_2$ follows from the Spectral Inclusion Theorem (see Proposition 3.6.2) below. □

This decomposition can be applied to the study of the asymptotic behavior of $\mathcal{T}$. In the situation of Corollary 3.3.5 assume

$$\sup\{|\lambda| : \lambda \in \tau_2\} < \alpha < \inf\{|\lambda| : \lambda \in \tau_1\}$$

for some $\alpha > 0$. If we set $\beta := (\log \alpha)/t_0$ and use Pazy [21, Chap.I, Theorem 6.5], we obtain $\omega_0(\mathcal{T}_2) < \beta$ and $\omega_0(\mathcal{T}_1^{-1}) < \beta$ by Proposition 3.1.1. Therefore we have constants m, M with $m \leqslant 1 \leqslant M$ such that

$$\begin{aligned} \|T(t)f\| &\leqslant M \cdot \mathrm{e}^{\beta t}\|f\| \quad \text{for } f \in E_2, \\ \|T(t)f\| &\geqslant m \cdot \mathrm{e}^{\beta t}\|f\| \quad \text{for } f \in E_1. \end{aligned}$$

As nice as they might look, results of this type are unsatisfactory. We need information on the semigroup in order to estimate its asymptotic behavior. In Chapter 4 we will try to obtain such results by exploiting information about the generator only.

Example 3.3.6 (Isolated singularities and poles)

In case that λ_0 is an isolated point of $\sigma(A)$ the holomorphic function $\lambda \mapsto R(\lambda, A)$ can be expanded as a Laurent series

$$R(\lambda, A) = \sum_{n=-\infty}^{+\infty} U_n(\lambda - \lambda_0)^n \text{ for } 0 < |\lambda - \lambda_0| < \delta \text{ and some } \delta > 0.$$

The coefficients U_n are bounded linear operators given by

$$U_n = \frac{1}{2\pi i} \int_\Gamma (z - \lambda_0)^{-(n+1)} R(z, A)\, dz,\ n \in \mathbb{Z}, \tag{3.3.1}$$

where $\Gamma = \{z \in \mathbb{C}\colon |z - \lambda_0| = \delta/2\}$. The coefficient U_{-1} is the spectral projection corresponding to the spectral set $\{\lambda_0\}$ (see Definition 3.3.1). It is called the *residue* of $R(\cdot, A)$ at λ_0 and will be denoted by P. From Eq. (3.3.1) one deduces

$$\begin{aligned} U_{-(n+1)} &= (A - \lambda_0)^n \circ P \quad \text{and} \\ U_{-(n+1)} \circ U_{-(m+1)} &= U_{-(n+m+1)} \text{ for } n, m \geqslant 0. \end{aligned} \tag{3.3.2}$$

If there exists $k > 0$ such that $U_{-k} \neq 0$ while $U_{-n} = 0$ for all $n > k$, the point λ_0 is called a *pole of $R(\cdot, A)$ of order k*, otherwise it is called an *essential singularity*. In view of Eq. (3.3.2) this is true if $U_{-k} \neq 0$ and $U_{-(k+1)} = 0$. In this case one can retrieve U_{-k} as

$$U_{-k} = \lim_{\lambda \to \lambda_0} (\lambda - \lambda_0)^k R(\lambda, A). \tag{3.3.3}$$

The dimension of PE (i.e., the dimension of the spectral subspace corresponding to $\{\lambda_0\}$) is called the *algebraic multiplicity* m_a of λ_0, while the *geometric multiplicity* is $m_g := \dim \ker(\lambda_0 - A)$. In case $m_a = 1$, we call λ_0 an *algebraically simple pole*. If k is the pole order ($k = \infty$ in case of an essential singularity), we have

$$\max\{m_g, k\} \leqslant m_a \leqslant k \cdot m_g \tag{3.3.4}$$

where $\infty \cdot 0 = \infty$.

These inequalities yield the following implications.

(i) $m_a < \infty$ if and only if λ_0 is a pole with $m_g < \infty$,

(ii) If λ_0 is a pole with order k, then $\lambda_0 \in P\sigma(A)$ and $PE = \ker(\lambda_0 - A)^k$.

If A has compact resolvent, then every point of $\sigma(A)$ is a pole of finite algebraic multiplicity. This is a consequence of Proposition 3.2.5 (iii) and the well-known Riesz-Schauder Theory for compact operators (see Dunford & Schwartz [5, VII.4.5]).

Example 3.3.7 (The essential spectrum) For an operator $T \in \mathcal{L}(E)$ the *Fredholm domain* $\varrho_F(T)$ is

$$\begin{aligned}\varrho_F(T) &:= \{\lambda \in \mathbb{C}\colon \lambda - T \text{ is a Fredholm operator}\}\\ &= \{\lambda \in \mathbb{C}\colon \ker(\lambda - T) \text{ and } E/\mathrm{im}(\lambda - T) \text{ are finite dimensional}\}.\end{aligned} \tag{3.3.5}$$

An equivalent characterization of $\varrho_F(T)$ is obtained through the *Calkin algebra* $\mathcal{L}(E)/\mathcal{K}(E)$, where $\mathcal{K}(E)$ stands for the closed ideal of all compact operators. In fact, $\varrho_F(T)$ coincides with the resolvent set of the canonical image of T in the Calkin algebra. The complement of $\varrho_F(T)$ is called *essential spectrum* of T and denoted by $\sigma_{\mathrm{ess}}(T)$. The corresponding spectral radius, called *essential spectral radius*, satisfies

$$r_{\mathrm{ess}}(T) := \sup\{|\lambda|\colon \lambda \in \sigma_{\mathrm{ess}}(T)\} = \lim_{n\to\infty} \|T^n\|_{\mathrm{ess}}^{1/n}, \tag{3.3.6}$$

where

$$\|T\|_{\mathrm{ess}} = \mathrm{dist}(T, \mathcal{K}(E)) := \inf\{\|T - K\|\colon K \in \mathcal{K}(E)\}$$

is the norm of T in $\mathcal{L}(E)/\mathcal{K}(E)$.

For every compact operator K we have $\|T - K\|_{\mathrm{ess}} = \|T\|_{\mathrm{ess}}$, hence

$$r_{\mathrm{ess}}(T - K) = r_{\mathrm{ess}}(T). \tag{3.3.7}$$

A detailed analysis of $\varrho_F(T)$ can be found in Section IV.5.6 of Kato [16]. In particular we recall that the poles of $R(\cdot, T)$ with finite algebraic multiplicity belong to $\varrho_F(T)$. Conversely, an element of the unbounded component of $\varrho_F(T)$ either belongs to $\varrho(T)$ or is a pole of finite algebraic multiplicity.

Thus $r_{\mathrm{ess}}(T)$ can be characterized as

$$\begin{aligned}&r_{\mathrm{ess}}(T) \text{ is the smallest } r \in \mathbb{R}_+ \text{ such that every } \lambda \in \sigma(T),\ |\lambda| > r\\ &\text{is a pole of finite algebraic multiplicity.}\end{aligned} \tag{3.3.8}$$

Now, if $\mathcal{T} = (T(t))_{t\geqslant 0}$ is a strongly continuous semigroup, then VIII.1, Lemma 4 of Dunford & Schwartz [5] applied to the function $t \mapsto \log \|T(t)\|_{\mathrm{ess}}$ ensures that

$$\omega_{\mathrm{ess}}(\mathcal{T}) := \lim_{t\to\infty} \frac{1}{t} \log \|T(t)\|_{\mathrm{ess}} = \inf\left\{\frac{1}{t} \log \|T(t)\|_{\mathrm{ess}}\colon t > 0\right\} \tag{3.3.9}$$

is well defined (possibly $-\infty$). By the definition of $\omega_{\mathrm{ess}}(\mathcal{T})$ and Eq. (3.3.6) we have

$$r_{\mathrm{ess}}(T(t)) = \exp(t\omega_{\mathrm{ess}}(\mathcal{T})), \quad t \geqslant 0. \tag{3.3.10}$$

Obviously, $\omega_{\mathrm{ess}} \leqslant \omega_0$ and equality occurs if and only if $r_{\mathrm{ess}}(T(t)) = r(T(t))$ for $t \geqslant 0$.

If $\omega_{\mathrm{ess}} < \omega_0$, there exists an eigenvalue λ of $T(t)$ satisfying $|\lambda| = r(T(t))$, hence by Theorem 3.6.3 below there exists $\lambda_1 \in P\sigma(A)$ such that $\mathrm{Re}(\lambda_1) = \omega_0$. Thus $\omega_{\mathrm{ess}} < \omega_0$ implies $s(A) = \omega_0(\mathcal{T})$, i.e., we have

$$\omega_0(\mathcal{T}) = \max\{\omega_{\mathrm{ess}}(\mathcal{T}), s(A)\}. \tag{3.3.11}$$

As a final observation we point out that

$$\omega_{\mathrm{ess}}(\mathcal{T}) = \omega_{\mathrm{ess}}(\mathcal{S}), \tag{3.3.12}$$

whenever $\mathcal{T}$ is generated by A and $\mathcal{S}$ is generated by $A+K$ for some compact operator K (see Part-II, Proposition 4.2.8 and Proposition 4.2.9).

3.4 The Spectrum of Induced Semigroups

In the previous section we tried to decompose a semigroup into the direct sum of two, hopefully simpler objects. Here we present other methods to reduce the complexity of a semigroup and its generator. Forming subspace or quotient semigroups as in Section 1.3.3, Section 1.3.4 are such methods. But also the constructions of new semigroups on canonically associated spaces such as the dual space, see Section 1.3.5, or the $\mathcal{F}$-product, see Section 1.3.7, might be helpful. We review these constructions under the spectral theoretical point of view and collect a number of technical properties for later use.

We start by studying the spectrum of subspace and quotient semigroups. To that purpose assume that the strongly continuous semigroup $\mathcal{T} = (T(t))_{t\geqslant 0}$ leaves invariant some closed subspace N of the Banach space E. There are canonically induced semigroups $\mathcal{T}_|$ on N, resp. $\mathcal{T}_/$ on E/N and their generators $A_|$, resp. $A_/$ are canonically obtained from the generator A of $\mathcal{T}$ (see Section 1.3). The following example shows that the spectra of A, $A_|$ and $A_/$ may differ quite drastically.

Example 3.4.1 As in the example in Section 1.2.4 we consider the translation semigroup on $E = L^1(\mathbb{R})$ and the invariant subspace

$$N := \{f \in E\colon f(x) = 0 \text{ for } x \geqslant 1\}.$$

Then $\sigma(A) = \mathrm{i}\mathbb{R}$, but $\sigma(A_|) = \{\lambda \in \mathbb{C}\colon \operatorname{Re}(\lambda) \leqslant 0\}$. Next we take the translation invariant subspace

$$M := \{f \in N\colon f(x) = 0 \text{ for } 0 \leqslant x \leqslant 1\}$$

and obtain $\sigma(A_{|/}) = \emptyset$ for the generator $A_{|/}$ of the quotient semigroup $\mathcal{T}_{|/}$ (use the fact that $\mathcal{T}_{|/}$ is nilpotent).

In the next proposition we collect the information on $\sigma(A)$ which in general can be obtained from the "subspace spectrum" $\sigma(A_|)$ and the "quotient spectrum" $\sigma(A_/)$.

Proposition 3.4.2 *Using the standard notations the following inclusions hold.*

(i) $\varrho(A) \subseteq [\varrho(A_|) \cap \varrho(A_/)] \cup [\sigma(A_|) \cap \sigma(A_/)]$,

(ii) $[\varrho(A_|) \cap \varrho(A_/)] \subseteq \varrho(A)$,

(iii) $\varrho_+(A) \subseteq [\varrho(A_|) \cap \varrho(A_/)]$,

where $\varrho_+(A)$ denotes the connected component of $\varrho(A)$ which is unbounded to the right.

Proof. (i) Assume $\lambda \in \varrho(A)$, i.e., $(\lambda - A)$ is a bijection from $D(A)$ onto E. Since N is $\mathcal{T}$-invariant, we have $D(A_|) = D(A) \cap N$ and $(\lambda - A)D(A_|) \subseteq N$. If $(\lambda - A)D(A_|) = N$, then $R(\lambda, A)N = D(A_|)$ and the induced operators $R(\lambda, A)_|$, resp. $R(\lambda, A)_/$ are the inverses of $(\lambda - A_|)$, resp. $(\lambda - A_/)$. If $(\lambda - A)D(A_|) \neq N$, then $\lambda \in \sigma(A_|)$.

In addition there exists $f \in D(A)\setminus N$ $f \in D(A) \setminus N$ such that $g := (\lambda - A)f \in N$. Hence for $\hat{f} := f + N$, $\hat{g} := g + N \in E_/$ $\hat{g} := g + N \in E/N$ it follows that $(\lambda - A_/)\hat{f} = \hat{g} = 0$, i.e., $\lambda \in \sigma(A_/)$.

(ii) Take $\lambda \in \varrho(A_|) \cap \varrho(A_/)$. Then $(\lambda - A)$ is injective since $(\lambda - A)f = 0$ implies $(\lambda - A_/)\hat{f} = 0$, hence $\hat{f} = 0$, i.e., $f \in N$ and therefore $f = 0$.

In addition, $(\lambda - A)$ is surjective: For $g \in E$ there exists $\hat{f} \in E_/$ $\hat{f} \in E/N$ such that $(\lambda - A_/)\hat{f} = \hat{g}$, i.e., there exists $h \in N$ such that $(\lambda - A)f - g = h = (\lambda - A)k$ for some $k \in D(A_|)$. Therefore we obtain $(\lambda - A)(f - k) = g$.

(iii) The integral representation of the resolvent for $\lambda > \omega_0(\mathcal{T})$ (see Proposition 1.1.11) shows that $R(\lambda, A)N \subseteq N$. By the power series expansion for holomorphic functions this extends to all $\lambda \in \varrho_+(A)$. Therefore the restriction $R(\lambda, A)_|$ coincides with the resolvent $R(\lambda, A_|)$. On the other hand $R(\lambda, A)_/$ is well defined on $E_/$ E/N and satisfies

$$R(\lambda, A)_/(f + N) = R(\lambda, A)f + N$$

(use again the integral representation). This proves that $R(\lambda, A)_/ = R(\lambda, A_/)$. □

Corollary 3.4.3 *Under the above assumptions take a point μ in the closure of $\varrho_+(A)$. Then*

(i) *$\mu \in \sigma(A)$ if and only if $\mu \in \sigma(A_|)$ or $\mu \in \sigma(A_/)$.*

(ii) *μ is a pole of $R(\cdot, A)$ if and only if μ is a pole of $R(\cdot, A_|)$ and of $R(\cdot, A_/)$.*

In that case,

$$\max\{k_|, k_/\} \leqslant k \leqslant k_| + k_/$$

for the respective pole orders. Note that hereby pole orders 0 are allowed.

Proof. (i) This follows from Proposition 3.4.2 (ii) and (iii).

(ii) By the previous assertion we may assume that for some $\delta > 0$ the pointed disc

$$\{\lambda \in \mathbb{C}\colon 0 < |\lambda - \mu| < \delta\}$$

is contained in $\varrho(A) \cap \varrho(A_|) \cap \varrho(A_/)$.

Call U_n the coefficients of the Laurent expansion of $R(\cdot, A)$. Since N is $R(\lambda, A)$-invariant for $\lambda \in \varrho_+(A)$, the same holds for each U_n. With the obvious notations we have $R(\lambda, A) = \sum_n U_n(\lambda - \mu)^n$, $R(\lambda, A)_| = \sum U_{n|}(\lambda - \mu)^n$ and $R(\lambda, A)_/ = \sum U_{n/}(\lambda - \mu)^n$

$$\mathsf{R}(\lambda, \mathsf{A}) = \sum_{\mathsf{n}} \mathsf{U}_{\mathsf{n}}(\lambda - \mu)^{\mathsf{n}},$$

$$\mathsf{R}(\lambda, \mathsf{A})_| = \sum_{\mathsf{n}} \mathsf{U}_{\mathsf{n}|}(\lambda - \mu)^{\mathsf{n}} \quad \text{and} \quad \mathsf{R}(\lambda, \mathsf{A})_/ = \sum_{\mathsf{n}} \mathsf{U}_{\mathsf{n}/}(\lambda - \mu)^{\mathsf{n}},$$

which shows $\max\{k_|, k_/\} \leqslant k$.

If $R(\cdot, A)_|$ has a pole in μ of order ℓ, then $U_{-(\ell+1)|} = 0$, i.e., $U_{-(\ell+1)}N = \{0\}$. Similarly, it follows that $U_{-(m+1)}E \subseteq N$ if $R(\cdot, A)_/$ has a pole in μ of order m. Therefore $U_{-(\ell+1)} \circ U_{-(m+1)} = 0$.

The relations Eq. (3.3.2) on page 84 imply $U_{-(m+\ell+1)} = 0$, hence the pole order of $R(\cdot, A)$ is dominated by $\ell + m$. □

3.4.1 Spectrum of the adjoint semigroup

We recall from Section 1.3.5 that to every strongly continuous semigroup $\mathcal{T} = (T(t))_{t\geqslant 0}$ there corresponds a strongly continuous adjoint semigroup $\mathcal{T}^* = (T(t)^*)_{t\geqslant 0}$ on the semigroup dual

$$E^* = \{\varphi \in E'\colon \lim_{t\to\infty} \|T(t)'\varphi - \varphi\| = 0\}.$$

$$\mathsf{E}^* = \{\varphi \in \mathsf{E}'\colon \lim_{\mathsf{t}\to 0} \|\mathsf{T}(\mathsf{t})'\varphi - \varphi\| = 0\}.$$

Its generator A^* is the maximal restriction of the adjoint A' to E^*. For these operators the spectra coincide, or more precisely:

(i) $\sigma(T(t)) = \sigma(T(t)') = \sigma(T(t)^*)$,
$R\sigma(T(t)) = P\sigma(T(t)') = P\sigma(T(t)^*)$,

(ii) $\sigma(A) = \sigma(A') = \sigma(A^*)$, $R\sigma(A) = P\sigma(A') = P\sigma(A^*)$,

(iii) $s(A) = s(A^*)$, $\omega_0(A) = \omega_0(A^*)$.

Proof. The left part of these equalities is either well known or has been stated in 3.2.2Proposition 3.2.2 (ii). The first statement of (iii) follows from (ii), while the second is an immediate consequence of the estimate $\|T(t)^*\| \leqslant \|T(t)\| \leqslant M \cdot \|T(t)^*\|$ given in Section 1.3.5.

As a sample for the remaining assertions we show that $0 \notin \sigma(A)$ if and only if $0 \notin \sigma(A^*)$: If A and therefore A' is invertible, it follows from Section 1.3.5 that A^* is a bijection from $D(A^*)$ onto E^*.

Conversely assume that A^* is invertible. Then A' must be injective by the Proposition in Section 1.3.5. Moreover $A'(D(A'))$ contains $A^*(D(A^*)) = E^*$ and is $\sigma(E', E)$-dense in E'. By standard duality arguments it follows that A is injective with dense image. Next we show that $A(D(A))$ is closed: For $f \in D(A)$ choose $\varphi \in D(A')$ such that $\|\varphi\| \leqslant 1$ and $|\langle f, \varphi\rangle| \geqslant \frac{1}{2}\|f\|$. Then

$$\begin{aligned}\|(A^*)^{-1}\|\|Af\| &\geqslant \|(A^*)^{-1}\||\langle Af, \varphi\rangle| \geqslant |\langle Af, (A^*)^{-1}\varphi\rangle| \\ &= |\langle f, \varphi\rangle| \geqslant \frac{1}{2}\|f\|,\end{aligned}$$

hence

$$\|Af\| \geqslant \frac{1}{2}\|(A^*)^{-1}\|^{-1}\|f\|,$$

and $A(D(A))$ is closed since A is closed. □

3.4.2 Spectrum of the $\mathcal{F}$-product semigroup

As stated in Section 1.3.7 the $\mathcal{F}$-product semigroup $\mathcal{T}_\mathcal{F} = (T_\mathcal{F}(t))_{t\geqslant 0}$ on $E_\mathcal{F}^\mathcal{T}$ of a strongly continuous semigroup $\mathcal{T}$ on E serves to convert sequences in E into points in $E_\mathcal{F}^\mathcal{T}$. In particular it can be used to convert approximate eigenvectors of the generator A into eigenvectors of $A_\mathcal{F}$.

Proposition 3.4.4 *Let A be the generator of a strongly continuous semigroup. Then the generator $A_\mathcal{F}$ of the $\mathcal{F}$-product semigroup satisfies:*

(i) $A\sigma(A) = A\sigma(A_\mathcal{F}) = P\sigma(A_\mathcal{F})$,

(ii) $\sigma(A) = \sigma(A_\mathcal{F})$.

Remark 3.4.5 In case A is bounded, then the canonical extension $A_\mathcal{F}$ is a generator and $E_\mathcal{F}^\mathcal{T} = E_\mathcal{F}$ (cf. Section 1.3.7). Thus the proposition applies to bounded linear operators and their canonical extensions to the $\mathcal{F}$-product $E_\mathcal{F}$.

Proof of the proposition. (i) The inclusion $P\sigma(A_\mathcal{F}) \subseteq A\sigma(A_\mathcal{F})$ holds trivially.

We show that $A\sigma(A_\mathcal{F}) \subseteq A\sigma(A)$: Take $\lambda \in A\sigma(A_\mathcal{F})$ and an associated approximate eigenvector $(\hat{f}^m)_{n\in\mathbb{N}}(\hat{f}^m)_{m\in\mathbb{N}}$, i.e., $\hat{f}^m \in D(A_\mathcal{F})$, $\|\hat{f}^m\| = 1$ and $(\lambda - A_\mathcal{F})\hat{f}^m \to 0$ as $m \to \infty$.

By the considerations in Section 1.3.7 we can represent each $\hat{f}^m$ as a normalized sequence $(f_n^m)_{n\in\mathbb{N}}$ in $D(A)$ such that

$$\lim_{m\to\infty} \limsup_{n\to\infty} \|(\lambda - A) f_n^m\| = 0.$$

Therefore we can find a sequence $g_k = f_k^{m(k)}$ satisfying

$$\lim_{k\to\infty} \|(\lambda - A) g_k\| = 0,$$

i.e., $\lambda \in A\sigma(A)$.

Finally we show $A\sigma(A) \subseteq P\sigma(A_{\mathcal{F}})$: For $\lambda \in A\sigma(A)$ take a corresponding approximate eigenvector (f_n). By Section 1.3.2 we have

$$\begin{aligned} \|T(t) f_n - f_n\| &\leqslant \|T(t) f_n - \mathrm{e}^{\lambda t} f_n\| + |\mathrm{e}^{\lambda t} - 1| \\ &= \left\| \int_0^t \mathrm{e}^{\lambda(t-s)} T(s)(\lambda - A) f_n \, \mathrm{d}s \right\| + |\mathrm{e}^{\lambda t} - 1| \end{aligned}$$

which converges to zero uniformly in n as $t \to 0$, i.e., $(f_n) \in m^{\mathcal{T}}(E)$. From the characterization of $D(A_{\mathcal{F}})$ given in Section 1.3.7 it follows that

$$\hat{f} := (f_n) + c_{\mathcal{F}}(E) \in D(A_{\mathcal{F}}) \quad \text{and} \quad A_{\mathcal{F}} \hat{f} = \lambda \hat{f}$$

i.e., $\lambda \in P\sigma(A_{\mathcal{F}})$.

(ii) The inclusion $A\sigma(A) \subseteq \sigma(A_{\mathcal{F}})$ follows from (i).

Now we show $R\sigma(A) \subseteq R\sigma(A_{\mathcal{F}})$. For $\lambda \in R\sigma(A)$ choose $f \in E$ such that $\|(\lambda - A)g - f\| \geqslant 1$ for every $g \in D(A)$. Then $\|(\lambda - A_{\mathcal{F}})\hat{g} - \hat{f}\| \geqslant 1$ for every $\hat{g} \in D(A_{\mathcal{F}})$ and $\hat{f} = (f, f, \ldots) + c_{\mathcal{F}}(E)$. Therefore $\lambda \in R\sigma(A_{\mathcal{F}})$.

We now show $\varrho(A) \subseteq \varrho(A_{\mathcal{F}})$: Assume $\lambda \in \varrho(A)$. By (i) $(\lambda - A_{\mathcal{F}})$ has to be injective. Choose $\hat{f} = (f_1, f_2, \ldots) + c_{\mathcal{F}}(E)$ such that $(f_n) \in m^{\mathcal{T}}(E)$. Then $(R(\lambda, A) f_n) \in m^{\mathcal{T}}(E)$ and $(\lambda - A_{\mathcal{F}})((R(\lambda, A) f_n) + c_{\mathcal{F}}(E)) = (f_n) + c_{\mathcal{F}}(E)$, i.e., $(\lambda - A_{\mathcal{F}})$ is surjective and $\lambda \in \varrho(A_{\mathcal{F}})$. □

Applying this proposition to a single operator $T(t)$, we obtain

$$A\sigma(T(t)) = P\sigma(T(t)_{\mathcal{F}}).$$

Note that in general $A\sigma(T(t)) \neq P\sigma(T_{\mathcal{F}}(t))$ (see the Examples 3.1.3 and 3.1.4 in combination with Theorem 3.6.3).

3.5 The Spectrum of Periodic Semigroups

In this section we determine the spectrum of a particularly simple class of strongly continuous semigroups and thereby achieve a rather complete description of the semi-

group itself. Besides being nice and simple these semigroups gain their importance as building blocks for the general theory.

Definition 3.5.1 A strongly continuous semigroup $\mathcal{T} = (T(t))_{t\geqslant 0}$ on a Banach space E is called *periodic* if $T(t_0) = \mathrm{Id}$ for some $t_0 > 0$.

The *period* τ of $\mathcal{T}$ is obtained as

$$\tau := \inf\{t_0 > 0 \colon T(t_0) = \mathrm{Id}\}.$$

We immediately observe that periodic semigroups are groups with inverses $T(t)^{-1} = T(n\tau - t)$ for $0 \leqslant t \leqslant n\tau$, τ the period of $\mathcal{T}$. Moreover, they are bounded, hence the growth bound is zero and $\sigma(A) \subseteq \mathrm{i}\mathbb{R}$.

Lemma 3.5.2 *Let $\mathcal{T}$ be a strongly continuous semigroup with period $\tau > 0$ and generator A. Then*

$$\sigma(A) \subseteq 2\pi\mathrm{i}/\tau \cdot \mathbb{Z}$$

and for $\mu \notin 2\pi\mathrm{i}/\tau \cdot \mathbb{Z}$,

$$R(\mu, A) = (1 - \mathrm{e}^{-\mu\tau})^{-1} \int_0^\tau \mathrm{e}^{-\mu s} T(s)\, \mathrm{d}s. \tag{3.5.1}$$

Proof. From the basic identities Eqs. (1.3.1) and (1.3.2) on page 19 for $t = \tau$, it follows that $(\mu - A)$ has a left and right inverse if $\mu \neq 2\pi\mathrm{i}n/\tau$, $n \in \mathbb{Z}$, and that the inverse is given by the above expression. □

The representation of $R(\mu, A)$ given in Proposition 1.1.11 shows that the resolvent of the generator of a periodic semigroup is a meromorphic function having only poles of order one and the residues

$$P_n := \lim_{\mu\to\mu_n} (\mu - \mu_n) R(\mu, A) \quad \text{in} \quad \mu_n := 2\pi\mathrm{i}n/\tau, \quad n \in \mathbb{Z},$$

are

$$P_n = \frac{1}{\tau} \int_0^\tau \exp(-\mu_n s) T(s)\, \mathrm{d}s. \tag{3.5.2}$$

Moreover, it follows that the spectrum of A consists of eigenvalues only and each P_n is the spectral projection belonging to μ_n (see Example 3.3.6). Another way of looking at P_n is given by saying that P_n is the n-th n-th Fourier coefficient of the τ-periodic function $s \mapsto T(s)$. From this it follows that no non-zero $\varphi \in E'$ vanishes on all $P_n E$ simultaneously. By the Hahn-Banach theorem we conclude that span $\bigcup_{n\in\mathbb{Z}} P_n E$ is dense in E.

Since $P_n E \subseteq D(A)$, we obtain again from Eq. (1.3.1) on page 19 that

$$AP_n f = \mu_n P_n f \tag{3.5.3}$$

for every $f \in E$, $n \in \mathbb{Z}$. This and Eq. (1.3.2) on page 19 imply

$$T(t)P_n f = \exp(\mu_n t) \cdot P_n f \tag{3.5.4}$$

for every $t \geqslant 0$. Therefore μ_n is an eigenvalue of A and $\exp(\mu_n t)$ is an eigenvalue of $T(t)$ if and only if $P_n \neq 0$. In that case, $P_n E$ is the corresponding eigenspace and we have proven the following lemma.

Lemma 3.5.3 *For a τ-periodic semigroup $\mathcal{T}$ we take $\mu_n := 2\pi i n/\tau$, $n \in \mathbb{Z}$, and consider*

$$P_n := \frac{1}{\tau} \int_0^\tau \exp(-\mu_n s) T(s)\, \mathrm{d}s.$$

Then the following assertions are equivalent.

(a) $P_n \neq 0$,

(b) $\mu_n \in P\sigma(A)$,

(c) $\exp(\mu_n t) \in P\sigma(T(t))$ *for every* $t > 0$.

The action of A, resp. $T(t)$ in the subspaces $P_n E$, $n \in \mathbb{Z}$, is determined by Eq. (3.5.3) on the previous page and Eq. (3.5.4). Moreover,

$$P_m P_n f = \frac{1}{\tau} \int_0^\tau \exp(-\mu_m s) T(s) P_n f\, \mathrm{d}s = \frac{1}{\tau} \int_0^\tau \exp((\mu_n - \mu_m)s) P_n f\, \mathrm{d}s = 0$$

for $n \neq m$, hence the subspaces $P_n E$ are "orthogonal". Since their union is total in E, one expects to be able to extend the representations Eq. (3.5.3) on the previous page and Eq. (3.5.4) of A and $T(t)$. This is possible if

$$\sum_{-\infty}^{+\infty} P_n = \mathrm{Id},$$

where the series should be summable for the strong operator topology.

Unfortunately, this is false in general since the family of projections

$$Q_H := \sum_{n \in H} P_n,$$

where H runs through all finite subsets of $\mathbb{Z}$, may be unbounded (see the example below). Nevertheless the following is true.

Theorem 3.5.4 *Let $\mathcal{T} = (T(t))_{t \geqslant 0}$ be a τ-periodic semigroup on a Banach space E with generator A and associated spectral projections*

$$P_n := \frac{1}{\tau} \int_0^\tau \exp(-\mu_n s) T(s)\, \mathrm{d}s, \quad \mu_n := 2\pi i n/\tau, \quad n \in \mathbb{Z}.$$

For every $f \in D(A)$ one has $f = \sum_{-\infty}^{+\infty} P_n f$ and therefore

(i) $T(t)f = \sum_{-\infty}^{+\infty} \exp(\mu_n t) P_n f \quad$ *if* $f \in D(A)$,

(ii) $Af = \sum_{-\infty}^{+\infty} \mu_n P_n f \quad$ *if* $f \in D(A^2)$.

Proof. It suffices to prove the first statement. Then (i) and (ii) follow by Eqs. (3.5.3) and (3.5.4) on the preceding page.

We assume $\tau = 2\pi$ and show first that $\sum_{-\infty}^{+\infty} P_n f$ is summable for $f \in D(A)$.

For $g := Af$ we obtain $P_n g = P_n A f = A P_n f = \mathrm{i}n P_n f$. Take H to be a finite subset of $\mathbb{Z} \setminus \{0\}$ and $\varphi \in E'$. Then

$$\left| \sum_{n\in H} \langle P_n f, \varphi \rangle \right| = \left| \sum_{n\in H} \frac{1}{\mathrm{i}n} \langle P_n g, \varphi \rangle \right| \leqslant \left(\sum_{n\in H} \frac{1}{n^2} \right)^{1/2} \left(\sum_{n\in H} |\langle P_n g, \varphi \rangle|^2 \right)^{1/2}.$$

From Bessel's inequality we obtain for the second factor

$$\sum_{n\in H} |\langle P_n g, \varphi \rangle|^2 \leqslant \frac{1}{2\pi} \cdot \int_0^{2\pi} |\langle T(s)g, \varphi \rangle|^2 \, \mathrm{d}s \leqslant \|\varphi\|^2 \cdot \frac{1}{2\pi} \cdot \int_0^{2\pi} \|T(s)g\|^2 \, \mathrm{d}s.$$

With the constant $c := \left(\frac{1}{2\pi} \cdot \int_0^{2\pi} \|T(s)g\|^2 \, \mathrm{d}s \right)^{1/2}$ we obtain

$$\left\| \sum_{n\in H} P_n f \right\| \leqslant \|P_0 f\| + c \left(\sum_{n\in H} n^{-2} \right)^{1/2}$$

for every finite subset H of $\mathbb{Z}$, i.e., $\sum_{-\infty}^{+\infty} P_n f$ is summable.

Next we set $h := \sum_{-\infty}^{+\infty} P_n f$ and observe that for every $\varphi \in E'$ the Fourier coefficients of the continuous, τ-periodic functions $s \mapsto \langle T(s)h, \varphi \rangle$ and $s \mapsto \langle T(s)f, \varphi \rangle$ coincide. Therefore these functions are identical for $s \geqslant 0$ and in particular for $s = 0$, i.e., $\langle h, \varphi \rangle = \langle f, \varphi \rangle$. By the Hahn-Banach Theorem we obtain $f = h$. □

The above theorem contains rather precise information on periodic semigroups. In particular, it characterizes periodic semigroups by the fact that $\sigma(A)$ is contained in $\mathrm{i}\alpha\mathbb{Z}$ for some $\alpha \in \mathbb{R}$ and the eigenfunctions of A form a total subset of E.

For a periodic semigroup with bounded generator only a finite number of spectral projections P_n are distinct from 0 and we have the following characterization.

Corollary 3.5.5 *Let $\mathcal{T} = (T(t))_{t\geqslant 0}$ be a semigroup with bounded generator on some Banach space E.*

This semigroup has period τ/k for some $k \in \mathbb{N}$ if and only if there exist finitely many pairwise orthogonal projections P_n, $-m \leqslant n \leqslant m$, $P_{-m} \neq 0$ or $P_m \neq 0$, such that

(i) $\sum_{-m}^{+m} P_n = \mathrm{Id}$,

(ii) $T(t) = \sum_{-m}^{+m} \exp(2\pi \mathrm{i} n t/\tau) P_n$,

(iii) $A = \sum_{-m}^{+m}(2\pi in/\tau)P_n$.

Example 3.5.6 From Section 1.2.5 we recall briefly the rotation group

$$R_\tau(t)f(z) := f(\exp(2\pi it/\tau)\cdot z)$$

on $E = C(\Gamma)$, resp. $E = L^p(\Gamma, m)$ for $1 \leqslant p < \infty$. The spectrum of the generator $Af(z) = (2\pi i/\tau)z \cdot f'(z)$ is $\sigma(A) = (2\pi i/\tau)\cdot\mathbb{Z}$. The eigenfunctions $\varepsilon_n(z) := z^n$ yield the projections

$$P_n = \frac{1}{2\pi i}\cdot\varepsilon_{-(n+1)}\otimes\varepsilon_n, \text{ i.e.,}$$

$$P_n f(z) = \frac{1}{2\pi i}\cdot\left(\int_\Gamma f(w)w^{-(n+1)}\,dw\right)\cdot z^n.$$

It is left as an exercise to compute the norms of $Q_m := \sum_{-m}^{+m} P_n$ $Q_N := \sum_{-N}^{+N} P_n$ in $L^p(\Gamma, m)$ for various p and then check the assertions of Theorem 3.5.4.

Clearly, this proves some classical convergence theorems for Fourier series (compare Davies [3, Chap.8.1]).

3.6 Spectral Mapping Theorems

We now return to the question posed in the introduction to this chapter: In which form and under which conditions is it true that the spectrum $\sigma(T(t))$ of the semigroup operators is obtained—via the exponential map—from the spectrum $\sigma(A)$ of the generator, or briefly

$$\textit{Do we have } \sigma(T(t)) = \exp(t\sigma(A)) \textit{ or at least } \sigma(T(t)) = \overline{\exp(t\sigma(A))}\,?$$

This and similar statements will be called *spectral mapping theorems* for the semigroup $\mathcal{T} = (T(t))_{t\geqslant 0}$ and its generator A. In addition, we saw in Proposition 3.1.1 that the validity of such a spectral mapping theorem implies

$$s(A) = \omega_0(A)$$

for the spectral- and growth bounds and therefore guarantees that the location of the spectrum of A determines the asymptotic behavior of $\mathcal{T}$. As we have seen in Examples 3.1.3 and 3.1.4 the last statement does not hold in general. We therefore present a detailed analysis, where and why it fails and what additional assumptions are needed for its validity. Before doing so, we have another look at the examples.

Example 3.6.1 (The counterexamples revisited)

(i) Take the nilpotent translation semigroup from Section 1.2.6. Then $\sigma(A) = \emptyset$ and $\sigma(T(t)) = \{0\}$ for every $t > 0$. By this trivial example and since $e^z \neq 0$ for every

$z \in \mathbb{C}$, it is natural to read the spectral mapping theorem modulo the addition of $\{0\}$, i.e.,

$$\sigma(T(t)) \setminus \{0\} = \exp(t\sigma(A)) \text{ for } t \geqslant 0.$$

(ii) The spectrum of the generator A of the τ-periodic rotation group $(R_\tau(t))_{t\geqslant 0}$ on $C(\Gamma)$ is $\sigma(A) = 2\pi i/\tau \cdot \mathbb{Z}$ and $\exp(2\pi i n t/\tau)$, $n \in \mathbb{Z}$, is an eigenvalue of $R_\tau(t)$ for every $t \geqslant 0$ (see Example 3.5.6). If t/τ is irrational, these eigenvalues form a dense subset of Γ. Since the spectrum is closed, we obtain $\sigma(T(t)) = \Gamma$ $\sigma(R_\tau(t)) = \Gamma$ for these t. Therefore in this example the spectral mapping theorem is valid only in the following "weak" form

$$\sigma(T(t)) = \overline{\exp(t\sigma(A))}, \quad t \geqslant 0.$$

(iii) By Example 3.1.3 there exists a semigroup $\mathcal{T} = (T(t))_{t\geqslant 0}$ with generator A such that $s(A) = -1$ and $\omega_0(\mathcal{T}) = 0$. This implies that for preassigned real numbers $\alpha < \beta$ there exists a semigroup $\mathcal{S} = (S(t))_{t\geqslant 0}$ with generator B such that $s(B) = \alpha$ and $\omega_0(\mathcal{S}) = \beta$. Indeed, take $S(t) = \mathrm{e}^{\beta t}T((\beta-\alpha)t)$ and observe that $B = (\beta-\alpha)A + \beta\,\mathrm{Id}$. In that case, $\exp(t\sigma(B))$ is contained in the circle about 0 with radius $\mathrm{e}^{\alpha t}$ while $\sigma(S(t))$ has spectral values satisfying $|\lambda| = r(S(t)) = \mathrm{e}^{\beta t} > \mathrm{e}^{\alpha t}$.

(iv) The Example 3.1.3 can be strengthened in order to yield a semigroup $\mathcal{T} = (T(t))_{t\geqslant 0}$ with generator A such that $\sigma(A) = \emptyset$, but $\|T(t)\| = r(T(t)) = 1$ for $t \geqslant 0$, i.e., $s(A) = -\infty$ and $\omega_0 = 0$. Take the translation semigroup on the Banach space

$$E := C_0(\mathbb{R}_+) \cap L^1(\mathbb{R}_+, \mathrm{e}^{x^2}\,\mathrm{d}x)$$

with $\|f\| := \sup\{|f(x)| : x \in \mathbb{R}_+\} + \int_0^\infty |f(x)|\mathrm{e}^{x^2}\,\mathrm{d}x$ (see Greiner et al. [11]).

(v) Another modification of Example 3.1.3 yields a group $\mathcal{T} = (T(t))_{t\in\mathbb{R}}$ satisfying $s(A) < \omega_0$. Therefore the spectral mapping theorem does not hold in the setting of groups (see Wolff [25]).

The next few theorems form the core of this chapter. We show that only one part of the spectrum and one inclusion is responsible for the failure of the spectral mapping theorem. The usefulness of this detailed analysis will become clear in the subsequent chapters on stability and asymptotics.

Proposition 3.6.2 (Spectral Inclusion Theorem) *Let A be the generator of a strongly continuous semigroup $\mathcal{T} = (T(t))_{t\geqslant 0}$ on some Banach space E. Then*

$$\exp(t\sigma(A)) \subseteq \sigma(T(t)) \text{ for } t \geqslant 0.$$

More precisely we have the following inclusions.

$$\exp(t \cdot P\sigma(A)) \subseteq P\sigma(T(t)), \tag{3.6.1}$$

$$\exp(t \cdot A\sigma(A)) \subseteq A\sigma(T(t)), \tag{3.6.2}$$

$$\exp(t \cdot R\sigma(A)) \subseteq R\sigma(T(t)). \tag{3.6.3}$$

Proof. Since $e^{\lambda t} - T(t) = (\lambda - A) \int_0^t e^{\lambda(t-s)} T(s)\, ds$ (see Eq. (1.3.1) on page 19), it follows that $(e^{\lambda t} - T(t))$ is not bijective if $(\lambda - A)$ fails to be bijective proving the main inclusion.

The inclusion Eq. (3.6.1) becomes evident from the following proof of Eq. (3.6.2). Take $\lambda \in A\sigma(A)$ and an associated approximate eigenvector (f_n) in $D(A)$. Then

$$g_n := e^{\lambda t} f_n - T(t) f_n = \int_0^t e^{\lambda(t-s)} T(s)(\lambda - A) f_n \, ds$$

converges to zero as $n \to \infty$. Consequently, $e^{\lambda t} \in A\sigma(T(t))$ and, in fact, the same approximate eigenvector (f_n) does the job for all $t \geqslant 0$.

For the proof of Eq. (3.6.3) we take $\lambda \in R\sigma(A)$ and observe that

$$(e^{\lambda t} - T(t)) f = (\lambda - A)\Big(\int_0^t e^{\lambda(t-s)} T(s) f \, ds\Big) \in (\lambda - A) D(A) \text{ for every } f \in E. \quad \square$$

As we know from the Examples 3.6.1, the converse inclusions do not hold in general, i.e., not every spectral value of a semigroup operator $T(t)$ comes—via the exponential map—from a spectral value of the generator. But at least this is true for some important parts of the spectrum.

Theorem 3.6.3 (Spectral Mapping Theorem for Point and Residual Spectrum) *Let A be the generator of a strongly continuous semigroup $\mathcal{T} = (T(t))_{t \geqslant 0}$. Then*

$$\exp(t \cdot P\sigma(A)) = P\sigma(T(t)) \setminus \{0\}, \tag{3.6.4}$$

$$\exp(t \cdot R\sigma(A)) = R\sigma(T(t)) \setminus \{0\} \text{ for } t \geqslant 0. \tag{3.6.5}$$

Proof. For the proof of Eq. (3.6.4), take $t_0 > 0$ and $0 \neq \lambda \in P\sigma(T(t_0))$.

After rescaling the semigroup $\mathcal{T} = (T(t))_{t \geqslant 0}$ to the semigroup

$$(\exp(-t \cdot \log \lambda / t_0) T(t))_{t \geqslant 0},$$

we may assume $\lambda = 1$. Then the closed, $\mathcal{T}$-invariant subspace

$$G := \{g \in E : T(t_0) g = g\}$$

is non-trivial. The restricted semigroup $\mathcal{T}_|$ is periodic with period $\tau \leqslant t_0$ and the spectrum of its generator $A_|$ contains at least one eigenvalue $\mu = 2\pi i n / t_0$ for some $n \in \mathbb{Z}$ (see Lemma 3.5.3). Since every eigenvalue of $A_|$ is an eigenvalue of A, we

obtain that $1 \in \exp(t_0 \cdot P\sigma(A))$. The converse inclusion has been proved in Eq. (3.6.1) on the preceding page.

In fact, even more can be said: Let $g \in G$ be an eigenvector of $T(t_0)$ corresponding to the eigenvalue $\lambda = 1$. For each $n \in \mathbb{Z}$ define

$$g_n := P_n g = 1/t_0 \cdot \int_0^{t_0} \exp(-2\pi i n s/t_0) T(s) g \, \mathrm{d}s \in G$$

as in Section 3.5. If $g_n \neq 0$, then g_n is an eigenvector of $A_{|}$, hence of A with eigenvalue $2\pi i n/t_0$ as soon as g_n is distinct from zero. Since $D(A_{|})$ is dense in G it follows from Theorem 3.5.4 that this holds for at least one $n \in \mathbb{Z}$. And from the proof of Eq. (3.6.1) on the facing page we know that this g_n is in fact an eigenvector for each $T(t)$, $t \geqslant 0$.

Since $R\sigma(A) = P\sigma(A^*)$ and $R\sigma(T(t)) = P\sigma(T(t)^*)$ (see Proposition 3.4.4Section 3.4.1) the assertion Eq. (3.6.5) on the preceding page follows from Eq. (3.6.4) on the facing page. □

Note that the proof is essentially an application of the structure theorem for periodic semigroups as given in Theorem 3.5.4. The information gained there can be reformulated into statements on the eigenspaces of A and $T(t)$.

Corollary 3.6.4 *For the eigenspaces of the generator A, resp. of the semigroup operators $T(t)$, $t > 0$, the following holds for $\mu \in \mathbb{C}$.*

(i) $\ker(\mu - A) = \bigcap_{s \geqslant 0} \ker(\mathrm{e}^{\mu s} - T(s))$,

(ii) $\ker(\mathrm{e}^{\mu t} - T(t)) = \overline{\operatorname{span}_{n \in \mathbb{Z}} \{\ker(\mu + 2\pi i n/t - A)\}}$.

We note that an analogous statement is valid for $\ker(\mu - A')$ and $\ker(\mathrm{e}^{\mu t} - T(t)')$ if we take in (ii) the $\sigma(E', E)$-closure.

Without proof (see Greiner [9, Proposition 1.10]) we add another corollary showing that poles of the resolvent of $T(t)$ correspond necessarily to poles of the resolvent of the generator. Again the converse is not true as shown by Example 3.5.6.

Corollary 3.6.5 *Assume that $\mathrm{e}^{\mu t}$ is a pole of order k of $R(\cdot, T(t))$ with residue P and Q as the k-th coefficient of the Laurent series. Then*

(i) *$\mu + 2\pi i n/t$ is a pole of $R(\cdot, A)$ of order $\leqslant k$—possibly zero—for every $n \in \mathbb{Z}$,*

(ii) *the residues P_n of $R(\cdot, A)$ in $\mu + 2\pi i n/t$ satisfy $PE = \overline{\operatorname{span}_{n \in \mathbb{Z}} \{P_n E\}}$ $PE = \overline{\operatorname{span}_{n \in \mathbb{Z}} \{P_n E\}}$,*

(iii) *the k-th coefficient of the Laurent series of $R(\cdot, A)$ at $\mu + 2\pi i n/t$ is*

$$Q_n = (t \cdot \mathrm{e}^{\mu t})^{1-k} \cdot Q \circ \frac{1}{t} \int_0^t \mathrm{e}^{-(\mu + 2\pi i n/t)s} T(s) \, \mathrm{d}s.$$

From Proposition 3.6.2 and Theorem 3.6.3 it follows that the approximate point spectrum is the trouble maker in the sense that not every approximate eigenvalue of $T(t)$ corresponds to an approximate eigenvalue of the generator A. Since nothing more can be said in general, we now look for additional hypotheses on the semigroup implying the spectral mapping theorem.

As a simple example we assume $T(t_0)$ to be compact for some $t_0 > 0$. Then

$$\sigma(T(t)) \setminus \{0\} = P\sigma(T(t)) \setminus \{0\}$$

for $t \geqslant t_0$ and the spectral mapping theorem is valid by Eq. (3.6.4) on page 96. A different class of semigroups verifying the spectral mapping theorem is given by the uniformly continuous semigroups (compare Corollary 3.1.2).

Both cases, and many more, are included in the following result.

Theorem 3.6.6 (Spectral Mapping Theorem for Eventually Norm Continuous Semigroups) *Let $\mathcal{T} = (T(t))_{t\geqslant 0}$ be an eventually norm continuous semigroup with generator A. Then the spectral mapping theorem is valid, i.e.,*

$$\sigma(T(t)) \setminus \{0\} = \mathrm{e}^{t\cdot\sigma(A)} \text{ for every } t \geqslant 0.$$

$$\sigma(\mathsf{T}(\mathsf{t})) \setminus \{\mathsf{0}\} = \mathrm{e}^{\mathsf{t}\cdot\sigma(\mathsf{A})} \text{ for every } \mathsf{t} \geqslant \mathsf{0}. \tag{3.6.6}$$

Proof. By the previous considerations it suffices to show that $A\sigma(T(t)) \setminus \{0\} \subseteq \mathrm{e}^{t\cdot\sigma(A)}$ for $t > 0$. This will be done by converting approximate eigenvectors into eigenvectors in the semigroup $\mathcal{F}$-product (see Subsection 3.4.2). The assertion then follows from Eq. (3.6.4) on page 96 and Proposition 3.4.4 (ii).

Assume $t \mapsto T(t)$ to be norm continuous for $t \geqslant t_0$. Moreover it suffices to consider $1 \in A\sigma(T(t_1))$ for some $t_1 > 0$, i.e., we have a normalized sequence $(f_n)_{n\in\mathbb{N}} \subseteq E$ such that

$$\lim_{n\to\infty} \|T(t_1)f_n - f_n\| = 0.$$

Choose $k \in \mathbb{N}$ such that $kt_1 > t_0$ and define $g_n := T(kt_1)f_n$. Then

$$\lim_{n\to\infty} \|g_n\| = \lim_{n\to\infty} \|T(t_1)^k f_n\| = \lim_{n\to\infty} \|f_n\| = 1$$

and

$$\lim_{n\to\infty} \|T(t_1)g_n - g_n\| = 0,$$

i.e., $(g_n)_{n\in\mathbb{N}}$ yields an approximate eigenvector of $T(t_1)$ with approximate eigenvalue 1. But the semigroup $\mathcal{T}$ is uniformly continuous on sets of the form $T(t_0)V$, V bounded in E. In particular, it is uniformly continuous on the sequence $(g_n)_{n\in\mathbb{N}}$, which therefore defines an element g in the semigroup $\mathcal{F}$-product $E_{\mathcal{F}}$.

Obviously, g is an eigenvector of $T_{\mathcal{F}}(t_1)$ with eigenvalue 1 and by Eq. (3.6.4) on page 96 we obtain an eigenvalue $2\pi i n/t_1$ of $A_{\mathcal{F}}$ for some $n \in \mathbb{Z}$. The coincidence of $\sigma(A)$ and $\sigma(A_{\mathcal{F}})$ proves the assertion. □

We point out that the above spectral mapping theorem implies the coincidence of spectral bound and growth bound for eventually norm continuous semigroups, hence we have generalized the Lyapunov Stability Theorem (see Corollary 3.1.2) to a much larger class of semigroups. As mentioned before, this will be of great use in many applications. Therefore we state explicitly the spectral mapping theorem for several important classes of semigroups all of which are eventually norm continuous (cf. the diagram preceding Example 2.1.26).

Corollary 3.6.7 *The Spectral Mapping Theorem 3.6.6 holds for each of the following classes of strongly continuous semigroups.*

(i) *eventually compact semigroups,*

(ii) *eventually differentiable semigroups,*

(iii) *holomorphic semigroups,*

(iv) *uniformly continuous semigroups.*

Another application of the above spectral mapping theorem can be made to tensor product semigroups (see Section 1.3.8). Let $\mathcal{T}_1 = (T_1(t))_{t\geqslant 0}$, $\mathcal{T}_2 = (T_2(t))_{t\geqslant 0}$ be strongly continuous semigroups on Banach spaces E_1, E_2 with generator A_1, A_2. The tensor product semigroup $\mathcal{T} = \mathcal{T}_1 \otimes \mathcal{T}_2$ on some (appropriate) tensor product $E := E_1 \otimes E_2$ has the generator $A = A_1 \otimes \mathrm{Id} + \mathrm{Id} \otimes A_2$, but in general the spectrum of A is not determined by the spectra of A_1, A_2. But with an additional hypothesis the following can be proved.

Corollary 3.6.8 *If $\mathcal{T}_1$ and $\mathcal{T}_2$ are eventually norm continuous, then*

$$\sigma(A) = \sigma(A_1) + \sigma(A_2),$$

where A is the generator of the tensor product semigroup

$$\mathcal{T}_1 \otimes \mathcal{T}_2 = (T_1(t) \otimes T_2(t))_{t\geqslant 0}.$$

Proof. Clearly, the tensor product semigroup is eventually norm continuous and hence the Spectral Mapping Theorem 3.6.6 is valid for all three semigroups $\mathcal{T}_1$, $\mathcal{T}_2$ and $\mathcal{T}$. Moreover the spectrum of the tensor product of bounded operators is the product of the spectra (Reed & Simon [23, XIII.9]). Therefore

$$\sigma(T_1(t) \otimes T_2(t)) = \sigma(T_1(t)) \cdot \sigma(T_2(t)), \quad t \geqslant 0.$$

Consequently we have the following identity for every $t \geqslant 0$

$$\begin{aligned} \mathrm{e}^{t\cdot\sigma(A)} &= \sigma(T_1(t)\otimes T_2(t))\setminus\{0\} \\ &= \sigma(T_1(t))\cdot\sigma(T_2(t))\setminus\{0\} \\ &= \mathrm{e}^{t\cdot\sigma(A_1)}\cdot\mathrm{e}^{t\cdot\sigma(A_2)} \\ &= \mathrm{e}^{t(\sigma(A_1)+\sigma(A_2))}. \end{aligned}$$

From this identity we want to deduce $\sigma(A) = \sigma(A_1) + \sigma(A_2)$.

"$\subseteq$" Take $\xi \in \sigma(A)$. Then for every $t > 0$ there exist $\mu_t \in \sigma(A_1)$, $\lambda_t \in \sigma(A_2)$ and $n_t \in \mathbb{Z}$ such that $\xi = \mu_t + \lambda_t + 2\pi \mathrm{i} n_t/t$.

Since the real parts of μ_t, λ_t are bounded above, they lie in some interval $[a, b]$. But $\sigma(A_k) \cap ([a, b] + \mathrm{i}\mathbb{R})$ is compact for $k = 1, 2$ since A_k is the generator of an eventually norm continuous semigroup (see Theorem 2.1.19). By taking t sufficiently small, we conclude that $n_{t'} = 0$ for some $t' > 0$, i.e., $\xi = \mu_{t'} + \lambda_{t'}$.

"$\supseteq$" Choose $\mu \in \sigma(A_1)$, $\lambda \in \sigma(A_2)$. For every $t > 0$ there exist $\eta_t \in \sigma(A)$, $m_t \in \mathbb{Z}$ such that $\mu + \lambda = \eta_t + 2\pi \mathrm{i} m_t/t$. Since $\mathrm{Re}(\mu) + \mathrm{Re}(\lambda) = \mathrm{Re}(\eta_t)$ and $\{\mathrm{Im}(\eta_t) : t > 0\}$ is bounded, $\mathcal{T} = (T_1(t) \otimes T_2(t))_{t\geqslant 0}$ being eventually norm continuous, it follows that $m_{t'} = 0$ for some $t' > 0$. □

3.7 Weak Spectral Mapping Theorems

In the previous section we showed under which hypotheses a spectral mapping theorem of the form

$$\sigma(T(t))\setminus\{0\} = \mathrm{e}^{t\cdot\sigma(A)}, \quad t \geqslant 0, \tag{3.7.1}$$

is valid for the generator A of a strongly continuous semigroup $(T(t))_{t\geqslant 0}$.

Among the various examples showing that Eq. (3.7.1) does not hold in general we recall the following. Take the Banach space $E = c_0$, the multiplication operator $A(x_n)_{n\in\mathbb{N}} = (\mathrm{i}n x_n)_{n\in\mathbb{N}}$ with maximal domain and the corresponding semigroup $T(t)(x_n)_{n\in\mathbb{N}} = (\mathrm{e}^{\mathrm{i}nt}x_n)_{n\in\mathbb{N}}$. Then $\sigma(A) = \{\mathrm{i}n : n \in \mathbb{N}\}$ and the spectral mapping theorem is valid only in the following weak form

$$\sigma(T(t)) = \overline{\exp(t\cdot\sigma(A))}, \quad t \geqslant 0. \tag{3.7.2}$$

In this section we prove similar weak spectral mapping theorems. We start with a generalization of the above example.

Consider the Banach space $E = C_0(X, \mathbb{C}^n)$ of all continuous $\mathbb{C}^n$-valued functions vanishing at infinity on some locally compact space X. In analogy to Section 1.2.3 we associate to every continuous function $q : X \to M(n)$, where $M(n)$ denotes the space of all complex $n \times n$-matrices, a "multiplication operator"

$$M_q\colon f \mapsto q \cdot f \text{ where } (q \cdot f)(x) = q(x) \cdot f(x) \text{ for } x \in X,$$

on the maximal domain $D(M_q) = \{f \in E\colon q \cdot f \in E\}$. If $\|\mathrm{e}^{tq(x)}\|$ is uniformly bounded for $0 \leqslant t \leqslant 1$ and $x \in X$, it follows that M_q generates the multiplication semigroup

$$(T(t)f)(x) = \mathrm{e}^{tq(x)} \cdot f(x), \quad f \in E, \quad x \in X, \quad t \geqslant 0.$$

Since M_q has a bounded inverse if and only if $q(x)^{-1}$ exists and is uniformly bounded for $x \in X$, it follows that the eigenvalues of each matrix $q(x)$ are always contained in $\sigma(M_q)$. In fact, much more can be said in case the function is bounded.

Lemma 3.7.1 *The spectrum of the matrix valued multiplication operator M_q, where $q\colon X \to M(n)$ is bounded, is given by $\sigma(M_q) = \overline{\bigcup_{x\in X} \sigma(q(x))}$.*

Proof. It remains to show that $0 \notin \overline{\bigcup_{x\in X} \sigma(q(x))}$ implies $0 \notin \sigma(M_q)$. Since $\det q(x)$ is the product of n eigenvalues (according to their multiplicity) of $q(x)$, the hypothesis implies that $d := \inf\{|\det q(x)|\colon x \in X\} > 0$. By Formula 4.12 in Chapter I of Kato [16] we obtain

$$\|q(x)^{-1}\| \leqslant \gamma \cdot \|q(x)\|^{n-1} \cdot |\det q(x)|^{-1} \leqslant \gamma/d \cdot \|M_q\|^{n-1}$$

for every $x \in X$ and a constant γ depending only on the norm chosen on $\mathbb{C}^n$. Therefore, $x \mapsto q(x)^{-1}$ defines a bounded continuous function on X which obviously yields the inverse of M_q, i.e., $0 \notin \sigma(M_q)$. □

Theorem 3.7.2 *Let $A = M_q$ be a matrix multiplication operator on $C_0(X, \mathbb{C}^n)$ generating a strongly continuous semigroup $(T(t))_{t\geqslant 0}$,*

$$T(t) := M_{\mathrm{e}^{tq}} \quad \textit{for } t \geqslant 0.$$

Then the following "Weak Spectral Mapping Theorem" holds:

$$\sigma(T(t)) = \overline{\exp(t \cdot \sigma(A))}.$$

Proof. By the Spectral Inclusion Proposition 3.6.2 we always have $\exp(t\sigma(A)) \subseteq \sigma(T(t))$. Since $T(t)$ is a matrix multiplication operator with a bounded function, we obtain from Lemma 3.7.1

$$\sigma(T(t)) = \overline{\bigcup_{x\in X} \sigma(\exp(tq(x)))} = \overline{\bigcup_{x\in X} \exp(t\sigma(q(x)))} \subseteq \overline{\exp(t\sigma(A))}$$

which proves the assertion. □

Corollary 3.7.3 *The growth bound $\omega_0(A)$ and the spectral bound $s(A)$ coincide for matrix multiplication semigroups.*

The above results remain valid for other Banach spaces of $\mathbb{C}^n$-valued functions such as $L^p(X, \mathbb{C}^n)$, $1 \leqslant p < \infty$.

The example given at the beginning of this section can be generalized in a different way. In fact, $A(x_n) := (inx_n)$ on $E = c_0$ generates a bounded group, and we will show that this property too ensures that the Weak Spectral Mapping Theorem 3.7.2 holds. Without any boundedness assumption on $(T(t))_{t\in\mathbb{R}}$ this result cannot be true (see Hille & Phillips [14, Sec.23.16] or Wolff [25]).

Theorem 3.7.4 *Let $\mathcal{T} = (T(t))_{t\in\mathbb{R}}$ be a strongly continuous group on some Banach space E such that $\|T(t)\| \leqslant p(t)$ for some polynomial p and all $t \in \mathbb{R}$. Then the Weak Spectral Mapping Theorem 3.7.2 holds, i.e.,*

$$\sigma(T(t)) = \overline{\exp(t \cdot \sigma(A))} \text{ for all } t \in \mathbb{R}.$$

From the proof we isolate a series of lemmas assuming always that the hypothesis made in Theorem 3.7.4 holds. Moreover we recall from Fourier analysis that the Fourier transformation $\varphi \mapsto \hat{\varphi}$,

$$\hat{\varphi}(\alpha) := \int_{-\infty}^{\infty} \varphi(x) \mathrm{e}^{-\mathrm{i}\alpha x}\, \mathrm{d}x$$

and its inverse $\Psi \mapsto \check{\Psi}$,

$$\check{\Psi}(x) := \frac{1}{2\pi} \int_{-\infty}^{\infty} \Psi(\alpha) \mathrm{e}^{\mathrm{i}\alpha x}\, \mathrm{d}\alpha$$

are topological isomorphisms of the Schwartz space $\mathcal{S}(= \mathcal{S}(\mathbb{R}))$. Since the subspace $\mathcal{D}$ of all functions having compact support is dense in $\mathcal{S}$, it follows that $\{\varphi \in \mathcal{S} \colon \check{\varphi} \in \mathcal{D}\}$ is also dense in $\mathcal{S}$.

Lemma 3.7.5 *For every function $\varphi \in \mathcal{S}$ we obtain an operator $T(\varphi) \in \mathcal{L}(E)$ by*

$$T(\varphi) f := \int_{-\infty}^{\infty} \varphi(s) T(s) f\, \mathrm{d}s, \quad f \in E.$$

This operator can be represented as

$$T(\varphi) f = \lim_{\varepsilon \to 0} \frac{1}{2\pi} \int_{-\infty}^{\infty} \check{\varphi}(\alpha) [R(\varepsilon - \mathrm{i}\alpha, A) f - R(-\varepsilon - \mathrm{i}\alpha, A) f]\, \mathrm{d}\alpha, \ f \in E. \quad (3.7.3)$$

Proof. That $T(\varphi)$ is well-defined follows from the polynomial boundedness of $(T(t))_{t\in\mathbb{R}}$. In fact, the mapping $\varphi \mapsto T(\varphi)$ is continuous from $\mathcal{S}$ into $(\mathcal{L}(E), \|\cdot\|)$. We obtain

$$\begin{aligned} T(\varphi)f &= \lim_{\varepsilon\to 0}\int_{-\infty}^{\infty}\varphi(s)\mathrm{e}^{-\varepsilon|s|}T(s)f\,\mathrm{d}s \\ &= \lim_{\varepsilon\to 0}\int_{-\infty}^{\infty}\frac{1}{2\pi}\int_{-\infty}^{\infty}\check{\varphi}(\alpha)\mathrm{e}^{\mathrm{i}\alpha s}\mathrm{e}^{-\varepsilon|s|}T(s)f\,\mathrm{d}\alpha\,\mathrm{d}s \\ &= \lim_{\varepsilon\to 0}\frac{1}{2\pi}\int_{-\infty}^{\infty}\check{\varphi}(\alpha)\int_{-\infty}^{\infty}\mathrm{e}^{\mathrm{i}\alpha s}\mathrm{e}^{-\varepsilon|s|}T(s)f\,\mathrm{d}s\,\mathrm{d}\alpha \\ &= \lim_{\varepsilon\to 0}\frac{1}{2\pi}\int_{-\infty}^{\infty}\check{\varphi}(\alpha)[R(\varepsilon-\mathrm{i}\alpha,A)f - R(-\varepsilon-\mathrm{i}\alpha,A)f]\,\mathrm{d}\alpha. \end{aligned}$$

Here we used that polynomially bounded groups have growth bound 0, hence $\omega_0(A) = \omega_0(-A) = 0$. Therefore the integral representation of $R(\varepsilon - \mathrm{i}\alpha, A)$ (cf. Proposition 1.1.11) exists for $\varepsilon \neq 0$. □

Lemma 3.7.6 *If $E \neq \{0\}$, then $\sigma(A) \neq \emptyset$.*

Proof. If $\sigma(A) = \emptyset$, then Eq. (3.7.3) on the preceding page implies $T(\varphi) = 0$ whenever $\check{\varphi}$ has compact support. Since these functions form a dense subspace of $\mathcal{S}$, we conclude that $T(\varphi) = 0$ for all $\varphi \in \mathcal{S}$. Choosing an approximate identity $(\psi_n)_{n\in\mathbb{N}} \subseteq \mathcal{D}$, we obtain

$$f = T(0)f = \lim_{n\to\infty} T(\psi_n)f = 0$$

for every $f \in E$. □

Proof of Theorem 3.7.4 (1st part). By the Spectral Inclusion (see Proposition 3.6.2), we have to show that every spectral value of $T(t)$ can be approximated by exponentials of spectral values of A. In view of the rescaling procedure it suffices to prove this when $-1 \in \varrho(T(\pi))$, provided that the following condition is satisfied.

$$\text{There exists } \varepsilon > 0 \text{ such that } \bigcup_{k\in\mathbb{Z}} \mathrm{i}\,[2k+1-2\varepsilon, 2k+1+2\varepsilon] \subseteq \varrho(A). \tag{3.7.4}$$

Assume now that Eq. (3.7.4) holds. Then each of the sets

$$\sigma_k := \{\mathrm{i}\alpha \in \sigma(A) \colon \alpha \in [2k-1, 2k+1]\}$$

is a spectral set of A with corresponding spectral projection P_k. If we choose $\varphi_0 \in \mathcal{D}$ such that

$$\begin{aligned} \operatorname{supp}(\varphi_0) &\subseteq [-1+\varepsilon, 1-\varepsilon], \\ \varphi_0(x) &= 1 \quad \text{for } x \in [-1+2\varepsilon, 1-2\varepsilon], \end{aligned}$$

it follows from Eq. (3.7.3) on the preceding page and the integral representation of P_k (cf. Eq. (3.3.1) on page 84) that $P_0 = T(\check{\varphi}_0)$.

More generally, since

$$\widehat{\left(e^{i2k}\check{\varphi}_0\right)}(\alpha) = \varphi_0(\alpha - 2k),$$

$$\widehat{\left(e^{i2k\cdot}\check{\varphi}_0\right)}(\alpha) = \varphi_0(\alpha - 2k),$$

the assertions Eq. (3.7.3) on page 102 and Eq. (3.7.4) on the preceding page imply

$$P_k = \int_{-\infty}^{\infty} e^{i2ks}\check{\varphi}_0(s)T(s)\,ds \text{ for } k \in \mathbb{Z}. \qquad \square \quad (3.7.5)$$

At this point we isolate another lemma.

Lemma 3.7.7 span $\bigcup_{k\in\mathbb{Z}} P_k E$ *is dense in* E.

Proof. The closure of span $\bigcup_{k\in\mathbb{Z}} P_k E$ is a $\mathcal{T}$-invariant subspace G of E. Consider the quotient group $(T_/(t))_{t\in\mathbb{R}}$ induced on E/G. The spectrum of its generator $A_/$ is contained in $\sigma(A)$ by Proposition 3.4.2 (ii). Moreover, the spectral projection corresponding to $\sigma(A_/) \cap \sigma_k$ is the quotient operator $P_{k/}$. Obviously $P_{k/} = 0$, hence $\sigma(A_/) \cap \sigma_k = \emptyset$ for every $k \in \mathbb{Z}$ and $\sigma(A_/) = \emptyset$. By Lemma 3.7.6 this implies $E/G = \{0\}$, i.e., $G = E$. $\square$

Proof of Theorem 3.7.4 (2nd part). We return to the situation of the first part of the proof. Using Eq. (3.7.5), the spectral projection P_k can be transformed into

$$\begin{aligned} P_k &= \int_{-\infty}^{\infty} e^{i2ks}\check{\varphi}_0(s)T(s)\,ds \\ &= \sum_{m\in\mathbb{Z}} \int_{(m-1/2)\pi}^{(m+1/2)\pi} e^{i2ks}\check{\varphi}_0(s)T(s)\,ds \\ &= \int_{-\pi/2}^{\pi/2} e^{i2ks} \sum_{m\in\mathbb{Z}} \check{\varphi}_0(s+m\pi)T(s+m\pi)\,ds, \end{aligned}$$

i.e., $P_k f$ is the k-th Fourier coefficient of the π-periodic, continuous function

$$\xi_f : s \mapsto \sum_{m\in\mathbb{Z}} \check{\varphi}_0(s+m\pi)T(s+m\pi)f,\ f \in E.$$

$$\xi_f : s \mapsto \sum_{m\in\mathbb{Z}} \check{\varphi}_0(s+m\pi)T(s+m\pi)f, \quad f \in E.$$

Since the projections P_k are mutually orthogonal, i.e., $P_k P_m = 0$ for $k \neq m$, it follows that $g = \sum_{n\in\mathbb{Z}} P_n g$ for every $g \in$ span $\bigcup_{k\in\mathbb{Z}} P_k E$. In particular, the Fourier coefficients of the function ξ_g are absolutely summable, hence the Fourier series of ξ_g converges to ξ_g.

For $s = 0$ we obtain for $g \in$ span $\bigcup_{k\in\mathbb{Z}} P_k E$

$$g = \sum_{n\in\mathbb{Z}} P_n g \cdot e^{-in0} = \sum_{m\in\mathbb{Z}} \check{\varphi}_0(0+m\pi)T(0+m\pi)g.$$

$$g = \sum_{n\in\mathbb{Z}} P_n g \cdot e^{-in\cdot 0} = \sum_{m\in\mathbb{Z}} \varphi_0(0+m\pi)T(0+m\pi)g.$$

Since span $\bigcup_{k\in\mathbb{Z}} P_k E$ is dense (Lemma 3.7.7), we conclude that

$$\sum_{m\in\mathbb{Z}} \check{\varphi}_0(m\pi)T(m\pi) = \mathrm{Id}\,. \tag{3.7.6}$$

As the final step we construct the inverse operator of $\mathrm{Id}+T(\pi)$ showing that $-1 \in \varrho(T(\pi))$. We define $\psi_0(\alpha) := \varphi_0(\alpha)\cdot(1+e^{i\pi\alpha})^{-1}$, $\alpha\in\mathbb{R}$. Then we have $\psi_0 \in \mathcal{S}$ and $\psi_0\cdot(1+e^{i\pi\cdot}) = \varphi_0$, hence $\check{\psi}_0(x)+\check{\psi}_0(x+\pi) = \check{\varphi}_0(x)$ for all $x\in\mathbb{R}$. Then Eq. (3.7.6) implies

$$\begin{aligned}\mathrm{Id} &= \sum_{m\in\mathbb{Z}} \check{\varphi}_0(m\pi)T(m\pi)\\ &= \sum_{m\in\mathbb{Z}} (\check{\psi}_0(m\pi)+\check{\psi}_0((m+1)\pi))T(m\pi)\\ &= \left[\sum_{m\in\mathbb{Z}} \check{\psi}_0(m\pi)T(m\pi)\right](\mathrm{Id}+T(\pi)).\end{aligned}$$

□

In the rest of this section we discuss the behavior of the single spectral values λ of $T(t)$, $t>0$. The goal is a characterization of $\sigma(T(t))$ involving only properties of the generator. By the rescaling procedure Section 1.3.2 we may assume $\lambda = 1$ and $t = 2\pi$.

From the Spectral Inclusion Theorem (Proposition 3.6.2) we know that $1\in\varrho(T(2\pi))$ implies $i\mathbb{Z}\subseteq\varrho(A)$. As seen in many examples the converse does not hold and we are now looking for additional conditions. Henceforth we assume $i\mathbb{Z}\subseteq\varrho(A)$ and define for $k\in\mathbb{Z}$

$$Q_k := \frac{1}{2\pi}\int_0^{2\pi} e^{-iks}T(s)\,ds = \frac{1}{2\pi}(1-T(2\pi))R(ik,A) \tag{3.7.7}$$

(cf. Eq. (1.3.1) on page 19).

Our approach is based on Fejér's Theorem (for Banach space valued functions). Let us recall this result. Suppose $\xi\colon[0,2\pi]\to E$ is a continuous function and let $\xi_k := \frac{1}{2\pi}\int_0^{2\pi} e^{-iks}\xi(s)\,ds$ be its k-th Fourier coefficient. Then the Fourier series is Cesàro summable to ξ in every point $t\in(0,2\pi)$. Moreover one has

$$\frac{1}{2}(\xi(0)+\xi(2\pi)) = C_1\text{-}\sum_{k\in\mathbb{Z}}\xi_n := \lim_{N\to\infty}\frac{1}{N}\sum_{n=0}^{N-1}\left(\sum_{k=-n}^{n}\xi_k\right).$$

$$\frac{1}{2}(\xi(0)+\xi(2\pi)) = C_1\text{-}\sum_{k\in\mathbb{Z}}\xi_k := \lim_{N\to\infty}\frac{1}{N}\sum_{n=0}^{N-1}\left(\sum_{k=-n}^{n}\xi_k\right). \tag{3.7.8}$$

This result enables us to prove the following proposition.

Proposition 3.7.8 *Let $(T(t))_{t\geqslant 0}$ be a semigroup on a Banach space E and denote its generator by A. Then the following conditions are equivalent.*

(a) $1 \in \varrho(T(2\pi))$,

(b) $\mathrm{i}\mathbb{Z} \subseteq \varrho(A)$ *and the series* $\sum_{k\in\mathbb{Z}} R(\mathrm{i}k, A)f$ *is Cesàro-summable for every* $f \in E$,

(c) $\mathrm{i}\mathbb{Z} \subseteq \varrho(A)$ *and the series* $\sum_{k\in\mathbb{Z}} R(\mathrm{i}k, A)Q_k f$ *is Cesàro-summable for every* $f \in E$.

Proof. (a) $\Rightarrow$ (b) The Spectral Inclusion Proposition 3.6.2 implies $\mathrm{i}\mathbb{Z} \subseteq \varrho(A)$. By Eq. (3.7.7) on the preceding page we have $R(\mathrm{i}k, A) = 2\pi \cdot (1 - T(2\pi))^{-1} Q_k$. Since $\sum_{k\in\mathbb{Z}} Q_k f$ is Cesàro-summable (towards $\frac{1}{2}(f + T(2\pi)f)$) (see Eq. (3.7.8)), it follows that $\sum_{k\in\mathbb{Z}} R(\mathrm{i}k, A)f$ is Cesàro-summable as well.

(b) $\Leftrightarrow$ (c) If we use Eq. (1.3.1) on page 19 and integrate by parts, we obtain

$$\begin{aligned} R(\mathrm{i}k, A)Q_k f &= \frac{1}{2\pi}\int_0^{2\pi} \mathrm{e}^{-\mathrm{i}ks} T(s) R(\mathrm{i}k, A) f \,\mathrm{d}s \\ &= \frac{1}{2\pi}\int_0^{2\pi} \left[R(\mathrm{i}k, A)f - \int_0^s \mathrm{e}^{-\mathrm{i}kt} T(t) f \,\mathrm{d}t \right] \mathrm{d}s \\ &= R(\mathrm{i}k, A)f - \frac{1}{2\pi}\int_0^{2\pi} \mathrm{e}^{-\mathrm{i}kt}(2\pi - t) T(t) f \,\mathrm{d}t. \end{aligned}$$

Fejér's theorem ensures that

$$\sum_{k\in\mathbb{Z}} \frac{1}{2\pi}\int_0^{2\pi} \mathrm{e}^{-\mathrm{i}kt}(2\pi - t) T(t) f \,\mathrm{d}t$$

is Cesàro summable. Hence $\sum_{k\in\mathbb{Z}} R(\mathrm{i}k, A)Q_k f$ is Cesàro-summable if and only if $\sum_{k\in\mathbb{Z}} R(\mathrm{i}k, A)f$ is.

(b) $\Rightarrow$ (a) We have $Q_k = \frac{1}{2\pi}(1 - T(2\pi))R(\mathrm{i}k, A)$. Furthermore we know by Eq. (3.7.7) on the previous page and Eq. (3.7.8) that $\sum_{k\in\mathbb{Z}} Q_k f$ is Cesàro-summable towards $\frac{1}{2}(f + T(2\pi)f)$. If we define $S\colon E \to E$ by

$$Sf := \frac{1}{2} f + \frac{1}{2\pi} \cdot C_1\text{-}\sum_k R(\mathrm{i}k, A) f.$$

Then we have

$$\begin{aligned} (1 - T(2\pi))Sf &= \frac{1}{2}(f - T(2\pi)f) + \frac{1}{2\pi} \cdot C_1\text{-}\sum_k (1 - T(2\pi)) R(\mathrm{i}k, A) f \\ &= \frac{1}{2}(f - T(2\pi)f) + \frac{1}{2}(f + T(2\pi)f) = f. \end{aligned}$$

Since S commutes with $T(2\pi)$, it follows that S is the inverse of $(1 - T(2\pi))$ thus $1 \in \varrho(T(2\pi))$. □

Based on the equivalence of (a) and (b), one can state a characterization of the spectrum of $T(t)$ in terms of the generator and its resolvent only. However, in general it is difficult to verify the summability condition stated in (b).

In Hilbert spaces the boundedness of the resolvents will suffice (see Theorem 3.7.10 below).

Lemma 3.7.9 *Let $(T(t))_{t\geqslant 0}$ be a semigroup on some Hilbert space H and assume $\mathrm{i}\mathbb{Z} \subseteq \varrho(A)$ for the generator A. Then we have*

(i) $(Q_k f)_{k\in\mathbb{Z}} \subseteq \ell^2(H)$ *for every $f \in H$, and*

(ii) *if* $\sup_{k\in\mathbb{Z}} \|R(\mathrm{i}k, A)\| < \infty$ *, then* , then $\sum_{k\in\mathbb{Z}} R(\mathrm{i}k, A) f_k$ *is summable whenever* is summable whenever $(f_k)_{k\in\mathbb{Z}} \in \ell^2(H)$.

Proof. (i) We consider the Hilbert space $L^2([0, 2\pi], H)$ and obtain

$$
\begin{aligned}
0 &\leqslant \left\| T(\cdot) f - \sum_{k=-n}^{n} Q_k f \cdot \mathrm{e}^{\mathrm{i}k\cdot} \right\|^2 \\
&= \int_0^{2\pi} \|T(s)f\|^2 \,\mathrm{d}s - \int_0^{2\pi} \sum_{k=-n}^{n} (T(s)f \mid \mathrm{e}^{\mathrm{i}ks} Q_k f) \,\mathrm{d}s \\
&\quad - \int_0^{2\pi} \sum_{k=-n}^{n} (\mathrm{e}^{\mathrm{i}ks} Q_k f \mid T(s)f) \,\mathrm{d}s \\
&\quad + \int_0^{2\pi} \left(\sum_{k=-n}^{n} \mathrm{e}^{\mathrm{i}ks} Q_k f \,\Big|\, \sum_{\ell=-n}^{n} \mathrm{e}^{\mathrm{i}\ell s} Q_\ell f \right) \mathrm{d}s \\
&= \int_0^{2\pi} \|T(s)f\|^2 \,\mathrm{d}s - 2\pi \sum_{k=-n}^{n} \|Q_k f\|^2 \quad \text{(use Eq. (3.7.5) on page 104).}
\end{aligned}
$$

$$
\begin{aligned}
0 &\leqslant \left\| T(\cdot)f - \sum_{k=-n}^{n} Q_k f \cdot e^{ik\cdot} \right\|^2 \\
&= \int_0^{2\pi} \|T(s)f\|^2 \, ds - \int_0^{2\pi} \sum_{k=-n}^{n} (T(s)f \mid e^{iks} Q_k f) \, ds \\
&\quad - \int_0^{2\pi} \sum_{k=-n}^{n} (e^{iks} Q_k f \mid T(s)f) \, ds \\
&\quad + \int_0^{2\pi} \left(\sum_{k=-n}^{n} e^{iks} Q_k f \,\middle|\, \sum_{\ell=-n}^{n} e^{i\ell s} Q_\ell f \right) ds \\
&= \int_0^{2\pi} \|T(s)f\|^2 \, ds - 2\pi \sum_{k=-n}^{n} \|Q_k f\|^2 \quad \text{(use Eq. (3.7.7) on page 105).}
\end{aligned}
$$

It follows that $\sum_{k\in\mathbb{Z}} \|Q_k f\|^2 \leqslant \frac{1}{2\pi} \int_0^{2\pi} \|T(s)f\|^2 \, ds < \infty$.

(ii) Fix $\lambda > 0$ sufficiently large and set

$$g_k := (1 + \lambda R(ik, A)) f_k, \, k \in \mathbb{Z}.$$

$$g_k := (1 + \lambda R(ik, A)) f_k, \quad k \in \mathbb{Z}.$$

Using the resolvent equation and then Eq. (1.3.1) on page 19, we obtain

$$R(ik, A) f_k = R(\lambda + ik, A) g_k = [1 - e^{-2\pi\lambda} T(2\pi)]^{-1} \int_0^{2\pi} e^{-\lambda s} e^{-iks} T(s) g_k \, ds.$$

This yields for every finite subset N of $\mathbb{Z}$ that

$$
\begin{aligned}
\left\| \sum_{k\in N} R(ik, A) f_k \right\| &\leqslant \|(1 - e^{-2\pi\lambda} T(2\pi))^{-1}\| \cdot \int_0^{2\pi} \|T(s)\| \left\| \sum_{k\in N} e^{-iks} g_k \right\| ds \\
&\leqslant \|(1 - e^{-2\pi\lambda} T(2\pi))^{-1}\| \cdot \left(\int_0^{2\pi} \|T(s)\|^2 \, ds \right)^{1/2} \cdot \left(\int_0^{2\pi} \left\| \sum_{k\in N} e^{-iks} g_k \right\|^2 dx \right)^{1/2} \\
&= c \Big(\sum_{k\in N} \|g_k\|^2 \Big)^{1/2} \leqslant c(1 + \lambda M) \left(\sum_{k\in N} \|f_k\|^2 \right)^{1/2}.
\end{aligned}
$$

$$\left\|\sum_{k\in N} R(ik,A)f_k\right\| \leqslant \|(1-e^{-2\pi\lambda}T(2\pi))^{-1}\| \cdot \int_0^{2\pi} \|T(s)\| \left\|\sum_{k\in N} e^{-iks}g_k\right\| ds$$

$$\leqslant \|(1-e^{-2\pi\lambda}T(2\pi))^{-1}\| \cdot \left(\int_0^{2\pi} \|T(s)\|^2\, ds\right)^{1/2} \cdot \left(\int_0^{2\pi} \left\|\sum_{k\in N} e^{-iks}g_k\right\|^2 ds\right)^{1/2}$$

$$= c\left(\sum_{k\in N} \|g_k\|^2\right)^{1/2} \leqslant c(1+\lambda M)\left(\sum_{k\in N} \|f_k\|^2\right)^{1/2}.$$

Here $c := \|(1-e^{-2\pi\lambda}T(2\pi))^{-1}\| \cdot \left(\int_0^{2\pi} \|T(s)\|^2\, ds\right)^{1/2}$ and $M := \sup_{k\in\mathbb{Z}} \|R(ik,A)\|$ do the job. □

Theorem 3.7.10 *Let A be the generator of a semigroup $(T(t))_{t\geqslant 0}$ on some Hilbert space H. Then the following form of the spectral mapping theorem is valid.*

> *$e^{\lambda t} \in \sigma(T(t)) \setminus \{0\}$ if and only if either $\mu_k := \lambda + 2\pi ik/t \in \sigma(A)$ for some $k \in \mathbb{Z}$ or if $(\|R(\mu_k, A)\|)_{k\in\mathbb{Z}}$ is unbounded.*

Proof. If $e^{\lambda t} \notin \sigma(T(t))$ $e^{\lambda t} \notin \sigma(T(t))$, it follows from the Spectral Inclusion Proposition 3.6.2 that $\mu_k \notin \sigma(A)$ $\mu_k \notin \sigma(A)$ for every $k \in \mathbb{Z}$ and from Formula (1.3.1), that $\|R(\mu_k, A)\|$ is bounded. For the converse inclusion it suffices to assume $t = 2\pi$ and $\lambda = 0$ (use the rescaling procedure Section 1.3.2). Assuming that $i\mathbb{Z} \subseteq \varrho(A)$ and $\|R(ik, A)\|$ is bounded, then $\sum_{k\in\mathbb{Z}} R(ik, A)Q_k f$ is summable by Lemma 3.7.9. Since every summable series is Cesàro-summable, condition (c) of Proposition 3.7.8 is satisfied and we conclude $1 \in \varrho(T(2\pi))$. □

As an immediate consequence we obtain an interesting characterization of the growth bound ω_0 of semigroups on Hilbert spaces.

Corollary 3.7.11 *The growth bound of a semigroup $(T(t))_{t\geqslant 0}$ on a Hilbert space H satisfies*

$$\omega_0 = \inf\{\lambda \in \mathbb{R}\colon \lambda + i\mathbb{R} \subseteq \varrho(A) \text{ and } \|R(\lambda + i\mu, A)\| \text{ is bounded for } \mu \in \mathbb{R}\}. \quad (3.7.9)$$

The Example 3.1.3 above in combination with Part-III, Corollary 3.1.3 shows that Eq. (3.7.9) is not valid in arbitrary Banach spaces.

Notes

Section 3.1: It was already known to Hille & Phillips [14] that for strongly continuous semigroups $(T(t))_{t\geqslant 0}$ with generator A the Spectral Mapping Theorem "$\sigma(T(t)) = \exp(t\sigma(A))$" may be violated in various ways [l.c.,Sec.23.16]. The simple

Examples 3.1.3 and 3.1.4 are due to Wolff (see Greiner et al. [11]) and Zabczyk [27]. A more sophisticated example of a positive group with "$s(A) < \omega_0(A)$" is given in Wolff [25].

Section 3.2: In Definition 3.2.1 we define the residual spectrum of A in such a way that it coincides with the point spectrum of the adjoint A' (see Proposition 3.2.2 (ii)). It therefore differs slightly from the one used, e.g., by Schaefer [24]. The spectral mapping theorem for the resolvent (Proposition 3.2.5) is well known and can, e.g., be deduced from Lemma 9.2 and Theorem 3.11 of Chap.VII in Dunford & Schwartz [5].

Section 3.3: The general theory of spectral decompositions can be found in Kato [16, Chap. III, §6.4]. Applications to isolated singularities like Example 3.3.6 are discussed extensively in [l.c., Chap. III, §6.5] and Yosida [26, Chap. VIII, Sec.8]. There are many ways to introduce an "essential spectrum" (see the footnote on page 243 of Kato [16]). However each one yields the same "essential spectral radius".

Section 3.4: These results are taken from Derndinger [4] and Greiner [9].

Section 3.5: Periodic semigroups are studied explicitly in Bart [1], but most of the results of this section seem to be well known.

Section 3.6: The *Spectral Inclusion Theorem* 3.6.2 and the *Spectral Mapping Theorem* 3.6.6 for eventually norm continuous semigroups date back to Hille & Phillips [14]. Davies [3] gives an elegant proof using Banach algebra techniques. Tensor products of operators and their spectral theory have been studied by Ichinose and others (see Chap. XIII.9 of Reed & Simon [23]). The spectral mapping theorem in Corollary 3.6.8 generalizes Theorem XIII.35 of Reed & Simon [23] (see also Herbst [12]).

Section 3.7: Matrix valued multiplication semigroups appear as solution, via Fourier transformation, of systems of partial differential equations. Kreiss initiated a systematic investigation (see, e.g., Kreiss [17], Kreiss [18], Miller & Strang [20]) and the Weak Spectral Mapping Theorem 3.7.2 must have been known to him. The direct proof of the Weak Spectral Mapping Theorem 3.7.4 for polynomially bounded groups seems to be new. The result can also be deduced from the theory of spectral subspaces of group representations (see, e.g., Combes et al. [2]), since the Arveson spectrum of a strongly continuous one-parameter group can be identified with the spectrum of the generator (see Evans [6]). The final part of this section is taken from Greiner [10] and yields a new approach to Gearhart's characterization of the spectrum of semigroups on Hilbert spaces (Gearhart [7]). Different proofs can be found in Herbst [13], Howland [15] and Prüss [22].

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Chapter 3

Spectral Theory of Positive Semigroups on $C_0(X)$

by
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It is known that for a single operator $T \in \mathcal{L}(C_0(X))$ the positivity of T has influence on the spectrum of T, mainly on the peripheral spectrum, i.e., the part of the spectrum containing all spectral values of maximal absolute value. This part of the spectrum is of interest because it determines the asymptotic behavior of the iterates T^n for large $n \in \mathbb{N}$. The spectral properties indicated above were first proved by Perron [19] and Frobenius [13] for positive square matrices, i.e., for positive operators on the Banach lattice $\mathbb{C}^n$. These results were later extended to the infinite dimensional setting, with important contributions from Jentzsch, Karlin, Krein, Krasnoselski'i, Lotz, Rota, Rutman, Schaefer and others (see Chapt.V of Schaefer [27]).

In this chapter we investigate the spectrum $\sigma(A)$ of the generator A of a positive semigroup $\mathcal{T} = (T(t))_{t\geqslant 0}$ on the Banach space $C_0(X)$. Throughout this chapter we assume that $C_0(X)$ is the space of all *complex-valued* functions on the locally compact space X. In case we restrict to compact spaces we write K instead of X.

3.1 The Spectral Bound

One of the basic results on the spectrum of a positive operator is the fact that its spectral radius is an element of the spectrum (Schaefer [27, V.Proposition 4.1]). We begin the investigation of the spectrum of positive semigroups with the analogous result. To that purpose we recall that the spectral bound $s(A)$ of a generator A is defined as the least upper bound of the real parts of all spectral values (cf. Eq. (3.1.2) on page 71).

Theorem 3.1.1 *If A is the generator of a positive semigroup $\mathcal{T}$ on $E = C_0(X)$, then $s(A) \in \sigma(A)$ unless $s(A) = -\infty$. In case X is compact, we always have $s(A) > -\infty$.*

Proof. We suppose $\sigma(A) \neq \emptyset$ (i.e., $s(A) > -\infty$) and assume $s(A) \notin \sigma(A)$. Then there exist $\varepsilon > 0$ and $\alpha_0, \beta_0 \in \mathbb{R}$ such that

$$[s(A) - \varepsilon, \infty) \subseteq \varrho(A), \quad \mu_0 := \alpha_0 + \mathrm{i}\beta_0 \in \sigma(A) \quad \text{and} \quad \alpha_0 > s(A) - \varepsilon. \tag{3.1.1}$$

Now we choose $\lambda_0 \in \mathbb{R}$ large enough such that

$$|\lambda_0 - \mu_0| < \lambda_0 - (s(A) - \varepsilon) \tag{3.1.2}$$

and, in addition, such that $\lambda_0 > \omega_0(A)$. Then the resolvent $R(\lambda_0, A)$ is a positive bounded operator, hence its spectral radius $r(R(\lambda_0, A))$ is a spectral value. From Part-I, Proposition 3.2.5 it follows that

$$\lambda_0 - r(R(\lambda_0, A))^{-1} \in \sigma(A) \quad \text{and} \quad r(R(\lambda_0, A)) \geqslant |\lambda_0 - \mu_0|^{-1}$$

$$\lambda_0 - \mathsf{r}(\mathsf{R}(\lambda_0, \mathsf{A}))^{-1} \in \sigma(\mathsf{A}) \quad \text{and} \quad \mathsf{r}(\mathsf{R}(\lambda_0, \mathsf{A})) \geqslant |\lambda_0 - \mu_0|^{-1}. \tag{3.1.3}$$

This and Eq. (3.1.2) imply that $\lambda_0 - r(R(\lambda_0, A))^{-1}$ is a real spectral value which is greater than $s(A) - \varepsilon$. We have derived a contradiction to Eq. (3.1.1) and thus have proved the first statement of the theorem.

To establish the second statement we recall that $\lim_{\lambda\to\infty} \lambda R(\lambda, A)f = f$ for every $f \in E$. In particular, for $f = \mathbf{1}_X$ we find a (large) $\lambda_0 \in \mathbb{R}$ such that

$$\lambda_0 R(\lambda_0, A)\mathbf{1}_X \geqslant 1/2 \cdot \mathbf{1}_X \quad \text{hence} \quad R(\lambda_0, A)\mathbf{1}_X \geqslant (2\lambda_0)^{-1} \cdot \mathbf{1}_X$$

$$\lambda_0 \mathsf{R}(\lambda_0, \mathsf{A})\mathbf{1}_X \geqslant 1/2 \cdot \mathbf{1}_X \quad \text{hence} \quad \mathsf{R}(\lambda_0, \mathsf{A})\mathbf{1}_X \geqslant (2\lambda_0)^{-1} \cdot \mathbf{1}_X. \tag{3.1.4}$$

We may assume $\lambda_0 > \omega_0(A)$ then $R(\lambda_0, A) \geqslant 0$, and iterating Eq. (3.1.4) we obtain

$$R(\lambda_0, A)^n \mathbf{1}_X \geqslant (2\lambda_0)^{-n} \cdot \mathbf{1}_X > 0 \quad \text{for every } n \in \mathbb{N}. \tag{3.1.5}$$

It follows that $\|R(\lambda_0, A)^n\| \geqslant (2\lambda_0)^{-n}$ and therefore

$$r(R(\lambda_0, A)) = \lim_{n\to\infty} \|R(\lambda_0, A)^n\|^{1/n} \geqslant (2\lambda_0)^{-1} > 0. \tag{3.1.6}$$

Thus $\sigma(R(\lambda_0, A))$ contains non-zero spectral values which in view of Part-I, Proposition 3.2.5 is equivalent to $\sigma(A) \neq \emptyset$. □

The following examples show that the spectrum may be empty in case X is not compact or if the semigroup is not positive.

Examples 3.1.2 (i) Consider $X = [0, 1)$ and $(T(t))$ on $C_0(X)$ given by

$$(T(t)f)(x) := \begin{cases} f(x + t) & \text{if } x + t < 1, \\ 0 & \text{if } x + t \geqslant 1. \end{cases} \tag{3.1.7}$$

$(T(t))_{t\geqslant 0}$ is nilpotent (we have $T(t) = 0$ for $t \geqslant 1$). It follows that $\sigma(T(t)) = \{0\}$ for all $t > 0$ and by Part-I, Proposition 3.6.2 we have $\sigma(A) = \emptyset$.

(ii) The operator A on $E := C_0[0,\infty)$ given by

$$(Af)(x) = f'(x) - xf(x), D(A) = \{f \in E\colon f \in C^1, Af \in E\}$$

$$(\mathsf{Af})(\mathsf{x}) = \mathsf{f}'(\mathsf{x}) - \mathsf{xf}(\mathsf{x}), \quad \mathsf{D}(\mathsf{A}) = \{\mathsf{f} \in \mathsf{E}\colon \mathsf{f} \in \mathsf{C}^1, \mathsf{Af} \in \mathsf{E}\} \tag{3.1.8}$$

has empty spectrum. It is the generator of a positive non-nilpotent semigroup which is given by

$$(T(t)f)(x) = \exp(-(t^2/2) - xt) \cdot f(x+t). \tag{3.1.9}$$

(iii) Taking into account that $C_0(\,[0,1)\,)$ as well as $C_0(\,[0,\infty)\,)$ both are topologically (but not isometrically) isomorphic to $C([0,1])$ $C[0,1]$ (see Semadeni [29, Section 21.5]), one obtains from (i) and (ii) (non-positive) semigroups on $C([0,1])$ $C[0,1]$ whose generators have empty spectrum.

The proof of Theorem 3.1.1 given above is based on the fact that the spectral radius of a bounded positive operator is an element of the spectrum. A direct proof not using this fact is given in Part-III, Corollary 3.1.4.

Corollary 3.1.3 *Suppose $\lambda_0 \in \varrho(A)$. Then $R(\lambda_0, A)$ is a positive operator if and only if $\lambda_0 > s(A)$. For $\lambda > s(A)$ we have $r(R(\lambda, A)) = (\lambda - s(A))^{-1}$.*

Proof. The second statement is an immediate consequence of Theorem 3.1.1 and Part-I, Proposition 3.2.5.

Given $\lambda_0 > s(A)$ we choose $\lambda_1 > \max\{\lambda_0, \omega_0(A)\}$. Since

$$|\lambda_1 - \lambda_0| < |\lambda_1 - s(A)| = r(R(\lambda_1, A))^{-1}$$

we have

$$R(\lambda_0, A) = \sum_{n=0}^{\infty} (\lambda_1 - \lambda_0)^n \cdot R(\lambda_1, A)^{n+1}. \tag{3.1.10}$$

Since $R(\lambda_1, A)$ is positive, it follows that $R(\lambda_0, A)$ is positive as well. On the other hand, assuming that $R(\lambda_0, A)$ is a positive operator, then λ_0 has to be a real number. Note that for $g \geqslant 0$ we have $f := R(\lambda_0, A)g \geqslant 0$ hence

$$\lambda_0 f - Af = g = \overline{g} = \overline{(\lambda_0 - A)f} = \overline{\lambda_0} f - Af.$$

As we have shown above, $R(\lambda, A)$ is positive for $\lambda > \max\{\lambda_0, s(A)\}$ hence an application of the resolvent equation yields

$$R(\lambda_0, A) = R(\lambda, A) + (\lambda - \lambda_0)R(\lambda, A)R(\lambda_0, A) \geqslant R(\lambda, A) \geqslant 0$$

$$\mathsf{R}(\lambda_0, \mathsf{A}) = \mathsf{R}(\lambda, \mathsf{A}) + (\lambda - \lambda_0)\mathsf{R}(\lambda, \mathsf{A})\mathsf{R}(\lambda_0, \mathsf{A}) \geqslant \mathsf{R}(\lambda, \mathsf{A}) \geqslant 0. \tag{3.1.11}$$

It follows that for all $\lambda > \max\{\lambda_0, s(A)\}$ we have

$$(\lambda - s(A))^{-1} = r(R(\lambda, A)) \leqslant \|R(\lambda, A)\| \leqslant \|R(\lambda_0, A)\| \tag{3.1.12}$$

which can be true only if λ_0 is greater than $s(A)$. □

Corollary 3.1.4 *Suppose X is compact and A has compact resolvent. Then there exists a real eigenvalue λ_0 admitting a positive eigenfunction such that* $\mathrm{Re}(\lambda) \leqslant \lambda_0$ *for every $\lambda \in \sigma(A)$.*

Proof. From Theorem 3.1.1 we conclude that $\lambda_0 := s(A)$ is a real number, contained in the spectrum and obviously $\mathrm{Re}(\lambda) \leqslant \lambda_0$ for every $\lambda \in \sigma(A)$. Since A has compact resolvent it follows that λ_0 is a pole of the resolvent. Let k be its order, then the highest coefficient in the Laurent series is given by

$$Q := \lim_{\lambda \to s(A)} (\lambda - s(A))^k R(\lambda, A). \tag{3.1.13}$$

It follows from Corollary 3.1.3 that Q is a positive operator. Since $Q \neq 0$, there exists $g \geqslant 0$ such that $h := Qg > 0$. Moreover, by Part-I, Example 3.3.6, we have

$$(\lambda_0 - A)h = (\lambda_0 - A)Qg = 0.$$ □

The example of the rotation semigroup (Part-I, Example 3.5.6) shows that the assumptions in Corollary 3.1.4 do not imply that $s(A)$ is dominant. Additional hypotheses ensuring this stronger property will be given below (see Corollary 3.2.11 and 3.2.12).

The following result is elementary. However, positivity is the crucial point in its proof. Note that it is not just a consequence of the spectral mapping theorem for the point spectrum.

Proposition 3.1.5 *Suppose A is the generator of the positive semigroup $(T(t))_{t \geqslant 0}$. Take $\tau > 0$, $r > 0$ and let $\alpha := \tau^{-1} \log(r)$.*

(i) *If r is an eigenvalue of $T(\tau)$ with positive eigenfunction h_0, then there is a positive $h \in D(A)$ such that $Ah = \alpha h$ and*

$$\{x \in X : h_0(x) > 0\} \subseteq \{x \in X : h(x) > 0\}.$$

(ii) *If r is an eigenvalue of $T(\tau)'$ with positive eigenvector φ_0, then there is a positive $\varphi \in D(A^*)$ such that $A^*\varphi = \alpha\varphi$ $\varphi \in D(A')$ such that $A'\varphi = \alpha\varphi$ and* $\mathrm{supp}\, \varphi_0 \subseteq \mathrm{supp}\, \varphi$.

Proof. Without loss of generality we may assume $r = 1$, i.e., $\alpha = 0$ and $T(\tau)h_0 = h_0$.

(i) Defining

$$h := \int_0^\tau T(s)h_0 \, \mathrm{d}s, \tag{3.1.14}$$

then for $0 \leqslant t \leqslant \tau$ we have

$$\begin{aligned} T(t)h &= \int_0^\tau T(s+t)h_0 \, \mathrm{d}s = \int_t^\tau T(s)h_0 \, \mathrm{d}s + \int_\tau^{\tau+t} T(s-\tau)T(\tau)h_0 \, \mathrm{d}s \\ &= \int_t^\tau T(s)h_0 \, \mathrm{d}s + \int_0^t T(s)T(\tau)h_0 \, \mathrm{d}s = h \end{aligned}$$

$$\begin{aligned} \mathsf{T}(t)h &= \int_0^\tau \mathsf{T}(s+t)h_0 \, \mathrm{d}s = \int_t^\tau \mathsf{T}(s)h_0 \, \mathrm{d}s + \int_\tau^{\tau+t} \mathsf{T}(s-\tau)\mathsf{T}(\tau)h_0 \, \mathrm{d}s \\ &= \int_t^\tau \mathsf{T}(s)h_0 \, \mathrm{d}s + \int_0^t \mathsf{T}(s)\mathsf{T}(\tau)h_0 \, \mathrm{d}s = h. \end{aligned}$$

It follows that

$$Ah = \lim_{t \to 0} t^{-1}(T(t)h - h) = 0.$$

So far, positivity was not used. The point is that in general, h may be zero. But if $(T(t))$ is positive and $h_0 \geqslant 0$, then $s \mapsto (T(s)h_0)(x)$ is a continuous positive function, hence

$$0 < h_0(x_0) = (T(0)h_0)(x_0) \text{ implies } h(x_0) = \int_0^\tau (T(s)h_0)(x_0) \, \mathrm{d}s > 0.$$

(ii) Defining $\varphi := \int_0^\tau T(s)'\varphi_0 \, \mathrm{d}s$, one can proceed as in (i) to obtain the desired result. □

We use Proposition 3.1.5 to prove an analogue of the famous Krein-Rutman result. For the sake of completeness we include the proof of this classical result, stating that the spectral radius of a positive operator T on $C(K)$ (or more generally on an order unit space) is an eigenvalue of the adjoint T' (see the Corollary of Theorem 2.6 in the appendix of Schaefer [25]).

Theorem 3.1.6 *Suppose K is compact and $(T(t))_{t \geqslant 0}$ is a positive semigroup on $C(K)$ with generator A. Then there exists a positive probability measure $\varphi \in D(A')$ such that $A'\varphi = \omega_0(A)\varphi$.*

Proof. Consider $T := T(1)$, $r := r(T) = \mathrm{e}^{\omega_0(A)}$. In view of Proposition 3.1.5 it is enough to show that r is an eigenvalue of T' with a positive eigenvector. Given $\lambda \in \mathbb{C}$, $|\lambda| > r$ and $f \in C(K)$ we have

$$|R(\lambda, T)f| = \left| \sum_{n=0}^\infty \lambda^{-n-1} T^n f \right| \leqslant \sum_{n=0}^\infty |\lambda|^{-n-1} T^n |f| = R(|\lambda|, T)|f|.$$

It follows that $\|R(\lambda, T)\| \leqslant \|R(|\lambda|, T)\|$ and therefore

$$\lim_{\lambda \downarrow r} \|R(\lambda, T)\| = \infty. \tag{3.1.15}$$

By the uniform boundedness principle there exist a sequence (λ_n), $\lambda_n \downarrow r$ and a positive $\psi \in M(K)$ such that $\|R(\lambda_n, T)'\psi\| \to \infty$. Defining

$$\psi_n := \|R(\lambda_n, T)'\psi\|^{-1} R(\lambda_n, T)'\psi$$

we have

$$\begin{aligned}(r - T')\psi_n &= \|R(\lambda_n, T)'\psi\|^{-1} \cdot ((r - \lambda_n) + (\lambda_n - T'))R(\lambda_n, T)'\psi \\ &= (r - \lambda_n)\psi_n + \|R(\lambda_n, T)'\psi\|^{-1}\psi \to 0.\end{aligned} \tag{3.1.16}$$

Since $(r - T')$ is $\sigma(M(K), C(K))$-continuous, equation Eq. (3.1.16) implies that every $\sigma(M(K), C(K))$ cluster point of (ψ_n) is a positive eigenvector, provided that it is non-zero. Because K is compact we have

$$\{\varphi \in M(K) \colon \varphi \geqslant 0, \|\varphi\| = 1\} = \{\varphi \in M(K) \colon \varphi \geqslant 0, \langle \varphi, 1\rangle = 1\},$$

$$\{\varphi \in \mathsf{M(K)} \colon \varphi \geqslant 0, \|\varphi\| = 1\} = \{\varphi \in \mathsf{M(K)} \colon \varphi \geqslant 0, \langle \varphi, \mathbf{1}_{\mathsf{K}}\rangle = 1\},$$

which shows that the set of probability measures is $\sigma(M(K), C(K))$-compact. Therefore the sequence (ψ_n) has non-zero cluster points. □

This theorem implies that for positive semigroups on $C(K)$ the growth and spectral bounds coincide (cf. Part-I, Section 4.1). Actually, this is true for locally compact spaces as well and can be proved directly (see Theorem 4.1.4). Using this result one can prove Theorem 3.1.6 by applying the classical Krein-Rutman theorem to any resolvent operator $R(\lambda, A)$ for $\lambda \in \mathbb{R}$ sufficiently large.

The theorem ensures that A' always has eigenvalues, but the generator itself may have no eigenvalue at all. Multiplication operators have no eigenvalues unless the multiplier is constant on an open subset. Theorem 3.1.6 fails to be true for locally compact spaces as the following example shows.

Example 3.1.7 Consider $E = C_0(\mathbb{R}^n)$ and the semigroup $(T(t))_{t \geqslant 0}$ generated by the Laplacian (cf. Section 1.2.8). From the explicit representation of $T(t)$,

$$(T(t)f)(x) = (4\pi t)^{-n/2} \int_{\mathbb{R}^n} \exp(-(x - y)^2/4t) \cdot f(y)\, \mathrm{d}y,$$

$$(\mathsf{T(t)f})(\mathsf{x}) = (4\pi \mathsf{t})^{-\mathsf{n}/2} \int_{\mathbb{R}^n} \exp(-|\mathsf{x} - \mathsf{y}|^2/(4\mathsf{t})) \cdot \mathsf{f(y)}\, \mathrm{dy}, \tag{3.1.17}$$

it follows that $\lim_{t \to \infty} T(t)f = 0$ for every $f \in C_0(\mathbb{R}^n)$. In fact, for f having compact support, we have

$$\|T(t)f\| \leqslant (4\pi t)^{-n/2} \int_{\mathbb{R}^n} |f(y)|\,\mathrm{d}y \to 0,$$

and since $\|T(t)\| = 1$ for all $t \geqslant 0$, convergence to 0 holds for all $f \in E$. If φ is an eigenvector of A' corresponding to $s(A) = \omega_0(A) = 0$, we have $T(t)'\varphi = \varphi$ for all $t \geqslant 0$, hence

$$\langle \varphi, f \rangle = \lim_{t\to\infty} \langle T(t)f, \varphi \rangle = 0 \text{ for every } f,$$

$$\langle \mathrm{f}, \varphi \rangle = \lim_{\mathrm{t}\to\infty} \langle \mathrm{T(t)f}, \varphi \rangle = 0 \text{ for every f},$$

i.e., $\varphi = 0$.

3.2 The Boundary Spectrum

In this section we restrict our attention to the *boundary spectrum* $\sigma_b(A)$ of a generator A, which, by definition, is the intersection of $\sigma(A)$ with the line

$$\{\lambda \in \mathbb{C}\colon\ \mathrm{Re}(\lambda) = s(A)\}.$$

Thus $\sigma_b(A)$ contains all spectral values of A which have maximal real part. Note that in general the boundary spectrum is a proper subset of the topological boundary of $\sigma(A)$. Our aim is to prove results ensuring that $\sigma_b(A)$ is a cyclic set (see Definition 3.2.5 below).

While most of the results of the preceding section were obtained by transforming the problem to a resolvent operator $R(\lambda, A)$ ($\lambda \in \mathbb{R}$ large enough), this procedure fails here. The reason is that there is no one-to-one correspondence between the boundary spectrum of A and the peripheral spectrum of $R(\lambda, A)$. Actually, from Theorem 3.1.1 and Part-I, Proposition 3.2.5 it follows that the peripheral spectrum of $R(\lambda, A)$ (i.e., the set of spectral values having maximal absolute value) is trivial since it only contains the spectral radius $r(R(\lambda, A)) = (\lambda - s(A))^{-1}$.

We begin our discussion with two lemmas.

Lemma 3.2.1 *Suppose K, L are compact and $T\colon C(K) \to C(L)$ is a linear operator satisfying $T\mathbf{1}_K = \mathbf{1}_L$. Then we have $T \geqslant 0$ if and only if $\|T\| \leqslant 1$.*

Proof. If T is positive, then

$$|Tf| \leqslant T|f| \leqslant T(\|f\| \cdot \mathbf{1}_K) = \|f\| \cdot T(\mathbf{1}_K), \quad f \in C(K), \tag{3.2.1}$$

hence $\|T\| = \|T\mathbf{1}_K\|$ whenever T is positive. This shows that $T \geqslant 0$ implies $\|T\| \leqslant 1$ whenever $T\mathbf{1}_K = \mathbf{1}_L$.

To prove the reverse direction, we first observe that for complex numbers and hence for complex-valued functions the following equivalence holds.

$$-1 \leqslant f \leqslant 1 \text{ if and only if } \|f - \mathrm{i}\cdot r\cdot 1\| \leqslant \varrho_r := (1+r^2)^{1/2} \text{ for every } r \in \mathbb{R}. \quad (3.2.2)$$

Now suppose $f \in C(K)$, $0 \leqslant f \leqslant 2\cdot \mathbf{1}_K$. Then we have $-\mathbf{1}_K \leqslant f - \mathbf{1}_K \leqslant \mathbf{1}_K$ hence by Eq. (3.2.2)

$$\|f - \mathbf{1}_K - \mathrm{i}\cdot r\cdot \mathbf{1}_K\| \leqslant \varrho_r \text{ for every } r \in \mathbb{R}.$$

From $T\mathbf{1}_K = \mathbf{1}_L$ and $\|T\| \leqslant 1$ it follows that

$$\|Tf - \mathbf{1}_L - \mathrm{i}\cdot r\cdot \mathbf{1}_L\| \leqslant \varrho_r \text{ for every } r \in \mathbb{R}.$$

Using Eq. (3.2.2) once again, we obtain

$$-\mathbf{1}_L \leqslant Tf - \mathbf{1}_L \leqslant \mathbf{1}_L \text{ or } 0 \leqslant Tf \leqslant 2\cdot \mathbf{1}_L. \qquad \square$$

Before we can formulate the second lemma we have to fix some notation.

Definition 3.2.2 (i) Given $h \in C_0(X)$ such that $h(x) \neq 0$ for all $x \in X$ then the operator S_h is defined to be the multiplication operator with sign h, i.e.,

$$S_h f = \frac{h}{|h|}\cdot f \quad \text{for } f \in C_0(X). \quad (3.2.3)$$

(ii) For $f \in C_0(X)$, $n \in \mathbb{Z}$ we define $f^{[n]} \in C_0(X)$ by

$$f^{[n]}(x) = \begin{cases} \left(\frac{f(x)}{|f(x)|}\right)^{n-1}\cdot f(x) & \text{if } f(x) \neq 0, \\ 0 & \text{if } f(x) = 0. \end{cases} \quad (3.2.4)$$

The following assertions are immediate consequences of the definition. They will be used frequently in the following.

$$\begin{aligned} &S_h \text{ is a linear isometry satisfying } |S_h f| = |f|, \\ &\text{its inverse being } S_{\overline{h}} \text{ where } \overline{h} \text{ is the complex conjugate of } h, \end{aligned} \quad (3.2.5)$$

$$f^{[1]} = f,\ f^{[0]} = |f|,\ f^{[-1]} = \overline{f},\ |f^{[n]}| = |f| \text{ for every } n \in \mathbb{Z},$$
$$\mathsf{f}^{[1]} = \mathsf{f},\quad \mathsf{f}^{[0]} = |\mathsf{f}|,\quad \mathsf{f}^{[-1]} = \bar{\mathsf{f}},\quad |\mathsf{f}^{[n]}| = |\mathsf{f}| \text{ for every } n \in \mathbb{Z}, \quad (3.2.6)$$

$$\text{If } h(x) \neq 0 \text{ for all } x \in X, \text{ then } h^{[n]} = S_h^n |h| = S_h^{n-1} h \text{ for every } n \in \mathbb{Z}. \quad (3.2.7)$$

Lemma 3.2.3 *Let T and R be bounded linear operators on $C_0(X)$ and assume that $h \in C_0(X)$ has no zeros. Suppose we have*

$$Rh = h, T|h| = |h| \text{ and } |Rf| \leqslant T|f| \text{ for every } f \in C_0(X).$$

$$\mathsf{Rh} = \mathsf{h}, \quad \mathsf{T}|\mathsf{h}| = |\mathsf{h}| \textit{ and } |\mathsf{Rf}| \leqslant \mathsf{T}|\mathsf{f}| \textit{ for every } f \in C_0(X). \tag{3.2.8}$$

Then R and T are similar, more precisely, $T = S_h^{-1} R S_h$. In particular, the spectra (and point spectra, resp.) of T and R coincide.

Proof. We first note that the assertion $|Rf| \leqslant T|f|$ ($f \in E$) implies that T is a positive operator. Therefore the assumption $T|h| = |h|$ implies that the principal ideal

$$E_h = \{f \in C_0(X) \colon |f| \leqslant n|h| \text{ for some } n \in \mathbb{N}\}$$

is an invariant subspace for T and for R as well. E_h is isomorphic to $C^b(X) \cong C(\beta X)$ (βX denotes the Stone-Čech compactification of X), an isomorphism is given by $f \mapsto f|h|$. Considering the restrictions $T_{|E_h}$ and $R_{|E_h}$ as operators on $C(\beta X)$ and denoting them $\tilde{T}$ and $\tilde{R}$ respectively, we have for all f .

$$\tilde{R}\tilde{h} = \tilde{h}, \quad \tilde{T}\mathbf{1} = \mathbf{1}, \quad \tilde{T} \geqslant 0, \quad |\tilde{R}f| \leqslant \tilde{T}|f|. \tag{3.2.9}$$

Here $\tilde{h}$ denotes the continuous extension of $h/|h|$ to βX.

Defining $T_1 := M_{\tilde{h}}^{-1} \tilde{R} M_{\tilde{h}}$ we have by Eq. (3.2.9)

$$T_1\mathbf{1} = M_{\tilde{h}}^{-1} \tilde{R}\tilde{h} = \mathbf{1} \tag{3.2.10}$$

and for all f

$$|T_1 f| = |M_{\tilde{h}}^{-1} \tilde{R} M_{\tilde{h}} f| = |\tilde{R} M_{\tilde{h}} f| \leqslant \tilde{T}|M_{\tilde{h}} f| = \tilde{T}|f|. \tag{3.2.11}$$

Hence we have $\|T_1\| \leqslant \|\tilde{T}\| = 1$ by Eq. (3.2.11). Then it follows from Eq. (3.2.10) and Lemma 3.2.1 that T_1 is a positive operator. Thus Eq. (3.2.11) implies that $0 \leqslant T_1 \leqslant \tilde{T}$ and therefore from Eqs. (3.2.9) and (3.2.10) we conclude

$$\|\tilde{T} - T_1\| = \|(\tilde{T} - T_1)\mathbf{1}\| = 0. \qquad \square$$

We are now able to prove a result which in some sense is the key to cyclicity results for the spectrum. These general results will be proved by reducing the problem in such a way that the following theorem can be applied.

Theorem 3.2.4 (i) *Let T be a positive linear operator on $C_0(X)$, let $h \in C_0(X)$ and $\lambda \in \mathbb{C}$, $|\lambda| = 1$. If $Th = \lambda h$ and $T|h| = |h|$, then we have*

$$Th^{[n]} = \lambda^n h^{[n]} \textit{ for every } n \in \mathbb{Z}$$

$$\mathsf{Th}^{[\mathsf{n}]} = \lambda^{\mathsf{n}} \mathsf{h}^{[\mathsf{n}]} \textit{ for every } \mathsf{n} \in \mathbb{Z}.$$

If h does not have zeros in X, then

$$\lambda T = S_h^{-1} T S_h.$$

(ii) *Suppose A is the generator of a positive semigroup, $h \in C_0(X)$, $\alpha, \beta \in \mathbb{R}$ such that $Ah = (\alpha + \mathrm{i}\beta)h$ and $A|h| = \alpha|h|$. Then we have*

$$Ah^{[n]} = (\alpha + \mathrm{i}n\beta)h^{[n]} \text{ for every } n \in \mathbb{Z}.$$

If h does not have zeros, then

$$S_h D(A) = D(A) \text{ and } \mathrm{i}\beta + A = S_h^{-1} A S_h.$$

$$\mathsf{S_h D(A) = D(A)} \text{ and } \mathrm{i}\beta + \mathsf{A} = \mathsf{S_h^{-1} A S_h}.$$

Proof. (i) The closed principal ideal $\overline{E_h}$, which is canonically isomorphic to $C_0(X_1)$ with $X_1 = \{x \in X : h(x) \neq 0\}$, is T-invariant. We give an object a tilde when we consider it as an element of $\overline{E_h} \cong C_0(X_1)$. Defining $\tilde{R} := \bar{\lambda}\tilde{T}$, then $\tilde{T}$, $\tilde{R}$, $\tilde{h}$ satisfy Eq. (3.2.8) on the previous page, hence we have

$$\tilde{T} = S_{\tilde{h}}^{-1} \circ \tilde{R} \circ S_{\tilde{h}} = \bar{\lambda} \cdot S_{\tilde{h}}^{-1} \circ \tilde{T} \circ S_{\tilde{h}} \tag{3.2.12}$$

which by iteration yields for all $n \in \mathbb{Z}$

$$\tilde{T} = \bar{\lambda}^n \cdot S_{\tilde{h}}^{-n} \circ \tilde{T} \circ S_{\tilde{h}}^n. \tag{3.2.13}$$

It follows that

$$\tilde{T}\tilde{h}^{[n]} = \tilde{T} \circ S_{\tilde{h}}^n |\tilde{h}| = \lambda^n \cdot S_{\tilde{h}}^n \circ \tilde{T}|\tilde{h}| = \lambda^n \cdot S_{\tilde{h}}^n \tilde{h} = \lambda^n \cdot \tilde{h}^{[n]},$$

which is precisely $Th^{[n]} = \lambda^n h^{[n]}$ for all $n \in \mathbb{Z}$ (see Eq. (3.2.7) on page 196 and Eq. (3.2.12)).

If h does not have zeros, then $\overline{E_h} = E$, hence $T = \tilde{T}$, $h = \tilde{h}$ and the remaining assertion follows from Eq. (3.2.12).

(ii) This can be deduced easily from (i) as follows. If

$$Ah = (\alpha + \mathrm{i}\beta)h \text{ and } A|h| = \alpha|h|,$$

then we have by Part-I, Corollary 3.6.4 for $t \geqslant 0$,

$$\mathrm{e}^{-\alpha t} T(t)h = \mathrm{e}^{\mathrm{i}\beta t} h \text{ and } \mathrm{e}^{-\alpha t} T(t)|h| = |h|.$$

By (i) we have for $t \geqslant 0$ and for $n \in \mathbb{Z}$

$$\mathrm{e}^{-\alpha t} T(t) h^{[n]} = \mathrm{e}^{\mathrm{i}n\beta t} h^{[n]},$$

which is equivalent to

$$Ah^{[n]} = (\alpha + \mathrm{i}n\beta)h^{[n]}.$$

If h does not have zeros, then

$$e^{-\alpha t}T(t) = e^{-\alpha t}e^{i\beta t}S_h^{-1}T(t)S_h \text{ for every } t \geqslant 0,$$

$$e^{-\alpha t}T(t) = e^{-\alpha t}e^{-i\beta t}S_h^{-1}T(t)S_h \text{ for every } t \geqslant 0,$$

which is equivalent to the final statement of (ii). □

Before we state a first cyclicity result we give the definition and illustrate it by some examples.

Definition 3.2.5 A subset $M \subseteq \mathbb{C}$ is called *imaginary additively cyclic* (or simply *cyclic*), if it satisfies the following condition.

$\alpha + i\beta \in M, \alpha, \beta \in \mathbb{R}$ implies that $\alpha + ik\beta \in M$ for every $k \in \mathbb{Z}$.

Every subset of $\mathbb{R}$ is cyclic. On the other hand, if M is cyclic and $M \not\subseteq \mathbb{R}$, then M has to be unbounded.

For a subset M of $i\mathbb{R}$ we give the following equivalent conditions.

(a) M is imaginary additively cyclic,

(b) M is the union of (additive) subgroups of $i\mathbb{R}$,

(c) $M = \bigcup_{\alpha \in S} i\alpha\mathbb{Z}$ for some set $S \subseteq \mathbb{R}$.

Here are some concrete cyclic subsets of $i\mathbb{R}$.

$$M_1 = \{0\}, \quad M_2 = i\mathbb{R}, \quad M_3 = i\alpha\mathbb{Z} \quad (\alpha > 0),$$
$$M_4 = i\alpha\mathbb{Z} + i\beta\mathbb{Z} = \{in\alpha + im\beta : n, m \in \mathbb{Z}\} \quad (\alpha, \beta \in \mathbb{R}),$$
$$M_5 = \{0\} \cup \{i\lambda : \lambda \in \mathbb{R}, |\lambda| \geqslant 1\},$$
$$M_6 = \cup_{n=0}^{\infty}\{i\lambda : \lambda \in \mathbb{R}, n\alpha \leqslant |\lambda| \leqslant n\beta\} \quad (0 < \alpha \leqslant \beta, \alpha, \beta \in \mathbb{R})$$

$$M_1 = \{0\}, \quad M_2 = i\mathbb{R}, \quad M_3 = i\alpha\mathbb{Z} \quad (\alpha > 0),$$
$$M_4 = i\alpha\mathbb{Z} + i\beta\mathbb{Z} = \{in\alpha + im\beta : n, m \in \mathbb{Z}\} \quad (\alpha, \beta \in \mathbb{R}),$$
$$M_5 = \{0\} \cup \{i\lambda : \lambda \in \mathbb{R}, |\lambda| \geqslant 1\},$$
$$M_6 = \bigcup_{n=0}^{\infty}\{i\lambda : \lambda \in \mathbb{R}, n\alpha \leqslant |\lambda| \leqslant n\beta\} \quad (0 < \alpha \leqslant \beta, \alpha, \beta \in \mathbb{R}).$$

In the following we consider the boundary spectrum of several semigroups. The notation M_k refers to the sets just defined.

Examples 3.2.6 (i) For the Laplacian Δ on $\mathbb{R}^n$ or the second derivative on $[0, 1]$ with Neumann boundary conditions the boundary spectrum is M_1.

(ii) The first derivative on $\mathbb{R}$ or $\mathbb{R}_+$ is an example where the boundary spectrum of the generator is M_2.

(iii) The rotation semigroup on $C(\Gamma)$ (see Part-I, Example 3.5.6) with period $2\pi/\alpha$ has boundary spectrum M_3.

(iv) For the semigroup on $C(\Gamma \times \Gamma)$ given by

$$(T(t)f)(z,w) = f(z \cdot \mathrm{e}^{\mathrm{i}\alpha t}, w \cdot \mathrm{e}^{\mathrm{i}\beta t}) \quad (f \in C(\Gamma \times \Gamma), (z,w) \in \Gamma \times \Gamma)$$

we have $P\sigma(A) = M_4$. If α/β is irrational, then this is a dense subset of $\mathrm{i}\mathbb{R}$ and $\sigma_b(A) = \sigma(A) = \mathrm{i}\mathbb{R}$.

(v) Consider

$$D := \{z \in \mathbb{C} \colon |z| \leqslant 1\} = \{r \cdot \mathrm{e}^{\mathrm{i}\omega} \colon r \in [0,1], \omega \in \mathbb{R}\}$$

and a strictly positive function $\kappa \in C[0,1]$. The flow on D governed by the differential equation $\dot{r} = 0$, $\dot{\omega} = \kappa(r)$ induces a strongly continuous semigroup on $C(D)$ (which is given by

$$(T(t)f)(z) = f(z \cdot \mathrm{e}^{\mathrm{i}\kappa(|z|)t}).$$

The boundary spectrum is M_6 with

$$\alpha := \inf_r \kappa(r) \quad \text{and} \quad \beta := \sup_r \kappa(r).$$

In particular, for $\kappa(r) = 1 + r$ we obtain as boundary spectrum the set M_5.

(vi) Suppose M is a closed cyclic subset of $\mathrm{i}\mathbb{R}$, $M = \bigcup_{\alpha \in S} \mathrm{i}\alpha\mathbb{Z}$ for a suitable $S \subseteq \mathbb{R}$ (e.g., $S = MS = -\mathrm{i}M$).

The space

$$E_1 := \{(f_\alpha)_{\alpha \in S} \colon f_\alpha \in C(\Gamma), \sup_\alpha \|f_\alpha\| < \infty\}$$

is a Banach space under the norm

$$\|(f_\alpha)\| := \sup_\alpha \|f_\alpha\|.$$

The closure of the linear subspace

$$E_0 := \{(f_\alpha) \in E_1 \colon f_\alpha \neq 0 \text{ only for finitely many } \alpha \in S\}$$

is isomorphic to $C_0(X)$ where X is the topological sum of $|S|$ copies of Γ.

Let $(T_\alpha(t))_{t \geqslant 0}$ denote the rotation semigroup on $C(\Gamma)$ with period $2\pi/\alpha$, then we define a semigroup $(T(t))_{t \geqslant 0}$ on $E := C_0(X)$ as

$$(T(t)(f_\alpha)) := (T_\alpha(t) f_\alpha) \quad ((f_\alpha)_{\alpha \in S} \in E).$$

This is a positive semigroup on $E = C_0(X)$ whose boundary spectrum is precisely the given closed cyclic set M. We leave the verification as an exercise.

Our first result concerns cyclicity of the eigenvalues contained in the boundary spectrum, i.e., of the set

$$P\sigma_b(A) := P\sigma(A) \cap \sigma_b(A) = \{\lambda \in P\sigma(A) : \operatorname{Re}(\lambda) = s(A)\}$$

$$\mathsf{P\sigma_b(A) := P\sigma(A) \cap \sigma_b(A) = \{\lambda \in P\sigma(A) : \operatorname{Re}(\lambda) = s(A)\}}.$$

It is an almost straightforward consequence of Theorem 3.2.4.

Proposition 3.2.7 *Assume that for some $t_0 > 0$ there is a strictly positive measure φ such that $T(t_0)'\varphi = \exp(s(A)t_0) \cdot \varphi$. Then $P\sigma_b(A)$ is imaginary additively cyclic.*

Proof. If $P\sigma_b(A)$ is empty, there is nothing to prove. Otherwise we have $s(A) > -\infty$. In view of the rescaling procedure we may assume $s(A) = 0$. By Proposition 3.1.5 (ii) there exists $\varphi \gg 0$ such that $T(t)'\varphi = \varphi$ for all $t \geqslant 0$. Given $\mathrm{i}\alpha \in P\sigma_b(A)$ then there is $h \in C_0(X)$, $h \neq 0$, such that $Ah = \mathrm{i}\alpha h$ or $T(t)h = \mathrm{e}^{\mathrm{i}\alpha t}h$ for all t (Part-I, Corollary 3.6.4).

Then we have

$$|h| = |\mathrm{e}^{\mathrm{i}\alpha t}h| = |T(t)h| \leqslant T(t)|h| \text{ or } T(t)|h| - |h| \geqslant 0$$

$$\mathsf{|h| = |e^{i\alpha t}h| = |T(t)h| \leqslant T(t)|h| \text{ or } T(t)|h| - |h| \geqslant 0,} \tag{3.2.14}$$

and

$$\langle T(t)|h| - |h|, \varphi\rangle = \langle |h|, T(t)'\varphi\rangle - \langle |h|, \varphi\rangle = 0. \tag{3.2.15}$$

Since φ is strictly positive, Eq. (3.2.14) and Eq. (3.2.15) imply that $T(t)|h| = |h|$ for $t \geqslant 0$ or equivalently $A|h| = 0$. Now Theorem 3.2.4 implies that $Ah^{[n]} = \mathrm{i}n\alpha h^{[n]}$ $(n \in \mathbb{Z})$. □

Concerning the hypothesis $T(t_0)'\varphi = \exp(s(A)t_0) \cdot \varphi \gg 0$ we recall that in case X is compact there are always positive linear forms such that $T(t)'\varphi = \mathrm{e}^{s(A)t}\varphi$ (see Theorem 3.1.6). If the semigroup is irreducible, then one also has $\varphi \gg 0$ (see Section 3.3 below).

In a second result we consider semigroups having compact resolvent. An important step of the proof is isolated as a lemma.

Before stating it, we recall that given a closed ideal $I \subseteq C_0(X)$, then I as well as $C_0(X)/I$ are spaces of continuous functions on a locally compact space vanishing at infinity. More precisely, if

$$I = \{f \in C_0(X) : f|_M = 0\}$$

for a suitable closed subset $M \subseteq X$, then $I \cong C_0(X\setminus M)$ $I \cong C_0(X \setminus M)$ and $C_0(X)/I \cong C_0(M)$ (cf. Chapter 1). It follows that given another closed ideal $J = \{f \in C_0(X) : f|_N = 0\}$ such that $I \subseteq J$ i.e., $N \subseteq M$, then J/I can be identified with $C_0(M\setminus N)C_0(M \setminus N)$.

We do not use this concrete representation of J/I. However, this shows that we stay within our setting of Banach spaces of continuous functions on locally compact spaces.

Lemma 3.2.8 *Suppose A is the generator of a positive semigroup $\mathcal{T}$ such that the spectral bound $s(A)$ is a pole of the resolvent of order k. Then there is a sequence*

$$I_{-1} := \{0\} \subseteq I_0 \subsetneq I_1 \subsetneq \ldots \subsetneq I_k := E$$

$$\mathsf{I}_{-1} := \{0\} \subseteq \mathsf{I}_0 \subsetneq \mathsf{I}_1 \subsetneq \cdots \subsetneq \mathsf{I}_k := \mathsf{E} \tag{3.2.16}$$

of $\mathcal{T}$-invariant closed ideals satisfying the following. Denoting by A_n $(n = 0, 1, \ldots, k)$ the generator of the semigroup on I_n/I_{n-1} which is induced by $\mathcal{T}$ we have

(i) $s(A_0) < s(A)$,

(ii) *If $n \geqslant 1$, then $s(A_n) = s(A)$ is a first order pole of the resolvent $R(\cdot, A_n)$. The corresponding residue is a strictly positive operator.*

Proof. We can assume that $s(A) = 0$ and we will denote the negative coefficients of the Laurent series of $R(\cdot, A)$ at 0 by Q_n. Thus the following relations hold (see Part-I, Example 3.3.6),

$$Q_n = \frac{1}{2\pi i}\int_\gamma z^{n-1}R(z,A)\,\mathrm{d}z \quad (n \in \mathbb{N}),$$
$$Q_n \neq 0 \text{ if } n \leqslant k \text{ and } Q_n = 0 \text{ for } n > k,$$
$$Q_n = A^{n-1}Q_1 \quad (n \in \mathbb{N}); \quad Q_k = \lim_{z\to 0} z^k \cdot R(z,A).$$

$$\begin{aligned} &Q_n = \frac{1}{2\pi \mathrm{i}}\int_\gamma z^{n-1}R(z,A)\,\mathrm{d}z \quad (n \in \mathbb{N}),\\ &Q_n \neq 0 \text{ if } n \leqslant k \text{ and } Q_n = 0 \text{ for } n > k,\\ &Q_n = A^{n-1}Q_1 \quad (n \in \mathbb{N}); \quad Q_k = \lim_{z\to 0} z^k \cdot R(z,A). \end{aligned} \tag{3.2.17}$$

We define I_n as follows $(n = 0, 1, \ldots, k-1)$.

$$I_n := \{f \in E : Q_{n+1}|f| = Q_{n+2}|f| = \ldots = Q_k|f| = 0\}$$

$$\mathsf{I}_n := \{\mathsf{f} \in \mathsf{E} : \mathsf{Q}_{n+1}|\mathsf{f}| = \mathsf{Q}_{n+2}|\mathsf{f}| = \cdots = \mathsf{Q}_k|\mathsf{f}| = 0\}.$$

At first we restrict our attention to I_{k-1}.

Since $R(\lambda, A)$ is positive if $\lambda > 0$ (see Corollary 3.1.3), it follows from Eq. (3.2.17) that Q_k is a positive bounded operator, hence

$$I_{k-1} = \{f \in E : Q_k|f| = 0\}$$

is a closed ideal. Since Q_k commutes with $R(\lambda, A)$ (see Eq. (3.2.17)), it follows that I_{k-1} is a $\mathcal{T}$-invariant ideal. By Part-I, Corollary 3.4.3 the generators $A_{|I_{k-1}}$ and A_k

induced by A on I_{k-1} and E/I_{k-1}, respectively, have a pole at 0. The coefficients of the Laurent series are the operators induced by Q_n on E/I_{k-1} and I_{k-1}, respectively.

Suppose that the pole order of $R(\cdot, A_k)$ is greater than 1, say m. Then $Q_{m/} = \lim_{z\to 0} z^m R(z, A_k)\, Q_{m/I_{k-1}} = \lim_{z\to 0} z^m R(z, A_k)$ is a positive non-zero operator, hence we find for every $x \in E_+$ an element $y \in I_{k-1}$ such that $Q_m x + y \geqslant 0$. Then we have

$$0 \leqslant Q_k|Q_m x + y| = Q_k Q_m x + Q_k y = Q_{k+m-1}x + Q_k y = 0 + Q_k y \leqslant Q_k|y| = 0$$

hence $Q_m x = (Q_m x + y) - y \in I_{k-1}$ $(x \in E_+)$. It follows that $Q_{m/} = 0\, Q_{m/I_{k-1}} = 0$ which is a contradiction.

So far we know that the resolvent of A_k has a pole of order $\leqslant 1$. Moreover, since $Q_{k|I_{k-1}} = 0$, the resolvent of $A_{|I_{k-1}}$ has a pole of order $\leqslant k-1$. From Part-I, Corollary 3.4.3 it follows that the pole order of A_k and $A_{|I_{k-1}}$ is precisely 1 and $k-1$, respectively. The residue $Q_{1/I_{k-1}} = \lim_{z\to 0} zR(z, A_k)\, Q_{1/I_{k-1}} = \lim_{z\to 0} zR(z, A_k)$ is positive since $R(z, A_k) \geqslant 0$ for $z > 0$ (see Corollary 3.1.3). It remains to prove that it is strictly positive. We assume $Q_{1/I_{k-1}}(|x + I_{k-1}|) = 0$ which means $Q_1|x| \in I_{k-1}$ hence $Q_k|x| = A^{k-1}Q_1|x| = 0$, that is, $x \in I_{k-1}$ or $x + I_{k-1} = 0$.

Applying what we have proved so far to I_{k-1} and $A_{|I_{k-1}}$ we obtain I_{k-2}, A_{k-1}, and so on. After k steps $(n = 1)$ we conclude that I_0 is $\mathcal{T}$-invariant and that the order of the pole of $R(\cdot, A_{|I_0})$ is 0, implying that $0 \in \varrho(A_{|I_0})$. Since $A_{|I_0}$ generates a positive semigroup and $R(\lambda, A_{|I_0}) = R(\lambda, A)_{|I_0}$ is positive for $\lambda > 0$, it follows from Corollary 3.1.3 that $s(A_0) = s(A_{|I_0}) < 0$. □

One can check the different steps of the proof by studying the following example. Consider this matrix as generator on $\mathbb{C}^4$

$$\begin{pmatrix} -1 & a & b & c \\ 0 & 0 & d & e \\ 0 & 0 & 0 & f \\ 0 & 0 & 0 & 0 \end{pmatrix} \qquad \text{where } a, b, c, d, e, f \geqslant 0.$$

The result is summarized in Table 3.1 on the next page where e_j are the unit vectors $(e_j := (\delta_{jk}))$. This example also shows that the operators $Q_{k-1}, \ldots, Q_1$ are not necessarily positive e.g., for $a > 0$, $b = c = 0$, $d = e = f = 2$.

A more general (and more interesting) example is the following. Suppose that A_i $(i = 1, \ldots, n)$ are generators of positive semigroups on $C_0(X)$ such that $s(A_i) = 0$ is a first order pole of the resolvent. And let A_{ij} $(1 \leqslant i < j \leqslant n)$ be positive bounded operators on $C_0(X)$. Then

$$A := \begin{pmatrix} A_1 & A_{12} & \cdots & A_{1n} \\ 0 & A_2 & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_n \end{pmatrix}$$

is the generator of a positive semigroup on

$$C_0(X, \mathbb{C}^n) \cong C_0(X) \times C_0(X) \times \cdots \times C_0(X),$$

and $s(A) = 0$ is a pole of the resolvent of order k where $1 \leqslant k \leqslant n$.

Table 3.1 Matrix Example

			pole order	I_0 I_0	I_2 I_1	I_2	I_3
$d > 0$,	$f > 0$,	$e \geqslant 0$	3	$\langle e_1 \rangle$	$\langle e_1, e_2 \rangle$	$\langle e_1, e_2, e_3 \rangle$	$\mathbb{C}^4$
$d = 0$,	$f > 0$,	$e \geqslant 0$	2	$\langle e_1 \rangle$	$\langle e_1, e_2, e_3 \rangle$	$\mathbb{C}^4$	
$d = 0$,	$f = 0$,	$e > 0$	2	$\langle e_1 \rangle$	$\langle e_1, e_2, e_3 \rangle$	$\mathbb{C}^4$	
$d > 0$,	$f = 0$,	$e > 0$	2	$\langle e_1 \rangle$	$\langle e_1, e_2 \rangle$	$\mathbb{C}^4$	
$d > 0$,	$f = 0$,	$e = 0$	2	$\langle e_1 \rangle$	$\langle e_1, e_2, e_4 \rangle$	$\mathbb{C}^4$	

Theorem 3.2.9 *Suppose A is the generator of a positive semigroup on $C_0(X)$ such that every point of $\sigma_b(A)$ is a pole of the resolvent. Then $P\sigma_b(A) = \sigma_b(A)$ is cyclic.*

Proof. If $\sigma(A) = \emptyset$, there is nothing to prove, thus we can assume that $s(A) = 0$. In view of Lemma 3.2.8 and Part-I, Corollary 3.4.3 (i) we can assume that $s(A)$ is a first order pole with strictly positive residue, we call it Q. We have $AQ = QA = s(A)A = 0$ $AQ = QA = s(A)Q = 0$ (see Part-I, Example 3.3.6), hence

$$QT(t) = T(t)Q = Q \text{ for all } t \geqslant 0. \tag{3.2.18}$$

If $Ah = \mathrm{i}\alpha h$ for some $\alpha \in \mathbb{R}$, $h \neq 0$, then $T(t)h = \mathrm{e}^{\mathrm{i}\alpha t} h$ (by Part-I, Corollary 3.6.4). Hence

$$|h| = |\mathrm{e}^{\mathrm{i}\alpha t} h| = |T(t)h| \leqslant T(t)|h|,$$

or equivalently, $T(t)|h| - |h| \geqslant 0$. By Eq. (3.2.18) we have $Q(T(t)|h| - |h|) = 0$. Since Q is strictly positive, it follows that

$$T(t)|h| = |h| \text{ or } A|h| = 0.$$

Now we can apply Theorem 3.2.4 and obtain

$$Ah^{[n]} = \mathrm{i}n\alpha h^{[n]} \text{ for every } n \in \mathbb{Z}.$$

This shows that $P\sigma_b(A) = \sigma(A) \cap \mathrm{i}\mathbb{R}$ is cyclic. □

If A has compact resolvent, then every point of $\sigma(A)$ is a pole of the resolvent (see Part-I, Example 3.3.6) hence we have

Corollary 3.2.10 *If A has compact resolvent, then $P\sigma_b(A) = \sigma_b(A)$ is cyclic.*

If it is known that the boundary spectrum of a generator is cyclic and nonvoid, the following alternative holds.

$$\textit{Either } \sigma_b(A) = \{s(A)\} \textit{ or else } \sigma_b(A) \textit{ is an infinite unbounded set.} \tag{3.2.19}$$

If one can exclude the second alternative, then there is a unique spectral value having maximal real part. A real spectral value λ_0 of a generator A is called *dominant* provided that $\mathrm{Re}(\lambda) < \lambda_0$ for every $\lambda \in \sigma(A)$, it is called *strictly dominant* if for some $\delta > 0$ one has $\mathrm{Re}(\lambda) \leqslant \lambda_0 - \delta$ for every $\lambda \in \sigma(A)$, $\lambda \neq \lambda_0$.

The assumptions of Corollary 3.2.10 do not imply that $s(A)$ is dominant, the rotation semigroup (Part-I, Example 3.5.6) is a counterexample.

Corollary 3.2.11 *Assume that for some $t_0 > 0$ one has*

$$r_{ess}(T(t_0)) < r(T(t_0)),$$

Then $s(A)$ is a strictly dominant eigenvalue.

This applies e.g., whenever $T(t_0)$ is compact and $r(T(t_0)) > 0$.

Proof. If $s(A)$ is not strictly dominant, then we have by Theorem 3.2.9 and Part-I, Corollary 3.6.5 that

$$\{\lambda \in \sigma(A) \colon \mathrm{Re}(\lambda) > s(A) - r\}$$

contains infinitely many eigenvalues for every $r > 0$. From Part-I, Corollary 3.6.4 it follows that $\{\lambda \in \sigma(T(t)) \colon |\lambda| > r\}$ $\{\lambda \in \sigma(T(t_0)) \colon |\lambda| > r\}$ contains infinitely many eigenvalues (counted according to their multiplicities) for every $r < \exp(s(A)t) = r(T(t))r < \exp(s(A)t_0) = r(T(t_0))$. This contradicts the assumption $r_{\mathrm{ess}}(T(t)) < r(T(t))$ $r_{\mathrm{ess}}(T(t_0)) < r(T(t_0))$ (see Eq. (3.3.7) on page 85). □

Corollary 3.2.12 *Suppose A has compact resolvent and non-empty spectrum. If the corresponding semigroup is eventually norm continuous (e.g., if it is holomorphic or differentiable), then there is a strictly dominant eigenvalue admitting a positive eigenfunction.*

Proof. Since $(T(t))_{t \geqslant 0}$ is eventually norm continuous,

$$\{\lambda \in \sigma(A) \colon \mathrm{Re}(\lambda) \geqslant s(A) - r\}$$

is compact for every $r > 0$ (see Part-I, Theorem 2.1.19) and this set does not have accumulation points because A has compact resolvent. In other words, it is a finite set. The assertion now follows from Theorem 3.2.9 and Corollary 3.1.4. □

We now consider some examples. The first one shows that there are positive semigroups with $P\sigma_b(A)$ being not cyclic. It is unknown whether there are positive semigroups where $\sigma_b(A)$ is not cyclic.

Example 3.2.13 Consider $E = C(\Gamma) \times C_0(\mathbb{R})$ ($\cong C_0(\Gamma \dot\cup \mathbb{R})$). We fix a positive function $k \in C_0(\mathbb{R})$ with compact support. The operator A given by

$$\begin{aligned} A(f,g) &:= (f', g' + \frac{1}{2\pi}\int_0^{2\pi} f(\vartheta)\ \mathrm{d}\vartheta \cdot k) \\ D(A) &:= \{(f,g) \in E\colon f, g \in C^1, g' \in C_0(\mathbb{R})\} \end{aligned} \tag{3.2.20}$$

generates a semigroup $(T(t))_{t\geqslant 0}$ which is given by

$$\begin{aligned} T(t)(f,g) &:= (f_t, g_t) \text{ with} \\ f_t(\vartheta) &:= f(\vartheta + t), \\ g_t(x) &:= g(x+t) + \frac{1}{2\pi}\int_0^{2\pi} f(\vartheta)\,\mathrm{d}\vartheta \cdot \int_x^{x+t} k(u)\ \mathrm{d}u. \end{aligned} \tag{3.2.21}$$

Then $(T(t))_{t\geqslant 0}$ is a positive semigroup and $\|T(t)\| \leqslant (1 + \|k\|_1)$. In particular, $s(A) \leqslant \omega_0(A) \leqslant 0$. It is easy to see that 0 is not an eigenvalue of A, while all $\mathrm{i}k$, $k \in \mathbb{Z}$, $k \neq 0$ $\mathrm{i}n$, $n \in \mathbb{Z}$, $n \neq 0$ are eigenvalues, the corresponding eigenfunctions being $(e_k, 0)$ with $e_k(\vartheta) = \mathrm{e}^{\mathrm{i}k\vartheta}$ $(e_n, 0)$ with $e_n(\vartheta) = \mathrm{e}^{\mathrm{i}n\vartheta}$.

Example 3.2.14 (i) *One-dimensional Schrödinger operator.*

Let $X = \mathbb{R}$, $E = C_0(X)$ and $V\colon \mathbb{R} \to \mathbb{R}$ be a continuous function such that $\inf_x V(x) > -\infty$.

If we define

$$\begin{aligned} (Af)(x) &:= f''(x) - V(x)f(x), \\ D(A) &:= \{f \in C_0(X)\colon f \in C^2, Af \in C_0(X)\}, \end{aligned} \tag{3.2.22}$$

then A is the generator of a positive semigroup.

In case $\lim_{|x|\to\infty} V(x) = \infty$, A has compact resolvent. Then, by Corollary 3.2.10, there exists a dominant real eigenvalue with corresponding positive eigenfunction. Actually, the eigenfunction f is strictly positive. In fact, if $f \in C^2$, $f \geqslant 0$ and $f(x_0) = 0$ for some x_0, then $f'(x_0) = 0$. Therefore the uniqueness theorem for ordinary differential equations implies that f is identically zero.

(ii) *A retarded linear differential equation.*

Consider $E = C[-1, 0]$ and define A_m and A_0 as follows,

$$A_m f := f', \quad f \in D(A_m) = C^1[-1, 0], \tag{3.2.23}$$

$$A_0 f := f', \quad f \in D(A_0) = \{f \in C^1[-1, 0]\colon f'(0) = 0\}. \tag{3.2.24}$$

Then A_0 generates a contraction semigroup $(T_0(t))_{t\geqslant 0}$ which is given by

$$(T_0(t)f)(x) = \begin{cases} f(x+t) & \text{if } x + t \leqslant 0, \\ f(0) & \text{if } x + t \geqslant 0. \end{cases} \tag{3.2.25}$$

This semigroup is positive, eventually norm continuous, $T_0(t) = \delta_0 \otimes \mathbf{1}$ for $t \geqslant 1$ and has compact resolvent. Given a linear functional φ on $C[-1,0]$, we consider

$$A_\varphi \coloneqq A_{m|D(A_\varphi)} \text{ with } D(A_\varphi) \coloneqq \{f \in C^1[-1,0] \colon f'(0) = \langle f, \varphi\rangle\}. \tag{3.2.26}$$

Denoting the exponential function $x \mapsto \mathrm{e}^{\lambda x}$ by e_λ, we have for real λ and $\lambda > \|\varphi\|$

$$\begin{aligned} &\mathrm{Id} - 1/\lambda \cdot \varphi \otimes e_\lambda \text{ is a bijection of } D(A_\varphi) \text{ onto } D(A_0) \\ &\qquad \text{and} \quad \lambda - A_\varphi = (\lambda - A_0)(\mathrm{Id} - 1/\lambda \cdot \varphi \otimes e_\lambda). \end{aligned} \tag{3.2.27}$$

Using the Neumann series expansion of $(\mathrm{Id} - 1/\lambda \cdot \varphi \otimes e_\lambda)^{-1}$ one obtains the estimate

$$\|(\mathrm{Id} - 1/\lambda \cdot \varphi \otimes e_\lambda)^{-1}\| \leqslant \lambda/(\lambda - \|\varphi\|) \quad \text{if } \lambda > \|\varphi\|. \tag{3.2.28}$$

It follows from Eq. (3.2.27) and Eq. (3.2.28) that for $\lambda > \|\varphi\|$, the resolvent $R(\lambda, A_\varphi)$ exists and satisfies

$$\|R(\lambda, A_\varphi)\| \leqslant \lambda/(\lambda - \|\varphi\|) \cdot 1/\lambda = 1/(\lambda - \|\varphi\|).$$

Then the Hille-Yosida Theorem (Part-I, Theorem 2.1.6) implies that A_φ generates a semigroup $(T(t))$ satisfying $\|T(t)\| \leqslant \exp(\|\varphi\|t)$. Moreover, this semigroup is eventually norm continuous (see Corollary 4.3.3).

By Example 2.1.22 we have the following equivalence.

$$A_\varphi \text{ generates a positive semigroup} \iff \varphi + r\delta_0 \geqslant 0 \text{ for some } r \in \mathbb{R}. \tag{3.2.29}$$

Thus Corollary 3.2.12 is applicable if $\varphi + r\delta_0 \geqslant 0$ for some $r \in \mathbb{R}$. Since every eigenvalue of A_φ is an eigenvalue of A_m and since

$$\ker(\lambda - A_m) = \{\alpha e_\lambda \colon a \in \mathbb{C}\},$$

$$\ker(\lambda - \mathsf{A_m}) = \{\alpha \mathsf{e}_\lambda \colon \alpha \in \mathbb{C}\},$$

the spectral bound $s(A_\varphi)$ is determined by the (unique) real $\lambda \in \mathbb{R}$ such that $e_\lambda \in D(A_\varphi)$ or equivalently, λ is a solution of the so-called *characteristic equation*

$$\lambda = \varphi(e_\lambda), \quad \lambda \in \mathbb{R}, \tag{3.2.30}$$

since the assumption $\varphi + r\delta_0 \geqslant 0$ implies that the function $\lambda \mapsto \varphi(e_\lambda)$ is strictly decreasing and $\lim_{\lambda\to\infty}\langle e_\lambda, \varphi\rangle > -\infty$, $\lim_{\lambda\to-\infty}\langle e_\lambda, \varphi\rangle = \infty$ $\lim_{\lambda\to\infty} \varphi(e_\lambda) > -\infty$, $\lim_{\lambda\to-\infty} \varphi(e_\lambda) = \infty$ unless $\varphi = r_0\delta_0$ for some $r_0 \in \mathbb{R}$.

We conclude this section with some additional remarks related to Theorem 3.2.9 and its corollaries.

Remarks 3.2.15 (i) If $s(A)$ is a pole of the resolvent, then for generators of positive semigroups one has the following equivalences.

(a) $s(A)$ is a first order pole.

(b) For every $0 < f \in \ker(s(A) - A)$ there exists $0 \leqslant \varphi \in \ker(s(A) - A')$ such that $\langle f, \varphi \rangle > 0$.

(c) For every $0 < \varphi \in \ker(s(A) - A')$ there exists $0 \leqslant f \in \ker(s(A) - A)$ such that $\langle f, \varphi \rangle > 0$.

In particular, if $\ker(s(A) - A)$ contains a strictly positive function or if $\ker(s(A) - A')$ contains a strictly positive measure, then $s(A)$ is a first order pole.

We sketch the proof of the equivalence of (a) and (b) assuming that $s(A) = 0$. If 0 is a first order pole, then the residue P is a positive projection satisfying $PE = \ker(A)$, $P'E' = \ker(A')$ (see Part-I, Example 3.3.6). Thus given $0 < f \in \ker(A)$ and any $0 \leqslant \psi \in E'$ such that $\langle f, \psi \rangle > 0$, we have for $\varphi := P'\psi \in \ker(A')$, and

$$\langle f, \varphi \rangle = \langle f, P'\psi \rangle = \langle Pf, \psi \rangle = \langle f, \psi \rangle > 0.$$

To prove the reverse direction, we first observe that the highest coefficient Q_k of the Laurent expansion is a positive operator. Thus if 0 is a pole of order $k \geqslant 2$, we choose $0 < h \in E$ such that $f := Q_k h > 0$. Then $Af = AQ_k h = 0$ and for every $\varphi \in \ker(A')$ we have

$$\langle f, \varphi \rangle = \langle Q_k h, \varphi \rangle = \langle h, Q_k' \varphi \rangle = \langle h, Q_{k-1}' A' \varphi \rangle = 0.$$

(ii) If a linear operator S on $C_0(X)$ is weakly compact, then S^2 is compact (see Proposition 4.2.4 (ii)). Therefore every non-zero spectral value of a weakly compact operator is a pole of the resolvent. This shows that Theorem 3.2.9 is applicable if either $T(t_0)$ is weakly compact for some t_0 or $R(\lambda, A)$ is weakly compact for some $\lambda \in \varrho(A)$. We quote two criteria for weak compactness.

> If $T \in \mathcal{L}(C(K))$, K compact, is positive, then it is weakly compact if and only if its biadjoint T'' maps bounded Borel functions into $C(K)$ (see Proposition 4.2.4). (3.2.31)

> A positive operator T on $C_0(X)$ which is dominated by a finite rank operator is weakly compact. (3.2.32)

Actually, its adjoint T' is dominated by a finite rank operator as well, hence it maps the unit ball in an order interval. Since the dual space has order continuous norm, order intervals are weakly compact (see Part-III, Section 1.5). It follows that T' is weakly compact, hence so is T.

(iii) Stronger results than Theorem 3.2.9 will be proved in Part-III, Chapter 3. Actually, assuming only that $s(A)$ is a pole of finite algebraic multiplicity one can show that $\sigma_b(A)$ contains only poles of finite multiplicity (Part-III, Lemma 3.3.13). In Part-III, Corollary 3.2.12 we will show that $\sigma_b(A)$ is cyclic whenever $s(A)$ is a pole of the resolvent.

(iv) Example 3.2.14 (ii) can be extended to systems of functional differential equations even the infinite dimensional case. For details we refer to Section 4.3.

3.3 Irreducible Semigroups

In the case of matrices it is well known that considerably stronger results are available if one considers positive matrices which are irreducible. The concept of irreducibility can be extended to our setting and in many cases one can check easily whether a given semigroup has this property (see Example 3.3.4). We will show that irreducible semigroups have many interesting properties. For example, the spectrum $\sigma(A)$ is always non-empty, positive eigenfunctions are strictly positive and if $s(A)$ is a pole, it is algebraically (and geometrically) simple (see Proposition 3.3.5). Moreover, in certain cases irreducibility ensures that $\sigma_b(A)$ and $P\sigma_b(A)$ are not only cyclic subsets but *subgroups* (see Theorem 3.3.6 and Theorem 3.3.11 for details).

We start the discussion with several, mutually equivalent, definitions of irreducibility.

Definition 3.3.1 A positive semigroup $\mathcal{T} = (T(t))$ on $E = C_0(X)$, X locally compact, with generator A is called *irreducible* if one of the following, mutually equivalent conditions is satisfied.

(a) There is no $\mathcal{T}$-invariant closed ideal except $\{0\}$ and E.

(b) Given $0 < f \in E$, $0 < \varphi \in E'$, then $\langle T(t_0)f, \varphi\rangle > 0$ for some $t_0 \geqslant 0$.

(c) For every $f > 0$ we have $\bigcup_{t\geqslant 0}\{x \in X\colon (T(t)f)(x) > 0\} = X$.

(d) For some (every) $\lambda > s(A)$ there exists no closed ideal which is invariant under $R(\lambda, A)$ except $\{0\}$ and E.

(e) For some (every) $\lambda > s(A)$, we have $R(\lambda, A)f$ is strictly positive whenever $f > 0$.

(f) $\bigcup_{t\geqslant 0} \operatorname{supp} T(t)'\delta_x$ is dense in X for every $x \in X$.

That these six conditions are actually equivalent can be seen as follows.

(a) $\Longrightarrow$ (b): Suppose there are $0 < f \in E$, $0 < \varphi \in E'$ such that $\langle T(t)f, \varphi\rangle = 0$ for every $t \geqslant 0$. Then the ideal I generated by $\{T(t)f\colon t \geqslant 0\}$ satisfy $T \neq \{0\}$ and $T \subseteq \{g \in E\colon \varphi(|g|) = 0\} \neq E$satisfies $I \neq \{0\}$ and $I \subseteq \{g \in E\colon \varphi(|g|) = 0\} \neq E$. Obviously I is $\mathcal{T}$-invariant.

(b) $\Longrightarrow$ (c): Given $0 < f \in E$, $x \in X$. Then by (b) there exists t_0 such that $(T(t_0)f)(x) = \langle T(t_0)f, \delta_x\rangle > 0$.

(c) $\Longrightarrow$ (f): Suppose that $\bigcup_{t\geqslant 0} \operatorname{supp} T(t)'\delta_y$ is not dense for some $y \in X$. Then there exists $f_0 \in E$, $f_0 > 0$ such that $\operatorname{supp} f_0 \cap \operatorname{supp} T(t)'\delta_y = \emptyset$ for every $t \geqslant 0$. Hence

$(T(t)f_0)(y) = \langle f_0, T(t)'\delta_y\rangle > 0$ $(T(t)f_0)(y) = \langle f_0, T(t)'\delta_y\rangle = 0$, i.e., $y \notin \bigcup_{t\geqslant 0}\{x \in X\colon (T(t)f_0)(x) > 0\}$.

(f) $\Longrightarrow$ (e): Given $0 < f \in E$, $\lambda > \omega_0(A)$, $y \in X$, there exists $t_0 \geqslant 0$ such that $\{x\colon f(x) > 0\} \cap \operatorname{supp} T(t_0)'\delta_y \neq \emptyset$. Hence, $(T(t_0)(f)(y) = \langle f, T(t_0)'\delta_y\rangle > 0$ $(T(t_0)f)(y) = \langle f, T(t_0)'\delta_y\rangle > 0$ and therefore

$$(R(\lambda, A)f)(y) = \int_0^\infty \mathrm{e}^{-\lambda t}(T(t)f)(y)\,\mathrm{d}t > 0.$$

Since $\lambda \mapsto R(\lambda, A)f$ is decreasing in the interval $(s(A), \infty)$ (use the resolvent equation and the fact that $R(\lambda, A)$ is positive) we have $R(\lambda, A)f \gg 0$ for all $\lambda > s(A)$.

(e) $\Longrightarrow$ (f)(d): If I is a $R(\lambda, A)$-invariant ideal and $0 < f \in I$, then $g := R(\lambda, A)f \in I$. By (e), g is strictly positive, thus I has to be dense (it contains all functions of compact support).

(d) $\Longrightarrow$ (a): At first we recall that a closed linear subspace which is invariant for $R(\lambda_0, A)$ ($\lambda_0 \in \varrho(A)$), is invariant for $R(\lambda, A)$ whenever λ and λ_0 belong to the same component of $\varrho(A)$. Hence, if $\lambda_0 \in \varrho_+(A)$ then by Part-I, ExampleSection 1.3.3 every $R(\lambda_0, A)$-invariant subspace is $\mathcal{T}$-invariant and vice versa.

Remark 3.3.2 Obviously, irreducibility of a semigroup $(T(t))_{t\geqslant 0}$ is implied by the following condition.

(g) $T(t)f \gg 0$ whenever $f > 0$ and $t > 0$.

The rotation semigroup (see Part-I, Example 1.2.5) is irreducible but it does not satisfy condition (g). However, assuming that the semigroup $(T(t))$ is holomorphic, then (g) is equivalent to irreducibility. We will give a proof of this result in the more general situation of Banach lattices (see Part-III, Theorem 3.3.2 (ii)).

A semigroup $(T(t))_{t\geqslant 0}$ is irreducible if and only if $(\mathrm{e}^{-\alpha t}T(t))_{t\geqslant 0}$, $\alpha \in \mathbb{R}$ is. More generally, irreducibility is invariant under perturbations by multiplication operators. In fact, we have the following result.

Proposition 3.3.3 *Suppose A generates a positive semigroup $\mathcal{T}$ on $C_0(X)$ and let h be a continuous, bounded real-valued function on X. Then the semigroup $\mathcal{S}$ generated by $B := A + M_h$ is irreducible if and only if $\mathcal{T}$ has this property.*

Proof. Since every closed ideal is of the form $\{f \in E\colon f|_M = 0\}$ where $M \subseteq X$ is a closed subset (cf. Section 1.1), it is clear that all closed ideals are invariant under the multiplication operator M_h and M_{-h}, respectively. Thus the assertion follows from the expansions which are true for λ sufficiently large

$$R(\lambda, B) = (1 - R(\lambda, A)M_h)^{-1} R(\lambda, A) = \sum_{n=0}^{\infty} (R(\lambda, A)M_h)^n R(\lambda, A)$$

$$R(\lambda, A) = (1 - R(\lambda, B)M_{-h})^{-1} R(\lambda, B) = \sum_{n=0}^{\infty} (R(\lambda, B)M_{-h})^n R(\lambda, B). \qquad \square$$

$$\mathsf{R}(\lambda, \mathsf{B}) = (\mathrm{Id} - \mathsf{R}(\lambda, \mathsf{A})\mathsf{M}_h)^{-1} \mathsf{R}(\lambda, \mathsf{A}) = \sum_{n=0}^{\infty} (\mathsf{R}(\lambda, \mathsf{A})\mathsf{M}_h)^n \mathsf{R}(\lambda, \mathsf{A})$$

$$\mathsf{R}(\lambda, \mathsf{A}) = (\mathrm{Id} - \mathsf{R}(\lambda, \mathsf{B})\mathsf{M}_{-h})^{-1} \mathsf{R}(\lambda, \mathsf{B}) = \sum_{n=0}^{\infty} (\mathsf{R}(\lambda, \mathsf{B})\mathsf{M}_{-h})^n \mathsf{R}(\lambda, \mathsf{B}). \qquad \square$$

Examples 3.3.4 (i) (cf. Section 2.3). Suppose $(T(t))_{t\geqslant 0}$ is governed by a continuous semiflow $\varphi\colon \mathbb{R}_+ \times X \to X$, i.e., $T(t)f = f \circ \varphi_t$ $(f \in C_0(X))$. Then the following assertions are equivalent.

(a) $(T(t))_{t\geqslant 0}$ is irreducible.

(b) There is no closed subset of X which is φ-invariant except $\emptyset$ and X.

(c) Every orbit $\{\varphi(t, x)\colon t \in \mathbb{R}_+\}$ is dense in X.

More generally, these equivalences are also valid if the semigroup $(T(t))$ is given by $T(t)f = h_t \cdot (f \circ \varphi_t)$ where h_t are suitable continuous, strictly positive, bounded functions on X.

(ii) Suppose that the semigroup $(T(t))_{t\geqslant 0}$ has the following form. There exist a strictly positive measure μ on X and a positive continuous function $k\colon (0, \infty) \times X \times X \to \mathbb{R}$ such that

$$(T(t)f)(x) = \int_X k(t, x, y) f(y)\, \mathrm{d}\mu(y) \quad (t > 0, f \in C_0(X), x \in X). \tag{3.3.1}$$

Then $(T(t))_{t\geqslant 0}$ is irreducible if and only if $\bigcup_{t>0} \operatorname{supp}\{k(t, x, .)\} \bigcup_{t>0} \operatorname{supp} k(t, x, \cdot)$ is dense in X for every $x \in X$.

(iii) We consider the first derivative $Af = f'$ (cf. Part-I, Section 1.2.4). If $E = C_0(\mathbb{R})$, then the corresponding semigroup $(T(t))$ is not irreducible. Note however, that there is no closed invariant ideal I with $\{0\} \neq I \neq E$ which is invariant under the group $(T(t))_{t\in\mathbb{R}}$ generated by A. For $E = C_0[0, \infty)$ and $E = C_0(-\infty, 0)$ the corresponding semigroups are reducible (i.e., not irreducible) as well. If $E = C_{2\pi}(\mathbb{R})$ (i.e., the 2π-periodic functions), then $Af = f'$ generates an irreducible semigroup on E. It is (isomorphic to) the semigroup of rotations on the unit circle.

(iv) (cf. Example 3.2.14 (ii)) Now we consider $Af = f'$ on $E = C[-1, 0]$ with $D(A_\varphi) = \{f \in C^1\colon f'(0) = \varphi(f)\}$ where the linear functional φ satisfies $\varphi + \alpha\delta_0 \geqslant 0$ for some $\alpha \in \mathbb{R}$ (see Example 2.1.22). The corresponding semigroup is irreducible if and only if $-1 \in \operatorname{supp} \varphi$.

(v) The second derivative $Af = f''$ generates an irreducible semigroup on $C_0(\mathbb{R})$ and on $C_0(0,1)$. With Neumann boundary conditions (or more generally, $f'(0) = \alpha_0 f(0)$, $f'(1) = \alpha_1 f(1)$ where $\alpha_0, \alpha_1 \in \mathbb{R}$) the second derivative generates an irreducible semigroup on $C([0,1])$ $C[0,1]$ (cf. Part-I, Section 1.2.7).

The operator $Af = f'' - Vf$ on $C_0(\mathbb{R})$, where V is continuous, real-valued with $\inf V(x) > -\infty$ (see Example 3.2.14 (i)) also generates an irreducible semigroup. This can be derived from the maximum principle as follows. For $\lambda > -\inf V(x)$, $f \in C_0(\mathbb{R})$, $g := R(\lambda, A)f$ we have $g \in C^2$ and $g'' - (\lambda + V)g = -f$. If $f > 0$, then $g > 0$, hence Protter & Weinberger [20, Chap.I, Theorem 3] implies that g is strictly positive.

(vi) The Laplacian Δ generates an irreducible semigroup on $C_0(\mathbb{R}^n)$ as can be seen easily from Part-I, Section 1.2.8. More general elliptic operators will be discussed below (see Example 3.3.10 (ii)).

We now return to the general situation and show that irreducible semigroups possess several interesting properties.

Proposition 3.3.5 *Suppose A is the generator of a strongly continuous semigroup on $C_0(X)$ which is irreducible. Then the following assertions are true.*

(i) $\sigma(A) \neq \emptyset$.

(ii) *Every positive eigenfunction of A is strictly positive.*

(iii) *Every positive eigenvector of A' is strictly positive.*

(iv) *If $\ker(s(A) - A')$ contains a positive element (e.g., if X is compact (cf. Theorem 3.1.6), then $\dim(\ker(s(A) - A) \leqslant 1$), then $\dim(\ker(s(A) - A)) \leqslant 1$.*

(v) *If $s(A)$ is a pole of the resolvent, then it is algebraically simple. The residue has the form $P = \varphi \otimes u$ where $\varphi \in E'$ and $u \in E$ are strictly positive eigenvectors of A' and A, respectively, satisfying $\langle u, \varphi \rangle = 1$.*

Proof. (i) Take any $f_0 \in C_0(X)$ which is positive and has compact support. If $\lambda > s(A)$, then $R(\lambda, A)f_0$ is strictly positive (by Definition 3.3.1 (e)), hence there exists $\varepsilon > 0$ such that $R(\lambda, A)f_0 \geqslant \varepsilon f_0$. It follows that $R(\lambda, A)^n f_0 \geqslant \varepsilon^n f_0 \geqslant 0$ for all $n \in \mathbb{N}$ and therefore

$$r(R(\lambda, A)) = \lim_{n\to\infty} \|R(\lambda, A)^n\|^{1/n} \geqslant \varepsilon > 0.$$

The assertion now follows from Part-I, Proposition 3.2.5.

(ii) Suppose $Ah = rh$ where $h \neq 0$ is positive. Then r has to be real and we have $T(t)h = \mathrm{e}^{rt}h$ (Part-I, Corollary 3.6.4). For $|f| \leqslant n \cdot h$ ($n \in \mathbb{N}$) we have

$$|T(t)f| \leqslant T(t)|f| \leqslant n \cdot T(t)h = n \cdot \mathrm{e}^{rt}h. \tag{3.3.2}$$

This shows that the ideal generated by h is invariant, hence dense by irreducibility. This is true if and only if h is strictly positive.

(iii) Suppose $A'\varphi = r\varphi$ for some $0 < \varphi \in E'$. Again r has to be real and $T(t)'\varphi = \mathrm{e}^{rt}\varphi$ $(t \geqslant 0)$. From

$$\langle |T(t)f|, \varphi\rangle \leqslant \langle T(t)|f|, \varphi\rangle = \langle |f|, \mathrm{e}^{rt}\varphi\rangle, f \in E$$

$$\langle |\mathsf{T(t)f}|, \varphi\rangle \leqslant \langle \mathsf{T(t)|f|}, \varphi\rangle = \langle |\mathsf{f}|, \mathrm{e}^{\mathsf{rt}}\varphi\rangle, \quad \mathsf{f} \in \mathsf{E} \tag{3.3.3}$$

it follows that $I := \{f \in E : \varphi(|f|) = 0\}$ is an invariant ideal. We have $I \neq E$ (because $\varphi \neq 0$), hence the irreducibility implies $I = \{0\}$, i.e., φ is strictly positive.

(iv) By (i) we know that $s(A) > -\infty$ hence we can assume without loss of generality that $s(A) = 0$. By (iii) there exists a strictly positive $\varphi \in E'$ such that $A'\varphi = 0$. It follows from Eqs. (3.2.14) and (3.2.15) on page 201 that

$$h \in \ker(A) \text{ implies } |h| \in \ker(A)$$

$$\mathsf{h} \in \ker(\mathsf{A}) \text{ implies } |\mathsf{h}| \in \ker(\mathsf{A}). \tag{3.3.4}$$

Assuming $\dim(\ker(A)) \geqslant 2$, then there is an eigenfunction $h \in \ker(A)$, $h \neq 0$ which has at least one zero in X ($h := h_1(x_0) \cdot h_2 - h_2(x_0) \cdot h_1$, where h_1, h_2 are linearly independent, $x_0 \in X$). By Eq. (3.3.4) $|h|$ is a positive eigenfunction but not strictly positive. This is a contradiction with (ii).

(v) If $s(A)$ is a pole, then there exists a corresponding positive eigenfunction (see the proof of Corollary 3.1.4). By (ii) it is even strictly positive, thus $s(A)$ is a first order pole by Remark 3.2.15 (i). The residue P is a positive operator satisfying

$$PE = \ker(s(A) - A) \text{ and } P'E' = \ker(s(A) - A'),$$

therefore the remaining assertion follows from (ii) and (iv). □

In the remainder of this section we focus our interest on the boundary spectrum of irreducible semigroups, more precisely, on the eigenvalues and the corresponding eigenfunctions of the boundary spectrum. In view of assertion (i) of Proposition 3.3.5 the assumption "$s(A) = 0$" is not crucial in the following theorem. However, it allows a simpler formulation. And in case X is compact, a $\varphi \geqslant 0$ satisfying $A'\varphi = 0$ always exists (see Theorem 3.1.6)

Theorem 3.3.6 *Suppose $(T(t))$ is an irreducible semigroup with generator A and spectral bound $s(A) = 0$. Assume that there exists a positive linear form $\varphi \neq 0$ such that $A'\varphi = 0$. If $P\sigma(A) \cap \mathrm{i}\mathbb{R}$ is non-empty, then the following assertions are true.*

(i) *$P\sigma(A) \cap \mathrm{i}\mathbb{R}$ is a (additive) subgroup of $\mathrm{i}\mathbb{R}$.*

(ii) *The eigenspaces corresponding to $\lambda \in P\sigma(A) \cap \mathrm{i}\mathbb{R}$ are one-dimensional.*

(iii) *If* $Ah = \mathrm{i}\alpha h$ $(h \neq 0, \alpha \in \mathbb{R})$, *then* h *has no zeros in* X. *In case* $\alpha = 0$ *then* $h(x)/|h(x)|$ *is constant; otherwise,* $\{h(x)/|h(x)| \colon x \in X\}$ *is a dense subset of the unit circle* $\Gamma := \{z \in \mathbb{C} \colon |z| = 1\}$.

(iv) *If* $Ah = \mathrm{i}\alpha h$ $(h \neq 0, \alpha \in \mathbb{R})$, *then*

$$S_h(D(A)) = D(A) \text{ and } S_h^{-1} \circ A \circ S_h = (A + \mathrm{i}\alpha). \tag{3.3.5}$$

In particular, spectrum and point spectrum of A *are invariant under translations by* $\mathrm{i}\alpha$.

(v) 0 *is the only eigenvalue admitting a positive eigenfunction.*

Proof. By Proposition 3.3.5 (iii) the invariant linear form φ is strictly positive and it satisfies $T(t)'\varphi = \varphi$ $(t \geqslant 0)$.

(iv) Supposing $Ah = \mathrm{i}\alpha h$ $(h \neq 0, \alpha \in \mathbb{R})$ then $A|h| = 0$ by Eqs. (3.2.14) and (3.2.15) on page 201. By Proposition 3.3.5 (ii) $|h|$ is strictly positive, thus Theorem 3.2.4 (ii) implies Eq. (3.3.5).

(ii) Assertion (iv) implies that S_h maps $\ker(\mathrm{i}\alpha + A)$ onto $\ker(A)$ whenever $\mathrm{i}\alpha \in P\sigma(A) \cap \mathrm{i}\mathbb{R}$. Moreover, we have seen in the proof of (iv) that $\ker(A) \neq \{0\}$ hence it is one-dimensional by Proposition 3.3.5 (iv).

(i) Assume that $Ah = \mathrm{i}\alpha h$, $Ag = \mathrm{i}\beta g$ $(\alpha, \beta \in \mathbb{R}, h \neq 0, g \neq 0)$. By Eq. (3.3.5) we have $S_{\bar{g}} A S_g = A + \mathrm{i}\beta$ and $S_h A S_{\bar{h}} = A - \mathrm{i}\alpha$, therefore

$$A + i(\beta - \alpha) = S_h(A + \mathrm{i}\beta)S_{\bar{h}} = S_h S_{\bar{g}} A S_g S_h^{-1}.$$

$$\mathsf{A} + \mathrm{i}(\beta - \alpha) = \mathsf{S_h}(\mathsf{A} + \mathrm{i}\beta)\mathsf{S_{\bar{h}}} = \mathsf{S_h S_{\bar{g}} A S_g S_h^{-1}}. \tag{3.3.6}$$

It follows that

$$\ker(A + i(\beta - \alpha)) = S_h S_{\bar{g}}(\ker(A)) \neq \{0\},$$

$$\ker(\mathsf{A} + \mathrm{i}(\beta - \alpha)) = \mathsf{S_h S_{\bar{g}}}(\ker(\mathsf{A})) \neq \{0\},$$

hence $\mathrm{i}(\beta - \alpha) \in P\sigma(A)\mathrm{i}(\alpha - \beta) \in P\sigma(A)$.

(v) If $Af = \lambda f$ where $f > 0$, then

$$\lambda\langle f, \varphi\rangle = \langle Af, \varphi\rangle = \langle f, A'\varphi\rangle = 0. \tag{3.3.7}$$

Since φ is strictly positive we have $\langle f, \varphi\rangle > 0$ hence $\lambda = 0$.

(iii) We already know that $Ah = \mathrm{i}\alpha h$ implies that $A|h| = 0$. It follows from Proposition 3.3.5 (ii) that h is strictly positive, i.e., h has no zeros in X. By Proposition 3.3.5 (iv) $\ker(A)$ is one-dimensional hence every eigenfunction corresponding to 0 is the scalar multiple of a strictly positive function. If $Ah = \mathrm{i}\alpha h$, $h \neq 0$, $\alpha \neq 0$, we consider $\tilde{h}(x) := h(x)/|h(x)|$. Assuming that $\tilde{h}(X)$ is not dense in Γ, there exists a sequence of polynomials $(p_n)_{n\in\mathbb{N}}$ such that

$$p_n(z) \to 1/z \text{ uniformly in } z \in \tilde{h}(X). \tag{3.3.8}$$

It follows that

$$h(x) \cdot p_n(\tilde{h}(x)) \to |h|(x)$$

uniformly in $x \in X$. Obviously, $h \cdot p_n(h)$ $h \cdot p_n(\tilde{h})$ is a linear combination of $h^{[1]}$, $h^{[2]}$, $h^{[3]}$, ..., that is, it is an element of

$$\operatorname{span}\{\ker(\mathrm{i}k\alpha - A) : k = 0, 1, 2, \ldots\}, \text{ (cf. Theorem 3.2.4).}$$

$$\operatorname{span}\{\ker(\mathrm{i}k\alpha - A) : k = 1, 2, 3, \ldots\}, \text{ (cf. Theorem 3.2.4).}$$

By Eq. (3.3.7) on the preceding page the linear form φ vanishes on $\ker(\lambda - A)$ whenever $\lambda \neq 0$. Therefore $\langle h \cdot p_n(h), \varphi\rangle = 0$ $\langle h \cdot p_n(\tilde{h}), \varphi\rangle = 0$ and we have

$$0 < \langle |h|, \varphi\rangle = \lim_{n\to\infty} \langle h \cdot p_n(h), \varphi\rangle = 0$$

$$0 < \langle |\mathsf{h}|, \varphi\rangle = \lim_{\mathsf{n}\to\infty} \langle \mathsf{h} \cdot \mathsf{p}_\mathsf{n}(\hbar), \varphi\rangle = 0$$

which is a contradiction. □

The group $P\sigma(A) \cap \mathrm{i}\mathbb{R}$ need not be discrete. For example, the semigroup described in Example 3.2.6 (iv) satisfies the assumptions of Theorem 3.3.6 if α/β is irrational. In this case $P\sigma(A) = \mathrm{i}\alpha\mathbb{Z} + \mathrm{i}\beta\mathbb{Z}$ is a dense subgroup of $\mathrm{i}\mathbb{R}$. Actually one can show that for every subgroup H of $\mathrm{i}\mathbb{R}$ there is an irreducible semigroup on $C(G)$, $G := (H_d)\tilde{}$ $\tilde{C}(G)$, $G := (H_d)\tilde{}$, such that $P\sigma(A) = H$. Here $(H_d)\tilde{}$ denotes the dual of the abelian group H equipped with the discrete topology. For details see Greiner [15, p.62].

An immediate consequence of assertion (iv) of Theorem 3.3.6 is the following corollary.

Corollary 3.3.7 *Suppose $(T(t))$ satisfies the hypotheses of Theorem 3.3.6 and let A be its generator. If k is a bounded continuous real-valued function, M_k the corresponding multiplication operator, then for $B := A + M_k$ we have $\sigma(B) + (P\sigma(A) \cap \mathrm{i}\mathbb{R}) = \sigma(B)$. In particular, $s(B) + (P\sigma(A) \cap \mathrm{i}\mathbb{R}) \subseteq \sigma(B)$.*

The next two corollaries are essentially consequences of assertion (iii) of Theorem 3.3.6. The statement of the first one can be summarized as follows. In case there are non-real eigenvalues in the boundary spectrum, then the semigroup *contains* the semigroup of rotations on the unit circle Γ.

Corollary 3.3.8 *Suppose that the hypotheses of Theorem 3.3.6 are satisfied and that there is an eigenvalue $\mathrm{i}\alpha$ of A with $\alpha > 0$ and let $\tau := 2\pi/\alpha$. Then there exists a continuous injective lattice homomorphism $j : C(\Gamma) \to C_0(X)$ such that the diagram commutes.*

$$\begin{array}{ccc} C(\Gamma) & \xrightarrow{R_\tau(t)} & C(\Gamma) \\ {\scriptstyle j}\downarrow & & \downarrow{\scriptstyle j} \\ C_0(X) & \xrightarrow[T(t)]{} & C_0(X) \end{array}$$

Here, $(R_\tau(t))$ denotes the rotation semigroup of period τ (see Part-I, Example 1.2.5). If X is compact, then j is a topological embedding.

Proof. Assume that $Ah = \mathrm{i}\alpha h$, $\alpha > 0$, and let $\tilde{h}(x) := h(x)/|h(x)|$. Then we define j by

$$j(f) := |h| \cdot f \circ \tilde{h} \quad \big(\text{i.e., } (j(f))(x) = |h(x)| \cdot f(\tilde{h}(x))\big). \tag{3.3.9}$$

Obviously, j is a lattice homomorphism and because h has no zeros and $\tilde{h}$ has a dense image in Γ (Theorem 3.3.6 (iii)), it follows that j is injective. For the functions $e_n \in C(\Gamma)$ given by $e_n(z) = z^n$ $(n \in \mathbb{Z})$ one has $j(e_n) = h^{[n]}$ $(n \in \mathbb{Z})$ and therefore by Theorem 3.2.4

$$T(t) \circ j(e_n) = T(t)h^{[n]} = \mathrm{e}^{\mathrm{i}n\alpha t} \cdot h^{[n]}$$

and

$$j \circ R_\tau(t)(e_n) = j(\mathrm{e}^{\mathrm{i}n\alpha t} e_n) = \mathrm{e}^{\mathrm{i}n\alpha t} \cdot h^{[n]}.$$

Since $\{e_n : n \in \mathbb{Z}\}$ is a total subset of $C(\Gamma)$, we have $T(t) \circ j = j \circ R_\tau(t)$ for every $t > 0$.

If X is compact, then $\tilde{h}(X)$ is closed, hence $\tilde{h}$ is onto, moreover, $|h| \geqslant \varepsilon$ for some $\varepsilon > 0$, thus the definition of j implies that $\|j(f)\| > \varepsilon\|f\| \; \|j(f)\| \geqslant \varepsilon\|f\|$ for every $f \in C(\Gamma)$. □

A consequence of Corollary 3.3.8 is the following. There exist positive functions f_1 and f_2 such that $T(t)f_1$ and $T(t)f_2$ have disjoint support for every $t \geqslant 0$ (consider the images under j of two disjoint functions on $C(\Gamma)$). This observation proves the following corollary.

Corollary 3.3.9 *Suppose that the hypotheses of Theorem 3.3.6 are satisfied and that for some $t_0 > 0$ we have $T(t_0)f \gg 0$ whenever $f > 0$. Then $P\sigma(A) \cap \mathrm{i}\mathbb{R} = \{0\}$.*

Corollary 3.3.9 can be applied if $T(t_0)$ is a kernel operator with strictly positive kernel. We give some examples.

Examples 3.3.10 (i) We assume that the semigroup $(T(t))$ satisfies the hypotheses of Theorem 3.3.6 and that it is given by

$$(T(t)f)(x) = \int_X k(t,x,y)f(y)\,\mathrm{d}\mu(y),$$

where μ is a positive measure and k is a positive continuous function (see Example 3.3.4 (ii)). We will show that $P\sigma(A) \cap (s(A) + i\mathbb{R}) = \{s(A)\}$.

Assuming the contrary, by Theorem 3.3.6 (iv) there exist $\alpha \neq 0$, $h \in C_0(X)$ such that

$$S_h^{-1} T(t) \circ S_h = e^{i\alpha t} \cdot T(t) \text{ for all } t \geqslant 0. \tag{3.3.10}$$

This implies that k satisfies

$$\frac{\overline{h(x)}}{|h(x)|} \cdot \frac{h(y)}{|h(y)|} \cdot k(t,x,y) = e^{i\alpha t} k(t,x,y) \quad (t > 0, x, y \in X). \tag{3.3.11}$$

It follows that for $0 < |s - t| < 2\pi/\alpha$ the kernel functions $k(t,\cdot,\cdot)$ and $k(s,\cdot,\cdot)$ have disjoint support. This is impossible if k is continuous.

(ii) Let Ω be a domain in $\mathbb{R}^n$ and define L_0 as

$$L_0 f := \sum_{i,j=1}^{n} a_{ij} f''_{ij} + \sum_{i=1}^{n} b_i f'_i + cf,$$

with domain

$$D(L_0) := \{f \in C_0(\Omega) \colon f \in C^\infty, L_0 f \in C_0(\Omega)\},$$

where f'_i stands for $\partial f/\partial x_i$, and $f''_{ij} = \partial^2 f/\partial x_i \partial x_j$.

Suppose that L_0 is elliptic, a_{ij}, b_i, c are real-valued C^∞-functions and $\sup_x c(x) < \infty$, assume further that the closure L of L_0 is the generator of a positive semigroup on $C_0(\Omega)$ which has compact resolvent. For example, this is true if the boundary $\partial\Omega$ is C^∞ and $a_{ij} \in C^\infty(\overline{\Omega})$ (cf. Theorem 4.8.3 of Fattorini [12]). We will show that

$$P\sigma(A) \cap (s(A) + i\mathbb{R}) = \{s(A)\}.$$

$$\mathsf{P}\sigma(\mathsf{L}) \cap (\mathsf{s}(\mathsf{L}) + i\mathbb{R}) = \{\mathsf{s}(\mathsf{L})\}.$$

In order to apply Theorem 3.3.6, we have to show that the corresponding semigroup $(T(t))$ is irreducible. Given $0 < f \in C_0(X) 0 < f \in C_0(\Omega)$, then there is $g \in D(L_0)$ such that $0 < g \leqslant f$. The function $h := R(\lambda, L)g$ is C^∞ (Weyl's Lemma) and satisfies

$$L_0 h - \lambda h = -g < 0.$$

Assuming that $\lambda > \lambda_0 := \sup_x c(x)$, then h is positive, even strictly positive by the maximum principle Protter & Weinberger [20, Chapter 2, Theorem 6]. Thus we have

$$R(\lambda, L) f \geqslant R(\lambda, L) g = h \gg 0 \text{ for every } f > 0$$

showing that $(T(t))$ is irreducible (see Definition 3.3.1 (e)).

Next we apply Theorem 3.3.6 (iv) in order to show that the spectral bound is a dominant eigenvalue. We can assume that $s(L) = 0$. If $s(L)$ is not dominant, then by Theorem 3.3.6 (iv) we have

$$L_0 h = i\alpha h,\ L_0|h| = 0,\ L_0\bar{h} = -i\alpha\bar{h} \text{ for some } h \neq 0,\ \alpha > 0.$$

$$\mathsf{L_0 h = i\alpha h,\quad L_0|h| = 0,\quad L_0\bar{h} = -i\alpha\bar{h} \text{ for some } h \neq 0,\ \alpha > 0.} \tag{3.3.12}$$

If we define $u := |h|$ and $w := h/|h|$, then Eq. (3.3.12) reads

$$L_0(uw) = \mathrm{i}\alpha uw, \quad L_0(u) = 0, \quad L_0(u/w) = -\mathrm{i}\alpha \cdot u/w. \tag{3.3.13}$$

Explicit calculation of $L_0(uw)$ and $L_0(u/w)$ using the product formula yields

$$L_0(uw) = wL_0(u) + u\sum_{i,j} a_{ij} w''_{ij} + \sum_i \left(ub_i + \sum_j a_{ij} u'_j \right) w'_i\,,$$

$$L_0(u/w) = \frac{1}{w} L_0(u) + u\sum_{i,j} a_{ij} (1/w)''_{ij} + \sum_i \left(ub_i + \sum_j a_{ij} u'_j \right) (1/w)'_i\,.$$

$$\begin{aligned} L_0(uw) &= wL_0(u) + u\sum_{i,j} a_{ij} w''_{ij} + \sum_i \left(ub_i + \sum_j a_{ij} u'_j \right) w'_i, \\ L_0(u/w) &= \frac{1}{w} L_0(u) + u\sum_{i,j} a_{ij} (1/w)''_{ij} + \sum_i \left(ub_i + \sum_j a_{ij} u'_j \right) (1/w)'_i. \end{aligned} \tag{3.3.14}$$

Observing that $(1/w)'_i = -w^{-2} \cdot w'_i$ and $(1/w)''_{ij} = w^{-3} \cdot (2w'_i w'_j - ww''_{ij})$, we obtain

$$L_0(uw) + w^2 L_0(u/w) = 2wL_0(u) + 2u/w \cdot \sum_{ij} a_{ij} w'_i w'_j\,.$$

$$\mathsf{L_0(uw) + w^2 L_0(u/w) = 2wL_0(u) + 2u/w \cdot \sum_{i,j} a_{ij} w'_i w'_j.} \tag{3.3.15}$$

This identity and Eq. (3.3.13) implies that $2u/w \cdot \sum_{ij} a_{ij} w'_i w'_j = 0$ $2u/w \cdot \sum_{i,j} a_{ij} w'_i w'_j = 0$. Since u has no zeros and (a_{ij}) is positive definite, we have grad $w = (w'_i) = 0$ in Ω, hence $w =$ const. Then, by Eq. (3.3.13), we have $\mathrm{i}\alpha uw = L_0(uw) = wL_0(u) = 0$, a contradiction.

The assumption that L_0 is elliptic, i.e., that (a_{ij}) is positive definite, is essential in order to show that there is only one eigenvalue in the boundary spectrum. In the following example (a_{ij}) is positive semi-definite and $P\sigma_b(A) = s(A) + \mathrm{i}\alpha\mathbb{Z} P\sigma_b(L) = s(L) + \mathrm{i}\alpha\mathbb{Z}$.

(iii) We consider

$$\Omega = \{(x, y) \in \mathbb{R}^2 \colon 1 < (x^2 + y^2)^{1/2} < 2\},$$

and the second order differential operator L_0 given by

$$(L_0 f)(x,y) = 1/(x^2+y^2)\cdot(x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy}) + (x f_y - y f_x).$$

The assertion concerning the boundary spectrum can be verified easily by using polar coordinates, $x = r\cdot\cos\omega$, $y = r\cdot\sin\omega$. Then L_0 becomes $L_0 f = f_{rr} + f_\omega$ on the space $C_0(1,2)\otimes C_{2\pi}(\mathbb{R})$.

In this section we have seen that the eigenvalues in the boundary spectrum of an irreducible semigroup form a subgroup of $\mathrm{i}\mathbb{R}$ (provided that $s(A) = 0$). We conclude this section mentioning an analogous statement for the whole boundary spectrum of Markov semigroups on $C(K)$, K compact. It seems to be unknown if this is true for irreducible semigroups in general. To prove this result one uses the proof of the analogous result for a single operator (cf. Schaefer [26, Theorem 7]) as a guideline.

Theorem 3.3.11 *Suppose that $(T(t))$ is an irreducible semigroup of Markov operators on $C(K)$, K compact. Then $\sigma_b(A)$ is a closed subgroup of $\mathrm{i}\mathbb{R}$, i.e.,*

$$\textit{either}\quad \sigma_b(A) = \{0\}\quad \textit{or}\quad \sigma_b(A) = \mathrm{i}\mathbb{R}\quad \textit{or}\quad \sigma_b(A) = \mathrm{i}\alpha\mathbb{Z}\ \textit{for some}\ \alpha > 0.$$

3.4 Semigroups of Lattice Homomorphisms

As we have seen in Section 3.2 the boundary spectrum of many positive semigroups is a cyclic set. However, there are hardly any restrictions on the set

$$\lambda \in \{\sigma(A):\ \mathrm{Re}(\lambda) < s(A)\},$$

$$\{\lambda \in \sigma(\mathsf{A}):\ \mathrm{Re}(\lambda) < \mathsf{s}(\mathsf{A})\},$$

except that it is symmetric with respect to the real axis. For semigroups of lattice homomorphisms the situation is quite different. We will show that the whole spectrum is an imaginary additively cyclic subset of $\mathbb{C}$ (see Definition 3.2.5). A complete proof of this result requires some facts of the theory of Banach lattices, therefore, we postpone it to Part-III, Theorem 3.4.2.

Theorem 3.4.1 *If A is the generator of a semigroup of lattice homomorphisms, then $\sigma(A)$, $A\sigma(A)$ and $P\sigma(A)$ are cyclic subsets of $\mathbb{C}$.*

First part of the proof. We prove the assertion concerning $A\sigma(A)$ and $P\sigma(A)$. Assume that

$$Ah = (\alpha + \mathrm{i}\beta)h,\ \alpha,\beta\in\mathbb{R},\ h\neq 0,$$

$$\mathsf{Ah} = (\alpha+\mathrm{i}\beta)\mathsf{h},\quad \alpha,\beta\in\mathbb{R},\quad \mathsf{h}\neq 0,$$

then by Part-I, Corollary 3.6.4

$$T(t)h = \mathrm{e}^{\alpha t}\mathrm{e}^{\mathrm{i}\beta t}h \text{ for all } t \geqslant 0.$$

Since $T(t)$ is a lattice homomorphism we have

$$T(t)|h| = |T(t)h| = \mathrm{e}^{\alpha t}|h| \ (t \geqslant 0) \quad \text{or} \quad A|h| = \alpha|h|,$$

hence by Theorem 3.2.4 (ii)

$$Ah^{[n]} = (\alpha + \mathrm{i}n\beta)h^{[n]} \text{ for all } n \in \mathbb{Z}.$$

We have shown that $P\sigma(A)$ is cyclic.

To prove that $A\sigma(A)$ is cyclic as well, one considers a semigroup $\mathcal{F}$-product $E_{\mathcal{F}}^{\mathcal{T}}$ of E (see Part-I, Section 1.3.7). It is easy to see that $E_{\mathcal{F}}^{\mathcal{T}}$ is a Banach lattice and $(T_{\mathcal{F}}(t))$ is a semigroup of lattice homomorphisms. Part-I, Proposition 3.4.4 implies $A\sigma(A) = P\sigma(A_{\mathcal{F}})$. Thus the assertion follows from the cyclicity of point spectrum. □

Performing a similar construction as in Example 3.2.6 (vi) one can show that every closed cyclic subset of $\mathbb{C}$ which is contained in a left halfplane is the spectrum of a suitable semigroup of lattice homomorphisms. For details see Derndinger & Nagel [10]. In the following we restrict ourselves to the case of compact spaces. Then a semigroup of lattice homomorphisms can be described explicitly by a semi-flow φ, and real-valued functions h and p (see Theorem 2.3.5 and Theorem 2.3.6). The function p has no influence on spectral properties (cf. Eq. (2.3.7) on page 168). Therefore we will assume that $(T(t))$ has the following form (cf. Theorem 2.3.5).

$$T(t)f = h_t \cdot f \circ \varphi_t \quad (t \geqslant 0, f \in C(K)) \tag{3.4.1}$$

where

$$\varphi = (\varphi_t) : \mathbb{R}_+ \times K \to K,\ \varphi_{t+s} = \varphi_t \circ \varphi_s \ (t, s \geqslant 0),$$
$$\varphi = (\varphi_{\mathsf{t}}) : \mathbb{R}_+ \times \mathsf{K} \to \mathsf{K},\ \varphi_{\mathsf{t+s}} = \varphi_{\mathsf{t}} \circ \varphi_{\mathsf{s}} \ (\mathsf{t}, \mathsf{s} \geqslant 0),$$

is a continuous semiflow and

$$h_t(x) := \exp\left(\int_0^t h(\varphi(s,x))\,\mathrm{d}s\right) \ (t \geqslant 0,\ x \in K)$$

are multipliers for some continuous function $h\colon K \to \mathbb{R}$. In the following we will describe the spectrum of the semigroup given by Eq. (3.4.1) in terms of φ and h.

At first we have to fix some notation. Let K, φ, h be as in Eq. (3.4.1). Define

$$K_t := \varphi_t(K) \quad \text{for } t < \infty, \quad \text{and} \quad K_\infty := \bigcap_{t<\infty} K_t. \tag{3.4.2}$$

Some properties of the sets K_t are listed in the following lemma. The proof is not difficult and will be omitted.

Lemma 3.4.2 *Every K_t $(0 \leqslant t \leqslant \infty)$ is a non-empty closed subset of K which is invariant under the semiflow φ. Moreover, the following assertions are true.*

(i) *For $s > t$ we have $K_s \subseteq K_t$. In case that $K_s = K_t$ then $K_t = K_\infty$.*

(ii) *$\varphi_t(K_\infty) = K_\infty$ for all $t \geqslant 0$.*

(iii) *If one partial mapping φ_t, $t > 0$, is injective (surjective), then all mappings φ_s are injective (surjective).*

We call a *semiflow φ injective (surjective)* if one and hence all of the partial mappings φ_t are injective (surjective). Studying the spectrum of the semigroup given by Eq. (3.4.1) on the facing page we divide the complex plane into these three sets

$$\begin{aligned} &\{\lambda \in \mathbb{C}\colon\ \mathrm{Re}(\lambda) < \underline{c}(h,\varphi)\},\\ &\{\lambda \in \mathbb{C}\colon\ \underline{c}(h,\varphi) \leqslant \mathrm{Re}(\lambda) \leqslant \overline{c}(h,\varphi)\},\\ &\{\lambda \in \mathbb{C}\colon\ \overline{c}(h,\varphi) < \mathrm{Re}(\lambda)\}. \end{aligned} \tag{3.4.3}$$

The quantities $\underline{c}(h,\varphi)$ and $\overline{c}(h,\varphi)$ are defined as follows.

$$\begin{aligned} \overline{c}(h,\varphi) &:= \lim_{t\to\infty} \overline{c}_t(h,\varphi) = \inf_{t>0} \overline{c}_t(h,\varphi),\\ &\text{where } \overline{c}_t(h,\varphi) := \sup_{x\in K}\left\{\frac{1}{t}\int_0^t h(\varphi(s,x))\,\mathrm{d}s\right\},\\ \underline{c}(h,\varphi) &:= \lim_{t\to\infty} \underline{c}_t(h,\varphi) = \sup_{t>0} \underline{c}_t(h,\varphi),\\ &\text{where } \underline{c}_t(h,\varphi) := \inf_{x\in K}\left\{\frac{1}{t}\int_0^t h(\varphi(s,x))\,\mathrm{d}s\right\}. \end{aligned} \tag{3.4.4}$$

It is easy to see that $\overline{c}_t(h,\varphi) = 1/t \cdot \log \|T(t)\|$, hence in the definition of $\overline{c}(h,\varphi)$, both the limit and the infimum exist and coincide with the growth bound (see Part-I, Eq. (1.1.1) on page 4). Furthermore, $\underline{c}_t(h,\varphi) = -\overline{c}_t(-h,\varphi)$. Therefore, $\underline{c}(h,\varphi)$ is well defined too.

First we will describe the part of $\sigma(A)$ which is contained in the left half-plane determined by $\underline{c}(h,\varphi)$. It turns out that either the whole half-plane is contained in $\sigma(A)$ or it has empty intersection with $\sigma(A)$. This depends only on properties of φ. Essentially there are three different cases. Before we state the general result (see Theorem 3.4.4) we give some typical examples.

Examples 3.4.3 (i) Consider on $K = [0,\infty]$ the semiflow defined by

$$\varphi(t,x) := x + t,\ (\infty + t = \infty).$$

Then $K_t = [t, \infty]$ and $K_\infty = \{\infty\}$. The spectrum of the corresponding semigroup $T(t)f = f \circ \varphi_t$ is

$$\sigma(A) = A\sigma(A) = \{\lambda \in \mathbb{C}\colon \operatorname{Re}(\lambda) \leqslant 0\}.$$

(ii) Consider again $K = [0, \infty]$ and define a semiflow by

$$\varphi(t,x) := \begin{cases} x - t & \text{if } x \geqslant t, \\ 0 & \text{if } x < t. \end{cases} \quad (\infty - t = \infty)$$

$$\varphi(\mathsf{t},\mathsf{x}) := \begin{cases} x - t & \text{if } x \geqslant t, \\ 0 & \text{if } x < t. \end{cases} \quad (\infty - \mathsf{t} = \infty).$$

Then $K_t = K$ for all t, hence $K_\infty = K$ and

$$\sigma(A) = \{\lambda \in \mathbb{C}\colon \operatorname{Re}(\lambda) \leqslant 0\},\ R\sigma(A) = \{\lambda \in \mathbb{C}\colon \operatorname{Re}(\lambda) < 0\} \cup \{0\}.$$

(iii) Consider on $K_1 := [-1, \infty)$ the equivalence relation defined by "$x \sim y$" if and only if

$$x,\ y \geqslant 0 \quad \text{and} \quad x - y \in \mathbb{Z}.$$

The semiflow φ_1 on K_1 given by $\varphi_1(t,x) = x + t$ induces a semiflow φ on $K := K_1/\!\sim$. For $0 < t < 1$ we have $K \neq K_t \neq K_\infty$ ($K_\infty = [0,1]/\!\sim\ \cong \Gamma$). The spectrum is $\sigma(A) = 2\pi\mathrm{i}\mathbb{Z}$.

(iv) Consider on $K = [-1, 1]$ the flow φ given by

$$\varphi(t,x) := \begin{cases} -1 & \text{if } x < 0 \text{ and } t > -\frac{x+1}{x}, \\ \frac{x}{1+tx} & \text{otherwise.} \end{cases}$$

Then $K_t = [-1, \frac{1}{1+t}]$, $K_\infty = [-1, 0]$ and

$$\sigma(A) = \{\lambda \in \mathbb{C}\colon \operatorname{Re}(\lambda) \leqslant 0\}, \quad \{\lambda \in \mathbb{C}\colon \operatorname{Re}(\lambda) < 0\} \subsetneq A\sigma(A) \cap R\sigma(A).$$

Theorem 3.4.4 *Suppose $\mathcal{T}$ is a semigroup of lattice homomorphisms given by Eq. (3.4.1) on page 220 with generator A. Considering $H := \{\lambda \in \mathbb{C}\colon \operatorname{Re}(\lambda) < \underline{c}(h,\varphi)\}$, where $\underline{c}(h,\varphi)$ is given by Eq. (3.4.4) on the previous page, then we have:*

(i) *If $K_t \neq K_\infty$ for every $t < \infty$, then $H \subseteq A\sigma(A)$.*

(ii) *If $\varphi_{|K_\infty}$ is not injective, then $H \subseteq R\sigma(A)$.*

(iii) *If $K_s = K_\infty$ for some $s < \infty$ and $\varphi_{|K_\infty}$ is injective, then $H \cap \sigma(A) = \emptyset$.*

Proof. For $\varepsilon > 0$ we define $H_\varepsilon = \{\lambda \in \mathbb{C}\colon \operatorname{Re}(\lambda) < \underline{c}(h,\varphi) - \varepsilon\}$. Obviously it is enough to prove assertion (i), (ii) and (iii) respectively for $H_{2\varepsilon}$, $\varepsilon > 0$ arbitrary, instead of H.

(i) By the definition given in Eq. (3.4.4) on page 221 there exists a $\tau > 0$ such that $\underline{c}_t(h,\varphi) \geqslant \underline{c}(h,\varphi) - \varepsilon$ for all $t \geqslant \tau$. It follows that

$$h_t(x) \geqslant \mathrm{e}^{(\alpha+\varepsilon)t} \text{ whenever } t \geqslant \tau, x \in K, \alpha < \underline{c}(h,\varphi) - 2\varepsilon. \tag{3.4.5}$$

Now we fix $\lambda = a + \mathrm{i}\beta \in H_{2\varepsilon}$ $(a, \beta \in \mathbb{R})$ and construct an approximate eigenvector (g_n) of A corresponding to λ. For $n \leqslant \tau + 1$ we choose an arbitrary function $g_n \neq 0$. Now suppose $n > \tau+1$. We choose $x_n \in K_{n+1/2} \setminus K_{n+1}$ (cf. Lemma 3.4.2 (i)), then there exists $y_n \in K$ such that $\varphi(n+1/2, y_n) = x_n$. We have $\varphi([0, n+1/2], y_n) \cap K_{n+1} = \emptyset$ and the mapping $t \mapsto \varphi(t, y_n)$ is a continuous injection, hence a homeomorphism from $[0, n+1/2]$ into K (this is true because $\varphi(n+1/2, y_n) \notin K_{n+1}$). By Tietze's Theorem there is $f_n \in C(K)$ such that

$$\begin{aligned} &\|f_n\| \leqslant 1, \quad f_n|_{K_{n+1}} = 0, \\ &f_n(\varphi(t, y_n)) = 0 \text{ for } 0 \leqslant t \leqslant n - (1+\delta) \text{ and } n + \delta \leqslant t \leqslant n+1, \\ &f_n(\varphi(t, y_n)) = \mathrm{e}^{\mathrm{i}\beta t} \text{ for } n - 1 \leqslant t \leqslant n. \end{aligned} \tag{3.4.6}$$

The constant $\delta \in (0, 1/2)$ will be determined later.

Considering $g_n := \int_0^{n+1} \mathrm{e}^{-\lambda t} T(t) f_n \, \mathrm{d}t$, then $g_n \in D(A)$ and

$$\begin{aligned} &(A - \lambda) g_n = (1 - \mathrm{e}^{-\lambda(n+1)} T(n+1)) f_n = f_n - \mathrm{e}^{-\lambda(n+1)} \cdot h_{n+1} \cdot f_n \circ \varphi_{n+1} = f_n. \\ &(\lambda - \mathsf{A}) \mathsf{g}_n = (\mathrm{Id} - \mathrm{e}^{-\lambda(n+1)} \mathsf{T}(n+1)) \mathsf{f}_n = \mathsf{f}_n - \mathrm{e}^{-\lambda(n+1)} \cdot \mathsf{h}_{n+1} \cdot \mathsf{f}_n \circ \varphi_{n+1} = \mathsf{f}_n. \end{aligned} \tag{3.4.7}$$

Moreover,

$$\|g_n\| \geqslant |g_n(y_n)| = \left| \int_0^{n+1} \mathrm{e}^{-\lambda t} h_t(y_n) f_n(\varphi(t, y_n)) \, \mathrm{d}t \right| \geqslant$$

$$\left| \int_{n-1}^{n} \mathrm{e}^{-\lambda t} h_t(y_n) \mathrm{e}^{\mathrm{i}\beta t} \, \mathrm{d}t \right| - \left[\int_{n-(1+\delta)}^{n-1} + \int_n^{n+\delta} \left| \mathrm{e}^{-\lambda t} | h_t(y_n) f_n(\varphi(t, y_n)) \right| \mathrm{d}t \right]$$

$$\geqslant \int_{n-1}^{n} \mathrm{e}^{-at} \mathrm{e}^{(a+\varepsilon)t} \, \mathrm{d}t - \left[\int_{n-(1+\delta)}^{n-1} + \int_n^{n+\delta} \mathrm{e}^{-at} |h_t(y_n)| \, \mathrm{d}t \right]$$

$$= \frac{1}{\varepsilon} \cdot \left(\mathrm{e}^{\varepsilon n} - \mathrm{e}^{\varepsilon(n-1)} \right) - \left(\int_{n-(1+\delta)}^{n-1} + \int_n^{n+\delta} \mathrm{e}^{-at} |h_t(y_n)| \, \mathrm{d}t \right).$$

$$\begin{aligned}\|g_n\| \geqslant |g_n(y_n)| &= \left|\int_0^{n+1} e^{-\lambda t} h_t(y_n) f_n(\varphi(t, y_n))\, dt\right| \\ &\geqslant \left|\int_{n-1}^{n} e^{-\lambda t} h_t(y_n) e^{i\beta t}\, dt\right| - \left[\int_{n-(1+\delta)}^{n-1} + \int_n^{n+\delta} \left|e^{-\lambda t} h_t(y_n) f_n(\varphi(t, y_n))\right| dt\right] \\ &\geqslant \int_{n-1}^{n} e^{-at} e^{(a+\varepsilon)t}\, dt - \left[\int_{n-(1+\delta)}^{n-1} + \int_n^{n+\delta} e^{-at} |h_t(y_n)|\, dt\right] \\ &= \frac{1}{\varepsilon} \cdot \left(e^{\varepsilon n} - e^{\varepsilon(n-1)}\right) - \left(\int_{n-(1+\delta)}^{n-1} + \int_n^{n+\delta} e^{-at} |h_t(y_n)|\, dt\right).\end{aligned}$$

The constant δ can be chosen such that

$$\|g_n\| \geqslant 1/2\varepsilon \cdot \left(e^{\varepsilon n} - e^{\varepsilon(n-1)}\right) \to \infty \text{ for } n \to \infty.$$

$$\|g_n\| \geqslant 1/(2\varepsilon) \cdot \left(e^{\varepsilon n} - e^{\varepsilon(n-1)}\right) \to \infty \text{ for } n \to \infty. \tag{3.4.8}$$

It follows from Eq. (3.4.8) and Eq. (3.4.7) on the preceding page that $g_n/\|g_n\|$ is an approximate eigenvector of A corresponding to λ. Thus (i) is proved.

The proofs of (ii) and (iii) will be handled simultaneously. First we show that we can restrict attention to the case where $K = K_\infty$.

Indeed, K_∞ is a φ-invariant subset, hence

$$I_\infty := \{f \in C(K) \colon f|_{K_\infty} = 0\}$$

is a $\mathcal{T}$-invariant ideal. Identifying $C(K)/I_\infty$ with $C(K_\infty)$ (cf. Section 1.1), then $(T_{/I_\infty}(t))$ is the semigroup governed by $\varphi_{|K_\infty}$ and $h_{|K_\infty}$.

Since one always has $R\sigma(A_{/}) \subseteq R\sigma(A) R\sigma(A_{/I_\infty}) \subseteq R\sigma(A)$, assertion (ii) is proved, when we can show that $H_{2\varepsilon} \subseteq R\sigma(A_{/}) H_{2\varepsilon} \subseteq R\sigma(A_{/I_\infty})$.

In case (iii) one has $K_s = K_\infty$ for some $s < \infty$, which implies $T(s)_{|I_\infty} = 0$. Hence we have $\sigma(A_{|I_\infty}) = \emptyset$ and therefore $\sigma(A) = \sigma(A_{/I_\infty})$ by Part-I, Proposition 3.4.2.

Henceforth we will assume that $K = K_\infty$, that is, φ is surjective (cf. Lemma 3.4.2 (ii)).

We choose $\tau > 0$ such that Eq. (3.4.5) on the previous page is true. Since φ is surjective, for every $f \in C(K)$ there is a $x_f \in K$ such that $\|f\| = \|f(\varphi(\tau, x_f))\|$ $\|f\| = |f(\varphi(\tau, x_f))|$ and we obtain for $\lambda \in H_{2\varepsilon}$, $\lambda = a + i\beta$, $\alpha, \beta \in \mathbb{R}\, a, \beta \in \mathbb{R}$

$$\begin{aligned}\|(e^{\lambda\tau} - T(\tau))f\| &\geqslant |h_\tau(x_f) f(\varphi(\tau, x_f)) - e^{\lambda\tau} f(x_f)| \\ &\geqslant h_\tau(x_f)\|f\| - e^{a\tau}|f(x_f)| \geqslant e^{(a+\varepsilon)\tau}\|f\| - e^{a\tau}\|f\| \\ &= e^{a\tau}(e^{\varepsilon\tau} - 1)\|f\|.\end{aligned} \tag{3.4.9}$$

It follows that the disc

$$D := \{\lambda \in \mathbb{C} \colon |\lambda| < \exp(\underline{c}(h, \varphi) - 2\varepsilon)\}$$

$$\mathsf{D} := \{\lambda \in \mathbb{C}\colon |\lambda| < \exp((\underline{\mathsf{c}}(\mathsf{h}, \varphi) - 2\varepsilon)\tau)\}$$

has an empty intersection with $A\sigma(T(\tau))$ and therefore $H_{2\varepsilon} \cap A\sigma(A) = \emptyset$ by Part-I, Proposition 3.6.2. Since every boundary point of the spectrum is an approximate eigenvalue (by Part-I, Proposition 3.2.2 (i)) we have the following alternative.

$$\begin{aligned} &\text{Either } D \subseteq \varrho(T(\tau)) \text{ and } H_{2\varepsilon} \subseteq \varrho(A) \\ &\text{or } D \subseteq R\sigma(T(\tau)) \text{ and } H_{2\varepsilon} \subseteq R\sigma(A). \end{aligned} \tag{3.4.10}$$

It is not difficult to see that $0 \in \varrho(T(\tau))$ whenever φ_t is bijective and that 0 is an eigenvalue of $T(\tau)'$ if φ_τ is not injective. Since we assumed that φ is surjective, assertions (ii) and (iii) of the theorem are immediate consequences of Eq. (3.4.10). □

The Examples 3.4.3 (i), (ii) and (iii) respectively are prototypes of the three different cases considered in Theorem 3.4.4. Example 3.4.3 (iii) also shows that there are semigroups whose spectrum is contained in a right half-plane, although they cannot be embedded in a group (compare Corollary 3.4.5 below). Example 3.4.3 (iv) shows that (i) and (ii) do not exclude each other.

Corollary 3.4.5 *If φ is injective or surjective, then the following assertions are equivalent.*

(a) *A is the generator of a strongly continuous group.*

(b) *$\sigma(A)$ is contained in a right half-plane.*

Proof. $(a) \Longrightarrow (b)$(a) $\Longrightarrow$ (b): holds true because $-A$ is a generator of a semigroup.

$(b) \Longrightarrow (a)$(b) $\Longrightarrow$ (a): We have to show that one (hence each) operator $T(t)$, $t > 0$ is invertible. Obviously this is true if φ is bijective. At first we assume that φ is surjective, that is, $K = K_\infty$. By Theorem 3.4.4 we have that $\varphi|_{K_\infty}$ $\varphi_{|K_\infty}$ is injective if (b) is true. Thus φ is bijective. Now we assume that φ is injective. We have to show that $K = K_\infty$. By Theorem 3.4.4 we have $K_\infty = K_s$ for some s, whenever (b) is true. Given $x \in K$ then by Lemma 3.4.2 (ii) there exists $y \in K_\infty$ such that $\varphi(s, x) = \varphi(s, y)$. If φ is injective we have $x = y \in K_\infty$. □

Example 3.4.6 Suppose $F\colon \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable. We denote the maximal flow corresponding to the differential equation $y' = F(y)$ by φ_0. In general, φ_0 is only defined on an open subset of $\mathbb{R} \times \mathbb{R}^n$ which contains $\{0\} \times \mathbb{R}^n$. For $x \in \mathbb{R}^n$ there exist $\underline{t}_x$ and $\bar{t}_x$ such that

$$\begin{aligned} &-\infty \leqslant \underline{t}_x < 0 < \bar{t}_x \leqslant \infty, \\ &\varphi_0(t, x) \text{ is defined if } \underline{t}_x < t < \bar{t}_x, \\ &\text{if } \bar{t}_x < \infty \ (\underline{t}_x > -\infty) \text{ then } |\varphi_0(t, x)| \to \infty \text{ as } t \uparrow \bar{t}_x \ (t \downarrow \underline{t}_x). \end{aligned} \tag{3.4.11}$$

For details see Section 18.2 of Dieudonné [11].

(i) If φ_0 is a global flow, i.e., if φ_0 is defined on $\mathbb{R}\times\mathbb{R}^n$, then one has a corresponding (semi-)group on $C_0(\mathbb{R}^n)$. If F is differentiable, its generator is the closure of A_1 which is defined as follows (cf. Example 2.3.15).

$$\begin{aligned} A_1 f &= (F|\operatorname{grad} f) := \sum_i F_i \cdot \partial_i f, \\ D(A_1) &:= \{f \in C^1 : \ \operatorname{supp}(f) \text{ is compact}\}. \end{aligned} \tag{3.4.12}$$

Then φ_0 can be uniquely extended to a flow $\tilde{\varphi}_0$ on $\mathbb{R}^n \cup \{\infty\}$ by defining $\tilde{\varphi}_0(t,\infty) := \infty$ for all $t \in \mathbb{R}$. φ_0 and $\tilde{\varphi}_0$ satisfy condition Theorem 3.4.4 (iii).

A global flow exists if F is globally Lipschitz continuous or uniformly bounded. For $\{x \in \mathbb{R}^n : (x|F(x)) > 0\}$ bounded in $\mathbb{R}^n$, a global semiflow always exists (cf. Deimling [7, Section 5.2]).

(ii) Now we do not assume that φ_0 is globally defined. Instead we consider a bounded domain $\Omega \subseteq \mathbb{R}^n$ with smooth boundary $\partial\Omega$ such that $(F(x)|\nu(x)) > 0$ for every $x \in \partial\Omega$, where $\nu(x)$ denotes the outward normal vector.

Then for $x \in \overline{\Omega}$ we have $\underline{t}_x = -\infty$. Moreover, either $\varphi_0(t,x) \in \Omega$ for all $t \geqslant 0$ or else there exists a unique s_x with $0 \leqslant s_x < \bar{t}_x$ such that $\varphi_0(s_x,x) \in \partial\Omega$. In the first case we write $s_x := \infty$. Then we define $\varphi : \mathbb{R}_+ \times \overline{\Omega} \to \overline{\Omega}$ $\varphi : \mathbb{R}_+ \times \overline{\Omega} \to \overline{\Omega}$ as

$$\varphi(t,x) := \begin{cases} \varphi_0(t,x) & \text{if } 0 \leqslant t < s_x, \\ \varphi_0(s_x,x) & \text{if } t \geqslant s_x. \end{cases}$$

$$\varphi(\mathsf{t},\mathsf{x}) := \begin{cases} \varphi_0(t,x) & \text{if } 0 \leqslant t < s_x, \\ \varphi_0(s_x,x) & \text{if } t \geqslant s_x. \end{cases}.$$

Then φ is a continuous semiflow on the compact set $K := \overline{\Omega}$. We have $K_\infty = K$ and $\varphi_{|K_\infty}$ is not injective.

In case F is differentiable, the generator of the corresponding semigroup is the closure of the operator A_2 defined by

$$A_2 f := (F|\operatorname{grad} f), \quad D(A_2) := \{f \in C^1(\overline{\Omega}) : (F|\operatorname{grad} f) = 0 \text{ on } \partial\Omega\}.$$

(iii) We consider Ω as in (ii) and assume that $(F(x)|\nu(x)) \leqslant 0$ for every $x \in \partial\Omega$. Then for every $x \in \overline{\Omega}$ we have $\bar{t}_x = \infty$. Thus $\varphi := \varphi_{0|\mathbb{R}_+\times\overline{\Omega}}$ is a continuous semiflow on $K := \overline{\Omega}$.

If $(F(x)|\nu(x)) < 0$ for some $x \in \partial\Omega$, we have $K_t \subsetneq K_s$ whenever $t > s$ and $\varphi_{|K_\infty}$ is injective. For a differentiable vector field F, the generator of the corresponding semigroup is the closure of A_3 defined as

$$A_3 f := (F|\operatorname{grad} f), \quad D(A_3) := C^1(\overline{\Omega}).$$

We conclude the discussion of semi-flows associated with ordinary differential equations by remarking that the ideas of (ii) and (iii) can be combined to obtain semigroups for more general subsets Ω.

We continue the discussion of the spectrum of semigroups of lattice homomorphisms on $C(K)$ given by Eq. (3.4.1) on page 220. Theorem 3.4.4 gives a good description of the part contained in

$$\{\lambda \in \mathbb{C}\colon \operatorname{Re}(\lambda) < \underline{c}(h,\varphi)\}.$$

The half-plane

$$\{\lambda \in \mathbb{C}\colon \operatorname{Re}(\lambda) > \overline{c}(h,\varphi)\}$$

is always a subset of the resolvent set (see Proposition 3.4.8 (i) below). The description of the remaining part

$$\{\lambda \in \sigma(A)\colon \underline{c}(h,\varphi) \leqslant \operatorname{Re}(\lambda) \leqslant \overline{c}(h,\varphi)\}$$

is more difficult. First we discuss some examples and then give a partial answer to this problem (see Proposition 3.4.8 (ii)–(v)).

Example 3.4.7 (i) Consider the flow on $[-\frac{\pi}{2}, \frac{\pi}{2}]$ defined by

$$\varphi(t,x) := \arctan(\tan x - t), x \in [-\pi/2,\pi/2], t \in \mathbb{R}$$

$$\varphi(\mathsf{t},\mathsf{x}) := \arctan(\tan \mathsf{x} - \mathsf{t}), \quad \mathsf{x} \in [-\pi/2,\pi/2], \mathsf{t} \in \mathbb{R}$$

and a continuous function $h\colon [-\frac{\pi}{2}, \frac{\pi}{2}] \to \mathbb{R}$ with $h(-\frac{\pi}{2}) \leqslant h(\frac{\pi}{2})$. The flow belongs to the differential equation $y' = -\cos^2 y$. Then we have $\underline{c}(h,\varphi) = h(-\frac{\pi}{2})$ and $\overline{c}(h,\varphi) = h(\frac{\pi}{2})$. The spectrum of the corresponding semigroup is given by

$$\sigma(A) = \{\lambda \in \mathbb{C}\colon h(-\pi/2) \leqslant \operatorname{Re}(\lambda) \leqslant h(\pi/2)\}.$$

(ii) Consider

$$K = \{z \in \mathbb{C}\colon 1 \leqslant |z| \leqslant 2\} = \{r \cdot \mathrm{e}^{\mathrm{i}\omega}\colon \omega \in \mathbb{R}, 1 \leqslant r \leqslant 2\}$$

$$\mathsf{K} = \{\mathsf{z} \in \mathbb{C}\colon 1 \leqslant |\mathsf{z}| \leqslant 2\} = \{\mathsf{r} \cdot \mathrm{e}^{\mathrm{i}\omega}\colon \omega \in \mathbb{R}, 1 \leqslant \mathsf{r} \leqslant 2\}$$

and a continuous function $\kappa : [1,2] \to \mathbb{R}_+ \kappa\colon [1,2] \to \mathbb{R}_+$.

Let $\tilde{\varphi}$ be the flow on K governed by the differential equation $\dot{\omega} = \kappa(r)$, $\dot{r} = 0$, hence $\tilde{\varphi}(t, r \cdot \mathrm{e}^{\mathrm{i}\omega}) = r \cdot \mathrm{e}^{\mathrm{i}(\omega+\kappa(r)t)}$.

For a continuous function $h : K \to \mathbb{R}\ h\colon K \to \mathbb{R}$ let $\hat{h}(r) := \frac{1}{2\pi}\int_0^{2\pi} h(r \cdot \mathrm{e}^{\mathrm{i}t})\,\mathrm{d}t$ $(1 \leqslant r \leqslant 2)$. The spectrum of the semigroup corresponding to $\varphi\ \tilde{\varphi}$ and h (cf. Eq. (3.4.1) on page 220) is given by

$$\sigma(A) = \{\hat{h}(r) + \mathrm{i}k\kappa(r)\colon k \in \mathbb{Z}, 1 \leqslant r \leqslant 2\} \cup \{h(z)\colon \kappa(|z|) = 0\}.$$

Proposition 3.4.8 *Suppose the semigroup $(T(t))_{t\geqslant 0}$ on $C(K)$ is given by Eq. (3.4.1) on page 220 and let $\underline{c}(h,\varphi)$, $\overline{c}(h,\varphi)$ be defined as in Eq. (3.4.4) on page 221. Then the following assertions hold.*

(i) $\{\lambda\in\mathbb{C}\colon\ \mathrm{Re}(\lambda)>\overline{c}(h,\varphi)\}\subseteq\varrho(A)$,

(ii) *$\overline{c}(h,\varphi)$ and $\underline{c}(h,\varphi)$ are spectral values,*

(iii) *If $\varphi(t,x_0)=x_0$ for every $t\geqslant 0$, then $h(x_0)\in R\sigma(A)$,*

(iv) *Assume x_0 has a finite orbit, i.e., $\varphi(\mathbb{R}_+,x_0)=\varphi([0,T],x_0)$ for some $T<\infty$ and*

$$\tau:=\inf\{t>0\colon\ \varphi(T+t,x_0)=\varphi(T,x_0)\}>0,$$

then

$$\hat{h}(x_0)+\frac{2\pi}{\tau}\mathrm{i}\mathbb{Z}\subseteq R\sigma(A)\quad \textit{where}\quad \hat{h}(x_0):=\frac{1}{\tau}\int_T^{T+\tau}h(\varphi(s,x_0))\,\mathrm{d}s$$

$$\hat{\mathsf{h}}(\mathsf{x}_0)+\frac{2\pi}{\tau}\mathrm{i}\mathbb{Z}\subseteq \mathsf{R}\sigma(\mathsf{A})\quad \textit{where}\quad \hat{\mathsf{h}}(\mathsf{x}_0):=\frac{1}{\tau}\int_{\mathsf{T}}^{\mathsf{T}+\tau}\mathsf{h}(\varphi(\mathsf{s},\mathsf{x}_0))\,\mathrm{d}\mathsf{s},$$

(v) *If x_0 has an infinite orbit and $\hat{h}:=\lim_{t\to\infty}h(\varphi(t,x_0))$ exists, then*

$$\hat{h}+\mathrm{i}\mathbb{R}\subseteq\sigma(A).$$

Proof. (i) and (ii): By Eq. (3.4.4) on page 221 we have $\overline{c}_t(h,\varphi)=1/t\cdot\log\|T(t)\|$ hence $\overline{c}(h,\varphi)=\omega_0(A)$ (cf. Eq. (1.1.1) on page 4). Consequently, we have $\{\lambda\in\mathbb{C}\colon\ \mathrm{Re}(\lambda)>\overline{c}(h,\varphi)\}\subseteq\varrho(A)$ and $\overline{c}(h,\varphi)\in\sigma(A)$ by Theorem 3.1.6.

To prove $\underline{c}(h,\varphi)\in\sigma(A)$, we can assume by Theorem 3.4.4 that $K_\infty=K_s$ for some s and that $\varphi_{|K_\infty}$ is injective. It is easy to see that $\underline{c}(h,\varphi)=\underline{c}(h_{|K_\infty},\varphi_{|K_\infty})$, moreover, we have $\sigma(A_{|I_\infty})=\emptyset$ hence $\sigma(A)=\sigma(A_{/I_\infty})$ by Part-I, Proposition 3.4.2. This shows that we also can assume that $K=K_\infty$, i.e., φ is bijective or A is the generator of a group. Now the assertion follows from

$$\underline{c}(h,\varphi)=\underline{c}(h,\varphi^{-1})=-\overline{c}(-h,\varphi^{-1})=-s(-A).$$

(iii) and (iv): One can check easily that in case of (iii) the Dirac functional δ_{x_0} is an eigenvector of A' corresponding to $h(x_0)$. A little bit more calculation is necessary to check that in case of (iv) the functional φ_k defined by

$$\varphi_k(f):=\int_T^{T+\tau}\exp\left(-\mathrm{i}\cdot\frac{2\pi k}{\tau}\cdot t\right)\cdot h_t(x_0)\cdot f(\varphi(t,x_0))\,\mathrm{d}t\ \ (k\in\mathbb{Z},f\in C(K))$$

is an eigenvector of A' corresponding to $\hat{h}(x_0)+\mathrm{i}\cdot\frac{2\pi k}{\tau}$.

(v) Given $\beta\in\mathbb{R}$ we will show that $\hat{h}+\mathrm{i}\beta\in A\sigma(A')\subseteq\sigma(A)$. For $n,m\in\mathbb{N}$ we define a linear functional φ_{nm} as

$$\varphi_{nm}(f) := \frac{1}{n}\int_0^n \exp(-(h^\star + \mathrm{i}\beta)t)\cdot h_t(\varphi(m,x_0))\cdot f(\varphi(m+t,x_0))\,\mathrm{d}t,\ f\in C(K).$$

$$\varphi_{\mathsf{nm}}(\mathsf{f}) := \frac{1}{\mathsf{n}}\int_0^{\mathsf{n}} \exp(-(\hat{\mathsf{h}} + \mathrm{i}\beta)\mathsf{t})\cdot \mathsf{h}_\mathsf{t}(\varphi(\mathsf{m},\mathsf{x}_0))\cdot \mathsf{f}(\varphi(\mathsf{m}+\mathsf{t},\mathsf{x}_0))\,\mathrm{d}\mathsf{t},\ \mathsf{f}\in \mathsf{C}(\mathsf{K}).$$

For $f \in D(A)$ we have

$$\big\langle(\hat{h}+\mathrm{i}\beta - A)f, \varphi_{nm}\big\rangle =$$
$$\frac{1}{n}\cdot\left(f(\varphi(m,x_0)) - \mathrm{e}^{-\mathrm{i}\beta n}\cdot\exp\left(\int_m^{m+n}(h(\varphi(s,x_0))-\hat{h})\,\mathrm{d}s\right)f(\varphi(m+n,x_0))\right).$$

It follows that $\varphi_{nm}\in D(A')$ and, since $\lim_{t\to\infty} h(\varphi(t,x_0)) = \hat{h}$,

$$\limsup_{m\to\infty}\|(\hat{h}+\mathrm{i}\beta - A')\varphi_{nm}\| \leqslant 1/n \text{ for every } n\in\mathbb{N}. \tag{3.4.13}$$

Because the orbit is infinite we have

$$\begin{aligned}\|\varphi_{nm}\| &= \frac{1}{n}\int_0^n \left|\mathrm{e}^{-(\hat{h}+\mathrm{i}\beta)t}h_t(\varphi(m,x_0))\right|\mathrm{d}t\\ &= \frac{1}{n}\int_0^n \exp\left(\int_m^{m+t}(h(\varphi(s,x_0))-\hat{h})\,\mathrm{d}s\right)\mathrm{d}t\end{aligned}$$

$$\begin{aligned}\|\varphi_{\mathsf{nm}}\| &= \frac{1}{\mathsf{n}}\int_0^{\mathsf{n}} \left|\mathrm{e}^{-(\hat{\mathsf{h}}+\mathrm{i}\beta)\mathsf{t}}\mathsf{h}_\mathsf{t}(\varphi(\mathsf{m},\mathsf{x}_0))\right|\mathrm{d}\mathsf{t}\\ &= \frac{1}{\mathsf{n}}\int_0^{\mathsf{n}} \exp\left(\int_{\mathsf{m}}^{\mathsf{m}+\mathsf{t}}(\mathsf{h}(\varphi(\mathsf{s},\mathsf{x}_0))-\hat{\mathsf{h}})\,\mathrm{d}\mathsf{s}\right)\mathrm{d}\mathsf{t}\end{aligned}$$

which shows that

$$\lim_{m\to\infty}\|\varphi_{nm}\| = 1 \text{ for every } n\in\mathbb{N}. \tag{3.4.14}$$

In view of Eq. (3.4.13) and Eq. (3.4.14) it is not difficult to find a subsequence $k(n)$ of $\mathbb{N}$ such that $(\varphi_{n,k(n)})$ is an approximate eigenvector of A' corresponding to $\hat{h}+\mathrm{i}\beta$. □

We are now going to apply the results obtained so far to the special case where $h = 0$, i.e., we consider semigroups of lattice homomorphisms which are Markov operators.

Theorem 3.4.9 *Suppose $\mathcal{T}$ is a semigroup of Markov lattice homomorphisms on $C(K)$ governed by the semiflow φ.*

(i) *If $\varphi_{|K_\infty}$ is not injective or if $K_t \neq K_\infty$ for every $t < \infty$, then*

$$\sigma(A) = \{\lambda\in\mathbb{C}:\ \mathrm{Re}(\lambda)\leqslant 0\}.$$

(ii) *If $K_\infty = K_s$ for some s and $\varphi_{|K_\infty}$ is injective, then $\sigma(A)$ is a cyclic closed subset of $\mathrm{i}\mathbb{R}$. Moreover, we have $\sigma(A) \neq \mathrm{i}\mathbb{R}$ if and only if there is a $T < \infty$ such that every orbit of φ has length less than T, i.e.,*

$$\varphi(\mathbb{R}_+, x) = \varphi([0, T], x) \text{ for every } x \in K.$$

Proof. (i) This is an immediate consequence of Theorem 3.4.4 and Proposition 3.4.8.

(ii) The first assertion follows from Theorem 3.4.4 and Theorem 3.4.1. Moreover, as in the proof of Theorem 3.4.4 (ii) and (iii) we can assume without loss of generality that $K = K_\infty$, hence φ is bijective. If there is no upper bound for the length of the orbits, then $\sigma(A) = \mathrm{i}\mathbb{R}$ by assertions (iv) and (v) of Proposition 3.4.8.

Now we assume that the lengths of the orbits are bounded by T. Because φ is bijective, for every $x \in K$ there exists a $r = r_x$ with $T/2 \leqslant r \leqslant T$ such that

$$\varphi(t, x) = \varphi(t + r, x) = \varphi(t + 2r, x) = \cdots = \varphi(t + kr, x) \quad (t \in \mathbb{R}_+, k \in \mathbb{N}).$$

Therefore we have for $\lambda \in \mathbb{C}$, $\operatorname{Re}(\lambda) > 0$, $f \in C(K)$, $x \in K$

$$\begin{aligned}(R(\lambda, A)f)(x) &= \int_0^\infty \mathrm{e}^{-\lambda t} f(\varphi(t, x))\,\mathrm{d}t = \\ &= \sum_{k=0}^{\infty} \mathrm{e}^{-\lambda kr} \int_{kr}^{(k+1)r} \mathrm{e}^{-\lambda(t-kr)} f(\varphi(t - kr, x))\,\mathrm{d}t = \\ &= (1 - \mathrm{e}^{-\lambda r})^{-1} \cdot \int_0^r \mathrm{e}^{-\lambda t} f(\varphi(t, x))\,\mathrm{d}t.\end{aligned}$$

$$\begin{aligned}(R(\lambda, A)f)(x) &= \int_0^\infty \mathrm{e}^{-\lambda t} f(\varphi(t, x))\,\mathrm{d}t \\ &= \sum_{k=0}^{\infty} \mathrm{e}^{-\lambda kr} \int_{kr}^{(k+1)r} \mathrm{e}^{-\lambda(t-kr)} f(\varphi(t - kr, x))\,\mathrm{d}t \qquad (3.4.15) \\ &= (1 - \mathrm{e}^{-\lambda r})^{-1} \cdot \int_0^r \mathrm{e}^{-\lambda t} f(\varphi(t, x))\,\mathrm{d}t.\end{aligned}$$

If $0 < \beta < 2\pi/T$, then the assumption $T/2 \leqslant r \leqslant T$ implies that there exists a neighborhood U of $\lambda_0 := \mathrm{i}\beta$ such that the functions $\lambda \mapsto (1 - \exp(-\lambda r_x))^{-1}$ are uniformly bounded on U, by M say. Then from Eq. (3.4.15) we can conclude that

$$\|R(\lambda, A)f\| \leqslant M\left(\int_0^r \left|\mathrm{e}^{-\lambda t}\right| \mathrm{d}t\right)\|f\| \text{ for } \lambda \in U,\ \operatorname{Re}(\lambda) > 0,$$

therefore $\lambda_0 = \mathrm{i}\beta \in \varrho(A)$. □

Remark 3.4.10 In case $\sigma(A) \neq \mathrm{i}\mathbb{R}$, then $\varphi_{|K_\infty}$ is bijective and has only finite orbits. Therefore every $x \in K_\infty$ has a well-defined period

$$\tau_x := \inf\{\tau > 0 \colon \varphi(\tau, x) = x\}.$$

A more detailed analysis yields the following description

$$\sigma(A) = \overline{\{\mathrm{i} \cdot 2\pi k/\tau_x \colon k \in \mathbb{Z}, x \in K_\infty, \tau_x > 0\}} \cup \{0\}. \tag{3.4.16}$$

The inclusion "$\subseteq$" can be derived from Theorem 3.4.11 which is stated below. The reverse inclusion follows from Proposition 3.4.8 (iv).

In our detailed discussion of the spectrum of lattice homomorphisms we restricted ourselves to the case where the space K is compact. The main reason is that there is no description as given in Eq. (3.4.1) on page 220 of the semigroups for locally compact spaces X. In general, it is difficult to define a semiflow on X because points may tend to infinity in a finite time. But even if one can find a flow on a suitable compactification of X, it may be impossible to find a multiplicator. This can be seen by studying the following example.

Suppose φ_1 is a semiflow on a compact space K_1 and K_0 is a closed φ_1-invariant subset, h a continuous function on K_1.

The semigroup $(T_1(t))$ on $C(K_1)$ corresponding to φ_1 and h leaves the ideal

$$I := \{f \in C(K_1) \colon f|_{K_0} = 0\}$$

invariant and induces via restriction a semigroup $(T(t))$ on $I = C_0(X)$, where $X = K_1 \setminus K_0$. In this case one can construct semi-flows associated with $(T(t))$ on $X \cup \{\infty\}$ or on $\overline{X}$ (closure in K_1), but in general one cannot find a corresponding multiplicator which is defined on one of these compactifications.

The situation is much nicer when groups of lattice homomorphisms instead of semigroups are considered. In this case there is an analogue of Eq. (3.4.1) on page 220 (cf. Theorem 2.3.14) and the spectrum can be described completely. For more details and the proof of the following result we refer to Arendt & Greiner [1].

Theorem 3.4.11 *Suppose X is a locally compact space and $(T(t))_{t\in\mathbb{R}}$ is a group of lattice homomorphisms governed by the flow φ and the multiplicator h. Then we have*

(i) $\sigma(A) = \sigma_1 \cup \sigma_2 \cup \sigma_3$ *where the sets σ_i are defined as follows.*

$$\begin{aligned}
\sigma_1 &:= \overline{\{\hat{h}(x) + \mathrm{i} \cdot 2\pi k/\tau_x \colon x \in X, 0 < \tau_x < \infty\}},\\
\sigma_2 &:= \overline{\{h(x) \colon x \in X, \tau_x = 0\}},\\
\sigma_3 &:= \{\lambda \in \mathbb{C} \colon \lambda + \mathrm{i}\mathbb{R} \subseteq \sigma(A)\}
\end{aligned}$$

$$\begin{aligned}
\sigma_1 &:= \overline{\{\hat{h}(x) + \mathrm{i} \cdot 2\pi k/\tau_x \colon k \in \mathbb{Z}, x \in X, 0 < \tau_x < \infty\}},\\
\sigma_2 &:= \overline{\{h(x) \colon x \in X, \tau_x = 0\}},\\
\sigma_3 &:= \{\lambda \in \mathbb{C} \colon \lambda + \mathrm{i}\mathbb{R} \subseteq \sigma(A)\}
\end{aligned}$$

(ii) $\sigma(T(t)) = \overline{\exp(t\sigma(A))}$ *for every $t \geqslant 0$.*

(iii) *Every isolated point of $\sigma(A)$ is a first order pole of the resolvent.*

Notes

Spectral theory for a single positive operator is an essential cornerstone for spectral theory of positive one-parameter semigroups. Many of the results we have presented in this chapter have analogues in the discrete case (i.e., for a single operator). This relation may serve as a guide. However, only in few cases can the continuous version be deduced directly from its discrete analogue. Therefore we will not try to trace back the roots of every result to the discrete version. Instead we refer to SCHAEFER [27] and the notes and references given there.

Many of the results we have presented in this chapter can be extended (more or less easily) to the more general setting of Banach lattices, which include the very important examples of L^p-spaces. Others are typical for $C_0(X)$ and allow no extension. We will discuss this fact in more detail in Part-III, Chapter 3. The more general setting considered there also allows us to prove results for $C_0(X)$ which we could not obtain staying within the framework of spaces of continuous functions.

Section 3.1: Theorem 3.1.1 was stated by KARLIN [16], but a complete proof is given in DERNDINGER [8]. Proposition 3.1.5 is taken from GREINER [15] and Theorem 3.1.6 is (implicitly) contained in DERNDINGER & NAGEL [10]. A generalization to (non-lattice) ordered Banach spaces can be found in Section 2.4 of BATTY & ROBINSON [2].

Section 3.2: Lemma 3.2.3 dates back to Rota (see SCHAEFER [27]). Our approach follows GREINER [14]. The notion "(imaginary) additively cyclic" was introduced by DERNDINGER [8] (and Schaefer (1980) respectively). Derndinger proves some cyclicity results for the boundary spectrum. A result similar to Proposition 3.2.7 is given in Section 7.4 of DAVIES [6]. Lemma 3.2.8 in combination with Part-III, Lemma 3.3.13 can be used to characterize semigroups whose spectral bound is a pole of finite algebraic multiplicity (see Eq. (3.3.19) on page 357). The hypothesis of Theorem 3.2.9 can be weakened, one only needs that $s(A)$ is a pole of the resolvent (see Part-III, Corollary 3.2.12). Further results on the cyclicity of the boundary spectrum will be given in Part-III, Chapter 3. In particular we refer to Part-III, Theorems 3.2.10, 3.3.12 and Lemma 3.3.13. The dichotomy stated in Eq. (3.2.19) on page 205 is probably the most interesting consequence of cyclicity results. It has far reaching consequences on the asymptotic behavior of positive semigroups. Example 3.2.13 is due to Davies (unpublished note). Example 3.3.43.2.14 (iiiii) will be discussed in more detail and more generality in Section 4.3. We return to Remark 3.2.15 (ii) in Section 4.2.

Section 3.3: The concept of irreducibility as defined in Definition 3.3.1 is closely related to various other notions: In topological dynamics flows inducing irreducible semigroups are called "minimal flows" (cf. Example 3.3.4 (i)). Moreover, "ergodicity" and "unique ergodicity" are closely related to irreducibility (see CORNFELD et al. [5]

or KRENGEL [17]). Irreducible semigroups are discussed to some extent in DAVIES [6] where he proves a special case of Theorem 3.3.6. Proposition 3.3.3 will be generalized in Part-III, Proposition 3.3.3. Assertion (i) of Proposition 3.3.5 was proven by SCHAEFER [28] while Theorem 3.3.6 is taken from GREINER [15]. Elliptic operators (more general than Example 3.3.4 3.3.10 (ii)) as generators on spaces of continuous functions, were investigated by many people, e.g., BONY et al. [3], KUHN [18], ROTH [21], ROTH [22] and STEWART [30].

Section 3.4: Theorem 3.4.1 is due to DERNDINGER [9]. The spectrum of semigroups of Markov lattice homomorphisms is investigated by DERNDINGER & NAGEL [10]. In particular they prove Theorem 3.4.4 for Markov semigroups. Earlier results are due to SCARPELLINI [24]. We indicated briefly in Example 3.4.6 that there is a relationship between spectral properties of lattice semigroups and differentiable dynamics. For more details we refer to CHICONE & SWANSON [4] and SACKER & SELL [23]. For example, the "annular hull theorem" is a special case of Theorem 3.4.11 (ii). The general result Theorem 3.4.11 was proven by ARENDT & GREINER [1].

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and

$$\|T(t)'\mu\| \leqslant \liminf_{n\to\infty} \|T(t)^*\mu_n\| \quad (t \geqslant 0).$$

Applying Fatou's Lemma we obtain

$$\begin{aligned}\int_0^\infty \|T(t)'\mu\|\,\mathrm{d}t &\leqslant \int_0^\infty \Big(\liminf_{n\to\infty} \|T(t)'\mu_n\|\Big)\,\mathrm{d}t \leqslant \\ &\leqslant \liminf_{n\to\infty} \int_0^\infty \|T(t)'\mu_n\|\,\mathrm{d}t = \\ &= \liminf_{n\to\infty} \|R(0,A^*)\mu_n\| \leqslant \\ &\leqslant \|R(0,A^*)\| \cdot \liminf_{n\to\infty} \|\mu_n\| < \\ &< \infty.\end{aligned}$$

(Observe that $t \to \|T(t)'\mu\| = \sup\{\langle T(t)f,\mu\rangle\colon \|f\| \leqslant 1\}$ is lower semi-continuous and hence measurable). Using Part-I, Proposition 4.1.10 we obtain $\omega_0(A^*) < 0$. But $\omega_0(A) = \omega_0(A^*)$ by Part-I, Proposition 3.4.4 (iii), contradicting $\omega_0(A) = 0$. □

Remark 4.1.5 If $(T(t))$ is a positive semigroup on an α-directed ordered Banach space E (see Asimow & Ellis [1], p.39), then the dual of E admits a reversion of the triangle inequality, i.e., $\sum \|\mu_i\| \leqslant \alpha\|\sum \mu_i\|$ for $\mu_i \in E'_+$, and Theorem 4.1.4 remains valid (see Batty & Davies [2]). The proof given above may be used with almost no modification.

We now close the discussion of the stability of positive semigroups on $C_0(X)$ and refer to Part-III, Section 4.1 and Part-IV, Section 4.4 respectively, where the stability of positive semigroups on arbitrary Banach lattices and on C*-algebras will be treated.

4.2 Compact and Quasi-Compact Semigroups

by Günther Greiner

Using the Riesz-Schauder Theory for compact operators (see, e.g., Chapter VII.4 in Dunford & Schwartz [9] or Section 26 of Pietsch [26]) and the results of Chapter 3, one can easily describe the asymptotic behavior of *eventually compact semigroups*. Since no positivity is involved, we state the fundamental result for arbitrary Banach spaces.

Theorem 4.2.1 *Let $(T(t))_{t\geqslant 0}$ be a strongly continuous semigroup on a Banach space G which is eventually compact (i.e., there is $t_0 > 0$ such that $T(t_0)$ is a compact operator). Then the spectrum of the generator A is a countable set (possibly finite or*

empty) and contains only poles of finite algebraic multiplicity. Furthermore, the set $\{\mu \in \sigma(A)\colon \operatorname{Re}(\mu) \geqslant r\}$ *is finite for every* $r \in \mathbb{R}$.

Thus

$$\sigma(A) = \{\lambda_1, \lambda_2, \lambda_3, \ldots\},$$

where the eigenvalues are ordered so that

$$\operatorname{Re}(\lambda_{n+1}) \leqslant \operatorname{Re}(\lambda_n) \qquad \textit{for all } n \in \mathbb{N}.$$

If the spectrum is infinite, then

$$\lim_{n\to\infty} \operatorname{Re}(\lambda_n) = -\infty.$$

$$\begin{aligned}
&T(t) = T_1(t) + T_2(t) + \cdots + T_m(t) + R_m(t),\\
&T_n(t) = \exp(\lambda_n t) \cdot \sum_{j=0}^{k(n)-1} \frac{1}{j!} t^j (A - \lambda_n)^j \circ P_n \quad (t \geqslant 0),\\
&\|R_m(t)\| \leqslant C \cdot \exp((\varepsilon + \operatorname{Re}(\lambda_m))t) \textit{ for } t \geqslant 0,\ \varepsilon > 0\\
&\qquad \textit{for } t \geqslant 0,\ \varepsilon > 0 \textit{ and a suitable constant } C = C(\varepsilon, m).
\end{aligned}$$

$$\begin{aligned}
&T(t) = T_1(t) + T_2(t) + \cdots + T_m(t) + R_m(t),\\
&T_n(t) = \exp(\lambda_n t) \cdot \sum_{j=0}^{k(n)-1} \frac{1}{j!} t^j (A - \lambda_n)^j \circ P_n \quad (t \geqslant 0),\\
&\|R_m(t)\| \leqslant C \cdot \exp((\varepsilon + \operatorname{Re}(\lambda_m))t)\\
&\qquad \textit{for } t \geqslant 0, \varepsilon > 0 \textit{ and a suitable constant } C = C(\varepsilon, m).
\end{aligned} \tag{4.2.1}$$

Proof. Fix $r \in \mathbb{R}$. By the Riesz-Schauder Theory we know that

$$\{z \in \sigma(T(t_0))\colon |z| \geqslant \exp(rt_0)\}$$

is a finite set and contains only poles of finite algebraic multiplicity. Thus the first assertion follows from Part-I, Corollary 3.6.5. To prove the remaining assertion we fix $m \in \mathbb{N}$ and apply the spectral decomposition as described in Section 3.3. For simplicity we assume $\operatorname{Re}(\lambda_{m+1}) < \operatorname{Re}(\lambda_m)$. Let P be the spectral projection of $T(t_0)$ corresponding to the spectral set $\{z \in \sigma(T(t_0))\colon |z| \geqslant \exp(\operatorname{Re}(\lambda_m) \cdot t_0)\}\{z \in \sigma(T(t_0))\colon |z| \geqslant \exp(\operatorname{Re}(\lambda_m) \cdot t_0)\}$. Then P reduces the semigroup and we have

$$\sigma(T(t_0)_{|\ker(P)} \subseteq \{z \in \mathbb{C}\colon |z| < \exp(\operatorname{Re}(\lambda_m) \cdot t_0)\}.$$

$$\sigma(\mathsf{T}(\mathsf{t}_0)_{|\ker(\mathsf{P})}) \subseteq \{\mathsf{z} \in \mathbb{C}\colon |\mathsf{z}| < \exp(\operatorname{Re}(\lambda_m) \cdot \mathsf{t}_0)\}.$$

Hence the type of $T(t_0)_{|\ker(P)}$ is less than $\operatorname{Re}(\lambda_m)$. Then there exists a constant C_0 such that

$$\|T(t)(\operatorname{Id} - P)\| \leqslant \|T(t)_{|\ker(P)}\| \cdot \|\operatorname{Id} - P\| \leqslant \|\operatorname{Id} - P\| \cdot C_0 \cdot \exp(\operatorname{Re}(\lambda_m) \cdot t).$$

We define $R_m(t) := T(t)(\mathrm{Id} - P)$ and $T_n(t) := T(t)P_n$ $(n \in \mathbb{N})$. Then $R_m(t)$ satisfies the estimate stated in Eq. (4.2.1) on the previous page, and we have $T(t) = \sum_{n=1}^{m} T_n(t) + R_m(t)$ because $P = \sum_{n=1}^{m} P_n$ by Part-I, Corollary 3.6.5 (ii). The family of projections $\mathrm{Id} - P, P_1, P_2, \ldots, P_m$ reduces the semigroup. Thus in order to prove the representation of $T_n(t)$ stated in Eq. (4.2.1) on the preceding page, we only have to consider elements $f \in P_nE = \ker(\lambda_n - A)f \in P_nG = \ker(\lambda_n - A)^{k(n)}$. Hence we can assume $E = P_nEG = P_nG$, $\sigma(A) = \{\lambda_n\}$, $P_n = \mathrm{Id}$ and for simplification we drop the index n, i.e., $\lambda = \lambda_n$, $k = k(n)$. Then A is a bounded operator satisfying $(\lambda - A)^k = 0$ and its resolvent is given by

$$R(\nu, A) = (\nu - \lambda)^{-1} \sum_{j=0}^{k-1} (\nu - \lambda)^{-j} (A - \lambda)^j$$

for $\nu \neq \lambda$. It follows that

$$R(\nu, A)^i = (\nu - \lambda)^{-i} \sum_{j=0}^{k-1} \binom{j+i-1}{i-1} (\nu - \lambda)^{-j} (A - \lambda)^j.$$

Hence we have

$$\left(\frac{i}{t} R\left(\frac{i}{t}, A\right)\right)^i = \left(1 - \lambda \frac{t}{i}\right)^{-i} \sum_{j=0}^{k-1} \binom{j+i-1}{i-1} (i - \lambda t)^{-j} t^j (A - \lambda)^j$$

for every $i \in \mathbb{N}$. Since

$$\lim_{i\to\infty} \left(1 - \lambda \frac{t}{i}\right)^{-i} = \mathrm{e}^{\lambda t} \quad \text{and} \quad \lim_{i\to\infty} \binom{j+i-1}{i-1} (i - \lambda t)^{-j} = \frac{1}{j!}$$

for every $j \in \mathbb{N}$, the assertion follows from Part-I, Eq. (2.1.3) on page 39. □

Combining Theorem 4.2.1 with the results of Chapter 3 one can describe the behavior of $T(t)$ as $t \to \infty$ provided that $(T(t))_{t\geqslant 0}$ is a positive semigroup. We give a typical example.

Corollary 4.2.2 *Let $(T(t))_{t\geqslant 0}$ be a positive semigroup on a space $C_0(X)$ which is irreducible and eventually compact. Then there exist a unique real number $r \in \mathbb{R}$, a strictly positive function h and a strictly positive bounded Borel measure ν such that for suitable constants $\delta > 0$, $M \geqslant 1$ one has*

$$\|\mathrm{e}^{-rt} \cdot T(t) - \nu \otimes h\| \leqslant M \cdot \mathrm{e}^{-\delta t} \text{ for all } t \geqslant 0. \tag{4.2.2}$$

In particular, for every $f \in C_0(X)$ and $t \geqslant 0$ one has

$$\left(\left|\int f \,\mathrm{d}\nu\right| - M \cdot \mathrm{e}^{-\delta t}\|f\|\right) \leqslant \mathrm{e}^{-rt}\|T(t)f\| \leqslant \left(\left|\int f \,\mathrm{d}\nu\right| + M \cdot \mathrm{e}^{-\delta t}\|f\|\right). \tag{4.2.3}$$

Proof. We take $r := s(A)$. By Proposition 3.3.5 (i) we have $r > -\infty$. Moreover, by assertion (v) of the same proposition we know that r is an algebraically simple pole and the corresponding residue P has the form $P = v \otimes h$ for strictly positive eigenvectors v of A' and h of A, respectively. We choose h such that $\|h\| = 1$. Corollary 3.2.11 of Chapter 3 implies that r is strictly dominant, i.e., enumerating the eigenvalues as described in Theorem 4.2.1 we have

$$\operatorname{Re}(\lambda_2) < \lambda_1 = r$$

Now Eq. (4.2.2) on the facing page follows from Eq. (4.2.1) on page 241 for $m = 1$. Applying the triangle inequality to

$$T(t)f = \mathrm{e}^{rt}\big(Pf + (\mathrm{e}^{-rt}T(t)f - Pf)\big)$$

and using Eq. (4.2.2) on the facing page one easily deduces Eq. (4.2.3) on the preceding page. □

Let us point out the following consequence of Corollary 4.2.2.

For every positive, non-zero initial value f the solution $T(\cdot)f$ of the abstract Cauchy problem $\dot{u} = Au$ decreases or increases exponentially in norm according to the sign of $r = s(A)$. If $s(A) = 0$, then $T(\cdot)f$ tends to an equilibrium state which is unique up to a constant and is non-zero whenever the initial value is positive and non-zero.

In order to apply Theorem 4.2.1 and its corollary to concrete problems one needs conditions which ensure that the semigroup is eventually compact. We discuss this problem for the spaces $C(K)$, K compact, in more detail. The crucial tool is the following characterization of weakly compact subsets in the dual space $M(K) = C(K)'$, due to Grothendieck [15].

Proposition 4.2.3 *Let K be a compact space. For a subset $D \subseteq M(K) = C(K)'$ the following assertions are equivalent.*

(a) *D is relatively compact for the weak topology $\sigma(M(K), M(K)')$.*

(b) *For each weak null sequence (f_n) in $C(K)$, $\lim_{n\to\infty}\langle f_n, \nu\rangle = 0$ uniformly for $\nu \in D$.*

(c) *For each sequence (U_n) of disjoint open subsets of K, $\lim_{n\to\infty} \nu(U_n) = 0$ uniformly for $\nu \in D$.*

For a proof of this result, see e.g., II.9.8 in Schaefer [29]. We use this proposition in order to describe weakly compact operators on spaces $C(K)$. As usual we naturally identify the bounded Borel functions on K with a subspace $B(K)$ of $M(K)' = C(K)''$, note that in general, $C(K) \subsetneq B(K) \subsetneq C(K)''$.

Proposition 4.2.4 *Let K be a compact space, G be a Banach space, and let $R\colon C(K) \to G$ be a bounded linear operator.*

(i) *The following assertions are equivalent.*

(a) *R is weakly compact,*

(b) *for every bounded Borel function g on K we have $R''g \in G$,*

(c) *for every Borel set $C \subseteq K$ we have $R''(\mathbf{1}_C) \in G$.*

In case $G = C(K)$ these conditions are equivalent to the following.

(d) *If $(f_n) \subseteq C(K)$ is a bounded sequence, then (Rf_n) has a subsequence which converges pointwise to a continuous function.*

(ii) *If R is weakly compact, then it maps weakly convergent sequences into norm convergent sequences. In particular, the square of a weakly compact operator $T\colon C(K) \to C(K)$ is a compact operator.*

Proof. (i) $(a) \Rightarrow (b)$ (a) $\Rightarrow$ (b) follows from the following characterization of weakly compact operators (see e.g., SCHAEFER [29, II. Proposition 9.4]).

An operator is weakly compact if and only if its second adjoint maps the bidual into the original space.

$(b) \Rightarrow (c)$ (b) $\Rightarrow$ (c) is trivial and it remains to show that $(c) \Rightarrow (a)$(c) $\Rightarrow$ (a).

On the Borel field $\mathcal{B}$ we define m by $m(C) := R''(\mathbf{1}_C)$. Then m is a G-valued additive set function. For $y' \in G'$ we have $y' \circ m = R'y' \in M(K)$. Hence $y' \circ m$ is a countably additive set function, for every $y' \in G'$, i.e., m is weakly countably additive. By Pettis' Theorem (see IV. Theorem 10.1 in DUNFORD & SCHWARTZ [9]) we have that m is countably additive with respect to the norm. In particular, for a sequence U_n of mutually disjoint Borel sets we have $\lim_{n\to\infty} \|m(U_n)\| = 0$. It follows that $\lim_{n\to\infty} y' \circ m(U_n) = 0$ uniformly for $y' \in G'$, $\|y'\| \leqslant 1$. Now condition (c) of Proposition 4.2.3 shows that $\{R'y' : y' \in G',\ \|y'\| \leqslant 1\}$ $\{R'y' : y' \in G',\ \|y'\| \leqslant 1\}$ is relatively weakly compact, i.e., R' is weakly compact. Thus R is weakly compact as well.

In case $G = C(K)$ the equivalence of (a) and (d) (a) and (d) is a consequence of two results. First, *Eberlein's Theorem* states that for the weak topology in any Banach space compactness and sequential compactness are equivalent. Second, *Lebesgue's Dominated Convergence Theorem* assures that a sequence $(f_n) \subseteq C(K)$ converges weakly to $f \in C(K)$ if and only if it is bounded and $f_n(x) \to f(x)$ for every $x \in K$.

(ii) Suppose that (f_n) is a sequence in $C(K)$ which converges to 0 for the weak topology. Since R is weakly compact, the same is true for the adjoint R', i.e.,$\{R'y' : y' \in G',$ $\|y'\| \leqslant 1\}$ $\{R'y' : y' \in G',\ \|y'\| \leqslant 1\}$ is relatively weakly compact in $M(K)$. From Proposition 4.2.3 $(a) \Rightarrow (b)$ (a) $\Rightarrow$ (b) we obtain that $\langle Rf_n, y'\rangle = \langle f_n, R'y'\rangle \to 0$ as $n \to \infty$ uniformly for $y' \in G'$, $\|y'\| \leqslant 1$. That is $\lim_{n\to\infty} \|Rf_n\| = 0$. The final assertion follows from the first and the characterization of weakly compact operators stated in (d) (d) of (i). □

The next result which is an immediate consequence of Theorem 4.2.1 and Proposition 4.2.4 is motivated by the theory of Markov processes. For a Markov operator (see Section 1.3) condition (b) of Proposition 4.2.4 (i) is called the strong *Feller property*.

Theorem 4.2.5 *Let $(T(t))_{t\geqslant 0}$ be a semigroup of Markov operators on $C(K)$, K compact, such that one operator $T(t_0)$ has the strong Feller property. Then there exists a positive projection P of finite rank such that $\|T(t) - P\| \leqslant M \cdot \mathrm{e}^{-\delta t}$ for suitable constants $\delta > 0$, $M \geqslant 1$.*

Proof. By Proposition 4.2.4 (i), $T(t_0)$ is weakly compact. Thus, by Proposition 4.2.4 (ii), $T(2t_0)$ is compact, i.e., $(T(t))_{t\geqslant 0}$ is eventually compact. Moreover, by Corollary 3.2.11 $s(A) = 0$ is strictly dominant and a first order pole of the resolvent by Remark 3.2.15 (i). The assertion now follows easily from Theorem 4.2.1. □

We close the discussion of eventually compact semigroups by describing a situation where Theorem 4.2.5 can be applied. A more detailed description of the connection between Markov processes and positive semigroups on $C(K)$ is given in Chapter 2 of Casteren [3].

Example 4.2.6 Let K be a compact space and $\{P_t : t > 0\}$ be a Markov transition function on K which satisfies the strong Feller property and which is stochastically continuous. That is, every P_t is a real-valued function defined on the product $K \times \mathcal{B}$, where $\mathcal{B}$ denotes the Borel field on K, such that

(i) for $x \in K$ and $t > 0$ fixed, $P_t(x,.)$ $P_t(x,\cdot)$ is a probability measure,

(ii) for $C \in \mathcal{B}$ and $t > 0$ fixed, $P_t(.,C)$ $P_t(\cdot,C)$ is a continuous function,

(iii) $P_{t+s}(x,C) = \int_K P_s(y,C)P_t(x,\mathrm{d}y)$ for all $s,t > 0$, $x \in K$, $C \in \mathcal{B}$,

(iv) $\lim_{t\downarrow 0} P_t(x,U) = 1$ for every open set U containing x.

Condition (ii) is the *strong Feller property*, (iii) is the *Chapman-Kolmogorov equation* and (iv) expresses stochastic continuity. Given $\{P_t\}$ as above, one defines a family of operators $(T(t))_{t>0}$ as follows

$$(T(t)f)(x) := \int_K f(y)P_t(x,\mathrm{d}y) \text{ for } f \in C(K) \text{ and } x \in K. \tag{4.2.4}$$

Then it is not difficult to verify that

- $T(t)f \in C(K)$,
- $T(t)$ is a Markov operator on $C(K)$,
- $(T(t))_{t\geqslant 0}$ — with $T(0) = \mathrm{Id}$ — is a one-parameter semigroup.

In fact, the first assertion is a consequence of (i) and (ii), the second follows from (i) and the semigroup property is implied by the Chapman-Kolmogorov equation

(iii). Moreover, the semigroup $(T(t))_{t\geqslant 0}$ is strongly continuous. This can be seen as follows. In view of Proposition 1.23 in Davies [5] we only have to show that $\lim_{t\downarrow 0}\langle T(t)f - f, \nu\rangle = 0$ for every $f \in C(K)$, $\nu \in M(K)$. Due to Lebesgue's Dominated Convergence Theorem this is true whenever $\lim_{t\downarrow 0}(T(t)f)(x) = f(x)$ for every $f \in C(K)$, $x \in K$. Given f, x and $\varepsilon > 0$ there exists an open neighborhood U of x such that $|f(x) - f(y)| < \varepsilon$ for every $y \in U$. Then we have

$$\begin{aligned}(T(t)f)(x) - f(x) &= \int_K f(y)P_t(x,\mathrm{d}y) - \int_K f(x)P_t(x,\mathrm{d}y)\\ &= \int_U (f(y)-f(x))P_t(x,\mathrm{d}y) + \int_{K\setminus U}(f(y)-f(x))P_t(x,\mathrm{d}y)\\ &\leqslant \varepsilon\cdot P_t(x,U) + 2\|f\|_\infty \cdot P_t(x,K\setminus U).\end{aligned}$$

$$\begin{aligned}(\mathsf{T(t)f})(\mathsf{x}) - \mathsf{f(x)} &= \int_{\mathsf{K}} \mathsf{f(y)P_t(x,\mathrm{d}y)} - \int_{\mathsf{K}} \mathsf{f(x)P_t(x,\mathrm{d}y)}\\ &= \int_{\mathsf{U}} (\mathsf{f(y)-f(x))P_t(x,\mathrm{d}y)} + \int_{\mathsf{K}\setminus\mathsf{U}} (\mathsf{f(y)-f(x))P_t(x,\mathrm{d}y)}\\ &\leqslant \varepsilon\cdot \mathsf{P_t(x,U)} + 2\|\mathsf{f}\|_\infty \cdot \mathsf{P_t(x,K\setminus U)}.\end{aligned}$$

Since $P_t(x,U) \leqslant 1$ and $\lim_{t\downarrow 0} P_t(x,U) = 1 = P_t(x,K)$, this estimate implies

$$\limsup_{t\downarrow 0}((T(t)f)(x) - f(x)) \leqslant \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary we have pointwise convergence, hence strong continuity of the semigroup.

Finally we observe that every operator $T(t)$ defined by Eq. (4.2.4) on the preceding page has the strong Feller property since $T(t)''\mathbf{1}_C = P_t(\cdot,C)$ $T(t)''\mathbf{1}_C = P_t(\cdot,C)$ for every Borel set $C \subseteq K$ (see Proposition 4.2.4 (i)). Thus Theorem 4.2.5 can be applied in this situation.

We now turn our interest from eventually compact semigroups to *quasi-compact semigroups*. While "eventually compact" means that the operators $T(t)$ with $t \geqslant t_0$ have to be compact, "quasi-compactness" only means that $T(t)$ approaches the compact operators as $t \to \infty$. To make this precise we introduce the following notations. For a Banach space G, the ideal of all compact linear operators on G is denoted by $\mathcal{K}(G)$. For $T \in \mathcal{L}(G)$ we define

$$\operatorname{dist}(T,\mathcal{K}(G)) := \inf\{\|T-K\| \colon K \in \mathcal{K}(G)\}.$$

Definition 4.2.7 A strongly continuous semigroup $(T(t))_{t\geqslant 0}$ on a Banach space G is called *quasi-compact* if $\lim_{t\to\infty}\operatorname{dist}(T(t),\mathcal{K}(G)) = 0$.

Quasi-compactness can be characterized in many ways. Two of them are stated in the following proposition. The first one uses the notion of the essential growth bound $\omega_{\mathrm{ess}}(\mathcal{T})$ of a semigroup $\mathcal{T}$ as introduced in Eq. (3.3.9) on page 85.

Proposition 4.2.8 *For a strongly continuous semigroup $\mathcal{T} = (T(t))_{t\geqslant 0}$ on a Banach space G the following conditions are equivalent.*

(a) *$\mathcal{T}$ is quasi-compact,*

(b) $\omega_{ess}(\mathcal{T}) < 0 \omega_{\mathrm{ess}}(\mathcal{T}) < 0$,

(c) *There exist $t_0 > 0$, $K \in \mathcal{K}(G)$ such that $\|T(t_0) - K\| < 1$.*

Proof. $(a) \Rightarrow (c)$ (a) $\Rightarrow$ (c) is obvious by the definition of quasi-compactness.

$(c) \Rightarrow (b)$(c) $\Rightarrow$ (b): Recalling the definition of the essential spectral radius from Eq. (3.3.6) on page 85, assertion (c) (c) implies $r_{\mathrm{ess}}(T(t_0)) \leqslant \|T(t_0)\|_{\mathrm{ess}} < 1$. Then $\omega_{\mathrm{ess}}(\mathcal{T}) < 0$ by Eq. (3.3.10) on page 85.

$(b) \Rightarrow (a)$(b) $\Rightarrow$ (a): By Eq. (3.3.10) on page 85 we have $r_{\mathrm{ess}}(T(1)) < 1$. Then Eq. (3.3.6) on page 85

$$\lim_{n\to\infty} \|T(n)\|_{\mathrm{ess}}^{1/n} < 1, \text{ where } \|T\|_{\mathrm{ess}} = \operatorname{dist}(T, \mathcal{K}(G)).$$

Thus for suitable $n_0 \in \mathbb{N}$, $a < 1$ we have $\|T(n)\|_{\mathrm{ess}} < a^n$ for $n \geqslant n_0$.

Choosing a sequence $K_n \in \mathcal{K}(G)$ such that $\|T(n) - K_n\| < a^n$ for $n \geqslant n_0$ and defining $M := \sup_{0\leqslant s\leqslant 1} \|T(s)\|$ we obtain for $t \in [n, n+1]$ $(n \geqslant n_0)$

$$\|T(t) - T(t-n)K_n\| \leqslant \|T(t-n)\|\|T(n) - K_n\| \leqslant M \cdot a^n.$$

This implies that $\lim_{t\to\infty} \operatorname{dist}(T(t), \mathcal{K}(G)) = 0$. □

A typical situation where quasi-compact semigroups occur is the following. If $\mathcal{T} = (T(t))_{t\geqslant 0}$ is a strongly continuous semigroup with $\omega_{\mathrm{ess}}(\mathcal{T}) < \omega_0(\mathcal{T})$, then the rescaled semigroup $(\exp(-\omega_0(\mathcal{T})t)\cdot T(t))_{t\geqslant 0}$ $(\exp(-\omega_0(\mathcal{T})t)\cdot T(t))_{t\geqslant 0}$ is quasi-compact. Obviously every semigroup with growth bound less than zero is quasi-compact. A more interesting situation is the following.

If $(T_0(t))_{t\geqslant 0}$ is a semigroup with growth bound less than zero and A_0 is its generator, then for every compact operator K the perturbed operator $A := A_0 + K$ generates a quasi-compact semigroup.

More generally we have the following result.

Proposition 4.2.9 *If $(T(t))_{t\geqslant 0}$ is a quasi-compact semigroup on a Banach space G with generator A and K is a compact operator, then $A + K$ generates a quasi-compact semigroup.*

Proof. If $(T(t))_{t\geqslant 0}$ and $(S(t))_{t\geqslant 0}$ are the semigroups generated by A and $A+K$, respectively, we have $S(t) = T(t)+\int_0^t T(t-s)KS(s)\,\mathrm{d}s$. In view of Proposition 4.2.8 (c) it is enough to show that $\int_0^t T(t-s)KS(s)\,\mathrm{d}s$ is a compact operator.

Since the mapping $(t,x)\mapsto T(t)x$ is jointly continuous on $\mathbb{R}_+\times G$ and since K is compact, the set $M_t := \{T(s)Kx\colon 0\leqslant s\leqslant t, \|x\|\leqslant 1\}$ is relatively compact in G. Having in mind that $\int_0^t T(t-s)KS(s)x\,\mathrm{d}s$ $(x\in G)$ is the norm limit of Riemann sums, one observes that $(ct)^{-1}\int_0^t T(t-s)KS(s)x\,\mathrm{d}s$ is an element of the closed convex hull $\overline{\mathrm{co}(M_t)}$ of M_t, where $c := \sup\{\|S(s)\|\colon 0\leqslant s\leqslant t\}$ and $\|x\|\leqslant 1$. Since $\overline{\mathrm{co}(M_t)}$ is compact (see II.4.3 in Schaefer [28]) the assertion follows. □

We will now show that for quasi-compact semigroups the asymptotic behavior is similar to the one stated for eventually compact semigroups in Theorem 4.2.1. One obtains a representation as in Eq. (4.2.1) on page 241 with a remainder of exponential decay. However, the rate of decay cannot be chosen to be arbitrarily large.

Theorem 4.2.10 *Let $\mathcal{T} = (T(t))_{t\geqslant 0}$ be a quasi-compact semigroup on a Banach space G with generator A. Then $\{\lambda\in\sigma(A)\colon \mathrm{Re}(\lambda)\geqslant 0\}$ is a finite set (possibly empty) and contains only poles of finite algebraic multiplicity. Denoting the eigenvalues with nonnegative* nonnegative *real part by $\lambda_1,\dots,\lambda_m$, the corresponding residues $P_1,\dots,P_m$, and the orders of the poles $k_1,\dots,k_m$ $k(1),\dots,k(m)$, we have*

$$T(t) = T_1(t)+\cdots+T_m(t)+R(t), \quad \text{where}$$
$$T_n(t) = e^{\lambda_n t}\sum_{j=0}^{k_n-1}\frac{t^j}{j!}(A-\lambda_n)^j P_n \quad (n=1,\dots,m), \quad \text{and}$$
$$\|R(t)\|\leqslant Ce^{-\varepsilon t} \quad \text{for suitable constants } \varepsilon>0,\ C\geqslant 1 \qquad (t\geqslant 0).$$

$$T(t) = T_1(t)+\cdots+T_m(t)+R(t), \quad \textit{where}$$
$$T_n(t) = \mathrm{e}^{\lambda_n t}\sum_{j=0}^{k(n)-1}\frac{t^j}{j!}(A-\lambda_n)^j P_n \quad (n=1,\dots,m), \quad \textit{and} \tag{4.2.5}$$
$$\|R(t)\|\leqslant C\mathrm{e}^{-\varepsilon t} \quad \textit{for suitable constants } \varepsilon>0,\ C\geqslant 1 \qquad (t\geqslant 0).$$

Proof. We have $\omega_{ess}(\mathcal{T})<0$, hence $r_{ess}(T(1))<1$ $\omega_{\mathrm{ess}}(\mathcal{T})<0$, hence $r_{\mathrm{ess}}(T(1))<1$ (see Eq. (3.3.10) on page 85). Therefore $\{z\in\sigma(T(1))\colon |z|\geqslant 1\}$ is a finite set and contains only poles of finite algebraic multiplicity (cf. Eq. (3.3.8) on page 85).

Let P denote the spectral projection of $T(1)$ corresponding to $\{z\in\sigma(T(1))\colon |z|\geqslant 1\}$. Then Part-I, Corollary 3.6.5 implies that $\{\lambda\in\sigma(A)\colon \mathrm{Re}(\lambda)\geqslant 0\}$ is a finite set, contains only poles of $R(\cdot,A)$ of finite algebraic multiplicity and $P = P_1+P_2+\ldots+P_m P = P_1+P_2+\cdots+P_m$. One can now prove the representation of $T(t)$ stated in Eq. (4.2.5) in the same way as statement Eq. (4.2.1) on page 241. □

In case we consider positive quasi-compact semigroups on $C_0(X)$ one can combine Theorem 4.2.10 with the results of Chapter 3. For example, Corollary 3.2.11 assures

that, in case there is at least one eigenvalue with nonnegative real part, the generator has a strictly dominant eigenvalue $r \in \mathbb{R}$. Thus in Eq. (4.2.5) on the preceding page the operators $T_j(t)$ belonging to $\lambda_j = r$ will determine the long-term behavior of $(T(t))$. More precisely, one has the following.

Corollary 4.2.11 *Let $\mathcal{T} = (T(t))_{t\geqslant 0}$ be a positive semigroup on $C_0(X)$ which is quasi-compact and let A be its generator.*

(i) *Let r be an eigenvalue of A admitting a strictly positive eigenfunction and satisfying $\operatorname{Re}(r) \geqslant 0$. Then $r = \omega_0(\mathcal{T}) = s(A)$ and there is a positive projection P of finite rank such that for suitable constants $\delta > 0$, $M \geqslant 1$ we have*

$$\|\mathrm{e}^{-rt}T(t) - P\| \leqslant M \cdot \mathrm{e}^{-\delta t} \quad \text{for all} \quad t \geqslant 0. \tag{4.2.6}$$

(ii) *If $(T(t))_{t\geqslant 0}$ is irreducible and $\omega_0(\mathcal{T}) \geqslant 0$, there exist a strictly positive function $h \in C_0(X)$ and a strictly positive bounded measure $\nu \in M(X)$ such that for suitable constants $\delta > 0$, $M \geqslant 1$ one has*

$$\|\exp(-\omega_0(\mathcal{T})t) \cdot T(t) - \nu \otimes h\| \leqslant M \cdot \mathrm{e}^{-\delta t} \quad \text{for all} \quad t \geqslant 0. \tag{4.2.7}$$

In both cases (i) and (ii) (i) and (ii) the estimates Eq. (4.2.3) on page 242 for $\|T(t)f\|$ hold true (in case (i) (i) one has to replace $|\int f \,\mathrm{d}\nu|$ by $\|Pf\|$).

Proof. (i) By Corollary 3.2.11 we know that $s(A)$ is a strictly dominant eigenvalue of A. By Theorem 4.2.10 both $s := s(A)$ and r are poles of the resolvent. Moreover, there exists a positive measure ν such that $A^*\nu = s\nu$. Denoting the strictly positive eigenfunction corresponding to r by h we have $\langle h, \nu\rangle > 0$. Hence

$$s\langle h, \nu\rangle = \langle h, A^*\nu\rangle = \langle Ah, \nu\rangle = r\langle h, \nu\rangle$$

implies $r = s$. By Remark 3.2.15 we know that s is a first order pole of the resolvent. Since s is strictly dominant, Eq. (4.2.6) follows from Eq. (4.2.5) on the facing page.

(ii) can be proved in the same way as Corollary 4.2.2. We omit the details. □

Corollary 4.2.11 can be used to describe the asymptotic behavior as $t \to \infty$ of certain semigroups if only its generator is known. We explain this by discussing a concrete example.

Example 4.2.12 Let $X := [0, \infty)$ and define on $E := C_0(X)$ the operator A as

$$\begin{aligned} Af &:= -f' + mf, \quad \text{with domain } D(A) \text{ given by} \\ D(A) &:= \big\{f \in C_0(X) \colon f \text{ is differentiable, } f' \in C_0(X), \\ &\qquad \text{and } f'(0) = \alpha f(0) - \int_0^\infty f(x)\ \mathrm{d}\nu(x)\big\}. \end{aligned} \tag{4.2.8}$$

Here α is a real number, ν is a bounded positive Borel measure with $\nu(\{0\}) = 0$ and m is a continuous function on X such that $m(\infty) := \lim_{x\to\infty} m(x)$ exists. It is not difficult to see that A generates a positive semigroup. Moreover, one can show that it is quasi-compact if (and only if) $m(\infty) < 0$.

In order to find eigenvalues and eigenfunctions, one has to solve the ordinary differential equation $f' = mf - \lambda f$. Any solution has (up to a constant) the following form

$$g_\lambda(x) = \exp\left(\int_0^x (m(y) - \lambda)\,\mathrm{d}y\right) = \mathrm{e}^{-\lambda x} \cdot \exp\left(\int_0^x m(y)\,\mathrm{d}y\right). \tag{4.2.9}$$

We assume that $m(\infty) < 0$ and $r \geqslant 0$. Then g_r is differentiable with $g_r, g_r' \in C_0(X)$. Thus, $g_r \in D(A)$ if and only if

$$g_r'(0) = \alpha g(0) - \int_0^\infty g_r(y)\,\mathrm{d}\nu(y).$$

$$\mathsf{g}_\mathsf{r}'(0) = \alpha \mathsf{g}_\mathsf{r}(0) - \int_0^\infty \mathsf{g}_\mathsf{r}(\mathsf{y})\,\mathrm{d}\nu(\mathsf{y}).$$

Inserting Eq. (4.2.9), then this condition becomes

$$m(0) - r = \alpha - \int_0^\infty \mathrm{e}^{-ry} \cdot \exp\left(\int_0^y m(z)\ \mathrm{d}z\right)\mathrm{d}\nu(y).$$

By monotonicity this equation has a unique solution $r \geqslant 0$ if and only if

$$m(0) + \int_0^\infty \exp\left(\int_0^y m(z)\,dz\right)\mathrm{d}\nu(y) \geqslant \alpha.$$

$$\mathsf{m}(0) + \int_0^\infty \exp\left(\int_0^\mathsf{y} \mathsf{m}(\mathsf{z})\ \mathrm{dz}\right)\mathrm{d}\nu(\mathsf{y}) \geqslant \alpha. \tag{4.2.10}$$

In case α, ν and m satisfy Eq. (4.2.10) and $m(\infty) < 0$, then g_r is a strictly positive eigenfunction of A corresponding to $r \geqslant 0$. Thus all assumptions of Corollary 4.2.11 (i) are satisfied. In addition, the semigroup is irreducible if (and only if) the support of ν is an unbounded subset of $[0, \infty)$.

Similar examples will be discussed in the next section and in Part-III, Section 4.3.

We finally give a criterion for quasi-compactness of positive semigroups on spaces $C(K)$. It is based on a criterion given by Doeblin for operators on spaces of bounded measurable functions and can be easily deduced from Proposition 3 in Lotz [25].

Proposition 4.2.13 *Let $\mathcal{T} = (T(t))_{t\geqslant 0}$ be a semigroup of Markov operators on $C(K)$, K compact, satisfying the following condition.*

where $k \in C[a,1]$.

It is easy to verify that L is positive and bounded, and that B is the generator of the positive semigroup $(S(t))_{t\geqslant 0}$ defined by

$$[S(t)f](x) = \begin{cases} f(x-t) & \text{if } x-t \geqslant a \\ 0 & \text{if } x-t < a \end{cases} \qquad (x \in [a,1]).$$

Furthermore B has compact resolvent. Define $\Phi f := L(f(-r))$ for $f \in E$ such that Eq. (4.3.12) on the facing page can be written as retarded Cauchy problem (RCP).

As before (see Formula Eq. (4.3.3) on page 256) Φ_λ is defined by $\Phi_\lambda x := \Phi(e_\lambda \otimes x)$ for $x \in F$. Gyllenberg and Heijmans have shown that $s(B+\Phi_\lambda) > -\infty$. Thus we can apply Theorem 4.3.7 and obtain that $s(A) = \lambda$ if and only if $\lambda = s(B+\Phi_\lambda)$.

Notes

Section 4.1: The coincidence of spectral bound and growth bound for positive semigroups on $C(K)$ was first observed by Derndinger [7] and then generalized to $C_0(X)$ and non-commutative C*-algebras by Batty & Davies [2] and Groh & Neubrander [14]. The stability Theorem 4.1.1 is a continuous version of a result of Choquet-Foias (see Schaefer [29, V.8.8].

Section 4.2: For the Riesz-Schauder Theory of compact operators we refer to Dunford & Schwartz [9], Section VII.4 and Pietsch [26], Section 26. Theorem 4.1.1 seems to be folklore. Proposition 4.2.3 is due to Grothendieck [15] and can be found in Section II.9 of Schaefer [29]. Proposition 4.2.4 is due to Dieudonné (see §3 of Grothendieck [15] and Schaefer [29], II.Exc.27). The notion *strong Feller property* used in Theorem 4.2.5 is due to Girsanov (see Dynkin [10]) while the theorem itself was proven by Davies [6]. It is well known that there is a close relationship between Markov processes and Markov semigroups. This relation more detailed than in Example 4.2.6, can be found, e.g., in Dynkin [10], in Chapter 2 of Casteren [3] or in Chapter 7 of Lamperti [23]. The notion *quasi-compact* for a single operator dates back to Eberlein [11] (see also Yosida & Kakutani [33] and Section 26.4 of Pietsch [26]). Quasi-compactness for strongly continuous semigroups and its relationship to uniform ergodicity is investigated in Lin [24]. Proposition 4.2.9 is due to Voigt [32], a special case was proven by Vidav [31]. Corollaries 4.2.2 and 4.2.11 can be found in Greiner [13]. The criterion stated in Eq. Proposition 4.2.13 is known as *Doeblin's condition* (see, e.g., Yosida & Kakutani [33]). It is sufficient and necessary for quasi-compactness of the Markov operators. A new proof of this result is given in Lotz [25].

Section 4.3: The standard reference to retarded differential equations is Hale [18], where it is shown that the solutions of (RCP), with values in a finite dimensional

space F, yield an operator semigroup. The extension to arbitrary Banach spaces F was first made by Travis & Webb [30]. Plant [27] showed the translation property (T) for the solution semigroup. Among the many papers pursuing this functional analytic investigation of partial differential equations with delay we quote Di Blasio et al. [8] and Kunisch & Schappacher [22].

Our approach is essentially due to Kerscher [20]. We show that the first derivative with an appropriate domain is the generator of a one-parameter semigroup on an abstract function space. Due to the translation property this semigroup yields the solutions of (RCP).

The aspect of positivity in (RCP) and its influence on the stability of the solutions was first studied in Section 4 of Kerscher & Nagel [21]. In Kerscher [20] this is pursued by showing how Theorem 4.3.7 in combination with the domination of semigroups (see Part-III, Section 3.4) can be used to derive many of the known "stability independent of the delay"- results (e.g., Cooke & Ferreira [4]).

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Chapter 3

Spectral Theory of Positive Semigroups on Banach Lattices

by
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In Part-II, Chapter 3 we showed that positive semigroups on spaces $C_0(X)$ possess several interesting spectral properties. In this chapter we will extend many of these results to the more general setting of Banach lattices, improving some considerably and giving the complete proof of Theorem 3.4.1. Throughout this chapter, we will assume that $E \neq \{0\}$ is a complex Banach lattice.

3.1 The Spectral Bound

The fact that the spectral bound of a positive semigroup is always contained in the spectrum (provided the latter is non-empty) also holds true in the general setting of Banach lattices. The proof given in Part-II, Theorem 3.1.1 for spaces $C_0(X)$ works in the general case too. Another proof is given below (cf. Corollary 3.1.4). Furthermore, Part-II, Corollary 3.1.3 and Proposition 3.1.5 are true in the setting of Banach lattices and their proofs can be extended to the general case. For the sake of completeness we summarize these results in the following theorem.

Theorem 3.1.1 *Let A be the generator of a positive semigroup $(T(t))_{t\geqslant 0}$ on a Banach lattice E.*

(i) $s(A) \in \sigma(A)$ *unless* $\sigma(A) = \emptyset$.

(ii) *For $\lambda_0 \in \varrho(A)$ we have $R(\lambda_0, A)$ is positive if and only if $\lambda_0 > s(A)$. In this case $r(R(\lambda_0, A)) = (\lambda_0 - s(A))^{-1}$.*

(iii) *If $T(1)$ has a positive fixed vector h_0, then* $\ker(A)$ *contains a positive element h such that $h_0 \in E_{|h|}$.*

(iv) *If* $T(1)'\varphi_0 = \varphi_0$ *for some* for some $\varphi_0 \in E'_+$, *then there exists* $\varphi \in D(A^*)_+$ *with* $\{f \in E\colon \langle |f|, \varphi\rangle = 0\} \subseteq \{f \in E\colon \langle |f|, \varphi_0\rangle = 0\}$ *such that* $A^*\varphi = 0$.

The fact that $s(A)$ is always an eigenvalue of the adjoint (cf. Part-II, Theorem 3.1.6) is characteristic for spaces $C(K)$, K compact, as can be seen by considering the Laplacian on $L^p(\mathbb{R}^n)$, where $1 < p < \infty$, or on $C_0(\mathbb{R}^n)$ (see Part-II, Example 3.1.7). Another result which cannot be extended to arbitrary Banach lattices is that spectral bound and growth bound coincide (cf. Part-II, Theorem 4.1.4); an example is given in Part-I, Example 3.1.3. Despite this, the resolvent $R(\lambda, A)$ of a positive semigroup is given as the Laplace transform of the semigroup in the half-plane $\{z \in \mathbb{C}\colon \operatorname{Re}(z) > s(A)\}$ (even in case that $\omega_0(A) > s(A)$). Note however, that the integral exists only as an improper Riemann integral. By Datko's Theorem (Part-I, Theorem 4.1.11) the function $t \mapsto \mathrm{e}^{-\lambda t}T(t)f$ cannot be Bochner integrable for all $f \in E$ in the case $\operatorname{Re}(\lambda) \leqslant \omega_0(A)$.

Theorem 3.1.2 *Suppose* A *is the generator of a positive semigroup* $(T(t))_{t\geqslant 0}$.

For $\operatorname{Re}(\lambda) > s(A)$ *we have*

$$R(\lambda, A)f = \lim_{t\to\infty} \int_0^t \mathrm{e}^{-\lambda s}T(s)f\,\mathrm{d}s \quad \textit{for all} \quad f \in E. \tag{3.1.1}$$

Moreover, the operators $\int_0^t \mathrm{e}^{-\lambda s}T(s)\,\mathrm{d}s$ *tend to* $R(\lambda, A)$ *with respect to the operator norm as* $t \to \infty$.

Proof. We fix $\lambda_0 > \omega_0(A)$. Then, by Part-I, Proposition 1.1.11,

$$R(\lambda_0, A)^{n+1}f = \frac{1}{n!}\int_0^\infty s^n \exp(-\lambda_0 s)T(s)f\,\mathrm{d}s \quad (n \in \mathbb{N}_0, f \in E). \tag{3.1.2}$$

Given μ such that $s(A) < \mu < \lambda_0$, $f \in E_+$ and $\varphi \in E'_+$, then

$$\begin{aligned}
\langle R(\mu, A)f, \varphi\rangle &= \sum_{n=0}^\infty (\lambda_0 - \mu)^n \langle R(\lambda_0, A)^{n+1}f, \varphi\rangle = \\
&= \sum_{n=0}^\infty \int_0^\infty \frac{1}{n!}(\lambda_0 - \mu)^n \exp(-\lambda_0 s)\langle T(s)f, \varphi\rangle\,\mathrm{d}s = \\
&= \int_0^\infty \sum_{n=0}^\infty \frac{1}{n!}(\lambda_0 - \mu)^n \exp(-\lambda_0 s)\langle T(s)f, \varphi\rangle\,\mathrm{d}s = \\
&= \int_0^\infty \exp((\lambda_0 - \mu)s)\exp(-\lambda_0 s)\langle T(s)f, \varphi\rangle\,\mathrm{d}s = \\
&= \int_0^\infty \exp(-\mu s)\langle T(s)f, \varphi\rangle\,\mathrm{d}s = \\
&= \lim_{t\to\infty}\left\langle \int_0^t \exp(-\mu s)T(s)f\,\mathrm{d}s, \varphi\right\rangle.
\end{aligned} \tag{3.1.3}$$

Note that one can interchange summation and integration because all the integrands are positive functions.

It follows from Eq. (3.1.3) on the facing page that the net

$$\left(\int_0^t \mathrm{e}^{-\mu s} T(s) f \,\mathrm{d}s\right)_{t\geqslant 0}$$

converges weakly to $R(\mu, A)f$ for $t \to \infty$. Because it is monotone increasing ($f \geqslant 0$), we have strong convergence (see the corollary to II.Theorem 5.9 in Schaefer [32]).

If $\lambda = \mu + \mathrm{i}\nu$ with μ, ν real and $\mu > s(A)$, we have for arbitrary $f \in E$, $\varphi \in E'$

$$\left|\left\langle \int_r^t \mathrm{e}^{-\lambda s} T(s) f \,\mathrm{d}s, \varphi \right\rangle\right| \leqslant \int_r^t \mathrm{e}^{-\mu s} \langle T(s)|f|, |\varphi|\rangle \,\mathrm{d}s$$

hence

$$\left\| \int_r^t \mathrm{e}^{-\lambda s} T(s) f \,\mathrm{d}s \right\| \leqslant \left\| \int_r^t \mathrm{e}^{-\mu s} T(s)|f| \,\mathrm{d}s \right\|$$

$$\left|\left\langle \int_r^t \mathrm{e}^{-\lambda s} T(s) f \,\mathrm{d}s, \varphi \right\rangle\right| \leqslant \int_r^t \mathrm{e}^{-\mu s} \langle T(s)|f|, |\varphi|\rangle \,\mathrm{d}s$$

hence

$$\left\| \int_r^t \mathrm{e}^{-\lambda s} T(s) f \,\mathrm{d}s \right\| \leqslant \left\| \int_r^t \mathrm{e}^{-\mu s} T(s)|f| \,\mathrm{d}s \right\|,$$

which shows that $\lim_{t\to\infty} \int_0^t \mathrm{e}^{-\lambda s} T(s) f \,\mathrm{d}s$ exists. Thus, by Part-I, Proposition 1.1.11 we have $R(\lambda, A)f = \int_0^\infty \mathrm{e}^{-\lambda s} T(s) f \,\mathrm{d}s$.

It remains to prove that the net $\left(\int_0^t \exp(-\mu s) T(s) \,\mathrm{d}s\right)_{t\geqslant 0} \left(\int_0^t \mathrm{e}^{-\lambda s} T(s) \,\mathrm{d}s\right)_{t\geqslant 0}$ converges with respect to the operator norm. We fix μ such that $s(A) < \mu < \mathrm{Re}(\lambda)$. As we have seen above, the map $s \mapsto \mathrm{e}^{-\mu s}\langle T(s) f, \varphi\rangle$ is Lebesgue integrable for every $(f, \varphi) \in E \times E'$, thus defining a bilinear map $b\colon E \times E' \to L^1(\mathbb{R}_+)$. Using the closed graph theorem, it is easy to see that b is separately continuous, hence jointly continuous by Schaefer [31, III.Theorem 5.1]. Thus there is a constant M such that

$$\int_0^\infty \mathrm{e}^{-\mu s} |\langle T(s) f, \varphi\rangle| \,\mathrm{d}s = \|b(f, \varphi)\| \leqslant M \|f\| \|\varphi\| \quad (f \in E, \varphi \in E'). \tag{3.1.4}$$

Given $0 \leqslant t < r$ and setting $\varepsilon := \mathrm{Re}(\lambda) - \mu$ we have

$$\left|\int_t^r \mathrm{e}^{-\lambda s}\langle T(s)f,\varphi\rangle\,\mathrm{d}s\right| \leqslant \int_t^r \exp(-(\mathrm{Re}(\lambda)-\mu)s)\mathrm{e}^{-\lambda s}|\langle T(s)f,\varphi\rangle|\,\mathrm{d}s$$
$$\leqslant \mathrm{e}^{-\varepsilon t}\int_t^r \mathrm{e}^{-\lambda s}|\langle T(s)f,\varphi\rangle|\,\mathrm{d}s \leqslant \mathrm{e}^{-\varepsilon t}M\|f\|\|\varphi\|.$$

$$\left|\int_t^r \mathrm{e}^{-\lambda s}\langle T(s)f,\varphi\rangle\,\mathrm{d}s\right| \leqslant \int_t^r \exp(-(\mathrm{Re}(\lambda)-\mu)s)\mathrm{e}^{-\mu s}|\langle T(s)f,\varphi\rangle|\,\mathrm{d}s$$
$$\leqslant \mathrm{e}^{-\varepsilon t}\int_t^r \mathrm{e}^{-\mu s}|\langle T(s)f,\varphi\rangle|\,\mathrm{d}s \leqslant \mathrm{e}^{-\varepsilon t}M\|f\|\|\varphi\|.$$

It follows that $\left\|\int_t^r \mathrm{e}^{-\lambda s}T(s)\,\mathrm{d}s\right\| \leqslant M\mathrm{e}^{-\varepsilon t}$, hence $\left(\int_0^t \mathrm{e}^{-\lambda s}T(s)\,\mathrm{d}s\right)_{t\geqslant 0}$ is a Cauchy net with respect to the operator norm. □

Theorem 3.1.2 has many consequences. In particular, we can conclude that $s(A) \in \sigma(A)$ whenever $s(A) > -\infty$ (without using the analogous result for bounded operators, cf. Corollary 3.1.4 below).

In each of the following corollaries we assume that A is the generator of a positive semigroup on a Banach lattice E.

Corollary 3.1.3 *If* $\mathrm{Re}(\lambda) > s(A)$*, then we have*

$$|R(\lambda,A)f| \leqslant R(\mathrm{Re}(\lambda),A)|f| \quad (f \in E). \tag{3.1.5}$$

The proof is an immediate consequence of Theorem 3.1.2.

Corollary 3.1.4 *We have* $s(A) \in \sigma(A)$ *unless* $s(A) = -\infty$.

Proof. Assume that $s(A) > -\infty$ and $s(A) \notin \sigma(A)$, then it follows from Eq. (3.1.5) that $\{R(\lambda,A)\colon\ \mathrm{Re}(\lambda) > s(A)\}$ is uniformly bounded in $\mathcal{L}(E)$, by M say. Then

$$\{z \in \mathbb{C}\colon\ \mathrm{Re}(z) = s(A)\} \subseteq \varrho(A) \text{ and } \|R(z,A)\| \leqslant M \text{ for } z \text{ with } \mathrm{Re}(z) = s(A).$$

It follows that $\{z \in \mathbb{C}\colon |\mathrm{Re}(z) - s(A)| < M^{-1}\} \subseteq \varrho(A)$, which is absurd by the definition of $s(A)$. □

Corollary 3.1.5 *Suppose that* $s(A)$ *is a pole of order* m *of the resolvent* $R(\lambda,A)$*. Then* m *is a majorant for the order of any other pole on the line* $s(A)+\mathrm{i}\mathbb{R}$*.*

Proof. Without loss of generality we may assume that $s(A) = 0$. By Eq. (3.1.5) we have $\|R(\varepsilon+\mathrm{i}\beta,A)\| \leqslant \|R(\varepsilon,A)\|$ for every $\beta \in \mathbb{R}$, $\varepsilon > 0$. Therefore

$$\lim_{\varepsilon\to 0}\|\varepsilon^k R(\varepsilon+\mathrm{i}\beta,A)\| \leqslant \lim_{\varepsilon\to 0}\|\varepsilon^k R(\varepsilon,A)\| = 0 \text{ for } k > m. \qquad \square$$

The spectrum of a positive semigroup may be empty (see Part-II, Example 3.1.2(i)) and the spectrum of a general group may be empty as well (see Hille & Phillips [19, Section 23.16]). However, for positive groups this cannot occur. More precisely, we have the following result

Corollary 3.1.6 *If A is the generator of a positive group, then $\sigma(A) \cap \mathbb{R} \neq \emptyset$.*

Proof. Both A and $-A$ are generators of positive semigroups, hence if $\sigma(A) = \emptyset$, then $s(A) = s(-A) = -\infty$ and Eq. (3.1.5) on the preceding page implies that

$$\{R(\lambda, A) : \operatorname{Re}(\lambda) \geqslant 0\} \cup \{R(\lambda, -A) : \operatorname{Re}(\lambda) \geqslant 0\}$$

is bounded in $\mathcal{L}(E)$, i.e., $\{R(\lambda, A) : \lambda \in \mathbb{C}\}$ is bounded. By Liouville's Theorem the function $\lambda \mapsto R(\lambda, A)$ is constant, hence identically zero because $\lim_{\lambda \to \infty} R(\lambda, A) = 0$. Thus we arrive at a contradiction. □

We conclude this section by indicating possible extensions and further consequences of the results stated above.

Remarks 3.1.7 (i) Many of the results of this section remain true for positive semigroups on ordered Banach spaces more general than Banach lattices. The interested reader is referred to Greiner et al. [17].

(ii) From Theorem 3.1.2 one can easily deduce that for positive semigroups on L^1-spaces spectral bound and growth bound coincide. To prove the analogous result for L^2-spaces one makes use of Corollary 3.1.3. For details we refer to Theorem 4.1.1.

3.2 The Boundary Spectrum

In Part-II, Chapter 3 we have seen that under suitable assumptions the boundary spectrum $\sigma_b(A)$, consisting of all spectral values with maximal real part is a cyclic set (cf. Part-II, Definition 3.2.5). In the main theorem of this section we prove a result which is more general and which is true for arbitrary Banach lattices.

We first want to extend some of the notions used in Part-II, Chapter 3 to the more general setting considered here. If E is a Banach lattice and $f, g \in E$ such that $g \in E_{|f|}$, then $(\operatorname{sign} f)g$ is well-defined (cf. Section 1.8). Thus the following definition makes sense

Definition 3.2.1 If E is a Banach lattice then for $f \in E$, $n \in \mathbb{Z}$ we define $f^{[n]}$ recursively as follows

$$\begin{aligned} f^{[0]} &:= |f|, \\ f^{[n]} &:= (\operatorname{sign} f) f^{[n-1]} \text{ if } n > 0, \\ f^{[n]} &:= (\operatorname{sign} \overline{f}) f^{[n+1]} \text{ if } n < 0. \end{aligned} \tag{3.2.1}$$

Obviously, for $E = C_0(X)$ this amounts to the same as Part-II, Definition 3.2.2. Moreover, in case E is an L^p-space, then $f^{[n]}$ is the function given by

$$f^{[n]}(x) = \begin{cases} (f(x)/|f(x)|)^{n-1} f(x) & \text{if } f(x) \neq 0, \\ 0 & \text{otherwise.} \end{cases} \tag{3.2.2}$$

The following properties are immediate consequences of Definition 3.2.1

$$f^{[0]} = |f|, f^{[1]} = f, f^{[-1]} = \overline{f}, |f^{[n]}| = |f| \quad (n \in \mathbb{Z})$$

$$\mathsf{f}^{[0]} = |\mathsf{f}|, \mathsf{f}^{[1]} = \mathsf{f}, \mathsf{f}^{[-1]} = \bar{\mathsf{f}}, |\mathsf{f}^{[n]}| = |\mathsf{f}| \quad (\mathsf{n} \in \mathbb{Z}); \tag{3.2.3}$$

$$f^{[n]} = (\operatorname{sign} f) f^{[n-1]} = (\operatorname{sign} \overline{f}) f^{[n+1]} \text{ for all } n \in \mathbb{Z}; \tag{3.2.4}$$

$$(\alpha f)^{[n]} = \alpha(\alpha/|\alpha|)^{n-1} f^{[n]} \text{ for } n \in \mathbb{Z}, \alpha \in \mathbb{C}, \alpha \neq 0. \tag{3.2.5}$$

Next we show that Part-II, Theorem 3.2.4 is true for arbitrary Banach lattices. For definition and simple properties of the sign operator S_h see Section 1.8.

Theorem 3.2.2 *Let $(T(t))_{t\geqslant 0}$ be a positive semigroup on a Banach lattice E with generator A and suppose that for $h \in E$, $\alpha, \beta \in \mathbb{R}$ we have*

$$Ah = (\alpha + \mathrm{i}\beta)h, \quad A|h| = \alpha|h|. \tag{3.2.6}$$

Then the following holds true

$$Ah^{[n]} = (\alpha + \mathrm{i}n\beta)h^{[n]} \text{ for all } n \in \mathbb{Z}. \tag{3.2.7}$$

In case $|h|$ is a quasi-interior point of E_+, then

$$S_h D(A) = D(A) \text{ and } A + \mathrm{i}\beta \operatorname{Id} = S_h^{-1} A S_h.$$

Proof. Without loss of generality we may assume that $\alpha = 0$. Then the assumption Eq. (3.2.6) implies that $T(t)|h| = |h|$ and $T(t)h = \mathrm{e}^{\mathrm{i}\beta t}h$ for $t \geqslant 0$ (see Part-I, Corollary 3.6.4). In particular, the principal ideal $E_{|h|}$ is invariant under every operator $T(t)$. By the Kakutani-Krein Theorem (Section 1.4) we can identify $E_{|h|}$ with a space $C(K)$, K compact. Then the restrictions $\overline{T}(t) := T(t)|_{E_{|h|}}$ are positive operators on $C(K)$ satisfying $\overline{T}(t)|\overline{h}| = |\overline{h}|$ and $\overline{T}(t)\overline{h} = \mathrm{e}^{\mathrm{i}\beta t}\overline{h}$.

From Part-II, Theorem 3.2.4(i) we conclude $\overline{T}(t)\overline{h}^{[n]} = \mathrm{e}^{\mathrm{i}\beta t}\overline{h}^{[n]}$ $\overline{T}(t)\overline{h}^{[n]} = \mathrm{e}^{\mathrm{i}n\beta t}\overline{h}^{[n]}$ for all $t \geqslant 0$, $n \in \mathbb{Z}$. Translating this back to $T(t)$ and E this means precisely $T(t)h^{[n]} = \mathrm{e}^{\mathrm{i}n\beta}h^{[n]}T(t)h^{[n]} = \mathrm{e}^{\mathrm{i}n\beta t}h^{[n]}$, hence $h^{[n]} \in D(A)$ and $Ah^{[n]} = \mathrm{i}n\beta h^{[n]}$.

Moreover, by Part-II, Theorem 3.2.4(i) we have $e^{i\beta t}\overline{T}(t) = S_h^{-1}\overline{T}(t)S_h$. If $|h|$ is a quasi-interior point, this relation extends by continuity from the dense subspace $E_{|h|}$ to the whole space E, i.e., we have $e^{i\beta t}T(t) = S_h^{-1}T(t)S_h$ for all $t \geqslant 0$. □

In the proof above we could not apply assertion (ii) of Part-II, Theorem 3.2.4 because the semigroup $(\overline{T}(t))$ on $E_{|h|} \cong C(K)$ need not be strongly continuous with respect to the sup-norm.

As a first application of Theorem 3.2.2 we prove a cyclicity result for the point spectrum of contraction semigroups on a class of Banach lattices which includes the L^p-spaces.

Corollary 3.2.3 *Suppose E is a Banach lattice such that the norm is strictly monotone on E_+ (i.e., $0 \leqslant f < g \Rightarrow \|f\| < \|g\|$). If $(T(t))$ is a positive contraction semigroup on E with $s(A) = 0$, then $P\sigma_b(A) = P\sigma(A) \cap i\mathbb{R}$ is imaginary additively cyclic.*

Proof. Suppose that $Ah = i\beta h$ ($\beta \in \mathbb{R}, h \in E$). Then we have $T(t)h = e^{i\beta t}h$ ($t \geqslant 0$) and $|h| = |T(t)h| \leqslant T(t)|h|$ since $T(t)$ is positive. Moreover, $\|h\| \leqslant \||T(t)|h|\| \leqslant \|h\|$ since $\|T(t)\| \leqslant 1$. The assumption on the norm of E implies that $T(t)|h| = |h|$ for all $t \geqslant 0$, equivalently $A|h| = 0$. Now we can apply Theorem 3.2.2 in order to obtain the desired result. □

A more general result on cyclicity of the eigenvalues in the boundary spectrum will be proved in Corollary 3.4.3. In the remaining part of this section we focus our interest on the entire boundary spectrum. We will prove that it is cyclic provided that the resolvent $R(\lambda, A)$ does not grow too fast as $\lambda \to s(A)$. We start with some preparations. An important tool in the proof is the notion of pseudo-resolvents.

Definition 3.2.4 Let D be an open (non-empty) subset of $\mathbb{C}$ and let G be a Banach space. A mapping $R\colon D \to \mathcal{L}(G)$ which satisfies the *resolvent equation*

$$R(\lambda) - R(\mu) = -(\lambda - \mu)R(\lambda)R(\mu) \quad (\lambda, \mu \in D) \tag{3.2.8}$$

is called a *pseudo-resolvent* on G.

An equivalent (often quite useful) version of Eq. (3.2.8) is the following

$$(1 - (\lambda - \mu)R(\lambda))(1 - (\mu - \lambda)R(\mu)) = 1 \quad (\lambda, \mu \in D)$$

$$(\mathrm{Id} - (\lambda - \mu)\mathsf{R}(\lambda))(\mathrm{Id} - (\mu - \lambda)\mathsf{R}(\mu)) = \mathrm{Id} \quad (\lambda, \mu \in \mathsf{D}) \tag{3.2.9}$$

Obviously, the resolvent of a closed linear operator A on G is a pseudo-resolvent on $D = \varrho(A)$. In general a pseudo-resolvent need not be the resolvent of an operator. Further information can be found in Hille & Phillips [19], Pazy [28] or Yosida [39]. For our purposes the following examples are of particular interest.

Example 3.2.5 (i) Suppose A is a densely defined linear operator on G with $\varrho(A) \neq \emptyset$ and let $G_{\mathcal{F}}$ be an $\mathcal{F}$-product of G (cf. Part-I, Section 1.3.7). Then the canonical extensions $R(\lambda, A)_{\mathcal{F}}$ of $R(\lambda, A)$ form a pseudo-resolvent $R_{\mathcal{F}}$ on $G_{\mathcal{F}}$ with $\varrho(A)$ as domain of definition. If A is unbounded, then $0 \in A\sigma(R(\lambda, A))$ hence $0 \in P\sigma(R_{\mathcal{F}}(\lambda, A))$ $0 \in P\sigma(R_{\mathcal{F}}(\lambda))$ (cf. Part-I, Proposition 3.4.2). It follows that $R_{\mathcal{F}}$ is not the resolvent of an operator on $G_{\mathcal{F}}$.

(ii) If $\{R(\lambda)\}_{\lambda \in D}$ is a pseudo-resolvent on G, then $\{R(\lambda)'\}_{\lambda \in D}$ is a pseudo-resolvent on G'. Moreover, if H is a closed linear subspace of G which is $\{R(\lambda)\}_{\lambda \in D}$-invariant ($R(\lambda)H \subseteq H$ for all $\lambda \in D$), then the operators on H and G/H induced by $R(\lambda)$ in the canonical way form a pseudo-resolvent on H and G/H, respectively.

We list several simple properties below. We assume that $R\colon D \to \mathcal{L}(G)$ is a pseudo-resolvent on a Banach space G

$$\begin{aligned} &\text{Given } \lambda_0 \in D, \mu \in \mathbb{C} \text{ there exists at most one operator } S \in \mathcal{L}(G) \\ &\text{such that} \quad R(\lambda_0) - S = -(\lambda_0 - \mu)R(\lambda_0)S = -(\lambda_0 - \mu)SR(\lambda_0). \\ &\text{In case such an operator exists we have} \\ &R(\lambda) - S = -(\lambda - \mu)R(\lambda)S = -(\lambda - \mu)SR(\lambda) \text{ for all } \lambda \in D. \end{aligned} \tag{3.2.10}$$

$$\begin{aligned} &\text{Given } \lambda_0 \in D, \text{ then for } \mu \in D \text{ with } |\mu - \lambda_0| < \|R(\lambda_0)\|^{-1} \\ &\text{we have } R(\mu) = \sum_{n=0}^{\infty} (\lambda_0 - \mu)^n R(\lambda_0)^{n+1}. \end{aligned}$$

$$\begin{aligned} &\text{Given } \lambda_0 \in D, \text{ then for } \mu \in D \text{ with } |\mu - \lambda_0| < \|R(\lambda_0)\|^{-1} \\ &\text{we have } R(\mu) = \sum_{n=0}^{\infty} (\lambda_0 - \mu)^n R(\lambda_0)^{n+1}. \end{aligned} \tag{3.2.11}$$

$$\begin{aligned} &\text{The map } \lambda \mapsto R(\lambda) \text{ is a locally holomorphic function defined on } D \subseteq \mathbb{C} \\ &\text{with values in } \mathcal{L}(G). \end{aligned} \tag{3.2.12}$$

We only sketch the proof of these assertions. Eq. (3.2.12) follows from Eq. (3.2.11) and the latter is a consequence of Eq. (3.2.10). The identity stated in Eq. (3.2.10) can be rewritten as follows

$$\begin{aligned} (\mathrm{Id} - (\lambda_0 - \mu)R(\lambda_0))(\mathrm{Id} - (\mu - \lambda_0)S) &= \mathrm{Id} \\ &= (\mathrm{Id} - (\mu - \lambda_0)S)(\mathrm{Id} - (\lambda_0 - \mu)R(\lambda_0)). \end{aligned}$$

Thus

$$S = (\mu - \lambda_0)^{-1}(1 - (1 - (\lambda_0 - \mu)R(\lambda_0))^{-1})$$

$$\mathsf{S} = (\mu - \lambda_0)^{-1}(\mathrm{Id} - (\mathrm{Id} - (\lambda_0 - \mu)\mathsf{R}(\lambda_0))^{-1})$$

has to be unique.

It follows from Eq. (3.2.11) and Eq. (3.2.12) on the preceding page that every pseudo-resolvent has a unique maximal extension. Further properties of pseudo-resolvents are given in the following two propositions.

Proposition 3.2.6 *Suppose G is a Banach space, $D \subseteq \mathbb{C}$ and $R\colon D \to \mathcal{L}(G)$ is a pseudo-resolvent.*

(i) *Given $\alpha \in \mathbb{C}$, $x \in G$ one has $(\lambda - \alpha)R(\lambda)x = x$ either for all $\lambda \in D$ or for none.*

(ii) *Suppose $\mu \in \overline{D} \setminus D$ $\mu \in \overline{D} \setminus D$. Then R can be extended to an open set containing μ if and only if there exists a sequence $(\lambda_n) \subseteq D$ converging to μ such that $\|R(\lambda_n)\|$ is bounded.*

Proof. (i) Suppose that $(\lambda - \alpha)R(\lambda)x = x$ for some fixed $\lambda \in D, x \in G$. Then using Eq. (3.2.8) on page 337 we have for $\mu \in D$

$$(\mu - \alpha)(\mu - \lambda)R(\mu)R(\lambda)x = (\lambda - \alpha)(R(\lambda)x - R(\mu)x) = x - (\lambda - \alpha)R(\mu)x.$$

It follows that $(\mu - \alpha)R(\mu)x = x$ for all $\mu \in D$.

(ii) If there exists an extension, then $\|R(\lambda_n)\|$ is bounded for every sequence (λ_n) converging to μ by Eq. (3.2.12) on the preceding page. On the other hand assuming that $\|R(\lambda_n)\|$ is bounded by M for a fixed sequence $(\lambda_n) \subseteq D$ with $\lambda_n \to \mu$ $(M \geqslant 0)$, we have

$$\|R(\lambda_n) - R(\lambda_m)\| = |\lambda_n - \lambda_m|\|R(\lambda_n)R(\lambda_m)\| \leqslant M^2|\lambda_n - \lambda_m|$$

which shows that $(R(\lambda_n))$ is a Cauchy sequence in $\mathcal{L}(G)$, hence $S := \lim_{n\to\infty} R(\lambda_n)$ exists. The assertion now follows from Eq. (3.2.10) on the facing page and Eq. (3.2.11) on the preceding page. □

In the next proposition, we consider a *positive pseudo-resolvent* R on a Banach lattice E, i.e., we assume that the domain D of R contains the positive real axis and that $R(\mu)$ is a positive operator for every $\mu > 0$. Applying Pringsheim's Theorem (see Theorem 2.1 in the appendix of Schaefer [31]) to the expansion given in Eq. (3.2.11) on the facing page one can conclude that R has an extension to the half-plane $\{z \in \mathbb{C}\colon \operatorname{Re}(z) > 0\}$.

This shows that without loss of generality one can assume that the domain of a positive pseudo-resolvent contains the half-plane $\{z \in \mathbb{C}\colon \operatorname{Re}(z) > 0\}$.

Proposition 3.2.7 *Suppose $R\colon \Delta \to \mathcal{L}(E)$ is a positive pseudo-resolvent on a Banach lattice E and $\Delta := \{z \in \mathbb{C}\colon \operatorname{Re}(z) > 0\}$.*

If for some $\beta \in \mathbb{R}$, $h \in E$ we have $\lambda R(\lambda + \mathrm{i}\beta)h = h$ and $\lambda R(\lambda)|h| = |h|$ $(\lambda \in \Delta)$, then

$$\lambda R(\lambda + \mathrm{i}n\beta)h^{[n]} = h^{[n]} \text{ for all } n \in \mathbb{Z}, \lambda \in \Delta.$$

Proof. At first we prove the following domination property which is the extension of Eq. (3.1.5) on page 334 to positive pseudo-resolvents

$$|R(\lambda)f| \leqslant R(\mathrm{Re}(\lambda))|f| \quad \text{for every} \quad \lambda \in \Delta, f \in E. \tag{3.2.13}$$

To do this we fix $\lambda \in \Delta$. Then there exists $r_0 > 0$ such that $|r - \lambda| < r$ whenever $r > r_0$. Therefore $R(\lambda) = \sum_{n=0}^{\infty}(r-\lambda)^n R(r)^{n+1}$ for $r > r_0$, which implies for $f \in E$

$$\begin{aligned} |R(\lambda)f| &\leqslant \sum_{n=0}^{\infty} |r-\lambda|^n R(r)^{n+1}|f| = \\ &= \sum_{n=0}^{\infty} (r-(r-|r-\lambda|))^n R(r)^{n+1}|f| = R(r-|\lambda - r|)|f|. \end{aligned}$$

Since $\lim_{r\to\infty}(r - |\lambda - r|) = \mathrm{Re}(\lambda)$, we obtain Eq. (3.2.13).

As a consequence of Eq. (3.2.13) and the assumption $rR(r)|h| = |h|$ we have that the principal ideal $E_{|h|}$ is $\{R(\lambda)\}_{\lambda\in\Delta}$-invariant. Identifying, according to the Kakutani-Krein Theorem, $E_{|h|}$ with a space $C(K)$, K compact, and by restricting the operators $R(\lambda)$ to $E_{|h|} \cong C(K)$, we obtain a positive pseudo-resolvent $\tilde{R}\colon \Delta \to \mathcal{L}(C(K))$. Then we have for every $\alpha > 0$ and $f \in E$

$$\alpha\tilde{R}(\alpha + \mathrm{i}\beta)h = h, \alpha\tilde{R}(\alpha)|h| = |h| = \mathbf{1}_K, \alpha|\tilde{R}(\alpha+\mathrm{i}\beta)f| \leqslant \alpha\tilde{R}(\alpha)|f|.$$

Applying Part-II, Lemma 3.2.3 we obtain $\tilde{R}(\alpha) = S_h^{-1}\tilde{R}(\alpha+\mathrm{i}\beta)S_h$ for every $\alpha > 0$ and using the uniqueness theorem for holomorphic functions we obtain

$$\tilde{R}(z) = S_h^{-n}\tilde{R}(z + \mathrm{i}n\beta)S_h^n \quad \text{for all} \quad z \in \Delta \quad \text{and} \quad n \in \mathbb{Z}. \tag{3.2.14}$$

In particular, $S_h^n|h| = S_h^{-n}z\tilde{R}(z)|h| = z\tilde{R}(z+\mathrm{i}n\beta)S_h^n|h| S_h^n|h| = S_h^n z\tilde{R}(z)|h| = z\tilde{R}(z + \mathrm{i}n\beta)S_h^n|h|$.

In terms of the initial space this means precisely $h^{[n]} = zR(z+\mathrm{i}n\beta)h^{[n]}$, and the proposition is proved. □

We will prove cyclicity of the boundary spectrum under a *growth condition* which is stated in the following definition.

Definition 3.2.8 Let A be the generator of a positive semigroup $(T(t))_{t\geqslant 0}$ with spectral bound $s(A) > -\infty$. The resolvent is said to *grow slowly* if one of the following (equivalent) conditions is satisfied.

$$\begin{aligned} &\text{(a)} \quad \{(\lambda - s(A))R(\lambda, A)\colon \lambda > s(A)\} \text{ is bounded in } \mathcal{L}(E). \\ &\text{(b)} \quad \left\{\frac{1}{t}\int_0^t \exp(-\tau s(A))T(\tau)\,\mathrm{d}\tau\colon t > 0\right\} \text{ is bounded in } \mathcal{L}(E). \end{aligned} \tag{3.2.15}$$

Before proving the equivalence of the two assertions we make some remarks.

Remark (i) Since one always has

$$\lambda R(\lambda, A) \to \mathrm{Id} \text{ for } \lambda \to \infty,$$

the set

$$\{(\lambda - s(A))R(\lambda, A) : \lambda > s(A) + \varepsilon\}$$

is bounded for every $\varepsilon > 0$. Thus in condition (a) of Eq. (3.2.15) on the facing page (a) the essential fact is boundedness near $s(A)$. On the other hand, the set of operators

$$\left\{\frac{1}{t}\int_0^t \exp(-\tau s(A))T(\tau)\,\mathrm{d}\tau : 0 \leqslant t \leqslant t_0\right\}$$

is bounded for every $t_0 \geqslant 0$, hence in condition (b) of Eq. (3.2.15) on the preceding page (b) the boundedness for $t \to \infty$ is important.

(ii) By Corollary 3.1.4 we have

$$\|R(\lambda, A)\| \geqslant r(R(\lambda, A)) = (\lambda - s(A))^{-1}.$$

Hence $\|R(\lambda, A)\|$ grows at least as fast as $(\lambda - s(A))^{-1}$. Thus if condition (a) of Eq. (3.2.15) on the facing page (a) is satisfied, the resolvent grows as slowly as it possibly can for $\lambda \to s(A)$.

(iii) We do not assume in Definition 3.2.8 that spectral bound and growth bound coincide. A slight modification of Part-I, Example 3.1.3 shows that there are semigroups with slowly growing resolvent and $s(A) < \omega_0(A)$.

Proof. To prove equivalence of (a) and (b) of Eq. (3.2.15) on the preceding page, we assume $s(A) = 0$ and write $C(t) := \frac{1}{t}\int_0^t T(\tau)\,\mathrm{d}\tau$.

(a) $\Rightarrow$ (b): Consider $\lambda > 0, t > 0$ such that $\lambda t = 1$. Then we have

$$\lambda \cdot \mathrm{e}^{-\lambda s} \geqslant \begin{cases} (\mathrm{e}t)^{-1} & \text{if } s \leqslant t, \\ 0 & \text{if } s > t. \end{cases}$$

Now Eq. (3.1.1) on page 332 implies

$$\lambda R(\lambda, A) = \int_0^\infty \lambda \exp(-\lambda s)T(s)\,\mathrm{d}s \geqslant \mathrm{e}^{-1}C(t) = \mathrm{e}^{-1} \cdot C(\frac{1}{\lambda}) \geqslant 0.$$

Thus $C(t)$ is bounded for $t \to \infty$ whenever $\lambda R(\lambda, A)$ is bounded for $\lambda \downarrow 0$.

(b) $\Rightarrow$ (a): Let $M := \sup\{\|C(t)\| : t > 0\}$. Given $f \in E$, $\lambda > 0$ and $r > 0$ then integration by parts yields

$$\lambda \int_0^r \mathrm{e}^{-\lambda s}T(s)f\,\mathrm{d}s = \lambda \mathrm{e}^{-\lambda r}\int_0^r T(\sigma)f\,\mathrm{d}\sigma + \lambda^2 \int_0^r s\mathrm{e}^{-\lambda s}\left(\frac{1}{s}\int_0^s T(\sigma)f\,\mathrm{d}\sigma\right)\mathrm{d}s.$$

It follows that

$$\left\|\lambda \int_0^r e^{-\lambda s} T(s) f \, ds\right\| \leqslant \left(r\lambda e^{-r} + \lambda^2 \int_0^r s e^{-\lambda s} \, ds\right) M\|f\| = (1 - e^{-\lambda r}) M\|f\|.$$

$$\left\|\lambda \int_0^r e^{-\lambda s} T(s) f \, ds\right\| \leqslant \left(r\lambda e^{-\lambda r} + \lambda^2 \int_0^r s e^{-\lambda s} \, ds\right) M\|f\| = (1 - e^{-\lambda r}) M\|f\|.$$

Letting $r \to \infty$, we obtain by Eq. (3.1.1) on page 332 $\|\lambda R(\lambda, A) f\| \leqslant M\|f\|$ $(f \in E, \lambda > 0)$ hence $\|\lambda R(\lambda, A)\| \leqslant M$. □

Two sufficient conditions for a resolvent to grow slowly are stated in the following proposition. Its simple proof is omitted.

Proposition 3.2.9 *Suppose $(T(t))_{t \geqslant 0}$ is a positive semigroup with generator A. Each of the following conditions guarantees that the resolvent grows slowly.*

(i) *$(T(t))_{t \geqslant 0}$ is bounded and $s(A) = 0$;*

(ii) *$s(A)$ is a first order pole of the resolvent.*

In case $s(A)$ is a pole of order greater than 1, the resolvent does not grow slowly. We will treat this case in Corollary 3.2.12.

Theorem 3.2.10 *The boundary spectrum of a positive semigroup with slowly growing resolvent is cyclic.*

Proof. Without loss of generality we can assume that $s(A) = 0$.

Given $i\beta \in \sigma(A)$ $(\beta \in \mathbb{R})$, then $i\beta \in A\sigma(A)$ (Part-I, Proposition 3.2.2) and $(\lambda - i\beta)^{-1} \in A\sigma(R(\lambda, A))$ (Part-I, Proposition 3.2.5). We consider an $\mathcal{F}$-product $E_{\mathcal{F}}$ of E and for convenience write E_1 instead of $E_{\mathcal{F}}$. The canonical extensions of $R(\lambda, A)$ to E_1 form a positive pseudo-resolvent $\{R_1(\lambda)\}_{\mathrm{Re}(\lambda) > 0}$ on E_1. By Proposition 3.2.6 (i) and Part-I, Section 3.1 there exists $h_1 \in E_1$, $h_1 \neq 0$ such that

$$\lambda R_1(\lambda + i\beta) h_1 = h_1 \quad \text{for } \mathrm{Re}(\lambda) > 0. \tag{3.2.16}$$

By Eq. (3.2.13) on page 340 we have

$$|h_1| = |r R_1(r + i\beta) h_1| \leqslant r R_1(r) |h_1| \ (r > 0). \tag{3.2.17}$$

Next, we choose any $\varphi \in E_1'$ such that $\langle h_1, \varphi \rangle \neq 0$. Since $\|R_1(\lambda)'\| = \|R_1(\lambda)\| = \|R(\lambda, A)\|$, the assumption of slow growth implies that $\{\lambda R_1(\lambda)' \varphi \colon \lambda > 0\}$ is bounded

in E_1', hence $\sigma(E_1', E_1)$-relatively compact by *Alaoglu's Theorem*. Thus, there exist $\varphi_1 \in \cap_{\varepsilon>0}\{\tau R_1(\tau)'\varphi \colon 0 < \tau < \varepsilon\}^{-\sigma(E_1', E_1)}$.

Using the resolvent equation Eq. (3.2.8) on page 337 we obtain for $r > 0\tau > 0$, $\varepsilon > 0$

$$(1 - \tau R_1(\tau)')\varepsilon R_1(\varepsilon)'\varphi = \varepsilon(\tau - \varepsilon)^{-1}(\tau R_1(\tau)'\varphi - \varepsilon R_1(\varepsilon)'\varphi).$$

$$(\mathrm{Id} - \tau \mathsf{R}_1(\tau)')\varepsilon \mathsf{R}_1(\varepsilon)'\varphi = \varepsilon(\tau - \varepsilon)^{-1}(\tau \mathsf{R}_1(\tau)'\varphi - \varepsilon \mathsf{R}_1(\varepsilon)'\varphi).$$

Since the right hand side tends to 0 as $\varepsilon \to 0$, we have $(1 - \tau R_1(\tau)')\varphi_1 = 0$ $(\mathrm{Id} - \tau R_1(\tau)')\varphi_1 = 0$ or

$$\lambda R_1(\lambda)'\varphi_1 = \varphi_1 \quad \text{for } \mathrm{Re}(\lambda) > 0. \tag{3.2.18}$$

Moreover, from

$$0 < |\langle h_1, \varphi\rangle| \leqslant \langle |h_1|, |\varphi|\rangle \leqslant \langle R_1(\tau)|h_1|, |\varphi|\rangle = \langle |h_1|, \tau R_1(\tau)'\varphi\rangle$$

it follows that

$$\langle |h_1|, \varphi_1\rangle > 0. \tag{3.2.19}$$

For arbitrary $f_1 \in E_1$, $\mathrm{Re}(\lambda) > 0$ we have $\langle |R_1(\lambda)f_1|, \varphi_1\rangle \leqslant \langle R_1(\mathrm{Re}(\lambda))|f_1|, \varphi_1\rangle = \langle |f_1|, R_1(\mathrm{Re}(\lambda))'\varphi_1\rangle = (\mathrm{Re}(\lambda))^{-1}\langle |f_1|, \varphi_1\rangle$.

Therefore the ideal $I := \{f_1 \in E_1 \colon \langle |f_1|, \varphi_1\rangle = 0\}$ is invariant under $\{R_1(\lambda)\}_{\mathrm{Re}(\lambda)>0}$. Furthermore we have (see Eq. (3.2.17) on the facing page, Eq. (3.2.18))

$$\begin{aligned}\langle |rR_1(r)|h_1| - |h_1||, \varphi_1\rangle &= \langle rR_1(r)|h_1| - |h_1|, \varphi_1\rangle = \\ &= \langle |h_1|, rR_1(r)'\varphi_1 - \varphi_1\rangle = 0\end{aligned}$$

$$\begin{aligned}\langle |\mathsf{r}\mathsf{R}_1(\mathsf{r})|\mathsf{h}_1| - |\mathsf{h}_1||, \varphi_1\rangle &= \langle \mathsf{r}\mathsf{R}_1(\mathsf{r})|\mathsf{h}_1| - |\mathsf{h}_1|, \varphi_1\rangle = \\ &= \langle |\mathsf{h}_1|, \mathsf{r}\mathsf{R}_1(\mathsf{r})'\varphi_1 - \varphi_1\rangle = 0,\end{aligned}$$

which implies

$$rR_1(r)|h_1| - |h_1| \in I \text{ for } r > 0).$$

$$\mathsf{r}\mathsf{R}_1(\mathsf{r})|\mathsf{h}_1| - |\mathsf{h}_1| \in \mathsf{I} \text{ for } \mathsf{r} > 0. \tag{3.2.20}$$

Denote by E_2 the quotient space E_1/I and by $\{R_2(\lambda)\}_{\mathrm{Re}(\lambda)>0}$ the pseudo-resolvent on E_2 induced by $\{(R_1(\lambda)\}_{\mathrm{Re}(\lambda)>0}$ $\{R_1(\lambda)\}_{\mathrm{Re}(\lambda)>0}$ in the canonical way. Then $h_2 := h_1 + I \neq 0$ (by Eq. (3.2.19)). Moreover, $\lambda R_2(\lambda + \mathrm{i}\beta)h_2 = h_2$ (by Eq. (3.2.16) on the preceding page) and $\lambda R_2(\lambda)|h_2| = |h_2|$ (by Eq. (3.2.20) and Proposition 3.2.6 (i)).

Now we apply Proposition 3.2.7 (b) and obtain

$$\lambda R_2(\lambda + \mathrm{i}n\beta)h_2^{[n]} = h_2^{[n]} \quad \text{for} \quad \mathrm{Re}(\lambda) > 0, n \in \mathbb{Z}. \tag{3.2.21}$$

In particular, we have $\|R_2(r + \mathrm{i}n\beta)\| \geqslant \frac{1}{r}$, thus

$$\|R(r+\mathrm{i}n\beta, A)\| = \|R_1(r+\mathrm{i}n\beta)\| \geqslant \|R_2(r+\mathrm{i}n\beta)\| \geqslant \frac{1}{r} \quad \text{for } r > 0.$$

This finally implies that $\mathrm{i}n\beta \in \sigma(A)$ for $n \in \mathbb{Z}$. □

To prove cyclicity of the boundary spectrum in case $s(A)$ is a pole (of arbitrary order), one applies Part-II, Lemma 3.2.8 to reduce the problem to the case of first order poles. Actually, Part-II, Lemma 3.2.8 is true for arbitrary Banach lattices and the proof given in Part-II, Chapter 3 works in the general case as well. For completeness we recall this result.

Proposition 3.2.11 *Let A be the generator of a positive semigroup $\mathcal{T}$ on a Banach lattice E and suppose that the spectral bound $s(A)$ is a pole of the resolvent of order k. Then there is a sequence*

$$I_{-1} := \{0\} \subseteq I_0 \subsetneq I_1 \subsetneq \ldots \subsetneq I_k := E$$

$$\mathsf{I}_{-1} := \{0\} \subseteq \mathsf{I}_0 \subsetneq \mathsf{I}_1 \subsetneq \cdots \subsetneq \mathsf{I}_k := \mathsf{E} \tag{3.2.22}$$

of $\mathcal{T}$-invariant closed ideals with the following properties. If A_n is the generator of the semigroup induced by $\mathcal{T}$ on the quotient I_n/I_{n-1}, then we have

(i) $s(A_0) < s(A)$,

(ii) *If $n \geqslant 1$, then $s(A_n) = s(A)$ is a first order pole of the resolvent $R(\cdot, A_n)$. The corresponding residue is a strictly positive operator.*

Combining Theorem 3.2.10 and Proposition 3.2.11 one obtains the following generalization of Part-II, Theorem 3.2.9. In contrast with the former result we do not assume that every point of $\sigma_b(A)$ is a pole.

Corollary 3.2.12 *If A is the generator of a positive semigroup such that $s(A)$ is a pole of the resolvent, then $\sigma_b(A)$ is cyclic .*

Proof. Considering the sequence of ideals as described in Proposition 3.2.11 and the corresponding generators A_n $(0 \leqslant n \leqslant k)$, then we have by Part-I, Proposition 3.4.2 that

$$\sigma_b(A) = \bigcup_{n=1}^{k} \sigma_b(A_n).$$

By Theorem 3.2.10 each set $\sigma_b(A_n)$ is cyclic, hence so is $\sigma_b(A)$. □

The proof of the preceding corollary indicates a possible generalization of Theorem 3.2.10. Instead of assuming that the resolvent grows slowly one merely needs that there exist a finite chain of closed $\mathcal{T}$-invariant ideals as described in Eq. (3.2.22) such that the semigroups induced on the corresponding quotient spaces have slowly growing resolvents.

In case that $\sigma_b(A)$ is cyclic one has the alternative (cf.Part-II, Eq. (3.2.19) on page 205)

$$\text{Either } \sigma_b(A) = \{s(A)\} \text{ or else } \sigma_b(A) \text{ is an unbounded set.}$$

Obviously one can use this fact to prove the existence of a dominant spectral value (cf. the explanation preceding Part-II, Corollary 3.2.11). We give a typical example.

Corollary 3.2.13 *Let A be the generator of a positive, eventually norm-continuous semigroup. Suppose either that the resolvent grows slowly or that $s(A)$ is a pole of the resolvent. Then $s(A)$ is a dominant spectral value.*

Proof. The boundary spectrum $\sigma_b(A)$ is cyclic (Theorem 3.2.10 and Corollary 3.2.12 resp.) and compact (Part-I, Theorem 2.1.19). Hence $\sigma_b(A) = \{s(A)\}$. □

A situation in which Corollary 3.2.13 can be applied is described in the following example. For more details and proofs we refer to Amann [2].

Example 3.2.14 Let Ω be a bounded domain in $\mathbb{R}^n$ of class C^2.

Assume that $\partial\Omega = \Gamma_0 \cup \Gamma_1$ where Γ_0 and Γ_1 are disjoint open and closed subsets of $\partial\Omega$. On $E = L^p(\Omega)$ $(1 \leqslant p < \infty)$ we consider a differential operator $L_{p,o}$ $L_{p,0}$ which is defined as follows

$$\begin{aligned} L_{p,0}f &:= \sum_{i,j=1}^{n} a_{ij} f''_{ij} + \sum_{i=1}^{n} b_i f'_i + c\, f, \quad \text{with domain} \\ D(L_{p,0}) &:= \{f \in C^2(\bar{\Omega}) \colon f(x) = 0 \text{ for } x \in \Gamma_0 \\ &\qquad \text{and } \partial f/\partial \nu(x) + \gamma(x) f(x) = 0 \text{ for } x \in \Gamma_1\} \end{aligned} \tag{3.2.23}$$

Here f'_i stands for $\partial f/\partial x_i$, thus $f''_{ij} = \partial^2 f/\partial x_i \partial x_j$. We assume that $L_{p,0}$ is elliptic and that all coefficients are real-valued satisfying $a_{ij} \in C^2(\bar{\Omega})$, $b_i \in C^1(\bar{\Omega})$, $\gamma \in C^1(\bar{\Omega})$, $c \in C^1(\bar{\Omega})$.

Then $L_{p,0}$ is closable and its closure L_p is the generator of a holomorphic semigroup of positive operators. Moreover, the resolvent is compact. Thus Corollary 3.2.13 is applicable and one obtains that $s(A)$ $s(L_p)$ is strictly dominant provided that $\sigma(A) \neq \emptyset \sigma(L_p) \neq \emptyset$. Using the results of Section 3.3 one can show that $\sigma(A) \neq \emptyset$ and that $s(A)$ $\sigma(L_p) \neq \emptyset$ and that $s(L_p)$ is an algebraically simple eigenvalue (see Theorem 3.3.7 and Proposition 3.3.5).

We conclude with some remarks.

Remarks 3.2.15 (i) In the proof of Theorem 3.2.10 we did not use the assumption that R is the resolvent of a semigroup. In fact one can state this theorem for

closed operators having positive resolvent. In this case one has to assume that $\{(\lambda - s(A))R(\lambda, A) : s(A) < \lambda < s(A) + 1\}$ $\{(\lambda - s(A))R(\lambda, A) : s(A) < \lambda < s(A) + 1\}$ is bounded in $\mathcal{L}(E)$.

One can go even further and consider positive pseudo-resolvents $\{R(\lambda)\}_{\lambda \in D}$. This is also possible with respect to Corollary 3.2.12.

(ii) If $s(A)$ is a pole, then the criteria stated in Part-II, Remark 3.2.15 (a) for $s(A)$ to be a first order pole are valid in the setting of arbitrary Banach lattices as well. In particular, one has a first order pole provided that $\ker(s(A) - A)$ contains a quasi-interior point or in case that $\ker(s(A) - A')$ contains a strictly positive linear form.

(iii) It is not difficult to give examples of semigroups whose resolvent do not grow slowly or cannot be reduced by a finite chain of invariant ideals as described after Corollary 3.2.12. For example, one can take a bounded positive operator A which is not nilpotent and satisfies $\sigma(A) = \{0\}$. However, the following question is still unanswered.

Does every positive semigroup have cyclic boundary spectrum?

3.3 Irreducible Semigroups

The concept of irreducibility is very natural in the general setting of Banach lattices. However, some of the (equivalent) assertions stated in Part-II, Definition 3.3.1 do not make sense here, others need a slightly different formulation.

Definition 3.3.1 A positive semigroup $(T(t))_{t \geqslant 0}$ on a Banach lattice E with generator A is called *irreducible* if one of the following (mutually equivalent) conditions is satisfied.

(a) There is no $(T(t))$-invariant closed ideal except $\{0\}$ and E.

(b) Given $f \in E$, $\varphi \in E'$ such that $f > 0$, $\varphi > 0$ then $\langle T(t_0)f, \varphi \rangle > 0$ for some $t_0 \geqslant 0$.

(c) For some (every) $\lambda > s(A)$ there is no closed ideal other than $\{0\}$ or E which is invariant under $R(\lambda, A)$.

(d) For some (every) $\lambda > s(A)$ we have $R(\lambda, A)f$ is a quasi-interior point of E_+ whenever $f > 0$.

Equivalence of the conditions above is obtained by a slight modification of the arguments given in Part-II, Definition 3.3.1, so we omit a detailed proof. Obviously, a semigroup is irreducible if one single operator $T(t_0)$ is irreducible. In general the converse does not hold (see p.65 in Greiner [13]). The situation is different when holomorphic semigroups are considered. Then an even stronger assertion holds. In

fact irreducibility of a holomorphic semigroup implies that every single operator maps the positive elements into quasi-interior points. This is the second statement of the following theorem (see also Part-II, Remark 3.3.2).

Theorem 3.3.2 (i) *If $(T(t))$ is an irreducible semigroup, then every operator $T(t)$ is strictly positive, i.e., given $f > 0$, $t \geqslant 0$, then $T(t)f > 0$.*

(ii) *Suppose $(T(t))_{t\geqslant 0}$ is a holomorphic positive semigroup. If $(T(t))$ is irreducible, then $T(t)f$ is a quasi-interior point of E_+ whenever $f > 0$ and $t > 0$. Equivalently, given $f \in E$, $\varphi \in E'$ such that $f > 0$, $\varphi > 0$, then $\langle T(t)f, \varphi\rangle > 0$ for all $t > 0$.*

Proof. (i) Given $t > 0$ and $f > 0$ such that $T(t)f = 0$ and $\lambda > s(A)$, then we have

$$T(t)(R(\lambda, A)f) = R(\lambda, A)T(t)f = 0.$$

Since $R(\lambda, A)f$ is a quasi-interior point, it follows that $T(t) = 0$. Thus for fixed $t \in \mathbb{R}_+$ we have either $T(t)$ is strictly positive or else $T(t) = 0$. Then strong continuity and $T(0) = \mathrm{Id} \neq 0$ implies that there exists $\tau > 0$ such that $T(t)$ is strictly positive for $0 \leqslant t \leqslant \tau$. For arbitrary $t \in \mathbb{R}_+$ we find $n \in \mathbb{N}$ such that $\frac{t}{n} \leqslant \tau$. Then $T(t) = T(\frac{t}{n})^n$ is also strictly positive.

(ii) We prove that for an arbitrary holomorphic positive semigroup $(T(t))_{t\geqslant 0}$ the following holds.

> Given $f > 0$ and $\varphi > 0$. Then either $\langle T(t)f, \varphi\rangle = 0$ for all $t \geqslant 0$ or $\langle T(t)f, \varphi\rangle > 0$ for all $t > 0$.

Then it follows from Definition 3.3.1 (b) that for irreducible semigroups always the second case occurs.

Assume that $\langle T(t_0)f, \varphi\rangle = 0$ for some $t_0 > 0$. We consider a null sequence (t_n), $0 < t_n < t_0$, such that $\|T(t_n)f - f\| \leqslant 2^{-n}$ and define

$$f_n := T(t_n)f \quad \text{and} \quad g_n := f - \sum_{k=n}^{\infty}(f - f_k)^+.$$

Then for $g_n \leqslant f$, $f = \lim_{n\to\infty} g_n$ and $m \geqslant n$, we have

$$g_n \leqslant f - (f - f_m)^+ = \inf\{f, f_m\} \leqslant f_m.$$

For $n \in \mathbb{N}$ fixed and $m \geqslant n$ we obtain

$$0 \leqslant \langle T(t_0 - t_m)g_n^+, \varphi\rangle \leqslant \langle T(t_0 - t_m)f_m, \varphi\rangle = \langle T(t_0)f, \varphi\rangle = 0.$$

Thus the function $t \mapsto \langle T(t)g_n^+, \varphi\rangle$ is identically zero by the uniqueness theorem for analytic functions. Since $f = \lim_{n\to\infty} g_n^+$, we have $\langle T(t)f, \varphi\rangle = 0$ for all $t \in \mathbb{R}_+$. □

The next result can be used to check irreducibility for a semigroup whose generator is a bounded perturbation of a known semigroup. It is a generalization (and an extension to Banach lattices) of Part-II, Proposition 3.3.3.

Proposition 3.3.3 *Suppose that A is the generator of a positive semigroup $\mathcal{T}$, further assume that K is a bounded positive operator and M is a bounded real multiplier (cf. Section 1.8). Let $\mathcal{S}$ be the semigroup generated by $B := A + K + M$.*

For a closed ideal $I \subseteq E$ the following assertions are equivalent.

(a) *I is $\mathcal{S}$-invariant.*

(b) *I is invariant both under $\mathcal{T}$ and K.*

Proof. We recall that a closed subspace $I \subseteq E$ is invariant for a semigroup generated by C if and only if $C(D(C) \cap I) \subseteq I$ and the restriction $C_{|I}$ of C with domain $D_{|I} := D(C) \cap I$ generates a semigroup on I (see Part-I, Section 1.3.3). Now let I be a closed ideal of E.

$(b) \Rightarrow (a)$(b) $\Rightarrow$ (a). If I is $\mathcal{T}$-invariant then $A_{|I}$ generates a semigroup on I. The restrictions $K_{|I}$ and $M_{|I}$ of K and M respectively are bounded linear operators on I. Note that each closed ideal is invariant for M, cf. Section 1.8. Thus $B_{|I} = A_{|I} + M_{|I} + K_{|I}$ with domain $D(A_{|I}) = D(A) \cap I = D(B) \cap I$ is the generator of a semigroup on I. It follows that I is invariant for the semigroup generated by B.

$(a) \Rightarrow (b)$(a) $\Rightarrow$ (b). Without loss of generality we assume $M \geqslant 0$. Then $0 \leqslant T(t) \leqslant S(t)$ for all $t \geqslant 0$. It follows that I is $\mathcal{T}$-invariant. Thus for $x \in D(A) \cap I = D(B) \cap I$, we have $Kx = Bx - Ax - Mx$. This shows that $K(D(B) \cap I) \subseteq I$. Since $B_{|I}$ is a generator $D(B) \cap I$ is dense in I. Then, by continuity, we have $KI \subseteq I$, i.e., I is K-invariant. □

Next we consider some concrete examples.

Examples 3.3.4 (i) Suppose that on $E = L^p(\mu)$ $(1 \leqslant p < \infty)$ the semigroup $(T(t))$ is given by

$$(T(t)f)(x) = \int_X k(t,x,y) f(y)\, \mathrm{d}\mu(y) \quad (x \in X,\ t > 0) \tag{3.3.1}$$

for some measurable function $k : \mathbb{R}_+ \times X \times X \to \mathbb{R}_+$. Then $(T(t))$ is irreducible if and only if for any two measurable sets M and N with $0 < \mu(M) < \infty$, $0 < \mu(N) < \infty$ and $\mu(M \cap N) = 0$ there exists $t_0 > 0$ such that

$$\int_M \int_N k(t_0,x,y)\, \mathrm{d}\mu(x)\, \mathrm{d}\mu(y) > 0.$$

(ii) Consider the first derivative on $\mathbb{R}$, $\mathbb{R}_+$ or $\mathbb{R}_{2\pi} = \Gamma$ as operator on the corresponding L^p-space (with respect to the Lebesgue measure.) Then the statements made in

Part-II, Example 3.3.4 (iii) are true. The same is true for Part-II, Example 3.3.4 (v) and (vi) (second order differential operator) when the corresponding L^p-spaces are considered.

(iii) Let $E = L^1[-1,0]$ and for $g \in L^\infty$ consider the operator A_g given by

$$A_g f := f', \quad D(A_g) := \left\{ f \in W^1[-1,0] : f(0) = \int_{-1}^{0} f(x)g(x)\,dx \right\} \tag{3.3.2}$$

If $g \geqslant 0$ then A_g generates a positive semigroup. In case there exists $\varepsilon > 0$ such that g vanishes a.e. on $[-1, -1+\varepsilon]$, then $I := \{f \in L^1 : f|_{[-1+\varepsilon,0]} = 0\}$ is a non-trivial closed ideal which is invariant under the semigroup. It is not difficult to see that the condition on g stated above is also necessary for $(T(t))$ to be reducible (i.e., not irreducible.)

(iv) Let $E = L^1([0,1] \times [-1,1])$ and consider the semigroup $(T(t))_{t\geqslant 0}$ defined as follows

$$(T(t)f)(x,v) := \begin{cases} f(x - vt, v) & \text{for } 0 \leqslant x - vt \leqslant 1, \\ 0 & \text{otherwise.} \end{cases} \tag{3.3.3}$$

$(T(t))_{t\geqslant 0}$ is a positive semigroup on E, describing the linear transport in an interval. The unbounded operator

$$D_0 := \{f \in C^1([0,1]\times[-1,1]) : f(0,v) = f_x(0,v) = 0 \text{ if } v \geqslant 0, \\ f(1,v) = f_x(1,v) = 0 \text{ if } v \leqslant 0\}$$

$$\mathsf{D_0} := \{\mathsf{f} \in \mathsf{C}^1([0,1]\times[-1,1]) : \mathsf{f(0,v) = f_x(0,v) = 0 \text{ if } v \geqslant 0}, \\ \mathsf{f(1,v) = f_x(1,v) = 0 \text{ if } v \leqslant 0}\}$$

is a core for its generator A (cf. Part-I, Corollary 2.1.33). We have

$$(Af)(x,v) = -v\frac{\partial f}{\partial x}(x,v) \quad (f \in D_0). \tag{3.3.4}$$

The Laplace transform of $(T(t))$ is the resolvent of A. An explicit calculation yields

$$(R(\lambda, A)f)(x,v) = \int_0^1 r_\lambda(x,x',v) f(x',v)\,dx' \quad (\lambda > 0) \tag{3.3.5}$$

where $r_\lambda : [0,1] \times [0,1] \times [-1,1] \to \mathbb{R}$ is given by

$$r_\lambda(x,x',v) = \begin{cases} |v|^{-1} \exp(-\lambda(x-x')v^{-1}) & \begin{cases} \text{if either } v > 0 \text{ and } x' \leqslant x \\ \text{or } v < 0 \text{ and } x' \geqslant x, \end{cases} \\ 0 & \text{otherwise.} \end{cases}$$

Let $\sigma\colon [0,1]\times[-1,1]\to\mathbb{R}_+$ and $\kappa\colon [0,1]\times[-1,1]\times[-1,1]\to\mathbb{R}_+$ be bounded measurable functions and consider the operators M and K given by

$$Mf := \sigma f, \quad Kf := \int_{-1}^{1} \kappa(\cdot,\cdot,v')f(\cdot,v')\,\mathrm{d}v'. \tag{3.3.6}$$

Then $B := A - M + K$ with domain $D(B) := D(A)$ is the generator of a positive semigroup.

Using Proposition 3.3.3 we can prove the following irreducibility criterion for the semigroup $(S(t))_{t\geqslant 0}$ generated by B.

$$\text{If } \kappa \text{ is strictly positive, then } (S(t))_{t\geqslant 0} \text{ is irreducible.} \tag{3.3.7}$$

Actually, in view of Proposition 3.3.3, we have to show that a closed ideal which is invariant under $R(\lambda, A)$ and K has to be $\{0\}$ or E.

We recall that closed ideals of E are uniquely determined (up to sets of measure zero) by measurable subsets Y of $[0,1]\times[-1,1]$, i.e., every closed ideal has the form

$$I_Y = \{f\in E\colon f \text{ vanishes (a.e.) on } [0,1]\times[-1,1]\setminus Y\}.$$

Since we assumed that κ is strictly positive, I_Y is K-invariant if and only if $Y = X\times[-1,1]$ for some measurable set $X\subseteq[0,1]$. If we assume that X has positive measure and define

$$\alpha := \sup\left\{x\in[0,1]\colon \int_0^x \mathbf{1}_X(s)\,\mathrm{d}s = 0\right\},$$
$$\beta := \inf\left\{x\in[0,1]\colon \int_x^1 \mathbf{1}_X(s)\,\mathrm{d}s = 0\right\}.$$

Then we have $\alpha<\beta$ and the support of the function $h := R(\lambda,A)\mathbf{1}_Y$, where $(Y := X\times[-1,1])$ is given by $\operatorname{supp} h = [\alpha,1]\times[0,1]\cup[0,\beta]\times[-1,0]$. Since we assumed that I_Y is $R(\lambda,A)$-invariant, we have $h\in I_Y$, i.e., $\operatorname{supp} h\subseteq Y = X\times[-1,1]$. Obviously, this is true only if $Y = [0,1]\times[-1,1]$ or $I_Y = E$.

A weaker condition than Eq. (3.3.7) entailing irreducibility is the following.

$$\begin{aligned}&\text{There exists } \delta>0 \text{ such that } \kappa \text{ is strictly positive}\\ &\text{on the sets } [0,\delta]\times[-1,1] \text{ and } [1-\delta,1]\times[-1,1].\end{aligned} \tag{3.3.8}$$

For details we refer to Greiner [15].

In the following proposition we list some properties which are consequences of irreducibility. This extends Theorem 2.3.5 Part-II, Proposition 3.3.5 to the setting of Banach lattices. The first assertion of the latter proposition is no longer true in the general setting (see Example 3.3.6 and Theorem 3.3.7).

Proposition 3.3.5 *Suppose A to be the generator of an irreducible, positive semigroup on a Banach lattice E. Then the following assertions are true.*

(i) *Every positive eigenvector of A is a quasi-interior point.*

(ii) *Every positive eigenvector of A' is strictly positive.*

(iii) *If* $\ker(s(A) - A')$ *contains a positive element, then* $\dim(\ker(s(A) - A)) \leqslant 1$.

(iv) *If $s(A)$ is a pole of the resolvent, then it has algebraic (and geometric) multiplicity* 1. *The corresponding residue has the form $P = \varphi \otimes u$, where $\varphi \in E'$ is a positive eigenvector of A', $u \in E$ is a positive eigenvector of A and $\langle u, \varphi \rangle = 1$.*

Proof. To prove (i), (ii) and (iv) one can proceed as in the case $C_0(X)$ (see Part-II, Proposition 3.3.5). We only prove (iii) and assume $s(A) = 0$. By assumption and by assertion (ii) there exists $\varphi \gg 0$ such that $T(t)'\varphi = \varphi$ $(t \geqslant 0)$. Given $f \in \ker A$, then $T(t)f = f$ hence $|f| = |T(t)f| \leqslant T(t)|f|$.

Since φ is strictly positive and $\langle |f|, \varphi \rangle \leqslant \langle T(t)|f|, \varphi \rangle = \langle |f|, \varphi \rangle$, it follows that $|f| = T(t)|f|$. We have shown that $\ker A$ is a sublattice.

Then for $f \in \ker A$, f real, i.e., $f = \overline{f}$, we have that f^+ and f^- are elements of $\ker A$. Hence the principal ideals generated by f^+ and f^- are $\mathcal{T}$-invariant. Since these ideals are orthogonal, the irreducibility of $\mathcal{T}$ implies that either f^+ or f^- is zero. We have shown that $\ker A \cap E_{\mathbb{R}}$ is totally ordered, hence at most one-dimensional (see Proposition 3.4 of Schaefer [32]). □

In arbitrary Banach lattices it is no longer true that an irreducible semigroup has necessarily nonvoid spectrum. We indicate how an irreducible semigroup having empty spectrum can be constructed.

Example 3.3.6 Consider the Banach lattice $E = L^p(\Gamma \times \Gamma)$. For (fixed) positive numbers α, β such that $\frac{\alpha}{\beta}$ is irrational we define a positive semigroup $(T_0(t))_{t \geqslant 0}$ as follows

$$(T_0(t)f)(z, w) := f(z \cdot e^{i\alpha t}, w \cdot e^{i\beta t}) \quad (z, w \in \Gamma = \{\xi \in \mathbb{C} : |\xi| = 1\}).$$

$$(\mathsf{T}_0(\mathsf{t})\mathsf{f})(\mathsf{z}, \mathsf{w}) := \mathsf{f}(\mathsf{z} \cdot \mathrm{e}^{\mathrm{i}\alpha \mathrm{t}}, \mathsf{w} \cdot \mathrm{e}^{\mathrm{i}\beta \mathrm{t}}) \quad (\mathsf{z}, \mathsf{w} \in \grave{} = \{\xi \in \mathbb{C} \colon |\xi| = 1\}). \tag{3.3.9}$$

Next we define for a positive function $m \colon \Gamma \times \Gamma \to \mathbb{R}$ which is continuous on $\Gamma \times \Gamma \setminus (1,1)\ \Gamma \times \Gamma \setminus \{(1,1)\}$ functions m_t, $t \geqslant 0$, as follows

$$m_t(z, w) := \exp\left(-\int_0^t m(z \cdot e^{i\alpha s}, w \cdot e^{i\beta s})\, ds\right). \tag{3.3.10}$$

Then $(T(t))_{t \geqslant 0}$ defined by

$$T(t)f := m_t \cdot (T_0(t)f) \tag{3.3.11}$$

is a strongly continuous semigroup of positive contractions on E. Since $\frac{\alpha}{\beta}$ is irrational, the semigroup $(T_0(t))$ is irreducible. Moreover, each m_t is strictly positive (i.e., $m_t > 0$ a.e.) thus $(T(t))$ is irreducible as well. If one chooses m such that $m(z,w)$ tends to $+\infty$ sufficiently fast as $(z,w) \to (1,1)$, one obtains $\|T(t)\| = \|m_t\|_\infty \leqslant \exp(-t^2)$ for all $t \geqslant 0$. Obviously such an estimate of $\|T(t)\|$ implies $\omega_0(A) = -\infty$, hence $\sigma(A) = \emptyset$.

Theorem 3.3.7 *Suppose that $(T(t))_{t\geqslant 0}$ is an irreducible, positive semigroup on the Banach lattice E. Each of the following conditions on E and $(T(t))$, respectively, implies that $\sigma(A) \neq \emptyset$*

(i) *$E = C_0(X)$ where X is locally compact.*

(ii) *$E = \ell^p$ $(1 \leqslant p < \infty)$ (more generally, E contains atoms).*

(iii) *either $T(t_0)$ is compact for some t_0 or $R(\lambda_0, A)$ is compact for some $\lambda_0 \in \varrho(A)$.*

(iv) *E has order continuous norm and either $T(t_0)$ or $R(\lambda_0, A)$ is a kernel operator for some $t_0 \geqslant 0$ $(\lambda_0 \in \varrho(A))$.* [1]

(v) *E is reflexive and there exist $t_0 > 0$, $h \in E_+$ such that $T(t_0)E \subseteq E_h$.*

Proof. (i) is proved in Part-II, Proposition 3.3.5(i).

Assertions (ii)-(v) will be proved utilizing the analogous results for a single operator. In view of Part-I, Proposition 3.2.5 we have to show that $r(R(\lambda, A)) > 0$ for some $\lambda \in \varrho(A)$. Moreover, from Part-I, Eq. (1.3.1) on page 19 we deduce

$$\begin{aligned} T(t)R(\lambda, A) &= \mathrm{e}^{\lambda t} R(\lambda, A) - \mathrm{e}^{\lambda t} \int_0^t \mathrm{e}^{-\lambda s} T(s)\, \mathrm{d}s \\ &\leqslant \mathrm{e}^{\lambda t} R(\lambda, A) \ (t \geqslant 0, \lambda > s(A)). \end{aligned}$$

Since the spectral radius is an isotone function on the cone of positive operators, it is enough to show that

$$r(T(t)R(\lambda, A)) > 0 \text{ for some } t \geqslant 0,\ \lambda > s(A). \tag{3.3.12}$$

Using Theorem 3.3.2 (i) it is easy to see that $T(t)R(\lambda, A) = R(\lambda, A)T(t)$ is irreducible.

The assertions (ii),(iv) and (v) now follow from the corresponding results for a single operator as presented in Sect.V.6 of Schaefer [32] (see Proposition 6.1, Theorem 6.5 Corollary and Theorem 6.5 l.c.). (iii) follows from the recent result of de Pagter [8] which ensures that every positive operator on a Banach lattice which is compact and irreducible has positive spectral radius. □

[1] For a precise definition of a kernel operator we refer to Section IV.9 of Schaefer [32] or Chapter 13 of Zaanen [40].

The theorem can be used to prove that elliptic operators as described in Example 3.2.14 have non-empty spectrum. It is shown in Amann [2] that these operators have compact resolvent and generate irreducible semigroups. Thus the assumption (iii) is satisfied.

Concerning the eigenvalues of an irreducible semigroup which are contained in $\sigma_b(A)$ all statements established for spaces $C_0(X)$ in Part-II, Theorem 3.3.6 are true in the setting of Banach lattices. The proof can be translated without difficulties and will be omitted (see also Greiner [13, Theorem 2.6]).

Theorem 3.3.8 *Suppose $\mathcal{T}$ is an irreducible semigroup on the Banach lattice E and let A be its generator. Assume that $s(A) = 0$ and that there exists a positive linear form $\psi \in D(A')$ with $A'\psi \leqslant 0$.*

If $P\sigma(A) \cap \mathrm{i}\mathbb{R}$ is non-empty, then the following assertions are true.

(i) *$P\sigma(A) \cap \mathrm{i}\mathbb{R}$ is a (additive) subgroup of $\mathrm{i}\mathbb{R}$.*

(ii) *The eigenspaces corresponding to $\lambda \in P\sigma(A) \cap \mathrm{i}\mathbb{R}$ are one-dimensional.*

(iii) *If $Ah = \mathrm{i}\alpha h$ ($h \neq 0$, $\alpha \in \mathbb{R}$) then $|h|$ is a quasi-interior point and the following holds*

$$S_h(D(A)) = D(A) \text{ and } S_h^{-1} \circ A \circ S_h = (A + \mathrm{i}\alpha \,\mathrm{Id}). \tag{3.3.13}$$

(iv) *0 is the only eigenvalue of A admitting a positive eigenvector.*

One can apply the theorem in order to prove that the rotation semigroup on Γ is the only positive periodic semigroup which is irreducible (cf. Part-I, Section 1.2.5).

Corollary 3.3.9 *Let $(T(t))_{t\geqslant 0}$ be a positive irreducible semigroup on a Banach lattice E which is periodic of period τ. Assume that $\dim E > 1$. Then there exist continuous lattice homomorphisms $i\colon C(\Gamma) \to E$ and $j\colon E \to L^1(\Gamma)$, both injective with dense range, such that the diagram commutes for all $t \geqslant 0$. Hereby, $j \circ i$ is the canonical inclusion of $C(\Gamma)$ in $L^1(\Gamma)$.*

$$\begin{array}{ccccc} C(\Gamma) & \xrightarrow{\ i\ } & E & \xrightarrow{\ j\ } & L^1(\Gamma) \\ {\scriptstyle R_\tau(t)}\downarrow & & {\scriptstyle T(t)}\downarrow & & \downarrow{\scriptstyle R_\tau(t)} \\ C(\Gamma) & \xrightarrow[\ i\ ]{} & E & \xrightarrow[\ j\ ]{} & L^1(\Gamma) \end{array}$$

Proof. By Theorem 3.3.8 and Part-I, Theorem 3.5.4 we have $R\sigma(A) = P\sigma(A) = \sigma(A) = \mathrm{i}\alpha\mathbb{Z}$ with $\alpha := \frac{2\pi}{n\tau}$ for suitable $n \in \mathbb{N}$. We fix $h \in \ker(\mathrm{i}\alpha - A)$, $h \neq 0$. Then $|h| \in \ker A$ and there exists $\varphi \in \ker A'$ such that $\langle |h|, \varphi\rangle = 1$. According to

the Kakutani-Krein Theorem we identify $E_{|h|}$ with $C(K)$. Then h is a unimodular function onto Γ (use the argument given in the proof of Part-II, Theorem 3.3.6(iii)).

We define $j\colon C(\Gamma) \to E$ by $j(f) := f \circ h \in C(K) \cong E_h \subseteq E$, then j $i\colon C(\Gamma) \to E$ by $i(f) := f \circ h \in C(K) \cong E_{|h|} \subseteq E$, then i is injective. For the monomials $e_n(z) := z^n$ $(n \in \mathbb{Z})$ we have $j(e_n) = h^{[n]}$ thus $j\, i(e_n) = h^{[n]}$ thus i has dense image in E (by Part-I, Theorem 3.5.4). Moreover,

$$\langle h^{[n]}, \varphi\rangle = \langle j(e_n), \varphi\rangle = \int_0^{2\pi} e_n(\mathrm{e}^{\mathrm{i}t})\,\mathrm{d}t = 2\pi \cdot \delta_{n0} \text{ for all } n \in \mathbb{Z},$$

$$\langle \mathsf{h}^{[\mathsf{n}]}, \varphi\rangle = \langle \mathsf{i}(\mathsf{e}_\mathsf{n}), \varphi\rangle = \frac{1}{2\pi}\int_0^{2\pi} \mathsf{e}_\mathsf{n}(\mathrm{e}^{\mathrm{i}t})\,\mathrm{dt} = \delta_{\mathsf{n}0} \text{ for all } \mathsf{n} \in \mathbb{Z},$$

hence $\int_0^{2\pi} f(\mathrm{e}^{\mathrm{i}t})\,\mathrm{d}t = \langle j(f), \varphi\rangle$ $\frac{1}{2\pi}\int_0^{2\pi} f(\mathrm{e}^{\mathrm{i}t})\,\mathrm{d}t = \langle i(f), \varphi\rangle$ for all $f \in C(\Gamma)$.

It follows that $(E, \varphi) \cong L^1(\Gamma)$, and we define $\tilde{j}\, j$ to be the canonical mapping from E into $(E, \varphi) \cong L^1(\Gamma)$ (see Section 1.4). Then $\tilde{j}\, j$ has dense image and is injective since φ is strictly positive (cf. Proposition 3.3.5 (ii)). One easily verifies that the diagram commutes. □

Now we are going to prove the main result of this section. As in the proof of Theorem 3.2.10 we will utilize pseudo-resolvents on a suitable $\mathcal{F}$-product of the Banach lattice. To simplify the proof we isolate two lemmas.

Lemma 3.3.10 *Let $\mathcal{F}$ be a filter on $\mathbb{N}$ which is finer than the Fréchet filter and let $E_\mathcal{F}$ be the $\mathcal{F}$-product of the Banach lattice E. Given $R \in \mathcal{L}(E)$ and denoting its canonical extension to $E_\mathcal{F}$ by $R_\mathcal{F}$ the following is true.*

If $\alpha \in A\sigma(R) \setminus P\sigma(R)$, then $\ker(\alpha - R_\mathcal{F})$ *is infinite dimensional.*

Proof. Let $(f_n)_{n\geqslant 1}$ be a normalized approximate eigenvector of R corresponding to α. Since every accumulation point of (f_n) is an eigenvector of R, the assumption $\alpha \notin P\sigma(A)$ $\alpha \notin P\sigma(R)$ implies that (f_n) does not have any accumulation points. Then there exist an $\varepsilon > 0$ and a subsequence (g_n) of (f_n) such that

$$\|g_n - g_m\| \geqslant \varepsilon \text{ whenever } n \neq m. \tag{3.3.14}$$

Obviously, (g_n) is a normalized approximate eigenvector of R and so is every subsequence of (g_n). In particular, for $k \in \mathbb{N}$ the sequence $(g_{n+k})_{n\geqslant 1}$ is a normalized approximate eigenvector of R. Then the elements $\hat{g}^k \in E_\mathcal{F}$ given by $\hat{g}^k := ((g_{n+k})_{n\geqslant 1} + c_\mathcal{F}(E))$ are normalized eigenvectors of $R_\mathcal{F}$ corresponding to α. As a consequence of Eq. (3.3.14) we obtain

$$\|\hat{g}^k - \hat{g}^m\| = \mathcal{F}\text{-}\limsup \|g_{n+k} - g_{n+m}\| \geqslant \varepsilon$$

$$\|\hat{\mathsf{g}}^\mathsf{k} - \hat{\mathsf{g}}^\mathsf{m}\| = \mathcal{F}\text{-}\limsup \|\mathsf{g}_{\mathsf{n}+\mathsf{k}} - \mathsf{g}_{\mathsf{n}+\mathsf{m}}\| \geqslant \varepsilon$$

provided that $k \neq m$. This shows that the unit ball in $\ker(\alpha - R_{\mathcal{F}})$ is not relatively compact, hence $\ker(\alpha - R_{\mathcal{F}})$ has to be infinite dimensional. □

Lemma 3.3.11 *Let E be a Banach lattice and let M, L be two linear subspaces of E. Assume that $f \in M$ implies $|f| \in L$, then* $\dim L \geqslant \dim M$.

Proof. Let $\{g_1, g_2, \ldots, g_m\}$ $(m \geqslant 1)$ be any (finite) subset of M which is linearly independent. For $u := \sum_{n=1}^{m} |g_n|$ all vectors g_n are contained in the principal ideal E_u which (by the Kakutani-Krein Theorem) is isomorphic to a space $C(K)$. Considering $g_1, g_2, \ldots, g_m$ as continuous functions on K, there exist points $x_1, x_2, \ldots, x_m \in K$ and functions $h_1, h_2, \ldots, h_m \in \operatorname{span}\{g_1, g_2, \ldots, g_m\}$ such that $h_i(x_j) = \delta_{ij}$. Then $|h_i|(x_j) = \delta_{ij}$ hence $\{|h_j| \colon 1 \leqslant j \leqslant m\}$ is a subset of m linearly independent vectors which (by assumption) is contained in L. □

The surprising fact in the following theorem is the conclusion that every point in the boundary spectrum is a simple algebraic pole if only $s(A)$ is supposed to be a pole.

Theorem 3.3.12 *Let $\mathcal{T}$ be an irreducible semigroup on a Banach lattice and let A be its generator. If $s(A)$ is a pole of the resolvent, then $\sigma_b(A) = s(A) + \mathrm{i}\alpha\mathbb{Z}$ for some $\alpha \geqslant 0$. Moreover, $\sigma_b(A)$ contains only algebraically simple poles.*

Proof. We will assume that $s(A) = 0$. Assuming first that every element of $\sigma_b(A)$ is an eigenvalue of A, one can conclude the following. By Theorem 3.3.8 (i) we know that $\sigma_b(A)$ is an additive subgroup of $\mathrm{i}\mathbb{R}$. Since it is a closed subset and 0 is an isolated point, it follows that $\sigma_b(A) = \mathrm{i}\alpha\mathbb{Z}$ for some $\alpha \geqslant 0$. Moreover, as a consequence of Eq. (3.3.13) on page 353, for every $k \in \mathbb{Z}$ we obtain

$$R(\lambda + \mathrm{i}k\alpha, A) = S_h^{-k} \circ R(\lambda, A) \circ S_h^k \quad (\lambda \in \varrho(A),\ k \in \mathbb{Z}). \tag{3.3.15}$$

By Proposition 3.3.5 (iv) 0 is an algebraically simple pole. Then Eq. (3.3.15) implies that every point $\mathrm{i}k\alpha$ has the same property.

We now show that every element $\mathrm{i}\beta$ is an eigenvalue of A. By Proposition 3.3.5 (iv) the residue of $R(\cdot, A)$ in $\lambda = 0$ has the form $P = \varphi \otimes u$ with $\varphi(u) = 1$. Given an ultrafilter $\mathcal{U}$ on $\mathbb{N}$ which is finer than the Fréchet filter, then $\lim_{\mathcal{U}} \varphi(f_n)$ exists for every bounded sequence (f_n) in E. Using this fact it is easy to see that the canonical extension $P_{\mathcal{U}}$ of P to the $\mathcal{U}$-product $E_{\mathcal{U}}$ of E has the following form.

$$\begin{aligned} &P_{\mathcal{U}} = \hat{\varphi} \otimes \hat{u} \text{ where } \hat{u} := (u, u, u, \ldots) + c_{\mathcal{U}}(E) \in E_{\mathcal{U}} \text{ and } \hat{\varphi} \in (E_{\mathcal{U}})' \\ &\text{is given by } \hat{\varphi}((f_n) + c_{\mathcal{U}}(E)) := \lim_{\mathcal{U}} \varphi(f_n)\ ((f_n) + c_{\mathcal{U}}(E) \in E_{\mathcal{U}}). \end{aligned} \tag{3.3.16}$$

Given $\mathrm{i}\beta \in \sigma_b(A)$ then $\mathrm{i}\beta \in A\sigma(A)$ hence $1 \in A\sigma(\lambda R(\lambda + \mathrm{i}\beta, A))$. If we assume that $\mathrm{i}\beta \notin P\sigma(A)$, then $1 \notin P\sigma(\lambda R(\lambda + \mathrm{i}\beta, A))$. Then Lemma 3.3.10 implies that

$$M := \ker(1 - \lambda R(\lambda + \mathrm{i}\beta, A)_{\mathcal{U}})$$

is infinite dimensional (and independent of λ by Proposition 3.2.6 (i).)

For $\hat{f} \in M$ we have

$$|\hat{f}| = \gamma R(\gamma + \mathrm{i}\beta, A)_{\mathcal{U}} |\hat{f}| \leqslant \gamma R(\gamma, A)_{\mathcal{U}} |\hat{f}|$$

$$|\hat{\mathsf{f}}| = |\gamma \mathsf{R}(\gamma + \mathrm{i}\beta, \mathsf{A})_{\mathcal{U}} \hat{\mathsf{f}}| \leqslant \gamma \mathsf{R}(\gamma, \mathsf{A})_{\mathcal{U}} |\hat{\mathsf{f}}|$$

for every $\gamma > 0$. It follows that

$$\hat{\varphi}(|\hat{f}|) = P_{\mathcal{U}} |\hat{f}| = \lim_{\gamma \to 0} \gamma R(\gamma, A)_{\mathcal{U}} |\hat{f}| \geqslant |\hat{f}|.$$

$$\hat{\boldsymbol{\varphi}}(|\hat{\mathsf{f}}|)\hat{\mathfrak{u}} = \mathsf{P}_{\mathcal{U}} |\hat{\mathsf{f}}| = \lim_{\gamma \to 0} \gamma \mathsf{R}(\gamma, \mathsf{A})_{\mathcal{U}} |\hat{\mathsf{f}}| \geqslant |\hat{\mathsf{f}}|.$$

Thus considering the closed ideal $I := \{\hat{f} \in E_{\mathcal{U}} \colon \hat{\varphi}(|\hat{f}|) = 0\}$ we have

$$\tilde{\varphi}(|\hat{f}|) - |\hat{f}| \in I \text{ for every } \hat{f} \in M.$$

$$\hat{\boldsymbol{\varphi}}(|\hat{\mathsf{f}}|)\hat{\mathfrak{u}} - |\hat{\mathsf{f}}| \in \mathsf{I} \text{ for every } \hat{\mathsf{f}} \in \mathsf{M}. \tag{3.3.17}$$

This implies that $M \cap I = \{0\}$. Hence the canonical image $M_/$ of M in the quotient space $E_{\mathcal{U}/I}$ $E_{\mathcal{U}}/I$ is infinite dimensional as well.

By Eq. (3.3.16) on the preceding page and Eq. (3.3.17) the absolute value of an element $\tilde{f} \in M_/$ is a scalar multiple of $\tilde{u} := \hat{u} + I$. This is a contradiction by Lemma 3.3.11. □

In view of Part-I, Proposition 3.4.2 the result above has consequences for semigroups which can be reduced (by considering restrictions to invariant ideals or quotients) to semigroups which satisfy the hypothesis of Theorem 3.3.12. Semigroups having this property are precisely those for which $s(A)$ is a pole of the resolvent of finite algebraic multiplicity. The latter claim is a consequence of Proposition 3.2.11 and the following lemma.

Lemma 3.3.13 *Suppose that $\mathcal{T} = (T(t))_{t \geqslant 0}$ is a positive semigroup such that $s(A)$ is a first order pole of the resolvent. Moreover assume that the corresponding residue is a strictly positive operator of finite rank. Then there are closed $\mathcal{T}$-invariant ideals* $J_1, J_2, .. J_m$ $J_1, J_2, \ldots, J_m$ *which are mutually orthogonal. For the restrictions $\mathcal{T}_k$ of $\mathcal{T}$ to J_k the following is true.*

(i) *$\mathcal{T}_k$ is irreducible with spectral bound $s(A_k) = s(A)$.*

(ii) *$s(A_{/J}) < s(A)$ where* $J := J_1 \oplus J_2 \oplus .. \oplus J_m$. $J := J_1 \oplus J_2 \oplus \cdots \oplus J_m$.

Proof. We assume $s(A) = 0$. Since P is a strictly positive projection $PE = \ker A$ is a sublattice of E. Actually, if $x \in PE$ i.e., $Px = x$, then $P|x| \geqslant |Px| = |x|$. Hence

$$P(|P|x| - |x||) = P^2|x| - P|x| = 0$$

which implies that $P|x| - |x| = 0$ or $|x| \in PE$. Thus we know that $\ker A$ is a finite dimensional sublattice of E hence it is isomorphic to a space $\mathbb{C}^m$ for some $m \in \mathbb{N}$ (see Section II.4 of Schaefer [32]).

Then there exist vectors $e_j \in E_+$ $(1 \leqslant j \leqslant m)$ such that the following holds.

$$\ker A = \operatorname{span}\{e_1, e_2, \ldots, e_m\} \text{ and } \inf\{e_i, e_j\} = 0 \text{ for } i \neq j. \tag{3.3.18}$$

We have $T(t)e_k = e_k$ hence the closed ideal generated by e_k is $\mathcal{T}$-invariant. We denote this ideal J_k and define $J := J_1 \oplus J_2 \oplus \ldots \oplus J_m$. The ideal J is closed, $\mathcal{T}$-invariant and we have $PE = \ker A \subseteq J$ (see Schaefer [32, III Theorem 1.2]). Then $P_{/J} = 0$ hence the spectral bound $s(A_{/J})$ is strictly less than zero (by Theorem 3.1.1 (i)). Moreover, the residue corresponding to the resolvent of $\mathcal{T}_k$, we denote it P_k, is the restriction of P to J_k.

Then P_k is strictly positive and $P(J_k) = \operatorname{span}\{e_k\}$. To show that $\tau_k\, \mathcal{T}_k$ is irreducible, we consider an invariant ideal I. Then we have $R(\lambda, A_k)I \subseteq I$ for $\lambda > 0$ hence $P_k = \lim_{\lambda \to 0} \lambda R(\lambda, A_k)$ leaves I invariant. If $I \neq \{0\}$, then $P_k I \neq \{0\}$ since P_k is strictly positive. □

Combining the lemma with Proposition 3.2.11 one obtains the following.

If $s(A)$ is a pole of finite algebraic multiplicity, then there exists a finite chain of $\mathcal{T}$-invariant ideals $I_{-1} := \{0\} \subseteq I_0 \subseteq \ldots \subseteq I_N := E$ $(N \in \mathbb{N})$ such that the following is true.

$$\begin{aligned} &\textit{For the semigroup } \mathcal{T}_n \textit{ on } I_n/I_{n-1} \ \ (0 \leqslant n \leqslant N) \textit{ which is induced by } \mathcal{T} \\ &\textit{we have either } s(A_n) = s(A) \textit{ and } \mathcal{T}_n \textit{ is irreducible or } s(A_n) < s(A). \end{aligned} \tag{3.3.19}$$

The following theorem is an immediate consequence of Eq. (3.3.19), Theorem 3.3.12 and Part-I, Proposition 3.4.2.

Theorem 3.3.14 *Let $\mathcal{T}$ be a positive semigroup on a Banach lattice with generator A. If $s(A)$ is a pole of finite algebraic multiplicity, then $\sigma_b(A)$ is a finite union of discrete subgroups, i.e.,*

$$\sigma_b(A) = s(A) + \bigcup_{k=1}^{N} \mathrm{i}\alpha_k\mathbb{Z} \textit{ with } \alpha_k \in \mathbb{R}.$$

Moreover, $\sigma_b\, \sigma_b(A)$ contains only poles of finite algebraic multiplicity.

Here the assumption that the multiplicity of $s(A)$ is finite is essential as can be seen from the following example.

Example 3.3.15 Consider $X := [0, 1] \times V$, $V := \{v \in \mathbb{R} \colon v_1 < |v| < v_2\}$ $(0 < v_1 < v_2 < \infty)$. The flow in the phase space X which describes the free motion in the

interval $[0,1]$ with velocities in V assuming that the particles are reflected at the endpoints generates a positive semigroup on $L^p(X,\mu)$ (μ the Lebesgue measure). For the spectrum of the generator A one obtains

$$\sigma(A) = \{i\gamma\colon n\gamma_1 \leqslant |\gamma| \leqslant n\gamma_2 \text{ for some } n \in \mathbb{N}_0\}$$

with $\gamma_1 := \pi/v_1$, $\gamma_2 := \pi/v_2\gamma_1 := \pi v_1$, $\gamma_2 := \pi v_2$. Moreover, 0 is a first order pole of the resolvent, obviously the only pole in $\sigma_b(A) = \sigma(A)$. These statements can be verified by calculating the resolvent explicitly. This can be done using the integral representation. The semigroup is given as follows.

$$(T(t)f)(x,v) = \begin{cases} f(x - vt + k, v) & \text{if } k-1 \leqslant vt - x \leqslant k \text{ and } k \text{ even}, \\ f(1 - (x - vt + k), -v) & \text{if } k-1 \leqslant vt - x \leqslant k \text{ and } k \text{ odd}. \end{cases}$$

$$(\mathsf{T}(\mathsf{t})\mathsf{f})(\mathsf{x},\mathsf{v}) = \begin{cases} f(x - vt + k, v) & \text{if } k-1 \leqslant vt - x \leqslant k \text{ and } k \text{ even}, \\ f(1 - (x - vt + k), -v) & \text{if } k-1 \leqslant vt - x \leqslant k \text{ and } k \text{ odd}. \end{cases} \tag{3.3.20}$$

Obviously one can apply Theorem 3.3.12 and Theorem 3.3.14, respectively, in order to prove existence of strictly dominant eigenvalues. We consider two typical cases in the following corollaries. The meaning of $r_{ess}(T(t))$ and $\omega_{ess}(\mathcal{T})$ and its properties are explained in Part-I, Example 3.3.7.

Corollary 3.3.16 *Suppose that $\mathcal{T}$ is a positive semigroup such that $\omega_{ess}(\mathcal{T}) < \omega_0(\mathcal{T})$. Then $s(A) = \omega_0(\mathcal{T})$ is a strictly dominant eigenvalue. If, in addition, there exists an eigenfunction which is a quasi-interior point of E_+ (e.g., if $\mathcal{T}$ is irreducible), then $s(A)$ is a first order pole of $R(\cdot, A)R(\cdot, A)$.*

Proof. There exist $\varepsilon > 0$ such that $\{\lambda \in \sigma(T(t))\colon |\lambda| \geqslant \exp((s(A) - \varepsilon)t)\}$ contains only (finitely many) poles of $R(\cdot, T(t))$ each being of finite algebraic multiplicity. In view of Part-I, Corollary 3.6.5 the set $\{\lambda \in \sigma(A)\colon \operatorname{Re}(\lambda) > s(A) - \varepsilon\}$ is finite and contains only poles of $R(\cdot, A)$. Thus we can apply Theorem 3.3.14. It follows that $s(A)$ is strictly dominant. For the final assertion we refer to Remark 3.2.15 (ii). □

Corollary 3.3.17 *Suppose that $\mathcal{T}$ is an irreducible, eventually norm continuous semigroup having compact resolvent. Then $s(A) = \omega_0(\mathcal{T})$ is an algebraically simple pole and a strictly dominant eigenvalue.*

Proof. By Theorem 3.3.7 (iii) we know that $s(A) > -\infty$. Theorem 3.3.12 is applicable since we assumed that $\mathcal{T}$ is irreducible and has compact resolvent. Thus $s(A)$ is an algebraically simple pole and $\sigma_b(A) = s(A) + i\alpha\mathbb{Z}$ for some $\alpha \geqslant 0$. In addition $\{\lambda \in \sigma(A)\colon \operatorname{Re}(\lambda) \geqslant -1\}$ is compact since $\mathcal{T}$ is eventually norm-continuous (see Part-I, Theorem 2.1.19). It follows that $s(A)$ is strictly dominant. By Part-I, Theorem 3.6.6 we have $s(A) = \omega_0(\mathcal{T})$. □

In the following proposition we give a condition which ensures that Theorem 3.3.14 can be applied to certain perturbations. Moreover, we state a criterion ensuring the existence of a dominant eigenvalue.

Proposition 3.3.18 *Suppose that A is the generator of a positive semigroup and that $K \in \mathcal{L}(E)$ is a positive linear operator.*

(i) *If K is A-compact, i.e., $K \cdot R(\lambda_0, A)$ is compact for some $\lambda_0 \in \varrho(A)$, and if $s(A+K) > s(A)$, then $B := A+K$ satisfies the assumptions of Theorem 3.3.14.*

(ii) *If, in addition, K is irreducible, then $s(B)$ is a dominant eigenvalue and the semigroup generated by B is irreducible.*

Proof. The resolvent equation

$$R(\lambda, A) = R(\lambda_0, A)(\mathrm{Id} - (\lambda - \lambda_0)R(\lambda, A))$$

implies that $K \cdot R(\lambda, A)$ is a compact operator for every $\lambda \in \varrho(A)$. For $\lambda > s(A)$ we have

$$\lambda - B = (\mathrm{Id} - KR(\lambda, A))(\lambda - A)$$

and $(\mathrm{Id} - KR(\lambda, A))^{-1}$ exists for $\lambda > s(B)$.

Therefore Reed & Simon [29, Theorem XIII.13] implies that

$$R(\lambda, B) = R(\lambda, A)(1 - KR(\lambda, A))^{-1}$$

$$\mathsf{R}(\lambda, \mathsf{B}) = \mathsf{R}(\lambda, \mathsf{A})(\mathrm{Id} - \mathsf{K}\mathsf{R}(\lambda, \mathsf{A}))^{-1}$$

has only poles of finite algebraic multiplicity in $\{\lambda \in \mathbb{C} \colon \operatorname{Re}(\lambda) > s(A)\}$. This proves the first claim.

In order to prove the second, we denote the semigroup corresponding to A and B by $(T(t))$ and $(S(t))$, respectively. It follows from Proposition 3.3.3 that $(S(t))$ is irreducible and we have $S(t) = T(t) + \int_0^t T(t-s)KS(s)\,\mathrm{d}s$ (see Part-I, Eq. (2.1.7) on page 52). Iterating this identity we obtain for every $m \in \mathbb{N}$, $t \geqslant 0$

$$S(t) = \sum_{n=0}^{m-1} T_n(t) + R_m(t) \quad \text{where}$$

$$T_0(t) := T(t), \; T_n(t) := \int_0^t T(t-s)KT_{n-1}(s)\,\mathrm{d}s \text{ for } n \in \mathbb{N},$$

$$R_m(t) := \int_0^t \int_0^{t_1} \dots \int_0^{t_{m-1}} T(t-t_1)KT(t_1-t_2)K \dots T(t_{m-1}-t_m)KS(t_m)\,\mathrm{d}t_m \dots \mathrm{d}t_1.$$

$$
\begin{aligned}
S(t) &= \sum_{n=0}^{m-1} T_n(t) + R_m(t) \quad \text{where} \\
T_0(t) &:= T(t),\ T_n(t) := \int_0^t T(t-s)KT_{n-1}(s)\,\mathrm{d}s \text{ for } n \in \mathbb{N}, \\
R_m(t) &:= \int_0^t \int_0^{t_1} \dots \int_0^{t_{m-1}} T(t-t_1)KT(t_1-t_2)K\dots \\
&\qquad \dots T(t_{m-1}-t_m)KS(t_m)\,\mathrm{d}t_m \dots \mathrm{d}t_1.
\end{aligned}
\tag{3.3.21}
$$

We fix $0 < f \in E$, $0 < \varphi \in E'$, $t > 0$. By Theorem 3.3.2 (i), $S(t)f > 0$. Since K is irreducible, there exists $m \in \mathbb{N}$ such that $\langle K^m f, \varphi\rangle > 0$. Thus the integrand appearing in the representation Eq. (3.3.21) of $\langle R_m(t)f, \varphi\rangle$ is non-zero at $t_1 = t_2 = \dots = t_{m-1} = t$, $t_m = t$.

Since the integrand is positive and continuous, we conclude

$$
\langle S(t)f, \varphi\rangle \geqslant \langle R_m(t)f, \varphi\rangle > 0 \text{ for } 0 < f,\ 0 < \varphi \text{ and } t > 0. \tag{3.3.22}
$$

It follows that $(\mathrm{e}^{-ts(B)}S(t))_{t\geqslant 0}$ cannot contain the rotation semigroup on Γ. On the other hand, assuming that $s(B)$ is not dominant, then $\dim(\ker((\mathrm{e}^{\tau\cdot s(B)} - S(\tau))) > 1$ $\dim(\ker(\mathrm{e}^{\tau\cdot s(B)} - S(\tau))) > 1$ for some $\tau > 0$. Hence, the restriction $(\mathrm{e}^{-ts(B)}S(t)_{|F})_{t\geqslant 0}$, $F := \ker(\mathrm{e}^{\tau\cdot s(B)} - S(\tau))$, contains the rotation semigroup by Corollary 3.3.9. □

We conclude this section considering the linear transport equation once again, see Example 3.3.4 (iv). The generator considered there is $B = (A - M) + K$, where K is positive linear. From Eq. (3.3.5) and Eq. (3.3.6) on page 350 one deduces that

$$
(KR(\lambda, A)f)(x, v) = \int_0^1 \int_{-1}^1 k(x, v, x', v')f(x', v')\,\mathrm{d}x'dv'
$$

$$
(\mathsf{KR}(\lambda, \mathsf{A})\mathsf{f})(\mathsf{x}, \mathsf{v}) = \int_0^1 \int_{-1}^1 \mathsf{k}(\mathsf{x}, \mathsf{v}, \mathsf{x}', \mathsf{v}')\mathsf{f}(\mathsf{x}', \mathsf{v}')\,\mathrm{d}\mathsf{x}'\,\mathrm{d}\mathsf{v}'
$$

where the kernel k is given by $k(x, v, x', v') := \kappa(x, v, v')r_\lambda(x, x', v')$ (cf. Eq. (3.3.5) and Eq. (3.3.6) on page 350). Using this representation of $KR(\lambda, A)$ it follows that K is A-compact A-compact. Moreover for λ sufficiently large one has $R(\lambda, A - M) = R(\lambda, A)(1 - MR(\lambda, A))^{-1}$ $R(\lambda, A - M) = R(\lambda, A)(\mathrm{Id} - MR(\lambda, A))^{-1}$ which shows that K is also $(A-M)$-compact. In order to apply Theorem 3.3.14 one needs $s(B) > s(A-M)$ (see Proposition 3.3.18) which is difficult to verify. In case the function σ is continuous one can state a sufficient condition as follows. There exist $r \in \mathbb{R}$ and $g \in L^1([0, 1]\times[-1, 1])$, $g > 0$ such that $r < \inf\{\sigma(x, 0) : x \in [0, 1]\}$ and $Bg \geqslant -rg$. The additional assumption made in the second part of Proposition 3.3.18 is not satisfied in this example. Nevertheless one can show that $s(B)$ is strictly dominant in this situation (provided that $s(B) > s(A)$). For details we refer to Greiner [15] or Voigt [36] where the linear transport equation in higher dimensional spaces is discussed.

3.4 Semigroups of Lattice Homomorphisms

In Section 3.2 we proved that the boundary spectrum of certain positive semigroups is a cyclic set. For semigroups of lattice homomorphisms much more can be said. The whole spectrum is an imaginary additively cyclic subset of $\mathbb{C}$ (cf. Theorem 3.4.2). This result can be used to derive cyclicity results for the eigenvalues in the boundary spectrum of positive semigroups (cf. Corollary 3.4.3). In the last part of this section we discuss a spectral decomposition of positive groups (cf. Theorem 3.4.8).

Lemma 3.4.1 *Suppose that $(T(t))_{t\geqslant 0}$ is a semigroup of lattice homomorphisms on a Banach lattice E with generator A. In case $\mathrm{i}\alpha \in R\sigma(A)$, $\alpha \in \mathbb{R}$, then one of the following assertions are true*

(a) $\mathrm{i}\alpha\mathbb{Z} \subseteq R\sigma(A)$;

(b) $\{\lambda \in \mathbb{C}\colon \operatorname{Re}(\lambda) < 0\} \subseteq R\sigma(A)$.

Proof. There exists $\varphi \in E'$, $\varphi \neq 0$ such that $T(t)'\varphi = \mathrm{e}^{\mathrm{i}\alpha t}\varphi$ $(t \geqslant 0)$. Then we have $|\varphi| = |T(t)'\varphi| \leqslant T(t)'|\varphi|$ $(t \geqslant 0)$. If we fix $r > \omega_0(\mathcal{T})$ and define $\psi := rR(r, A)'|\varphi|$, we have

$$T(t)'\psi \leqslant \mathrm{e}^{rt}\psi \quad \text{and} \quad T(t)'\psi \geqslant \psi \ (t \geqslant 0) \quad \text{and} \quad |\varphi| \leqslant \psi. \tag{3.4.1}$$

In fact, Part-I, Eq. (1.3.1) on page 19 implies $(\mathrm{e}^{rt} - T(r))R(r, A) \geqslant 0\,(\mathrm{e}^{rt} - T(t))R(r, A) \geqslant 0$ hence

$$T(t)'\psi = rR(r, A)'T(t)'|\varphi| \leqslant r \cdot \mathrm{e}^{rt}R(r, A)'|\varphi| = \mathrm{e}^{rt}\psi.$$

Moreover, the inequality $T(t)'|\varphi| \geqslant |\varphi|$ $(t \geqslant 0)$ implies

$$T(t)'\psi = rR(r, A)'T(t)'|\varphi| \geqslant rR(r, A)'|\varphi| = \psi$$

and

$$\psi = rR(r, A)'|\varphi| = r\int_0^\infty \mathrm{e}^{-rt}T(t)'|\varphi|\,\mathrm{d}t \geqslant r\int_0^\infty \mathrm{e}^{-rt}|\varphi|\,\mathrm{d}t = |\varphi|.$$

Now, considering the AL-space (E, ψ) (see Section 1.4) the first inequality of Eq. (3.4.1) implies that $(T(t))_{t\geqslant 0}$ induces a strongly continuous semigroup $(T_1(t))_{t\geqslant 0}$ on (E, ψ). That is we have

$$T_1(t) \circ q_\psi = q_\psi \circ T(t) \ (t \geqslant 0). \tag{3.4.2}$$

$$\begin{array}{ccc} E & \xrightarrow{\quad T(t) \quad} & E \\ {\scriptstyle q_\psi}\downarrow & & \downarrow{\scriptstyle q_\psi} \\ (E, \psi) & \xrightarrow[T_1(t)]{} & (E, \psi)) \end{array}$$

Denoting by A_1 the generator of $(T_1(t))$ we have $R\sigma(A_1) \subseteq R\sigma(A)$. Indeed, assuming $A_1^* x = \lambda x$ then $T_1(t)'x = \mathrm{e}^{\lambda t}x$, hence $T(t)'q'_\psi(x) = \mathrm{e}^{\lambda t}q'_\psi(x)$ (by Eq. (3.4.2) on the previous page) or equivalently, $A^*(q'_\psi(x)) = q'_\psi(x)A^*(q'_\psi(x)) = \lambda q'_\psi(x)$. Thus it remains to show that either $\mathrm{i}\alpha\mathbb{Z}$ or $\{\lambda \in \mathbb{C}\colon \operatorname{Re}(\lambda) < 0\}$ is contained in $R\sigma(A_1)$.

Obviously, $(T_1(t))$ is a semigroup of lattice homomorphisms as well. The second inequality of Eq. (3.4.1) on the preceding page implies

$$\|T_1(t)f\|_\psi = \langle |T_1(t)f|, \psi\rangle = \langle |f|, T_1(t)'\psi\rangle \geqslant \langle |f|, \psi\rangle = \|f\|_\psi. \tag{3.4.3}$$

Then for $\lambda \in \mathbb{C}$ with $\operatorname{Re}(\lambda) < 0$ we have

$$\|(\mathrm{e}^{\lambda t} - T_1(t))f\|_\psi \geqslant \|T_1(t)f\|_\psi - \|\mathrm{e}^{\lambda t}f\|_\psi \geqslant (1 - |\mathrm{e}^{\lambda t}|)\|f\|_\psi \ (f \in (E,\psi))$$

and we obtain for the corresponding generator

$$\begin{aligned}\|(\lambda - A_1)f\|_\psi &= \lim_{t\to 0}\left\|\frac{1}{t}(\mathrm{e}^{-\lambda t}T_1(t)f - f)\right\|_\psi \geqslant \lim_{t\to 0}\frac{1}{t}(\mathrm{e}^{-t\operatorname{Re}(\lambda)} - 1)\|f\|_\psi \\ &= -\operatorname{Re}(\lambda)\cdot\|f\|_\psi \ \text{ for } \ \operatorname{Re}(\lambda) < 0 \ \text{ and } \ f \in (E,\psi).\end{aligned} \tag{3.4.4}$$

It follows from Eq. (3.4.3) and Eq. (3.4.4) that $A\sigma(T_1(t)) \cap \{z \in \mathbb{C}\colon |z| < 1\} = \emptyset$ and that $A\sigma(A_1) \cap \{\lambda \in \mathbb{C}\colon \operatorname{Re}(\lambda) < 0\} = \emptyset$. Since the topological boundary of the spectrum is always contained in the approximate point spectrum (see Part-I, Proposition 3.2.2) and $R\sigma(T(t))\setminus\{0\} = \exp(tR\sigma(A))$ $R\sigma(T(t)) \setminus \{0\} = \exp(tR\sigma(A))$ (see Part-I, Theorem 3.6.3), precisely one of the following two cases occurs.

(A) $\{\lambda \in \mathbb{C}\colon \operatorname{Re}(\lambda) < 0\} \subseteq \varrho(A_1)$ and $\{z \in \mathbb{C}\colon |z| < 1\} \subseteq \varrho(T_1(t))$;

(B) $\{\lambda \in \mathbb{C}\colon \operatorname{Re}(\lambda) < 0\} \subseteq R\sigma(A_1)$ and $\{z \in \mathbb{C}\colon |z| < 1\} \subseteq R\sigma(T_1(t))$.

We mentioned above that $R\sigma(A_1) \subseteq R\sigma(A)$. Thus we only have to analyze case (A). In this case each operator $T_1(t)$ is an invertible lattice homomorphism hence a lattice isomorphism. It follows that $T_1(t)'$ is a lattice isomorphism as well. The third inequality in Eq. (3.4.1) on the previous page implies that φ can be considered as an element of $(E,\psi)'$ and $T(t)'\varphi = \mathrm{e}^{\mathrm{i}\alpha t}\varphi$ $(t \geqslant 0)$ implies $T_1(t)'\varphi = \mathrm{e}^{\mathrm{i}\alpha t}\varphi$. Furthermore, we have $T_1(t)'|\varphi| = |T_1(t)'\varphi| = |\mathrm{e}^{\mathrm{i}\alpha t}\varphi|$ or equivalently $A_1^*|\varphi| = 0$. Now we can apply Theorem 3.2.2 and obtain $\mathrm{i}\alpha\mathbb{Z} \subseteq P\sigma(A_1^*) = R\sigma(A_1)$. □

Theorem 3.4.2 *Let A be the generator of a semigroup $(T(t))_{t\geqslant 0}$ of lattice homomorphisms on a Banach lattice E. Then $\sigma(A)$, $A\sigma(A)$ and $P\sigma(A)$ are imaginary additively cyclic subsets of $\mathbb{C}$.*

Proof. We first consider the point spectrum. If $\lambda \in P\sigma(A)$, $\lambda = \alpha + \mathrm{i}\beta$ $(\alpha,\beta \in \mathbb{R})$, then there exists $f \in E$, $f \neq 0$ such that $Af = \lambda f$. It follows that $T(t)f = \mathrm{e}^{\lambda t}f$ $(t \geqslant 0)$ hence $T(t)|f| = |T(t)f| = \mathrm{e}^{\alpha t}|f|$ $(t \geqslant 0)$, or equivalently, $A|f| = \alpha|f|$. Now Theorem 3.2.2 is applicable and we obtain $A(f^{[n]}) = (\alpha + \mathrm{i}n\beta)f^{[n]}$ for all $n \in \mathbb{Z}$.

To prove the assertion for $A\sigma(A)$ we consider an $\mathcal{F}$-product semigroup in order to reduce the problem to the point spectrum. We use the notation of Part-I, Section 1.3.7. Obviously the space $m(E)$ is a Banach lattice and every operator $\hat{T}(t)$ is a lattice homomorphism. We have

$$|T(t)|f| - |f|| = ||T(t)f| - |f|| \leqslant |T(t)f - f| \quad (f \in E),$$

hence $(|f_n|) \in m^{\mathcal{T}}(E)$ whenever $(f_n) \in m^{\mathcal{T}}(E)$. This proves that $m^{\mathcal{T}}(E)$ is a sublattice, hence a Banach lattice as well. Obviously, $c_{\mathcal{F}}(E) \cap m^{\mathcal{T}}(E)$ is an order ideal. Thus $E^{\mathcal{T}}_{\mathcal{F}}$ is a Banach lattice and $(T_{\mathcal{F}}(t))$ is a semigroup of lattice homomorphisms. It follows that $P\sigma(A_{\mathcal{F}})$ is cyclic, hence $A\sigma(A)$ is cyclic by Part-I, Proposition 3.4.2. Cyclicity of the entire spectrum follows from the cyclicity of $A\sigma(A)$ and Lemma 3.4.1. □

One can use Theorem 3.4.2 in order to prove cyclicity for the eigenvalues in the boundary spectrum of positive semigroups. We list some typical cases in the following corollary.

Corollary 3.4.3 *Let $\mathcal{T} = (T(t))_{t\geqslant 0}$ be a positive semigroup on a Banach lattice E which is bounded. Each of the following conditions implies that $P\sigma(A) \cap i\mathbb{R}$ $P\sigma(A) \cap \mathrm{i}\mathbb{R}$ is imaginary additively cyclic.*

(i) E *E is weakly sequentially complete (e.g., $E = L^p(\mu)$, $1 \leqslant p < \infty$);*

(ii) *Every operator $T(t)$ is mean ergodic (i.e., the Cesàro means $\frac{1}{n}\sum_{k=0}^{n-1} T(t)^k$ converge strongly as $n \to \infty$);*

(iii) *There is a strictly positive linear form which is $\mathcal{T}$-invariant.*

Proof. We will show that each of the conditions (i), (ii), (iii) implies that $\ker(1 - T(s))$ is a Banach lattice (not necessarily a sublattice of E) for every $s \geqslant 0$. Then one argues as follows. Given $\mathrm{i}\alpha \in P\sigma(A)$, $\alpha \in \mathbb{R}$, then $T(t)g = \mathrm{e}^{\mathrm{i}\alpha t}g$ for suitable $g \neq 0$. For $\tau := 2\pi|\alpha|^{-1}$ we have $g \in F := \ker(1 - T(\tau))$. Then the restriction $(T(t)_{|F})_{t\geqslant 0}$ is a τ-periodic positive semigroup on F. Since $\big(T(t)_{|F}\big)^{-1} = T(n\tau - t)_{|F} \geqslant 0$, it follows that $(T(t)_{|F})$ is a semigroup of lattice isomorphisms. Since $g \in F$, we have $\mathrm{i}\alpha \in P\sigma(A_{|F})$ hence $\mathrm{i}\alpha\mathbb{Z} \in P\sigma(A_{|F}) \subseteq P\sigma(A)$ $\mathrm{i}\alpha\mathbb{Z} \subseteq P\sigma(A_{|F}) \subseteq P\sigma(A)$ by Theorem 3.4.2.

Now we show that $\ker(1 - T(s))$ is a vector lattice for the induced order and a Banach lattice for an equivalent norm.

In case (iii), $\ker(1 - T(s))$ is even a sublattice of E. Indeed, assume $T(t)'\varphi = \varphi$ and $\varphi \gg 0$, $(t \geqslant 0)$ $(t \geqslant 0)$ and $\varphi \gg 0$, then $T(s)f = f$ implies $T(s)|f| \geqslant |f|$. Thus from

$$\langle T(s)|f| - |f|, \varphi\rangle = \langle |f|, T(s)'\varphi - \varphi\rangle = 0$$

it follows that $T(s)|f| = |f|$.

Now we assume that E is weakly sequentially complete, which is equivalent to (cf. Section 1.5)

$$\textit{Every increasing norm-bounded net of } E_+ \textit{ converges.} \tag{3.4.5}$$

We fix $s > 0$ and define $T := T(s)$ and $F := \ker(1 - T)$. Obviously $f \in F$ implies $\bar{f} \in F$ hence $F = F \cap E_{\mathbb{R}} + iF \cap E_{\mathbb{R}} F = F \cap E_{\mathbb{R}} + \mathrm{i}F \cap E_{\mathbb{R}}$. Thus we have to show that $F_{\mathbb{R}} = F \cap \mathbb{E}_{\mathbb{R}} F_{\mathbb{R}} = F \cap E_{\mathbb{R}}$ is a sublattice. Given $f \in F_{\mathbb{R}}$, then $Tf = f$ hence $|f| \leqslant T|f|$. Iterating this inequality we obtain $|f| \leqslant T|f| \leqslant T^2|f| \leqslant T^3|f| \leqslant \ldots$. By Eq. (3.4.5) $|f|_o := \lim_{n\to\infty} T^n|f|$ exists and we have $T|f|_o = \lim_{n\to\infty} T^{n+1}|f| = |f|_o$, i.e., $|f|_o \in F_{\mathbb{R}}$. And, for $g \in F_{\mathbb{R}}$ satisfying $\pm f \leqslant g$, we have $|f|_o \leqslant g$ thus $|f|_o = \sup_F\{f, -f\}$. Moreover, $\|f\|_o := \||f|_o\|$ $(f \in F)$ is an equivalent norm on F such that $(F, \|\cdot\|_o)$ is a Banach lattice.

(ii) If $T(s)$ is mean-ergodic, then we have $\ker(1 - T(s)) = PE$ where P is the mean-ergodic projection, i.e., $Pf = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} T(s)^k f$. Obviously P is positive, hence II.11.5 of Schaefer [32] implies that PE is a Banach lattice (for the induced order and an equivalent norm). □

The assumptions made in Corollary 3.4.3 can be weakened slightly (cf. Greiner [13]). However, one still cannot prove cyclicity of $P\sigma_b(A)$ for arbitrary positive semigroups.

Example 3.4.4 At first we recall Part-II, Example 3.2.13. There we constructed a bounded semigroup on $C(\Gamma) \times C_0(\mathbb{R})$ such that $P\sigma_b(A) = \{\mathrm{i}k : k \in \mathbb{Z}, k \neq 0\}$.

Let us perform the same construction on the Hilbert space $H := L^2(\Gamma) \times L^2(\mathbb{R})$. For a fixed positive, non-zero function $k \in C_c(\mathbb{R})$, we define $T(t)$ on H as follows.

$$\begin{aligned} T(t)([f,g]) &:= [f_t, g_t] \quad \text{with} \\ f_t(z) &:= f(z \cdot \mathrm{e}^{\mathrm{i}t}) \quad (z \in \Gamma) \quad \text{and} \\ g_t(x) &:= g(x+t) + \frac{1}{2\pi} \int_0^{2\pi} f(z \cdot \mathrm{e}^{\mathrm{i}s})\, \mathrm{d}s \cdot \int_x^{x+t} k(u)\, \mathrm{d}u. \end{aligned}$$

$$\begin{aligned} T(t)([f,g]) &:= [f_t, g_t] \quad \text{with} \\ f_t(z) &:= f(z \cdot \mathrm{e}^{\mathrm{i}t}) \quad (z \in \Gamma) \quad \text{and} \\ g_t(x) &:= g(x+t) + \frac{1}{2\pi} \int_0^{2\pi} f(\mathrm{e}^{\mathrm{i}s})\, \mathrm{d}s \cdot \int_x^{x+t} k(u)\, \mathrm{d}u. \end{aligned} \tag{3.4.6}$$

Then $(T(t))_{t\geqslant 0}$ is a positive semigroup on H and for the spectrum of the generator we obtain $\sigma(A) = i\mathbb{R}$, $P\sigma(A) = i\mathbb{R}\setminus\{0\}\sigma(A) = \mathrm{i}\mathbb{R}$, $P\sigma(A) = \mathrm{i}\mathbb{R} \setminus \{0\}$. In view of Corollary 3.4.3 (i) the semigroup cannot be bounded. (The explicit representation Eq. (3.4.6) only allows the estimate $\|T(t)\| \leqslant \sqrt{2} + t \cdot \|k\|_2$ $(t \geqslant 0)$.)

In the next proposition we show that for semigroups of lattice homomorphisms on L^1-spaces, there is a spectral mapping theorem for the real part of the spectrum.

Proposition 3.4.5 *Let $(T(t))_{t\geqslant 0}$ be a strongly continuous semigroup of lattice homomorphisms on an L^1-space and denote by A its generator. Then we have*

$$\exp(t\sigma(A)\cap\mathbb{R}) = \sigma(T(t))\cap(0,\infty) \quad \textit{for every} \quad t\geqslant 0. \tag{3.4.7}$$

Proof. In view of Part-I, Proposition 3.6.2 it is enough to prove that the left hand side contains the set on the right.

Fix $t > 0$ and assume $r \in \sigma(T(t))$, $r > 0$ and let $\alpha := \frac{1}{t}\log r$. At first we assume $r \in R(\sigma(T(t)))r \in R\sigma(T(t))$. Then by Part-I, Theorem 3.6.3 there exists $\beta\in\mathbb{R}$ such that $\alpha+\mathrm{i}\beta\in R\sigma(A)$. By Lemma 3.4.1 either $\alpha+\mathrm{i}\beta\mathbb{Z}\subseteq R\sigma(A)$ or $\{\lambda\in\mathbb{C}\colon\ \mathrm{Re}(\lambda)<\alpha\}\subseteq R\sigma(A)$. In both cases we have $\alpha\in\sigma(A)$.

Now we assume $r\in A\sigma(T(t))$. Then there exists a normalized sequence $(f_n)\subseteq E$ such that $\lim_{n\to\infty}\|T(t)f_n - rf_n\| = 0$. Since we have

$$|(T(t)|f| - r|f|)| = |(|T(t)f| - r|f|)| \leqslant |T(t)f - rf|,\ \ (f\in E)$$

we may assume that (f_n) is a sequence in E_+.

Defining $g_n := \int_0^t \mathrm{e}^{-\alpha s}T(s)f_n\,\mathrm{d}s$ we have $g_n\in D(A)$ and

$$(A-\alpha)g_n = (A-\alpha)\int_0^t \mathrm{e}^{-\alpha s}T(s)f_n\,\mathrm{d}s = \mathrm{e}^{-\alpha t}T(t)f_n - f_n = \frac{1}{r}(T(t)f_n - rf_n).$$

Therefore $\lim_{n\to\infty}\|(A-\alpha)g_n\| = 0$. It remains to prove $\liminf_{n\to\infty}\|g_n\| > 0$. The latter assertion is a consequence of the following facts.

(i) Since f_n is positive and the norm is additive on E_+, we have

$$\|g_n\| = \int_0^t \mathrm{e}^{-\alpha s}\|T(s)f_n\|\,\mathrm{d}s.$$

(ii) The inequality

$$\|T(t)f\| \leqslant \|T(t-s)\|\|T(s)f\|$$

implies

$$\|T(s)f\| \geqslant M^{-1}\|T(t)f\|$$

for $0\leqslant s\leqslant t$, $f\in E$ and $M := \sup_{0\leqslant s\leqslant t}\|T(s)\|$.

(iii) Since

$$\lim_{n\to\infty}\|T(t)f_n - rf_n\| = 0 \text{ and } \|f_n\| = 1$$

we have $\lim_{n\to\infty}\|T(t)f_n\| = r > 0$. □

For semigroups satisfying the assumption of Proposition 3.4.5, the spectrum $\sigma(A)$ is additively cyclic (by Theorem 3.4.2) and $\sigma(T(t))$ is multiplicatively cyclic (by Schaefer [32, V.Theorem 4.4]). Then the relation Eq. (3.4.7) implies that decompositions

of the spectrum by vertical lines allow a spectral decomposition of the semigroup (cf. Part-I, Definition 3.3.1) (One simply performs a spectral decomposition of a single operator $T(t)$).

In the following we will show that for positive groups (on arbitrary Banach lattices) spectral decompositions of this type always exist. Moreover, it will turn out that the decomposition is compatible with the lattice structure. The proof of this result uses Kato's equality (see Section 2.5). As a consequence of Corollary 2.5.8 we have the following.

Let E be a Banach lattice with order continuous norm and $(T(t))_{t\in\mathbb{R}}$ be a group of positive operators on E with generator A. Then the domain $D(A)$ is a sublattice of E and

$$A|f| = \mathrm{Re}[(\operatorname{sign} f)Af] \text{ for every } f \in D(A),$$
$$\mathsf{A}|\mathsf{f}| = \mathrm{Re}[(\operatorname{sign}\mathsf{f})\mathsf{A}\mathsf{f}] \text{ for every } \mathsf{f}\in\mathsf{D}(\mathsf{A}). \tag{3.4.8}$$

For real μ one has $\mu|f| = \mathrm{Re}[(\operatorname{sign} f)\mu f]$, hence

$$(\mu - A)|f| = \mathrm{Re}[(\operatorname{sign} f)(\mu - A)f] \quad \text{for } \mu\in\mathbb{R}, f\in D(A).$$

The relations $f^+ = \frac{1}{2}(|f| + f)$, $f^- = \frac{1}{2}(|f| - f)$ yield

$$(\mu - A)f^+ = \frac{1}{2}[(\operatorname{sign} f)(\mu - A)f + (\mu - A)f] \quad \text{and}$$
$$(\mu - A)f^- = \frac{1}{2}[(\operatorname{sign} f)(\mu - A)f - (\mu - A)f],$$

in case f is contained in the underlying real Banach lattice $E_{\mathbb{R}}$. For $\mu\in\varrho(A)\cap\mathbb{R}$, we can apply $R(\mu, A)$ on both sides and the substitution $f = R(\mu, A)g$ finally leads to

$$\begin{aligned}(R(\mu,A)g)^+ &= \frac{1}{2}R(\mu,A)[(\operatorname{sign} R(\mu,A)g)g + g]\\ (R(\mu,A)g)^- &= \frac{1}{2}R(\mu,A)[(\operatorname{sign} R(\mu,A)g)g - g]\end{aligned} \tag{3.4.9}$$

for all $g\in E_{\mathbb{R}}$. If we set

$$g_1 := \frac{1}{2}\cdot(g + (\operatorname{sign} R(\mu,A)g)g) \text{ and } g_2 := \frac{1}{2}\cdot(g - (\operatorname{sign} R(\mu,A)g)g),$$

then obviously $g = g_1 + g_2$. Moreover, g is positive if and only if both, g_1 and g_2 are positive. We summarize these considerations in the following lemma.

Lemma 3.4.6 *Let A be the generator of a positive group on a Banach lattice E which has order continuous norm. Given $\mu\in\varrho(A)\cap\mathbb{R}$, then every $g\in E_{\mathbb{R}}$ is representable as sum of two elements g_1 and g_2 such that*

(i) $g\geqslant 0$ *if and only if both g_1 and g_2 are positive,*

(ii) $R(\mu, A)g_1 = (R(\mu, A)g)^+$,

(iii) $R(\mu, A)g_2 = -(R(\mu, A)g)^-$.

We need another lemma. It can be formulated for arbitrary positive semigroups on Banach lattices.

Lemma 3.4.7 *Let $(T(t))_{t\geqslant 0}$ be a positive semigroup on a Banach lattice E with generator A. Given $\mu \in \varrho(A) \cap \mathbb{R}$ and $h \in E_+$, then the following assertions are equivalent.*

(a) $R(\mu, A)h \geqslant 0$,

(b) $\left\{\int_0^t \mathrm{e}^{-\mu s}T(s)h\,\mathrm{d}s : t \in \mathbb{R}_+\right\}$ *is bounded in E.*

Proof. $(a) \Rightarrow (b)$ (a) $\Rightarrow$ (b) We have $\int_0^t \mathrm{e}^{-\mu s}T(s)h\,\mathrm{d}s = (\mathrm{Id} - \mathrm{e}^{-\mu t}T(t))R(\mu, A)h$ (see Part-I, Eq. (1.3.2) on page 19). Since $R(\mu, A)h \geqslant 0$ and $T(t)$ is a positive operator we obtain

$$\int_0^t \mathrm{e}^{-\mu s}T(s)h\,\mathrm{d}s = R(\mu, A)h - \mathrm{e}^{-\mu t}T(t)R(\mu, A)h \leqslant R(\mu, A)h$$

which implies assertion (b).

$(b) \Rightarrow (a)$ (b) $\Rightarrow$ (a) The assumption implies that

$$\int_0^\infty \mathrm{e}^{-\nu s}T(s)h\,\mathrm{d}s := \lim_{t\to\infty} \int_0^t \mathrm{e}^{-\nu s}T(s)h\,\mathrm{d}s \quad \text{exists for} \quad \nu > \mu.$$

Using that A is a closed operator it follows that

$$(\nu - A)\left(\int_0^\infty \mathrm{e}^{-\nu s}T(s)h\,\mathrm{d}s\right) = h.$$

For ν sufficiently close to μ such that $\nu \in \varrho(A) \cap \mathbb{R}$ we have

$$R(\nu, A)h = \int_0^\infty \mathrm{e}^{-\nu s}T(s)h\,\mathrm{d}s \geqslant 0.$$

By continuity we conclude $R(\mu, A)h \geqslant 0$. □

By now we are prepared to prove the spectral decomposition for positive groups. Before we formulate the theorem we recall the following consequence of Theorem 3.4.2. For any $\mu \in \varrho(A) \cap \mathbb{R}$ the line $\mu + i\mathbb{R}$ $\mu + \mathrm{i}\mathbb{R}$ is a subset of the resolvent set and divides $\sigma(A)$ into disjoint sets. Both sets will be unbounded in general.

Theorem 3.4.8 *Let $(T(t))_{t\in\mathbb{R}}$ be a strongly continuous group of positive operators on a Banach lattice E with order continuous norm. If A is the generator and $\mu \in \varrho(A) \cap \mathbb{R}$,*

then $I_\mu := \{f \in E\colon R(\mu, A)|f| \geqslant 0\}$ *and* $J_\mu := \{f \in E\colon R(\mu, A)|f| \leqslant 0\}$ *are* $(T(t))_{t\in\mathbb{R}}$*-invariant projection bands, the direct sum of them is* E*, and the spectra of the restrictions satisfy*

$$\sigma(A_{|I_\mu}) = \sigma(A) \cap \{\lambda \in \mathbb{C}\colon \operatorname{Re}(\lambda) < \mu\},$$
$$\sigma(A_{|J_\mu}) = \sigma(A) \cap \{\lambda \in \mathbb{C}\colon \operatorname{Re}(\lambda) > \mu\}.$$

Proof. At first we consider I_μ. Obviously it is a closed subset. From Lemma 3.4.7 we deduce that it is a lattice ideal. Moreover, I_μ is $R(\mu, A)$-invariant and $(T(t))_{t\in\mathbb{R}}$-invariant as well (use Lemma 3.4.7 again).

Since $-A$ is the generator of the positive group $(T(-t))_{t\in\mathbb{R}}$ and

$$J_\mu = \{f \in E\colon R(-\mu, -A)|f| \geqslant 0\},$$

J_μ has the same properties.

If $f \in I_\mu \cap J_\mu$, then $R(\mu, A)|f| = 0$, hence $f = 0$ which shows that $I_\mu \cap J_\mu = \{0\}$. On the other hand, decomposing $0 \leqslant h = h_1 + h_2 \in E_+$ according to Lemma 3.4.6, then assertion (ii) of this lemma implies that $h_1 \in I_\mu$, while assertion (iii) ensures that $h_2 \in J_\mu$. Since the positive cone E_+ is generating, we have $E = I_\mu \oplus J_\mu$ and the first part of the theorem is proved.

Since I_μ is $R(\mu, A)$-invariant, we have $\mu \in \varrho(A_{|I_\mu})$ and $R(\mu, A_{|I_\mu}) = R(\mu, A)_{|I_\mu} \geqslant 0$. Theorem 3.1.1 (ii) then implies $\sigma(A_{|I_\mu}) \subseteq \{\lambda \in \mathbb{C}\colon \operatorname{Re}(\lambda) < \mu\}$. The same argument applied to $-A$ and $-\mu$ yields $\sigma(A_{|J_\mu}) \subseteq \{\lambda \in \mathbb{C}\colon \operatorname{Re}(\lambda) > \mu\}$. Now the assertion follows from Part-I, Proposition 3.4.2. □

The spectral projections corresponding to the spectral decomposition described above have the expected representation as an integral 'around' the spectral sets (see Corollary 3 in Greiner [14]).

Corollary 3.4.9 *Assume that the assumptions of the theorem are satisfied,* $\mu \in \varrho(A) \cap \mathbb{R}$*,* $\beta > s(A)$*,* $\alpha < -s(-A)$*. If we denote the projections corresponding to the decomposition* $E = I_\mu \oplus J_\mu$ *by* P_μ *and* Q_μ *(i.e.,* $P_\mu E = \ker Q_\mu = I_\mu$*,* $Q_\mu E = \ker P_\mu = J_\mu$*), then for* $f \in D(A^2)$ *we have*

$$\begin{aligned} P_\mu f &= \frac{1}{2\pi} \cdot \int_{-\infty}^{\infty} R(\mu + \mathrm{i}\tau, A) f \,\mathrm{d}\tau - \frac{1}{2\pi} \cdot \int_{-\infty}^{\infty} R(\alpha + \mathrm{i}\tau, A) f \,\mathrm{d}\tau, \\ Q_\mu f &= \frac{1}{2\pi} \cdot \int_{-\infty}^{\infty} R(\beta + \mathrm{i}\tau, A) f \,\mathrm{d}\tau - \frac{1}{2\pi} \cdot \int_{-\infty}^{\infty} R(\mu + \mathrm{i}\tau, A) f \,\mathrm{d}\tau. \end{aligned} \tag{3.4.10}$$

(The integrals appearing in Eq. (3.4.10) are improper Riemann integrals.)

We mention another consequence of Theorem 3.4.8. Just like Proposition 3.4.5 it is a spectral mapping theorem for the real part of the spectrum.

Corollary 3.4.10 *If $(T(t))_{t\in\mathbb{R}}$ is a positive group on a space L^2 or $C_0(X)$ with generator A, then*

$$\sigma(T(t)) \cap \mathbb{R}_+ = \exp(t\sigma(A) \cap \mathbb{R}) \text{ for every } t \geqslant 0. \tag{3.4.11}$$

Proof. We borrow from the next chapter that for positive semigroups on spaces L^1, L^2 or $C_0(X)$ spectral bound and growth bound coincide (see Theorem 4.1.1).

We only have to show that $\exp(t\varrho(A) \cap \mathbb{R}) \subseteq \varrho(T(t)) \cap \mathbb{R}_+$.

If we consider a positive semigroup on an L^2-space, Theorem 3.4.8 can be applied directly. Given $\mu \in \varrho(A) \cap \mathbb{R}$, then $E = I_\mu \oplus J_\mu$ according to Theorem 3.4.8. The result mentioned above implies $r(T(t)_{|I_\mu}) < \mathrm{e}^{\mu t}$ and $r(T(-t)_{|J_\mu}) < \mathrm{e}^{\mu t} r(T(-t)_{|J_\mu}) < \mathrm{e}^{-\mu t}$. Hence

$$\sigma(T(t)_{|I_\mu}) \subseteq \{\lambda \in \mathbb{C} \colon |\lambda| < \mathrm{e}^{\mu t}\}$$

and

$$\sigma(T(t)_{|J_\mu}) = (\sigma(T(-t)_{|J_\mu}))^{-1} \subseteq \{\lambda \in \mathbb{C} \colon |\lambda| > \mathrm{e}^{\mu t}\}.$$

Thus $\sigma(T(t)) = \sigma(T(t)_{|I_\mu}) \cup \sigma(T(t)_{|J_\mu})$ does not contain $\mathrm{e}^{\mu t}$.

In case $(T(t))$ is a positive group on $C_0(X)$, then the adjoint group $(T(t)')$ is a group of lattice homomorphisms on E'. It follows that E^* is a sublattice of $C_0(X)'$ which is isomorphic to $M_b(X)$, hence an L^1-space. The argument given for the L^2-space yields $\sigma(T(t)^*) \cap \mathbb{R}_+ = \exp(t\sigma(A^*) \cap \mathbb{R})$ for every $t \geqslant 0$. Thus the assertion follows from Part-I, Section 3.4.1. □

We conclude by describing a general situation of lattice semigroups. In Part-II, Section 3.4 we constructed semigroups of lattice homomorphisms on $C_0(X)$ starting with a continuous (semi-) flow on the locally compact space X and a multiplication operator. One can perform similar constructions on spaces $L^p(\mu)$ for $1 \leqslant p < \infty$ under certain conditions on the flow. We consider an example which shows where the problems are.

Define the semiflow φ on $\mathbb{R}_+$ as follows. $\varphi(t,x) := x - t$ for $x \geqslant t$ and $\varphi(t,x) := 0$ for $x < t$. For $f \in L^p(\mu)$ one has difficulties to define $f \circ \varphi_t$ properly since the preimage of the zero-set $\{0\}$ does not have measure zero. This problem does not arise in case every transformation φ_t is measure preserving, i.e., $\mu(\varphi_t^{-1}(C)) = \mu(C)$ for every Borel set C. A more general criterion is stated in the following proposition.

Proposition 3.4.11 *Let X be a locally compact space and let μ be a regular, positive Borel measure on X. Assume that the continuous semiflow $\varphi \colon \mathbb{R}_+ \times X \to X$ satisfies the condition*

$$\varphi_t^{-1}(K) \text{ is compact for every compact set } K \subseteq X, t \geqslant 0. \tag{3.4.12}$$

(i) *For every p, $1 \leqslant p < \infty$ the following assertions are equivalent.* (a)

(a) *The operators $T(t)$ defined by $T(t)f := f \circ \varphi_t$ for $f \in L^p(\mu)$, $t \geqslant 0$, are well-defined as bounded linear operator on $L^p(\mu)$ and $(T(t))_{t\geqslant 0}$ is a strongly continuous semigroup.*

(b) *There exist constants $t_0 > 0$, $M > 0$ such that $\mu(\varphi_t^{-1}(C)) \leqslant M \cdot \mu(C)$ for every open (compact) set $C \subseteq X$ and every $t \leqslant t_0$.*

(ii) *In case the conditions (a) and (b) are fulfilled, then $(T(t))_{t\geqslant 0}$ is a semigroup of lattice homomorphisms on $L^p(\mu)$ and $C_c(X) \cap D(A)$ is a core of the generator.*

Proof. (i) Since μ is assumed to be regular, the inequality stated in (b) holds true for all Borel sets provided it is true for all open sets (or all compact sets, respectively).

$(a) \Rightarrow (b)$:

(a) $\Rightarrow$ (b): Assume that $(T(t))$ is a strongly continuous semigroup on $L^p(\mu)$ with $1 \leqslant p < \infty$. For $t_0 > 0$ we define $M := (\sup\{\|T(t)\| : 0 \leqslant t \leqslant t_0\})^{1/p}$ $M := (\sup\{\|T(t)\| : 0 \leqslant t \leqslant t_0\})^p$. Given a Borel set $C \subseteq X$ we write $C(t) := \varphi_t^{-1}(C)$. Then we have $T(t)\mathbf{1}_C = \mathbf{1}_{C(t)}$, hence $\mu(\varphi_t^{-1}(C)) = \|\mathbf{1}_{C(t)}\|_p^p = \|T(t)\mathbf{1}_C\|_p^p \leqslant M \cdot \|\mathbf{1}_C\|_p^p = M \cdot \mu(C)$.

$(b) \Rightarrow (a)$:

(b) $\Rightarrow$ (a): Since the inequality in (b) holds for all Borel sets, $\varphi_t^{-1}(C)$ is a μ-null set whenever C is a μ-null set. Thus given Borel functions f, g such that $f = g$ μ-a.e., then $f \circ \varphi_t = g \circ \varphi_t$ μ-a.e.. Moreover, for $0 \leqslant f \in L^p(\mu)$, there exists an increasing sequence (h_n) of simple functions converging pointwise to f. Then $(h_n \circ \varphi_t)$ is a monotone sequence converging pointwise to $f \circ \varphi_t$. Using the fact that $\mathbf{1}_C \circ \varphi_t = \mathbf{1}_{C(t)}$, $C(t)$ as above, and the assumption $\mu(C(t)) \leqslant M \cdot \mu(C)$, it is easy to see that $\|h_n \circ \varphi_t\|_p^p \leqslant M \cdot \|h_n\|_p^p \leqslant M \cdot \|f\|_p^p$. Thus by the Monotone Convergence Theorem we have $f \circ \varphi_t \in L^p(\mu)$ and $\|f \circ \varphi_t\|_p \leqslant M^{1/p}\|f\|_p$. It follows that $T(t)$ is a bounded linear operator on $L^p(\mu)$ and $\|T(t)\| \leqslant M^{1/p}$ for $0 \leqslant t \leqslant t_0$. Since φ is a semiflow, we have $T(0) = \mathrm{Id}$ and $T(t+s) = T(s)T(t)$ $(0 \leqslant s,\, t < \infty)$.

It remains to prove strong continuity. Since φ is continuous and Eq. (3.4.12) on the previous page holds, we know that $T(t)(C_c(X)) \subseteq C_c(X)$ and that $T(t)f$ tends to f uniformly as $t \to 0$ provided that $f \in C_c(X)$. It follows that $\lim_{t\to 0} \|T(t)f - f\|_p = 0$ for $f \in C_c(X)$. Since $C_c(X)$ is dense in $L^p(\mu)$ and $\|T(t)\| \leqslant M^{1/p}$ for $0 \leqslant t \leqslant t_0$, the semigroup is strongly continuous.

(ii) Obviously every operator $T(t)$ defined in assertion (a) of (i) is a lattice homomorphism. Above we pointed out that $C_c(X)$ is invariant under $(T(t))$, then the intersection $D(A) \cap C_c(X)$ is invariant as well. It is dense because the elements of the form $\int_0^r T(s)f\,\mathrm{d}s$, $f \in C_c(X)$, $r > 0$ belong to $C_c(X)$ and to $D(A)$. Hence $D(A) \cap C_c(X)$ is a core (by Part-I, Proposition 1.1.9). □

Proposition 3.4.11 can be used to prove that flows corresponding to certain ordinary differential equations on $\mathbb{R}^n$ generate strongly continuous semigroups on $L^p(\mathbb{R}^n)$ (where $\mathbb{R}^n$ is equipped with the Lebesgue measure). One has to impose conditions on

the corresponding vector field. Note that for continuous flows condition Eq. (3.4.12) on page 369 is automatically satisfied because for a compact $K \subseteq X$ the set $\varphi_t^{-1}(K) = \varphi_{-t}(K)$ is compact as the continuous image of a compact set.

Example 3.4.12 Let $F\colon \mathbb{R}^n \to \mathbb{R}^n$ be a C^1-vector field and assume that the derivative DF is uniformly bounded on $\mathbb{R}^n$. Then the ordinary differential equation $y' = F(y)$ possesses a global flow $\varphi\colon \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$ which is C^1. Moreover, we have

$$\begin{aligned} &\|D\varphi_t(x)\| \leqslant \mathrm{e}^{M|t|} \text{ for all } x \in \mathbb{R}^n,\, t \in \mathbb{R} \text{ where} \\ &\qquad M := \sup\{\|DF(x)\| \colon x \in \mathbb{R}^n\}. \end{aligned} \tag{3.4.13}$$

All these properties were proven in Part-II, Example 2.3.15. We will show that φ satisfies condition (b) of Proposition 3.4.11, hence it induces a strongly continuous (semi-)group of lattice homomorphisms on $L(\mathbb{R}^n)$ $L^p(\mathbb{R}^n)$ $(1 \leqslant p < \infty)$ via $T(t)f = f \circ \varphi_t$. This is done using the change of variables formula as follows.

Let U be an open subset of $\mathbb{R}^n$, then $\varphi_t^{-1}(U) = \varphi_{-t}(U) =: U(-t)$. If λ denotes the Lebesgue measure, then

$$\begin{aligned} \lambda(\varphi_t^{-1}(U)) = \int_{U(-t)} \mathbf{1}\,\mathrm{d}x = \int_U \mathbf{1} \circ \varphi_{-t}(x) \cdot |\det D\varphi_{-t}(x)|\,\mathrm{d}x = \\ \int_U |\det D\varphi_{-t}(x)|\,\mathrm{d}x \leqslant \int_U \mathrm{e}^{nM|t|}\,\mathrm{d}x = \mathrm{e}^{nM|t|} \cdot \lambda(U). \end{aligned} \tag{3.4.14}$$

Here we used Eq. (3.4.13) and the fact that the determinant of an n×n-matrix $n \times n$-matrix B satisfies $|\det B| \leqslant \|B\|^n$.

In general, existence of a global flow does not ensure that one can associate a semigroup of bounded linear operators on $L^p(\mathbb{R}^n)$, even if the vector field is C^∞. For example the differential equation $y' = \sin(y^2)$ does not induce a semigroup on $L^p(\mathbb{R})$. There is another important class of differential equations, *Hamiltonian differential equations*, which do induce semigroups of lattice homomorphisms on L^p-spaces. In fact, Liouville's Theorem states that the flow corresponding to a Hamiltonian vector field preserves the volume (see Abraham & Marsden [1, Section 3.3]). Thus assertion (b) of Proposition 3.4.11 (i) is trivially satisfied. Further examples of flows which are measure preserving and therefore induce semigroups of lattice homomorphisms on L^p-spaces are billiard flows on compact convex subsets of $\mathbb{R}^n$ and geodesic flows on Riemannian manifolds (see Cornfeld et al. [6]).

Notes

Spectral theory for a single positive operator as developed in Chapter V of Schaefer [32] is an essential tool for this chapter. Various results on the spectral theory of

positive one-parameter semigroups can be found in Chapter 7 of Davies [7] and in the second part of Batty & Robinson [4].

Section 3.1: That the spectral bound is always an element of the spectrum was stated by Karlin [21], but a valid proof was given by Derndinger [9]. This assertion as well as assertion (ii) of Theorem 3.1.1 allow generalizations in various directions. They are valid for ordered Banach spaces (see Greiner et al. [17] and Klein [24]) and one only needs that A has positive resolvent (see Kato [22] or Nussbaum [27]). Theorem 3.1.2 as well as its corollaries are also valid in ordered Banach spaces. For the analogue in the theory of the Laplace transform we refer to Section 10.5 in Widder [38] and Voigt [34].

Section 3.2: Theorem 3.2.2 is the basis for the subsequent cyclicity results. Pseudoresolvents are discussed e.g., in Hille & Phillips [19] or Yosida [39]. For nonpositive semigroups the two assertions stated in Definition 3.2.8 are no longer equivalent. A special case of Theorem 3.2.10 was proven by Derndinger [9] while the general result is due to Greiner [12]. Instead of pseudo-resolvents on the whole $\mathcal{F}$-product Derndinger works with the semigroup on the semigroup $\mathcal{F}$-product. Therefore he can only consider eigenvalues. Elliptic differential operators as generators of positive semigroups are discussed by many authors, e.g., Amann [2], Fattorini [10], Friedman [11] or Pazy [28].

Section 3.3: There exist various notions which are (more or less closely) related to irreducibility, e.g., "positivity improving" in Reed & Simon [29], "u_0-positivity" in Krasnosel'skii [25] or "quasi-strictly positive" in Karlin [21]. Sawashima [30] uses "non-support" instead of irreducible. She also discusses several modifications ("semi-non-support", "strictly non-support", "strongly positive") and the interrelation between these notions. The notion of irreducibility can be extended to the (non-lattice) ordered setting (see Batty & Robinson [4]). Assertion (ii) of Theorem 3.3.2 is due to Majewski & Robinson [26] while special cases can be found in Section XIII.12 of Reed & Simon [29] and in Kishimoto & Robinson [23]. Proposition 3.3.3 is due to Voigt [35]. Retarded equations as discussed in Example 3.3.4 (iii) will be discussed in more detail in Part-III, Section 4.3. Example 3.3.4 (iv) is a one-dimensional version of the linear transport equation. The higher dimensional equation is more delicate but can be treated similarly (see e.g., Greiner [16], Kaper et al. [20], or Voigt [37]). A special case of Proposition 3.3.5 can be found in Davies [7]. Theorem 3.3.7 and Example 3.3.6 are taken from Schaefer [33]. The most interesting criterion of Theorem 3.3.7 seems to be condition (iii), since it gives a sufficient condition for the existence of eigenvalues for a sufficiently large class of semigroups. For semigroups induced by measure-preserving flows Theorem 3.3.8 and Corollary 3.3.9 are proven in Cornfeld et al. [6]. Corollary 3.3.9 is a special case of the Halmos-von Neumann Theorem which classifies irreducible semigroups having discrete spectrum (see Cornfeld et al. [6], Greiner [13] and Schaefer [32] for the general result).

Lemma 3.3.10 is taken from Groh [18] and Theorems 3.3.12 and 3.3.14 can be found (with slightly different proofs) in Greiner [12].

Section 3.4: It was Derndinger [9] who proved Theorem 3.4.2. In Corollary 3.4.3 one can replace boundedness of the semigroup by the assumption that the resolvent grows slowly (see Greiner [13]). Example 3.4.4 is due to Davies and Proposition 3.4.5 to H. Kellermann (both unpublished). The spectral decomposition for positive groups as described in Theorem 3.4.8 is valid in arbitrary Banach lattices (see Arendt [3] and Greiner [14]). This also holds for Corollaries 3.4.9 and 3.4.10. Proposition 3.4.11 and Example 3.4.12 indicate the relationship of positive groups to dynamical systems. For example, the "Annular Hull Theorem" (see Chicone & Swanson [5]) is closely related to the results of this section.

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